EMBEDDED CLOSED MINIMAL SURFACES WITH EVERYWHERE NEGATIVE CURVATURE IN THE ROUND FOUR-SPHERE

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Allowed standard prerequisites (editorial).

- Classical research prerequisites cited by the authors. Complete original bibliography retained. Author-cited standard results are prerequisites; all source-local core constructions, estimates and explicit non-adjacent continuations retained.

1. Introduction

Understanding how the curvature of an ambient manifold constrains the intrinsic geometry of its minimal surfaces is a classical and fascinating problem in Riemannian geometry. By the Gauss equation, the Gaussian curvature of a minimal surface equals the ambient sectional curvature along its tangent plane minus one half of the squared norm of its second fundamental form. Nonpositive ambient curvature therefore forces nonpositive Gaussian curvature. On the other hand, positive ambient curvature presents a more subtle question, since the sign of the Gaussian curvature of a minimal surface depends on the balance between the ambient curvature and the second fundamental form. Round spheres, as the model spaces of constant positive sectional curvature, provide a natural setting in which to study this interaction.

Let Σ be a smooth Riemannian surface and let $\iota : \Sigma \to \mathbb{S}^n$ be a smooth minimal immersion, where $g_{\mathbb{S}^n}$ denotes the round metric of sectional curvature one. Write $g_\iota := \iota^* g_{\mathbb{S}^n}$ for its induced metric, K_ι for the Gaussian curvature of g_ι, and A_ι for its normal-bundle-valued second fundamental form in $\mathbb{S}^n$. The Gauss equation gives

$$K_\iota = 1 - \frac{|A_\iota|^2_{g_\iota}}{2},$$

where the norm uses g_ι on the tangent bundle and the restriction of $g_{\mathbb{S}^n}$ on the normal bundle. Thus minimality gives the upper bound $K_\iota \leq 1$, while everywhere negative Gaussian curvature requires $|A_\iota|^2_{g_\iota} > 2$ at every point. If Σ is closed, connected and orientable, of genus γ, the Gauss–Bonnet theorem gives

$$\int_\Sigma K_\iota \, d\,\mathrm{vol}_{g_\iota} = 4\pi(1-\gamma).$$

For $\gamma \geq 2$, this forces the curvature to be negative somewhere, but gives no pointwise upper bound by a negative number. The existence of minimal surfaces of large genus therefore does not by itself settle the existence of examples with everywhere negative curvature.

The classical theory provides several different types of intrinsic geometry. Borůvka constructed minimal two-spheres of constant positive Gaussian curvature in even-dimensional round spheres [Bor33], and Calabi developed a systematic theory of minimal two-spheres in spheres of arbitrary dimension [Cal67]. The Clifford torus gives a closed flat minimal surface in $\mathbb{S}^3$. Allowing the curvature to vary, Lawson [Law70] constructed closed embedded minimal surfaces in $\mathbb{S}^3$ of every orientable genus. These results show that neither constant positive curvature, flatness, nor arbitrarily large genus is incompatible with minimality in a round sphere. The requirement of everywhere negative curvature is more restrictive.

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In his 1982 problem list, Yau asked whether a round sphere contains a closed minimal surface whose Gaussian curvature is everywhere negative [Yau82, Problem 101]. A subsequent theorem of Bryant [Bry85, Theorem 2.3] shows that a surface of constant negative Gaussian curvature cannot admit a minimal isometric immersion into any finite-dimensional round sphere, even locally. Consequently, any solution of Yau's problem must have nonconstant Gaussian curvature. The question is whether one can allow the curvature to vary while keeping it strictly negative at every point.

An analogous existence problem has been solved in complex projective geometry. Mohsen [Moh22] proved that every smooth complex projective manifold of complex dimension at least three, equipped with a prescribed Hermitian metric, contains a smooth closed complex curve whose induced Gaussian curvature is everywhere negative. When the prescribed metric is Kähler, the inclusion of this curve is minimal, since complex submanifolds of Kähler manifolds are minimal. His construction therefore provides embedded examples for every Kähler metric on such a projective manifold. The curves are constructed as complete intersections using high tensor powers of an ample line bundle and the asymptotic methods of Donaldson and Auroux; see [Don96], [Aur97], [Aur01].

For round spheres, recent progress has come from the connection between harmonic maps, unitary representations and random matrix theory developed by Song [Son24]. In particular, [Son24, Theorem 0.3] constructs closed branched minimal immersions $\iota_j : \Sigma_j \to \mathbb{S}^{n_j}$, with $n_j \to +\infty$, such that

$$\lim_{j\to\infty} \frac{1}{\operatorname{Area}(\iota_j)} \int_{\Sigma_j} |K_{\iota_j} + 8| \, d\operatorname{vol}_{g_{\iota_j}} = 0.$$

The constant -8 corresponds to the normalization in which the ambient spheres have sectional curvature one. For every fixed $\varepsilon > 0$, the proportion of the area on which $|K_{\iota_j} + 8| \geq \varepsilon$ therefore tends to zero. However, this does not imply negativity at every point, as the convergence of the Gaussian curvatures to -8 can be shown to be locally smooth only on compact subsets away from the cusp regions; no analogous uniform statement holds near the images of the cusps.

Building on Song's method, Ancona–Labourie–Roig-Sanchis–Toulisse [ALRST25] obtained closed minimal immersions into all sufficiently high-dimensional round spheres with everywhere negative Gaussian curvature. More precisely, their curvatures converge uniformly to -8 as the ambient dimension tends to infinity. Their construction uses compact hyperbolic orbifolds with trivial Teichmüller space in place of the thrice-punctured sphere used in Song's construction. This permits uniform geometric control over the entire surface. Their argument does not provide an explicit dimension threshold or establish embeddedness.

In this paper we construct closed embedded examples in the round four-sphere, the smallest possible ambient dimension. Throughout the paper, all round spheres have sectional curvature one, and the curvature and area of an embedded surface refer to its induced metric. When a surface is specified as a submanifold rather than by a parametrization, we write K_Σ for the Gaussian curvature associated with its inclusion.

Theorem 1.1. *There exists $m_0 \in \mathbb{N}$ such that, for every integer $m \geq m_0$, the unit round sphere $\mathbb{S}^4$ contains a smooth closed connected embedded minimal surface Σ_m of genus $m^2 + 1$ satisfying*

$$K_{\Sigma_m} < 0 \quad \text{everywhere on } \Sigma_m \qquad \text{and} \qquad \lim_{m\to\infty} \operatorname{Area}(\Sigma_m) = 4\pi^2.$$

These surfaces are obtained by deforming two nearby toroidal sheets joined by m^2 small necks modeled on catenoids. Their areas tend to twice the area of the flat Clifford torus, while their genera tend to infinity. In particular, the Gauss–Bonnet theorem gives

$$\int_{\Sigma_m} K_{\Sigma_m} \, d\operatorname{vol}_{\Sigma_m} = -4\pi m^2.$$

The construction and the estimates that preserve negative curvature under the deformations are described below.

Corollary 1.2. *Every unit round sphere $\mathbb{S}^n$, $n \geq 4$, contains a smooth closed connected embedded minimal surface with everywhere negative Gaussian curvature.*

Proof. Compose an embedding supplied by Theorem 1.1 with a totally geodesic inclusion $\mathbb{S}^4 \hookrightarrow \mathbb{S}^n$. This preserves the induced metric, minimality and embeddedness. □

The ambient dimension in Theorem 1.1 is the smallest possible. Indeed, a closed minimal immersion $\iota : \Sigma \to \mathbb{S}^3$ cannot have $K_\iota < 0$ everywhere; see [Law70, Proposition 1.5]. For an orientable surface, the Hopf quadratic differential of its second fundamental form is holomorphic, and its zeros are precisely the umbilic points. Since the immersion is minimal, an umbilic point satisfies $A_\iota = 0$ and therefore $K_\iota = 1$. Everywhere negative curvature would thus require the Hopf differential to be nowhere zero. On the other hand, Gauss–Bonnet would force the genus γ to be at least two, and a nonzero holomorphic quadratic differential on such a surface has $4\gamma - 4$ zeros counted with multiplicity. This is a contradiction. A nonorientable example would give an orientable one by passing to its orientable double cover. Finally, the metric induced by an immersion of a surface into $\mathbb{S}^2$ has Gaussian curvature one.

Twistor geometry provides a different approach to minimal surfaces in even-dimensional spheres. Bryant [Bry82] proved that every compact Riemann surface admits a conformal superminimal immersion into $\mathbb{S}^4$. In [Can26], we constructed families of linearly full branched superminimal immersions of arbitrarily large dimension, with both the conformal structure of the domain and the target sphere $\mathbb{S}^{2n}$, $n \geq 2$, fixed. Here linearly full means that the image is not contained in any proper totally geodesic subsphere. In $\mathbb{S}^4$, superminimality means that, at every regular point, the image of the unit tangent circle under $v \mapsto A_\iota(v, v)$ is a circle, possibly reduced to a point, in the normal plane. This additional condition imposes an obstruction to negative curvature: a nonconstant branched superminimal immersion of a closed Riemann surface into $\mathbb{S}^4$ cannot have nonpositive Gaussian curvature throughout its regular locus [Can26, Remark 1.4]. Thus the surfaces of Theorem 1.1 lie outside the superminimal class.

1.1. **Outline of the proof.** We construct our surfaces in a suitable minimal hypersurface of $\mathcal{N} \subset \mathbb{S}^4$, by joining two nearby copies of the Clifford torus with small catenoidal necks and then correcting their mean curvature. We use the doubling construction of Kapouleas–Yang [KY10] and the linearized doubling theory of Kapouleas–McGrath [KM23].

The choice of $\mathcal{N}$ serves two purposes. First, the Clifford torus is totally geodesic within $\mathcal{N}$, and its scalar Jacobi operator is $\Delta_{\mathbb{T}} + 2$. We use this operator to choose the heights of the two sheets so that their mean curvature within $\mathcal{N}$ vanishes to first order. Second, we must control the mean-curvature component normal to $\mathcal{N}$: minimality within $\mathcal{N}$ alone would not imply minimality in $\mathbb{S}^4$. For every constant-height torus, this additional component vanishes. For a sheet defined by a height function over the central torus, the absolute value of the mean-curvature component normal to $\mathcal{N}$ is bounded by a constant times the squared norm of the gradient of that function. The gradient and its norm are computed using the flat metric on $\mathbb{T}$. Thus, sheets whose height functions vary slowly have a small additional mean-curvature component. Together with separate estimates near the necks, this bound allows us to control the displacement normal to $\mathcal{N}$ needed to obtain minimality in $\mathbb{S}^4$.

For each sufficiently large integer m, we choose a square lattice $L_m \subset \mathbb{T}$ with m^2 points. We choose $\psi = \tau\Phi_m$ as the principal term of the height function defining the upper sheet, where

$$(\Delta_{\mathbb{T}} + 2)\Phi_m = 2\pi \sum_{p \in L_m} \delta_p.$$

Here, δ_p is the unit Dirac measure at p. Since the Dirac measures are supported at the lattice points, the height function ψ satisfies $(\Delta_{\mathbb{T}} + 2)\psi = 0$ away from L_m. This cancels the linear term in the scalar mean curvature within $\mathcal{N}$ of the graph with height ψ. The nonlinear terms will be estimated

and corrected later. Near a lattice point $p \in L_m$, the Green function Φ_m has the expansion

$$\Phi_m(q) = \log r + A_m + o(1), \qquad r := d_{\mathbb{T}}(q,p) \to 0.$$

Thus the corresponding terms in the sheet's height function $\psi = \tau\Phi_m$ are $\tau \log r + \tau A_m$. For a Euclidean catenoid whose smallest circle has radius τ and lies at height zero, the height of its upper end is approximately $\tau \log r + \tau \log(2/\tau)$ when r is large compared with τ. The coefficients of $\log r$ therefore already agree. To make the additive constants agree as well, we choose the reference radius τ_0 so that

$$A_m = \log(2/\tau_0), \qquad \text{or equivalently} \qquad \tau_0 = 2e^{-A_m}.$$

The Green-function estimates give $A_m = m^2/(2\pi) + \log m + O(1)$, and hence they force

$$\tau_0 \asymp m^{-1} \exp\left(-\frac{m^2}{2\pi}\right),$$

with comparison constants independent of m. To encode this multiplicative freedom in our construction, we set the common minimum radius of the catenoidal necks to be $\tau_\zeta := e^\zeta \tau_0$, with $\zeta \in [-1,1]$. Since, by construction, $\pm\tau_\zeta\Phi_m$ are the principal terms of the two sheet height functions, ζ varies the neck radius and the separation of the sheets together. This parameter will be chosen when solving the remaining scalar mean-curvature equation, to cancel the coefficient not determined by the scalar correction alone.

In Section 3, we join the two sheets to the catenoids at distances comparable to τ_ζ^α, for a fixed sufficiently small $\alpha > 0$. These distances are much larger than τ_ζ and much smaller than the lattice spacing, which is of order m^{-1}. The resulting surface $M_\zeta \subset \mathcal{N}$ is smooth, closed, connected and embedded, with genus $m^2 + 1$.

Two functions are needed to determine suitable normal deformation of M_ζ which will make it minimal. Writing ν_ζ for the outward unit normal of M_ζ within $\mathcal{N}$, we first move each point within $\mathcal{N}$ by a normal displacement ϕ, and then move the resulting point in the spherical normal direction n by a displacement b:

$$\begin{aligned} G_{\zeta,\phi}(x) &:= \exp_x^{\mathcal{N}}\big(\phi(x)\nu_\zeta(x)\big), \\ \widehat{X}_{\zeta,\phi,b}(x) &:= \exp^{\mathbb{S}^4}_{G_{\zeta,\phi}(x)}\big(b(x)n(G_{\zeta,\phi}(x))\big) \qquad \forall\, x \in M_\zeta. \end{aligned}$$

The final surface is the image of $\widehat{X}_{\zeta,\phi,b}$ for a suitable choice of ζ, ϕ, b.

We impose the symmetries defined in Section 2 on the initial surfaces and all their deformations to make the linear problem solvable with suitable estimates. On the central torus, lattice invariance excludes the kernel of $\Delta_{\mathbb{T}} + 2$, making Φ_m uniquely determined. On each model neck, the conditions on b force zero average around every circular cross-section. The resulting angular Poincaré inequality is essential to obtaining inverse estimates independent of the neck radius. The conditions on ϕ allow a constant angular component, which requires a separate linear argument.

To test minimality after deformation, we parallel transport the two normals of $G_{\zeta,\phi}(M_\zeta)$ along the spherical displacement geodesics and project onto the new normal planes. For small deformations, these projections remain independent. Minimality is therefore equivalent to the vanishing of the mean-curvature vector's inner products with both normal fields.

For every $\zeta \in [-1,1]$ and every ϕ satisfying the bounds in Section 4, we first choose b to vanish the component associated with n. The angular inequality, metric comparisons and elliptic estimates give a uniformly bounded inverse for the linearized equation. A contraction argument then solves the nonlinear equation, with

$$\|b\|_{L^\infty(M_\zeta)} \le C\tau_\zeta^2.$$

Section 4 also controls the derivatives of b up to order too, which appear in the curvature estimates.

The displacement b changes the other mean-curvature component. Section 5 handles this change when solving for ϕ and ζ. The scalar linear construction determines a correction up to an error proportional to one prescribed function. Adjusting ζ cancels the remaining coefficient. Since b depends on (ϕ, ζ), these choices are made by solving a fixed-point problem. The estimates give a continuous map preserving a compact convex set of corrections and parameters; the polynomial losses in m are absorbed by positive powers of τ_ζ. Schauder's fixed-point theorem supplies a choice for which both normal components vanish. The resulting surface $\Sigma_m := \widehat{X}_{\zeta,\phi,b}(M_\zeta)$ is therefore minimal in $\mathbb{S}^4$.

Strictly negative curvature requires a further comparison. Write $\tau := \tau_\zeta$ for the final neck parameter. For a sheet defined by a height function w, the quadratic curvature expression is

$$\mathcal{K}_2(w) := \det D^2 w + w\Delta_{\mathbb{T}} w + |Dw|^2,$$

where derivatives, norms and determinants use the flat metric on $\mathbb{T}$. For $\psi = \tau\Phi_m$, the equation $\Delta_{\mathbb{T}}\psi = -2\psi$ and the Green-function estimates give

$$\mathcal{K}_2(\psi) = |D\psi|^2 - \frac{1}{2}|D^2\psi|^2 \le -c\tau^2\big(m^4 + \rho^{-4}\big), \qquad \rho(q) := d_{\mathbb{T}}(q, L_m),$$

for $q \in \mathbb{T} \smallsetminus L_m$, with $c > 0$ independent of m. Thus the Hessian of the chosen height function provides the negative curvature bound. At distances comparable to the lattice spacing, this bound has size $m^4\tau^2 \to 0$. Absolute smallness of the corrections is therefore insufficient: the curvature errors must be smaller than the negative bound.

Section 6 proves precisely this comparison on the sheet portions outside a sufficiently large fixed multiple of the neck radius. The interpolation, the nonlinear curvature terms and both deformations together produce an error smaller than half the negative bound, once m is sufficiently large.

On the remaining neck portions, we divide lengths by τ. Both deformations tend to zero in C^2 in the rescaled coordinates, and the final necks converge to the Euclidean catenoid of minimum radius one. The catenoid has curvature bounded above by a strictly negative constant on each fixed bounded portion. Consequently, the final necks satisfy $K_{\Sigma_m} \le -c_R\tau^{-2}$ there. The sheet and neck estimates cover the entire surface.

Finally, the neck and interpolation regions have total area $O(m^2\tau^{2\alpha}) \to 0$. Each remaining sheet has area tending to the Clifford torus area $2\pi^2$, and the corrections preserve these limits. Hence $\mathrm{Area}(\Sigma_m) \to 4\pi^2$.

1.2. **Open problems.** The results above leave several open questions concerning general ambient metrics, the possible genera of the surfaces, and the existence of negatively curved examples within more specific subclasses of minimal surfaces.

In particular, Song has proposed the following extension of Yau's question to general closed Riemannian manifolds [Son26, p. 4].

Question (Song). *Let (M, g) be a closed Riemannian manifold with $\dim P \ge 4$. Does there exist a closed connected surface Σ and a smooth minimal immersion $\iota : \Sigma \to (M, g)$ such that the Gaussian curvature of $g_\iota := \iota^* g$ is strictly negative at every point?*

Corollary 1.2 answers this question in full generality for round spheres. Mohsen's theorem gives another class of examples, namely complex projective manifolds of complex dimension at least three equipped with Kähler metrics. For a general minimal immersion into (M, g), the Gauss equation reads

$$K_\iota(x) = \sec_g\big(d\iota_x(T_x\Sigma)\big) - \frac{|A_\iota(x)|^2_{g_\iota}}{2} \qquad \forall\, x \in \Sigma,$$

where $\sec_g$ denotes ambient sectional curvature and A_ι is the second fundamental form in (M, g). Thus one must construct the immersion while controlling its second fundamental form relative to

the curvature of its tangent planes in the ambient manifold. Whether the immersion can always be chosen to be an embedding is a stronger question.

A more restricted problem is to establish the same existence result for every smooth metric in some C^∞-neighborhood of the round metric on $\mathbb{S}^4$. Strict negativity of Gaussian curvature persists under sufficiently small C^2 perturbations of a fixed immersion and the ambient metric. However, one must also construct a minimal immersion for the perturbed metric, as curvature continuity alone does not provide this.

Even for the round metric, the present construction does not determine all genera for which such examples exist. Theorem 1.1 gives the genera m^2+1 for all sufficiently large integers m.

Question. *Which integers $\gamma \geq 2$ occur as the genus of a smooth closed connected orientable embedded minimal surface in $\mathbb{S}^4$ with everywhere negative Gaussian curvature? In particular, does every sufficiently large integer occur?*

The lower bound $\gamma \geq 2$ follows from Gauss–Bonnet. Determining the smallest possible genus is already a separate problem.

Finally, the superminimal existence problem discussed in [Can26, Conjecture 1.3] remains distinct from the existence result proved here. Its first possible ambient sphere is $\mathbb{S}^6$. In particular, one can ask for an unbranched embedded example as follows.

Question. *Does $\mathbb{S}^6$ contain a smooth closed connected embedded linearly full superminimal surface with everywhere negative Gaussian curvature?*

The examples obtained from Corollary 1.2 lie in a totally geodesic $\mathbb{S}^4 \subset \mathbb{S}^6$ and therefore do not answer this question. A construction within the superminimal class would have to impose the curvature inequality together with the holomorphicity and horizontality conditions on the corresponding twistor lifts.

On the use of AI. The author used ChatGPT-6 Astra solely to improve the clarity and style of the prose, to assist in identifying potential errors or gaps in the mathematical arguments, and to create the figures appearing in the manuscript. The research ideas and all the proofs were developed without the use of AI tools. All mathematical arguments were independently checked by the author, who takes full responsibility for the content of this paper.

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2. Geometry and symmetries near the Clifford torus

We first embed the Clifford torus inside an explicit minimal hypersurface $\mathcal{N}$ of the round four-sphere and compute the geometric quantities needed in the construction. For a surface in $\mathcal{N}$, we then express the two components of its mean-curvature vector within $\mathbb{S}^4$ and their normal linearization, and provide suitable tubular coordinates for deformations out of the hypersurface. Finally, we specify the finite group of symmetries used in the construction and derive the resulting restrictions on the normal variations and their angular Fourier terms near each neck.

2.1. A minimal hypersurface containing the Clifford torus.

The hypersurface $\mathcal{N}$ constructed below will contain the initial surfaces used in the perturbation argument. Its central Clifford torus is totally geodesic within it, and the scalar Jacobi operator of that torus within $\mathcal{N}$ has a particularly simple expression.

We regard the unit round four-sphere as

$$\mathbb{S}^4 = \{(w_1, w_2, x_5) \in \mathbb{C}^2 \oplus \mathbb{R} : \, |w_1|^2 + |w_2|^2 + x_5^2 = 1\}.$$

Set $\ell := \sqrt{2}\pi$ and let $\mathbb{T} := (\mathbb{R}/\ell\mathbb{Z})^2$, equipped with the flat metric $g_{\mathbb{T}} := ds^2 + dt^2$. Define $f : \mathbb{T} \to \mathbb{S}^4$ by

$$f(s,t) := \frac{1}{\sqrt{2}}\big(e^{i\sqrt{2}s}, e^{i\sqrt{2}t}, 0\big) \qquad \forall\, (s,t) \in \mathbb{T}.$$

Note that f is an isometric embedding of $\mathbb{T}$ into $\mathbb{S}^4$ and $\mathrm{Area}(\mathbb{T}) = 2\pi^2$. Fix $z_0 \in (0, \pi/4)$ and define the smooth embedding $\mathcal{F} : \mathbb{T} \times (-z_0, z_0) \to \mathbb{S}^4$ by

$$\mathcal{F}(s,t,z) := \cos z\, f(s,t) + \sin z\, e_5 \qquad \forall\, (s,t,z) \in \mathbb{T} \times (-z_0, z_0),$$

where $e_5 = (0,0,1) \in \mathbb{C}^2 \oplus \mathbb{R}$. We let

$$\mathcal{N} := \mathcal{F}\big(\mathbb{T} \times (-z_0, z_0)\big)$$

and we define the projection $\pi : \mathcal{N} \to \mathbb{T}$ as

$$\pi(\mathcal{F}(s,t,z)) := (s,t) \qquad \forall (s,t,z) \in \mathbb{T} \times (-z_0, z_0).$$

Let $g_{\mathbb{S}^4}$ denote the unit round metric on $\mathbb{S}^4$, let $\iota_{\mathcal{N}} : \mathcal{N} \hookrightarrow \mathbb{S}^4$ denote the inclusion, and equip $\mathcal{N}$ with the induced metric $g_{\mathcal{N}} := \iota_{\mathcal{N}}^* g_{\mathbb{S}^4}$. Note that

$$g_{\mathcal{N}} = (\mathcal{F}^{-1})^*\big(dz^2 + \cos^2 z\, (ds^2 + dt^2)\big). \tag{2.1}$$

Define $N : \mathbb{T} \to \mathbb{C}^2 \oplus \mathbb{R}$ by

$$N(s,t) := \frac{1}{\sqrt{2}}\big(e^{i\sqrt{2}s}, -e^{i\sqrt{2}t}, 0\big) \qquad \forall\, (s,t) \in \mathbb{T}.$$

The vector $N(s,t)$ has unit Euclidean norm and is orthogonal to $\mathcal{F}(s,t,z)$ and to each of $\partial_s \mathcal{F}$, $\partial_t \mathcal{F}$ and $\partial_z \mathcal{F}$ at (s,t,z). Consequently, the field n defined along $\mathcal{N}$ by

$$n\big(\mathcal{F}(s,t,z)\big) := N(s,t) \qquad \forall\, (s,t,z) \in \mathbb{T} \times (-z_0, z_0)$$

is a smooth global unit normal to $\mathcal{N}$ within $\mathbb{S}^4$.

We call $\mathcal{N}$ the *suspension band* of the Clifford torus and refer to $f(\mathbb{T}) = \mathcal{F}(\mathbb{T} \times \{0\})$ as its *central torus*.

Remark 2.1. Note that the function on $\mathcal{N}$ defined by $\mathcal{F}(s,t,z) \mapsto z$ is the signed distance to the central torus in $(\mathcal{N}, g_{\mathcal{N}})$, i.e.

$$\mathrm{dist}_{g_{\mathcal{N}}}\big(\mathcal{F}(s,t,z), f(\mathbb{T})\big) = |z| \qquad \forall\, (s,t,z) \in \mathbb{T} \times (-z_0, z_0).$$

Indeed, the coordinate segment joining $\mathcal{F}(s,t,z)$ to $\mathcal{F}(s,t,0)$ has length $|z|$, while every curve joining $\mathcal{F}(s,t,z)$ to the central torus has length at least $|z|$, by (2.1).

We fix the following conventions for immersions, which will also be used in the subsequent sections. Let $\iota : M \to (P, \overline{g})$ be a smooth immersion and equip M with the induced metric $g := \iota^* \overline{g}$. We denote the Levi-Civita connection of g by ∇^g. For a smooth function u on M, the notation $\nabla^g u$ denotes its gradient with respect to g. We use the convention

$$\Delta_g u := \mathrm{div}_g(\nabla^g u) \qquad \forall\, u \in C^\infty(M)$$

for the Laplace–Beltrami operator. We omit the metric subscript when it is clear from the context.

Denote by $\overline{\nabla}$ the connection on $\iota^* TP$ induced by the Levi-Civita connection of $(P, \overline{g})$. Let

$$T_\iota^\perp M := \big(d\iota(TM)\big)^\perp \subset \iota^* TP$$

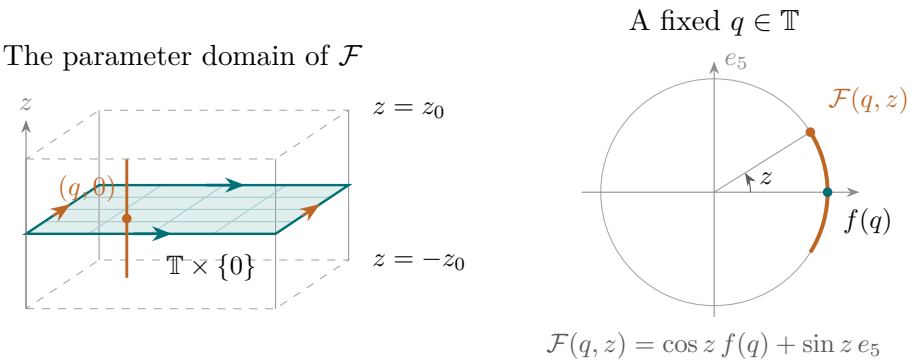


FIGURE 1. The hypersurface $\mathcal{N}$ is the image under $\mathcal{F}$ of the parameter domain on the left. Opposite vertical faces are identified in the s and t directions; the central square, with its opposite edges identified, represents $\mathbb{T} \times \{0\}$. The boundary heights $\pm z_0$ are not included. The orange segment keeps q fixed. On the right, its image is an arc of the unit circle in the plane spanned by $f(q)$ and e_5.

be the normal bundle of ι, where $\cdot^\perp$ indicates the $\overline{g}$-orthogonal complement of $d\iota(TM)$ in ι^*TP. For a vector in ι^*TP, the superscript $^\perp$ denotes its $\overline{g}$-orthogonal projection onto $T^\perp_\iota M$. The *second fundamental form* of ι is the tensor $A_\iota \in \Gamma(\mathrm{Sym}^2 T^*M \otimes T^\perp_\iota M)$ defined by

$$A_\iota(X, Y) := \big(\overline{\nabla}_X(d\iota(Y))\big)^\perp \qquad \forall\, X, Y \in \Gamma(TM).$$

For a smooth unit normal field $\eta \in \Gamma(T^\perp_\iota M)$, its scalar component in the direction η is the bilinear form $h^\eta_\iota \in \Gamma(\mathrm{Sym}^2 T^*M)$, defined by

$$h^\eta_\iota(X, Y) := \overline{g}\big(A_\iota(X, Y), \eta\big) \qquad \forall\, X, Y \in \Gamma(TM).$$

The *shape operator* of ι in the direction η is the g-self-adjoint endomorphism $S^\eta_\iota \in \Gamma(T^*M \otimes TM)$ characterized by

$$g(S^\eta_\iota(X), Y) = h^\eta_\iota(X, Y) \qquad \forall\, X, Y \in \Gamma(TM).$$

In particular, $|S^\eta_\iota|_g = |h^\eta_\iota|_g$. The *mean-curvature vector* of ι and its scalar component in the direction η are defined using the trace convention:

$$\mathbf{H}_\iota := \mathrm{tr}_g\, A_\iota = \sum_{a=1}^{\dim M} A_\iota(e_a, e_a),$$
$$H_{\iota,\eta} := \mathrm{tr}_g\, h^\eta_\iota = \sum_{a=1}^{\dim M} h^\eta_\iota(e_a, e_a) = \overline{g}(\mathbf{H}_\iota, \eta),$$

where $(e_a)_{a=1}^{\dim M}$ is any local g-orthonormal frame of TM.

The *normal connection* of ι is given by

$$\nabla^{\perp,\iota}_X \xi := (\overline{\nabla}_X \xi)^\perp \qquad \forall\, X \in \Gamma(TM), \quad \xi \in \Gamma(T^\perp_\iota M).$$

With these conventions, the Weingarten identity reads

$$\overline{\nabla}_X \eta = -d\iota(S^\eta_\iota(X)) + \nabla^{\perp,\iota}_X \eta \qquad \forall\, X \in \Gamma(TM). \tag{2.2}$$

Indeed, differentiating $\overline{g}(\eta, d\iota(Y)) = 0$ gives

$$\begin{aligned}\overline{g}(\overline{\nabla}_X \eta, d\iota(Y)) &= -\overline{g}(\eta, \overline{\nabla}_X(d\iota(Y))) \\ &= -h^\eta_\iota(X, Y) = -g(S^\eta_\iota(X), Y) \qquad \forall\, X, Y \in \Gamma(TM),\end{aligned}$$

which determines the tangential component of $\overline{\nabla}_X \eta$; its normal component is $\nabla^{\perp,\iota}_X \eta$ by definition. In codimension one, this normal component vanishes because η has unit length and spans the normal bundle.

For the fixed metrics on $\mathbb{S}^4$ and $\mathcal{N}$, we use the abbreviations

$$\nabla^{\mathbb{S}^4} := \nabla^{g_{\mathbb{S}^4}}, \qquad \nabla^{\mathcal{N}} := \nabla^{g_{\mathcal{N}}},$$

for their respective Levi-Civita connections.

Lemma 2.2. *Write $S := S^n_{\iota_{\mathcal{N}}} \in \Gamma(T^*\mathcal{N} \otimes T\mathcal{N})$ for the shape operator of the inclusion $\iota_{\mathcal{N}} : \mathcal{N} \hookrightarrow \mathbb{S}^4$ with respect to the unit normal n, using the preceding conventions. Define $E_1, E_2, E_3 \in \Gamma(T\mathcal{N})$ by*

$$\begin{aligned} E_1(\mathcal{F}(s,t,z)) &:= \frac{1}{\cos z}\partial_s \mathcal{F}(s,t,z), \\ E_2(\mathcal{F}(s,t,z)) &:= \frac{1}{\cos z}\,\partial_t \mathcal{F}(s,t,z), \\ E_3(\mathcal{F}(s,t,z)) &:= \partial_z \mathcal{F}(s,t,z), \end{aligned}$$

for every $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$. Then (E_1, E_2, E_3) is a global $g_{\mathcal{N}}$-orthonormal frame of $T\mathcal{N}$. For every $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$, writing $q := \mathcal{F}(s,t,z)$, one has

$$\begin{aligned} S_q(E_1(q)) &= -\sec z\, E_1(q), \\ S_q(E_2(q)) &= \sec z\, E_2(q), \\ S_q(E_3(q)) &= 0. \end{aligned} \tag{2.3}$$

In particular, $\mathcal{N}$ is a minimal hypersurface of $\mathbb{S}^4$.

The central torus $f(\mathbb{T}) \subset \mathcal{N}$ is totally geodesic as a hypersurface of $(\mathcal{N}, g_{\mathcal{N}})$. The scalar Jacobi operator $L_{\mathbb{T}} : C^\infty(\mathbb{T}) \to C^\infty(\mathbb{T})$ of the isometric embedding $f : (\mathbb{T}, g_{\mathbb{T}}) \to (\mathcal{N}, g_{\mathcal{N}})$, with respect to the unit normal field $E_3 \circ f$, is given by

$$L_{\mathbb{T}} u = \Delta_{\mathbb{T}} u + 2u \qquad \forall\, u \in C^\infty(\mathbb{T}), \tag{2.4}$$

where $\Delta_{\mathbb{T}} := \Delta_{g_{\mathbb{T}}}$.

Finally, there exists a constant $C = C(z_0) > 0$ such that, for every $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$ and $q := \mathcal{F}(s,t,z)$,

$$|S_q|_{g_{\mathcal{N}}} \le C, \qquad |(\nabla^{\mathcal{N}} S)_q|_{g_{\mathcal{N}}} \le C|z|. \tag{2.5}$$

Here $\nabla^{\mathcal{N}} S$ is the covariant derivative of the tensor field S, and both norms are induced by $g_{\mathcal{N}}$.

Proof. Fix $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$ and set $q := \mathcal{F}(s,t,z)$. Differentiating $\mathcal{F}$ and using the definitions of the frame fields gives

$$\begin{aligned} E_1(q) &= \partial_s f(s,t), \\ E_2(q) &= \partial_t f(s,t), \\ E_3(q) &= -\sin z\, f(s,t) + \cos z\, e_5. \end{aligned}$$

These three vectors are orthonormal in $\mathbb{R}^5$, and therefore also with respect to the induced metric $g_{\mathcal{N}}$. All inner products below are Euclidean inner products in $\mathbb{R}^5$. Viewing n as a map from $\mathcal{N}$ to $\mathbb{R}^5$, the spherical covariant derivative satisfies

$$\nabla^{\mathbb{S}^4}_v n = dn_q(v) + \langle v, n(q)\rangle q = dn_q(v) \qquad \forall\, v \in T_q\mathcal{N}.$$

The last equality follows from the orthogonality of $n(q)$ to $T_q\mathcal{N}$. Consequently,

$$\begin{aligned} \big(\nabla^{\mathbb{S}^4}_{E_1} n\big)(q) = dn_q(E_1(q)) &= \frac{1}{\cos z}\,\partial_s (n \circ \mathcal{F})(s,t,z) \\ &= \frac{1}{\cos z}\,\partial_s N(s,t) = \frac{1}{\cos z}\,\partial_s f(s,t) = \sec z\, E_1(q). \end{aligned}$$

Similarly, using $\partial_t N = -\partial_t f$ and the independence of $n \circ \mathcal{F}$ from z, we obtain

$$\begin{aligned}\big(\nabla^{\mathbb{S}^4}_{E_2} n\big)(q) &= -\sec z\, E_2(q),\\ \big(\nabla^{\mathbb{S}^4}_{E_3} n\big)(q) &= 0.\end{aligned}$$

By (2.2), this proves (2.3). In particular, $\operatorname{tr} S = 0$, so $\mathcal{N}$ is minimal in $\mathbb{S}^4$.

We next compute the coefficients of the Levi-Civita connection of $g_{\mathcal{N}}$ in the orthonormal frame (E_1, E_2, E_3). Viewing tangent vector fields on $\mathcal{N}$ as $\mathbb{R}^5$-valued maps, orthogonal projection of their Euclidean derivatives onto $T_q\mathcal{N}$ gives

$$\begin{aligned}\big(\nabla^{\mathcal{N}}_X Y\big)(q) = {} & dY_q(X(q)) + \langle X(q), Y(q)\rangle q\\ & - \langle S_q(X(q)), Y(q)\rangle n(q) \qquad \forall\, X, Y \in \Gamma(T\mathcal{N}).\end{aligned}$$

Indeed, the Euclidean normal space to $\mathcal{N}$ at q is spanned by the orthonormal vectors q and $n(q)$. For example, since $\partial_{ss} f = -f - N$, we obtain

$$\begin{aligned}\big(\nabla^{\mathcal{N}}_{E_1} E_1\big)(q) &= \frac{1}{\cos z}\,\partial_s(E_1 \circ \mathcal{F})(s,t,z) + q + \sec z\, n(q)\\ &= \sec z\,\partial_{ss} f(s,t) + q + \sec z\, N(s,t)\\ &= -\sec z\big(f(s,t) + N(s,t)\big) + q + \sec z\, N(s,t)\\ &= q - \sec z\, f(s,t)\\ &= \tan z\big(-\sin z\, f(s,t) + \cos z\, e_5\big)\\ &= \tan z\, E_3(q).\end{aligned}$$

Applying the same projection formula to the remaining pairs of frame fields gives

$$\begin{aligned}\big(\nabla^{\mathcal{N}}_{E_i} E_j\big)(q) &= \tan z\,\delta_{ij} E_3(q),\\ \big(\nabla^{\mathcal{N}}_{E_i} E_3\big)(q) &= -\tan z\, E_i(q),\\ \big(\nabla^{\mathcal{N}}_{E_3} E_i\big)(q) &= 0,\\ \big(\nabla^{\mathcal{N}}_{E_3} E_3\big)(q) &= 0,\end{aligned}$$

for every $i, j \in \{1, 2\}$. Hence, at $z = 0$ the normal components of $\nabla^{\mathcal{N}}_{E_i} E_j$ vanish for $i, j \in \{1, 2\}$. Thus, the central torus is totally geodesic in $\mathcal{N}$.

Since $S(E_3) = 0$, the Gauss equation for $\mathcal{N} \subset \mathbb{S}^4$ shows that each plane spanned by E_i and E_3, with $i \in \{1, 2\}$, has sectional curvature 1. Consequently,

$$\operatorname{Ric}_{g_{\mathcal{N}}}(E_3, E_3) = 2.$$

Let $\nu := E_3 \circ f$ be the unit normal field along the isometric embedding $f : (\mathbb{T}, g_{\mathbb{T}}) \to (\mathcal{N}, g_{\mathcal{N}})$. Since the central torus is totally geodesic in $\mathcal{N}$, we have $h^\nu_f \equiv 0$. The scalar Jacobi formula gives

$$\begin{aligned}L_{\mathbb{T}} u &= \Delta_{\mathbb{T}} u + \Big(|h^\nu_f|^2_{g_{\mathbb{T}}} + \big(\operatorname{Ric}_{g_{\mathcal{N}}}(E_3, E_3)\big) \circ f\Big) u\\ &= \Delta_{\mathbb{T}} u + 2u \qquad \forall\, u \in C^\infty(\mathbb{T}),\end{aligned}$$

which proves (2.4).

Finally, set

$$\lambda_1(z) := -\sec z, \qquad \lambda_2(z) := \sec z \qquad \forall\, z \in (-z_0, z_0).$$

Using the connection identities above and the formula

$$(\nabla^{\mathcal{N}}_X S)(Y) = \nabla^{\mathcal{N}}_X(S(Y)) - S(\nabla^{\mathcal{N}}_X Y) \qquad \forall\, X, Y \in \Gamma(T\mathcal{N}),$$

we obtain, for every $i, j \in \{1, 2\}$,

$$\big((\nabla^{\mathcal{N}}_{E_3} S)(E_i)\big)(q) = \lambda_i(z) \tan z\, E_i(q),$$

$$\begin{aligned}\big((\nabla^{\mathcal{N}}_{E_i}S)(E_j)\big)(q) &= \lambda_j(z)\tan z\,\delta_{ij}E_3(q),\\ \big((\nabla^{\mathcal{N}}_{E_i}S)(E_3)\big)(q) &= \lambda_i(z)\tan z\,E_i(q),\\ \big((\nabla^{\mathcal{N}}_{E_3}S)(E_3)\big)(q) &= 0.\end{aligned}$$

Since (E_1, E_2, E_3) is orthonormal, these identities and (2.3) imply

$$|S_q|^2_{g_{\mathcal{N}}} = 2\sec^2 z, \qquad |(\nabla^{\mathcal{N}}S)_q|^2_{g_{\mathcal{N}}} = 6\sec^2 z\,\tan^2 z.$$

For every $z \in (-z_0, z_0)$, one has $\sec z \le \sec z_0$ and $|\tan z| \le \sec^2 z_0\,|z|$. The estimates (2.5) follow. □

2.2. **Normal variations and tubular coordinates.** For a surface M contained in $\mathcal{N}$, minimality in $\mathbb{S}^4$ requires both its scalar mean curvature within $\mathcal{N}$ and the component of its mean-curvature vector within $\mathbb{S}^4$ in the direction n to vanish. We compute the latter component and the normal connection in terms of the shape operator S of $\mathcal{N} \subset \mathbb{S}^4$ and the chosen unit normal of M within $\mathcal{N}$, and specialize these formulas to height graphs over the central torus. We then represent the normal linearization of the mean curvature map as a 2×2 matrix of scalar differential operators, acting on the coefficients of a normal variation in these two directions. Its off-diagonal entries describe how a variation in either direction affects the mean-curvature component in the other direction. Finally, we construct convenient tubular coordinates around $\mathcal{N}$ in $\mathbb{S}^4$.

Let $M \subset \mathcal{N}$ be a smooth embedded two-sided surface in $\mathcal{N}$, and let

$$\jmath : M \hookrightarrow \mathcal{N}, \qquad \iota : M \hookrightarrow \mathbb{S}^4$$

denote its inclusions in $\mathcal{N}$ and $\mathbb{S}^4$, respectively. Equip M with the induced metric $g_M := \jmath^* g_{\mathcal{N}} = \iota^* g_{\mathbb{S}^4}$, and fix a smooth global unit normal field ν to M within $\mathcal{N}$. No compactness or minimality assumption is imposed on M in this subsection. We write $\nabla^M := \nabla^{g_M}$ and $\Delta_M := \Delta_{g_M}$, and denote the restriction of n to M again by n. Then (ν, n) is a global orthonormal frame of the normal bundle $T^\perp_\iota M$.

Using the conventions above, the mean-curvature vectors of the two inclusions decompose as

$$\begin{aligned}\mathbf{H}_\jmath &= H_{\jmath,\nu}\nu,\\ \mathbf{H}_\iota &= H_{\iota,\nu}\nu + H_{\iota,n}n.\end{aligned}$$

The equality $h^\nu_\iota = h^\nu_\jmath$, verified in the proof of Lemma 2.3 below, implies $H_{\iota,\nu} = H_{\jmath,\nu}$. Thus minimality in $\mathcal{N}$ requires $H_{\iota,\nu} = 0$, while minimality in $\mathbb{S}^4$ requires both $H_{\iota,\nu} = 0$ and $H_{\iota,n} = 0$.

Inner products of vectors are taken with respect to $g_{\mathbb{S}^4}$; tensor norms and contractions use the corresponding induced metrics.

For the normal connection of ι, we abbreviate $\nabla^\perp := \nabla^{\perp,\iota}$. In the frame (ν, n), the connection $\nabla^\perp$ is determined by the one-form $\omega \in \Gamma(T^*M)$ defined by

$$\omega(X) := \langle \nabla^\perp_X \nu, n\rangle = \langle \nabla^{\mathbb{S}^4}_X \nu, n\rangle \qquad \forall\, X \in \Gamma(TM). \tag{2.6}$$

Indeed, differentiating $\langle \nu, \nu\rangle = \langle n, n\rangle = 1$ and $\langle \nu, n\rangle = 0$ gives

$$\nabla^\perp_X \nu = \omega(X)n, \qquad \nabla^\perp_X n = -\omega(X)\nu \qquad \forall\, X \in \Gamma(TM). \tag{2.7}$$

Thus ω records the coupling of the two normal directions. The following lemma expresses this one-form and the second fundamental form in the direction n in terms of S. To state the estimates, write

$$\nu_h(x) := \sum_{a=1}^{2} \langle \nu(x), E_a(x)\rangle E_a(x) \qquad \forall\, x \in M$$

for the component of ν in the plane spanned by E_1, E_2.

Lemma 2.3. *For every $X, Y \in \Gamma(TM)$, one has*

$$\begin{aligned} h^n_\iota(X,Y) &= \langle S(X), Y\rangle, \\ H_{\iota,n} &= -\langle S(\nu),\nu\rangle, \\ \omega(X) &= \langle S(X),\nu\rangle. \end{aligned} \tag{2.8}$$

Moreover,

$$(\nabla^M_X\omega)(Y) = \langle(\nabla^{\mathcal{N}}_X S)(Y),\nu\rangle + h^\nu_\iota(X,Y)\langle S(\nu),\nu\rangle - \langle S(Y), S^\nu_\iota(X)\rangle \qquad \forall\, X,Y\in\Gamma(TM). \tag{2.9}$$

There exists $C = C(z_0) > 0$ such that, for every $(s,t,z) \in \mathbb{T}\times(-z_0,z_0)$ with $x := \mathcal{F}(s,t,z) \in M$,

$$\begin{aligned} |(h^n_\iota)_x| &\le C, & |H_{\iota,n}(x)| &\le C|\nu_h(x)|^2, \\ |\omega_x| &\le C|\nu_h(x)|, & |(\nabla^M\omega)_x| &\le C\big(|z| + |(S^\nu_\iota)_x|\big). \end{aligned} \tag{2.10}$$

The norms of h^n_ι, ω, $\nabla^M\omega$ and S^ν_ι are induced by g_M, while $|\nu_h|$ is computed using $g_{\mathcal{N}}$.

Proof. For every $X, Y \in \Gamma(TM)$, differentiating $\langle Y, n\rangle = 0$ gives

$$h^n_\iota(X,Y) = \langle\nabla^{\mathbb{S}^4}_X Y, n\rangle = -\langle Y, \nabla^{\mathbb{S}^4}_X n\rangle = \langle S(X), Y\rangle.$$

Let (e_1,e_2) be a local g_M-orthonormal frame of TM. Since (e_1,e_2,ν) is an orthonormal frame of $T\mathcal{N}|_M$ and $\operatorname{tr} S = 0$, taking the trace yields

$$H_{\iota,n} = \sum_{i=1}^2 \langle S(e_i), e_i\rangle = \operatorname{tr} S - \langle S(\nu),\nu\rangle = -\langle S(\nu),\nu\rangle.$$

Likewise, differentiation of $\langle\nu, n\rangle = 0$ gives

$$\omega(X) = -\langle\nu, \nabla^{\mathbb{S}^4}_X n\rangle = \langle S(X),\nu\rangle \qquad \forall\, X\in\Gamma(TM).$$

This proves (2.8).

To differentiate ω, we also need the geometry of M within $\mathcal{N}$. The difference between $\nabla^{\mathbb{S}^4}_X Y$ and $\nabla^{\mathcal{N}}_X Y$ is a multiple of n, which is orthogonal to ν. Hence

$$h^\nu_\iota(X,Y) = \langle\nabla^{\mathcal{N}}_X Y,\nu\rangle = h^\nu_\jmath(X,Y) \qquad \forall\, X,Y\in\Gamma(TM).$$

Since both inclusions induce the metric g_M, it follows that $S^\nu_\iota = S^\nu_\jmath$ as well. The tangential and normal decompositions in $\mathcal{N}$ therefore give

$$\begin{aligned} \nabla^{\mathcal{N}}_X Y &= \nabla^M_X Y + h^\nu_\jmath(X,Y)\nu, \\ \nabla^{\mathcal{N}}_X \nu &= -S^\nu_\jmath(X) \qquad \forall\, X,Y\in\Gamma(TM). \end{aligned}$$

Substituting these identities into the covariant derivative of ω gives

$$\begin{aligned} (\nabla^M_X\omega)(Y) &= X\big(\langle S(Y),\nu\rangle\big) - \langle S(\nabla^M_X Y),\nu\rangle \\ &= \langle(\nabla^{\mathcal{N}}_X S)(Y),\nu\rangle + \langle S(\nabla^{\mathcal{N}}_X Y - \nabla^M_X Y),\nu\rangle + \langle S(Y), \nabla^{\mathcal{N}}_X\nu\rangle \\ &= \langle(\nabla^{\mathcal{N}}_X S)(Y),\nu\rangle + h^\nu_\iota(X,Y)\langle S(\nu),\nu\rangle - \langle S(Y), S^\nu_\iota(X)\rangle \end{aligned}$$

for every $X, Y \in \Gamma(TM)$. This proves (2.9).

We now deduce the estimates. Fix $(s,t,z) \in \mathbb{T}\times(-z_0,z_0)$ with $x := \mathcal{F}(s,t,z) \in M$. The self-adjointness of S and the identity $S(E_3) = 0$ imply

$$\begin{aligned} S_x(\nu(x)) &= S_x(\nu_h(x)), \\ \langle S_x(\nu(x)),\nu(x)\rangle &= \langle S_x(\nu_h(x)),\nu_h(x)\rangle. \end{aligned}$$

Therefore (2.8) gives

$$|(h^n_\iota)_x| \le |S_x|, \qquad |H_{\iota,n}(x)| \le |S_x|\,|\nu_h(x)|^2, \qquad |\omega_x| \le |S_x|\,|\nu_h(x)|.$$

Taking tensor norms in (2.9) and using $|(h^\nu_\iota)_x| = |(S^\nu_\iota)_x|$ also gives

$$|(\nabla^M \omega)_x| \le |(\nabla^{\mathcal{N}} S)_x| + 2|S_x|\,|(S^\nu_\iota)_x|.$$

The bounds (2.10) follow from (2.5). □

Remark 2.4. For a height graph, these geometric identities give explicit expressions in terms of its defining function. Let $U \subset \mathbb{T}$ be open and let $w : U \to (-z_0, z_0)$ be smooth. Define $F_w : U \to \mathcal{N}$ and its image M_w by

$$F_w(s,t) := \mathcal{F}(s,t,w(s,t)) \qquad \forall\,(s,t) \in U, \qquad M_w := F_w(U).$$

On U, set $w_s := \partial_s w$, $w_t := \partial_t w$, $Dw := (w_s, w_t)$, $c := \cos w$ and $Q := c^2 + |Dw|^2$, where $|Dw|^2 = w_s^2 + w_t^2$. Apply the preceding notation to M_w, choosing its unit normal ν within $\mathcal{N}$ so that $\langle \nu, E_3\rangle > 0$. Then the following identities hold on U:

$$\begin{aligned} \nu \circ F_w &= \frac{-w_s(E_1 \circ F_w) - w_t(E_2 \circ F_w) + c(E_3 \circ F_w)}{\sqrt{Q}}, \\ H_{\iota,n} \circ F_w &= \frac{w_s^2 - w_t^2}{cQ}, \\ F_w^* \omega &= \frac{w_s\,ds - w_t\,dt}{\sqrt{Q}}. \end{aligned} \tag{2.11}$$

To verify these formulas, fix $(s,t) \in U$ and put $x := F_w(s,t)$. The chain rule gives

$$\begin{aligned} d(F_w)_{(s,t)}(\partial_s) &= c(s,t)E_1(x) + w_s(s,t)E_3(x), \\ d(F_w)_{(s,t)}(\partial_t) &= c(s,t)E_2(x) + w_t(s,t)E_3(x). \end{aligned}$$

The numerator in the first line of (2.11) is orthogonal to both tangent vectors and has squared norm Q. Since $c > 0$, its E_3-component is positive. This proves the formula for ν. Applying (2.3) at height $z = w(s,t)$ then gives

$$\begin{aligned} S_x(\nu(x)) &= \frac{w_s(s,t)E_1(x) - w_t(s,t)E_2(x)}{c(s,t)\sqrt{Q(s,t)}}, \\ H_{\iota,n}(x) &= -\langle S_x(\nu(x)), \nu(x)\rangle = \frac{w_s(s,t)^2 - w_t(s,t)^2}{c(s,t)Q(s,t)}. \end{aligned}$$

Finally, using the definition of pullback and (2.8), we obtain

$$\begin{aligned} (F_w^*\omega)_{(s,t)}(\partial_s) &= \big\langle S_x\big(d(F_w)_{(s,t)}(\partial_s)\big), \nu(x)\big\rangle = \langle -E_1(x), \nu(x)\rangle = \frac{w_s(s,t)}{\sqrt{Q(s,t)}}, \\ (F_w^*\omega)_{(s,t)}(\partial_t) &= \big\langle S_x\big(d(F_w)_{(s,t)}(\partial_t)\big), \nu(x)\big\rangle = \langle E_2(x), \nu(x)\rangle = -\frac{w_t(s,t)}{\sqrt{Q(s,t)}}. \end{aligned}$$

These are the coefficients of the one-form in the last line of (2.11).

We next consider normal variations of a general surface $M \subset \mathcal{N}$ as above. The geometric identities already established will allow us to express the linearization of its mean-curvature vector in the frame (ν, n). For $V \in \Gamma(T^\perp_\iota M)$, fix an open set $U \subset M$ whose closure in M is compact. There exists $r_U > 0$ such that the maps

$$\begin{aligned} G_r : U &\to \mathbb{S}^4, \\ x &\mapsto \exp^{\mathbb{S}^4}_{\iota(x)}(rV(x)) \end{aligned}$$

are smooth immersions for every $r \in (-r_U, r_U)$. Indeed, $G_0 = \iota|_U$, and the differential of G_r depends smoothly on (x,r). Since ι is an immersion, compactness of $\overline{U}$ ensures that this differential remains

injective for all $x \in U$ when $|r|$ is sufficiently small. Moreover,

$$\frac{\partial}{\partial r} G_r(x)\Big|_{r=0} = V(x) \qquad \forall\, x \in U.$$

Thus these maps define a normal variation of ι near every point of M, although the interval for r may depend on U. The local variations agree on overlapping domains whenever both are defined, so their linearizations at $r = 0$ define an operator on all of M. If M is compact, we may take $U = M$. Define the *normal linearization of the mean curvature map* at ι to be the operator $J : \Gamma(T_\iota^\perp M) \to \Gamma(T_\iota^\perp M)$ by

$$(JV)(x) := \left(\frac{D}{dr}\Big|_{r=0} \mathbf{H}_{G_r}(x) \right)^\perp \qquad \forall\, x \in M,$$

where D/dr denotes covariant differentiation in $\mathbb{S}^4$ along the geodesic $r \mapsto G_r(x)$ and the projection is onto $(T_\iota^\perp M)_x$.

The general normal variation formula expresses J in terms of the Laplacian associated with the normal connection, given by

$$\Delta^\perp V := \sum_{i=1}^{2} \left(\nabla^\perp_{e_i} \nabla^\perp_{e_i} V - \nabla^\perp_{\nabla^M_{e_i} e_i} V \right) \qquad \forall\, V \in \Gamma(T_\iota^\perp M),$$

where (e_1, e_2) is any local g_M-orthonormal frame of TM. For a surface in the unit round sphere, this formula reads

$$JV = \Delta^\perp V + 2V + \sum_{i,j=1}^{2} \langle A_\iota(e_i, e_j), V\rangle A_\iota(e_i, e_j) \qquad \forall\, V \in \Gamma(T_\iota^\perp M); \tag{2.12}$$

see [LL26, Lemma 3.1]. This local identity applies without assuming that M is minimal. The term $2V$ is the ambient curvature contribution, with one copy of V for each orthonormal tangent direction.

In the next lemma, we compute the scalar operators representing J. In particular, we compare its component in the direction ν with the scalar mean-curvature linearization for $M \subset \mathcal{N}$.

Lemma 2.5. *Define $L_{M/\mathcal{N}}, B, B^* : C^\infty(M) \to C^\infty(M)$ and $q \in C^\infty(M)$ by*

$$\begin{aligned} L_{M/\mathcal{N}} u &:= \Delta_M u + \big(|S^\nu_\jmath|^2 + \mathrm{Ric}_{g_\mathcal{N}}(\nu, \nu)\big) u, \\ q &:= 2 + |S^n_\iota|^2 - |\omega|^2, \\ Bb &:= -2\langle \omega, db\rangle - (\mathrm{div}_M\, \omega) b + \langle h^\nu_\iota, h^n_\iota\rangle b, \\ B^* u &:= 2\langle \omega, du\rangle + (\mathrm{div}_M\, \omega) u + \langle h^\nu_\iota, h^n_\iota\rangle u \end{aligned} \tag{2.13}$$

for every $u, b \in C^\infty(M)$. Here

$$\begin{aligned} \mathrm{div}_M\, \omega &:= \sum_{i=1}^{2} (\nabla^M_{e_i} \omega)(e_i), \\ \langle h^\nu_\iota, h^n_\iota\rangle &:= \sum_{i,j=1}^{2} h^\nu_\iota(e_i, e_j) h^n_\iota(e_i, e_j), \end{aligned}$$

where (e_1, e_2) is any local g_M-orthonormal frame of TM. Then, for every $u, b \in C^\infty(M)$,

$$\begin{pmatrix} \langle J(u\nu + bn), \nu\rangle \\ \langle J(u\nu + bn), n\rangle \end{pmatrix} = \begin{pmatrix} L_{M/\mathcal{N}} + H^2_{\iota,n} & B \\ B^* & \Delta_M + q \end{pmatrix} \begin{pmatrix} u \\ b \end{pmatrix}, \tag{2.14}$$

where $H^2_{\iota,n}$ and q act by multiplication. If M is closed, then B^* is the formal $L^2(M, d\,\mathrm{vol}_M)$ adjoint of B, where $d\,\mathrm{vol}_M$ is the area measure induced by g_M.

Proof. We compute the two terms in (2.12) that depend on the geometry of M. Fix $u, b \in C^\infty(M)$ and put $V := u\nu + bn$. The connection identities (2.7) give

$$\nabla^\perp_X V = \big(Xu - b\omega(X)\big)\nu + \big(Xb + u\omega(X)\big)n \qquad \forall\, X \in \Gamma(TM).$$

To compute $\Delta^\perp V$ at $x \in M$, choose a local g_M-orthonormal frame (e_1, e_2) with $(\nabla^M_{e_i} e_j)(x) = 0$ for $i, j \in \{1, 2\}$. Set $\omega_i := \omega(e_i)$ for $i \in \{1, 2\}$ on the domain of this frame. Applying the preceding identity twice gives

$$\begin{aligned}\nabla^\perp_{e_i}\nabla^\perp_{e_i} V = {} & \big(e_i(e_i u - b\omega_i) - (e_i b + u\omega_i)\omega_i\big)\nu \\ & + \big(e_i(e_i b + u\omega_i) + (e_i u - b\omega_i)\omega_i\big)n \qquad \forall\, i \in \{1, 2\}.\end{aligned}$$

Expanding and summing at x yields

$$\begin{aligned}\Delta^\perp V = {} & \big(\Delta_M u - 2\langle \omega, db\rangle - b \operatorname{div}_M \omega - u|\omega|^2\big)\nu \\ & + \big(\Delta_M b + 2\langle \omega, du\rangle + u \operatorname{div}_M \omega - b|\omega|^2\big)n.\end{aligned}$$

Since x was arbitrary, this identity holds on M.

For the second-fundamental-form term, the decomposition of A_ι in the orthonormal frame (ν, n) gives

$$A_\iota(e_i, e_j) = h^\nu_\iota(e_i, e_j)\nu + h^n_\iota(e_i, e_j)n \qquad \forall\, i, j \in \{1, 2\}.$$

Consequently,

$$\sum_{i,j=1}^{2} \langle A_\iota(e_i, e_j), V\rangle A_\iota(e_i, e_j) = \big(|h^\nu_\iota|^2 u + \langle h^\nu_\iota, h^n_\iota\rangle b\big)\nu + \big(\langle h^\nu_\iota, h^n_\iota\rangle u + |h^n_\iota|^2 b\big)n.$$

Combining these computations with (2.12) and using $|h^\eta_\iota| = |S^\eta_\iota|$ for $\eta \in \{\nu, n\}$, we obtain

$$\begin{aligned}\langle JV, \nu\rangle &= \Delta_M u + \big(2 + |S^\nu_\iota|^2 - |\omega|^2\big)u + Bb, \\ \langle JV, n\rangle &= B^* u + (\Delta_M + q)b.\end{aligned}$$

It remains to express the first diagonal entry in terms of $L_{M/\mathcal{N}}$. The Gauss equation for $\mathcal{N} \subset \mathbb{S}^4$, together with $\operatorname{tr} S = 0$ and $|\nu| = 1$, gives

$$\operatorname{Ric}_{g_\mathcal{N}}(\nu, \nu) = 2 + (\operatorname{tr} S)\langle S(\nu), \nu\rangle - |S(\nu)|^2 = 2 - |S(\nu)|^2.$$

By (2.8) and the self-adjointness of S, the components of $S(\nu)$ in the orthonormal frame (e_1, e_2, ν) are

$$S(\nu) = \sum_{i=1}^{2} \omega(e_i)e_i - H_{\iota,n}\nu, \qquad |S(\nu)|^2 = |\omega|^2 + H^2_{\iota,n}.$$

Using $S^\nu_\iota = S^\nu_\jmath$, we conclude that

$$2 + |S^\nu_\iota|^2 - |\omega|^2 = |S^\nu_\jmath|^2 + \operatorname{Ric}_{g_\mathcal{N}}(\nu, \nu) + H^2_{\iota,n},$$

which proves (2.14).

Finally, if M is closed, integration by parts gives, for every $u, b \in C^\infty(M)$,

$$\begin{aligned}\int_M u\, Bb \, d\operatorname{vol}_M &= -2\int_M u\langle \omega, db\rangle \, d\operatorname{vol}_M - \int_M ub \operatorname{div}_M \omega \, d\operatorname{vol}_M \\ &\quad + \int_M ub\langle h^\nu_\iota, h^n_\iota\rangle \, d\operatorname{vol}_M \\ &= \int_M b\big(2\langle \omega, du\rangle + u \operatorname{div}_M \omega + u\langle h^\nu_\iota, h^n_\iota\rangle\big) \, d\operatorname{vol}_M \\ &= \int_M b\, B^* u \, d\operatorname{vol}_M .\end{aligned}$$

This proves the adjoint assertion. □

We finally construct an explicit parametrization of a neighbourhood of $\mathcal{N}$ in $\mathbb{S}^4$ by following the geodesics normal to $\mathcal{N}$. The following lemma provides a uniform range of the normal parameter on which this parametrization is an embedding.

Lemma 2.6. *Choose $b_0 \in (0, \pi/2)$ such that $\tan b_0 < \cos z_0$. Define $\Psi : \mathbb{T} \times (-z_0, z_0) \times (-b_0, b_0) \to \mathbb{S}^4$ by*

$$\Psi(s,t,z,b) := \cos b\, \mathcal{F}(s,t,z) + \sin b\, N(s,t) \tag{2.15}$$

for every (s,t,z,b) in its domain. Then Ψ is a smooth embedding, and its image is an open neighbourhood of $\mathcal{N}$ in $\mathbb{S}^4$. Moreover,

$$\Psi(s,t,z,b) = \exp^{\mathbb{S}^4}_{\mathcal{F}(s,t,z)}\big(b\, n(\mathcal{F}(s,t,z))\big)$$

for every (s,t,z,b) in its domain.

Proof. The vectors $\mathcal{F}(s,t,z)$ and $N(s,t)$ are orthonormal in $\mathbb{R}^5$. Hence the formula for Ψ defines a smooth map into $\mathbb{S}^4$. Written in the coordinates of $\mathbb{C}^2 \oplus \mathbb{R}$, this map has the form

$$\Psi(s,t,z,b) = \big(r_1 e^{i\sqrt{2}s}, r_2 e^{i\sqrt{2}t}, \cos b \sin z\big),$$
$$r_1 := \frac{\cos b \cos z + \sin b}{\sqrt{2}}, \qquad r_2 := \frac{\cos b \cos z - \sin b}{\sqrt{2}},$$

for every (s,t,z,b) in the domain. Since $|z| < z_0$ and $|b| < b_0 < \pi/2$, the choice of b_0 implies

$$r_1, r_2 \geq \frac{\cos b}{\sqrt{2}}\big(\cos z - |\tan b|\big) > \frac{\cos b}{\sqrt{2}}\big(\cos z_0 - \tan b_0\big) > 0.$$

Thus, if $(w_1, w_2, x_5) = \Psi(s,t,z,b)$, then $r_1 = |w_1|$ and $r_2 = |w_2|$. The phase factors $w_1/|w_1|$ and $w_2/|w_2|$ recover s and t uniquely as elements of $\mathbb{R}/\ell\mathbb{Z}$: the map $s \mapsto e^{i\sqrt{2}s}$ is a diffeomorphism from this quotient circle onto the unit circle in $\mathbb{C}$. The other two parameters are recovered by

$$b = \arcsin\left(\frac{|w_1| - |w_2|}{\sqrt{2}}\right),$$
$$z = \arcsin\left(\frac{x_5}{\cos b}\right),$$

using the branch of arcsin with values in $(-\pi/2, \pi/2)$. These formulas recover the original parameters because $\sin b = (r_1 - r_2)/\sqrt{2}$ and $x_5 = \cos b \sin z$, while both b and z belong to $(-\pi/2, \pi/2)$.

All the recovery maps are smooth on a neighbourhood of each point of the image: neither complex coordinate vanishes, $\cos b > 0$, and the arguments of arcsin lie strictly between -1 and 1. Their composition with Ψ is the identity on its domain. Consequently, $d\Psi$ is injective everywhere, and the recovered parameters give a continuous smooth inverse on the image. This proves that Ψ is an embedding. Its domain and target both have dimension four, so its image is open by the inverse function theorem. The image contains $\mathcal{N}$, since $\Psi(s,t,z,0) = \mathcal{F}(s,t,z)$.

Finally, for each $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$, the curve $b \mapsto \cos b\, \mathcal{F}(s,t,z) + \sin b\, N(s,t)$ is the unit-speed great circle with initial point $\mathcal{F}(s,t,z)$ and initial velocity $N(s,t) = n(\mathcal{F}(s,t,z))$. This is precisely the defining geodesic of the spherical exponential map in the last assertion. □

2.3. **Symmetries of the construction.** We will construct the initial surfaces inside $\mathcal{N}$ by joining two sheets on opposite sides of the central torus with small necks. Each neck will be parametrized using a Euclidean catenoid. The neck positions will be indexed by a square lattice $L_m \subset \mathbb{T}$, defined below: for each $p \in L_m$, we will insert one neck near the point $f(p)$ of the central torus $f(\mathbb{T})$. We will impose the same finite group of symmetries on the initial surfaces and on the maps used to deform them in $\mathbb{S}^4$.

The symmetry conditions have two uses in the later linear analysis. Invariance under lattice translations removes the kernel of $L_{\mathbb{T}} = \Delta_{\mathbb{T}} + 2$ on the functions used to construct the two sheets. Near each neck, the other symmetries impose different conditions on the two coefficients of a normal

variation. The coefficient in the direction n has no constant angular term and satisfies an angular Poincaré inequality with a constant independent of the neck size. This inequality is used to estimate the correction in the direction n, uniformly as the neck radii tend to zero. The coefficient in the normal direction within $\mathcal{N}$ can have a constant angular term; that term must be treated in the analysis of deformations within $\mathcal{N}$. We establish these facts below and verify that the normal linearization from the preceding subsection preserves the selected symmetry conditions.

Fix an integer $m \geq 2$, and define

$$L_m := \left\{ \left(\frac{j\ell}{m}, \frac{k\ell}{m} \right) : j, k \in \{0, \dots, m-1\} \right\} \subset \mathbb{T}.$$

The corresponding points of the central torus are $f(L_m)$. Define the following isometries of $\mathbb{S}^4$:

$$\begin{aligned}
R_1(w_1, w_2, x_5) &:= (e^{2\pi i/m} w_1, w_2, x_5), \\
R_2(w_1, w_2, x_5) &:= (w_1, e^{2\pi i/m} w_2, x_5), \\
C_1(w_1, w_2, x_5) &:= (\overline{w_1}, w_2, x_5), \\
C_2(w_1, w_2, x_5) &:= (w_1, \overline{w_2}, x_5), \\
\sigma(w_1, w_2, x_5) &:= (w_2, w_1, x_5), \\
Z(w_1, w_2, x_5) &:= (w_1, w_2, -x_5) \qquad \forall\, (w_1, w_2, x_5) \in \mathbb{S}^4.
\end{aligned}$$

Set

$$\Gamma_m^0 := \langle R_1, R_2, C_1, C_2, \sigma \rangle, \qquad \Gamma_m := \langle \Gamma_m^0, Z \rangle.$$

These groups are finite. Indeed, every element permutes the two complex coordinates, possibly conjugates each of them, and multiplies them by mth roots of unity; an element of Γ_m may also change the sign of the fifth coordinate. There are only finitely many maps of this form.

The definition of $\mathcal{F}$ gives the following actions on its parameters:

isometry	parameters of its value at $\mathcal{F}(s,t,z)$
R_1	$(s + \ell/m, t, z)$
R_2	$(s, t + \ell/m, z)$
C_1	$(-s, t, z)$
C_2	$(s, -t, z)$
σ	(t, s, z)
Z	$(s, t, -z)$

$\forall\, (s,t,z) \in \mathbb{T} \times (-z_0, z_0).$

For example,

$$R_1\big(\mathcal{F}(s,t,z)\big) = \frac{\cos z}{\sqrt{2}} \big(e^{i\sqrt{2}(s+\ell/m)}, e^{i\sqrt{2}t}, 0 \big) + \sin z\, e_5 = \mathcal{F}(s + \ell/m, t, z),$$

because $\sqrt{2}\ell/m = 2\pi/m$. The other entries follow by conjugating or interchanging the displayed coordinates, or by changing the sign of $\sin z$. In particular, every $g \in \Gamma_m$ preserves $\mathcal{N}$ and the central torus. Write $\widehat{g} : \mathbb{T} \to \mathbb{T}$ for the map induced by $g \in \Gamma_m$ on the first two parameters, i.e.

$$\hat{g} := f^{-1} \circ (g|_{f(\mathbb{T})}) \circ f.$$

Then

$$\pi \circ g = \widehat{g} \circ \pi, \qquad \widehat{g}(L_m) = L_m \qquad \forall\, g \in \Gamma_m.$$

The two translations induced by R_1, R_2 are denoted by

$$T_s(s,t) := (s + \ell/m, t), \qquad T_t(s,t) := (s, t + \ell/m) \qquad \forall\, (s,t) \in \mathbb{T}.$$

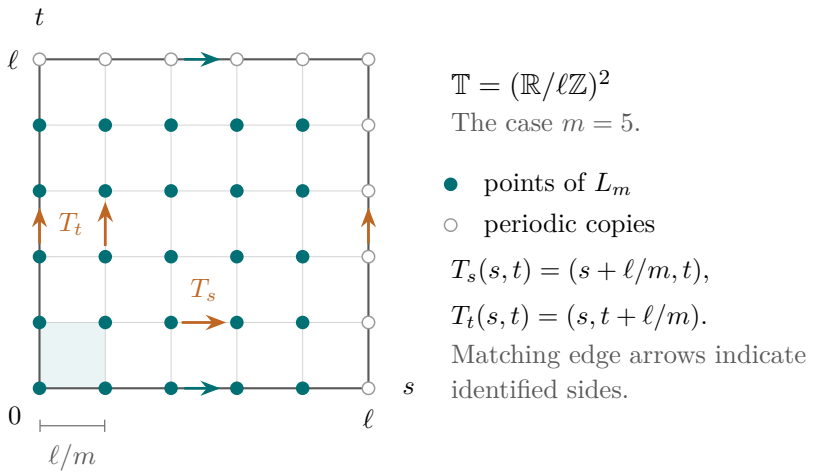


FIGURE 2. The square lattice L_m in a fundamental square of $\mathbb{T}$, with its two elementary translations. The top and right edges repeat points already represented on the bottom and left edges.

We next show that the scalar Jacobi operator of the central torus has trivial kernel on the space of smooth functions invariant under the lattice translations. For integers a, b, define the complex-valued Fourier function

$$e_{a,b}(s,t) := \exp\left(\frac{2\pi i}{\ell}(as + bt)\right) \qquad \forall\,(s,t) \in \mathbb{T}.$$

Since $\ell^2 = 2\pi^2$, differentiation gives

$$\begin{aligned} \Delta_{\mathbb{T}} e_{a,b} &= -2(a^2 + b^2)e_{a,b}, \\ L_{\mathbb{T}} e_{a,b} &= 2(1 - a^2 - b^2)e_{a,b} \qquad \forall\,(a,b) \in \mathbb{Z}^2. \end{aligned}$$

Moreover,

$$e_{a,b} \circ T_s = e^{2\pi i a/m} e_{a,b}, \qquad e_{a,b} \circ T_t = e^{2\pi i b/m} e_{a,b}.$$

Uniqueness of Fourier coefficients therefore implies that a smooth function invariant under both translations can have a nonzero coefficient at (a,b) only if $a, b \in m\mathbb{Z}$. For $m \geq 2$, no such pair satisfies $a^2 + b^2 = 1$. Hence

$$\left.\begin{array}{llll} v \in C^\infty(\mathbb{T}), & v \circ T_s = v, & v \circ T_t = v, & \\ & L_{\mathbb{T}} v = 0 & & \end{array}\right\} \quad \to \quad v = 0.$$

The constant functions are permitted by these symmetries, but they are not in the kernel, since $L_{\mathbb{T}} 1 = 2$. The later construction of the sheet heights uses the operator $L_{\mathbb{T}}$ on this translation-invariant space.

We next determine how the group Γ_m acts on the two normal directions of a surface in $\mathcal{N}$. Every $g \in \Gamma_m$ either preserves the two complex coordinate planes individually or interchanges them. Define

$$\chi(g) := \begin{cases} +1, & \text{if } g \text{ preserves the two planes individually,} \\ -1, & \text{if } g \text{ interchanges the two planes} \end{cases} \qquad \forall\, g \in \Gamma_m.$$

Interchanging the planes twice preserves them individually, so $\chi(gh) = \chi(g)\chi(h)$ for every $g, h \in \Gamma_m$. Thus $\chi : \Gamma_m \to \{1, -1\}$ is a group homomorphism, also called a *character*. It equals -1 on σ and $+1$ on each of the other five generators.

The value $\chi(g)$ determines whether the differential of g preserves or reverses the chosen unit normal field n: it sends $n(x)$ to $n(g(x))$ when $\chi(g) = +1$, and to $-n(g(x))$ when $\chi(g) = -1$, for every $x \in \mathcal{N}$. For every isometry $g : \mathbb{S}^4 \to \mathbb{S}^4$ and $x \in \mathbb{S}^4$, let $dg_x : T_x\mathbb{S}^4 \to T_{g(x)}\mathbb{S}^4$ denote its differential at

x. We first compute how the differentials of the six generators transform the normal field n. Fix $(s,t,z) \in \mathbb{T} \times (-z_0, z_0)$ and set $x := \mathcal{F}(s,t,z)$. Since $n(x) = N(s,t)$, differentiating the coordinate formulas for the generators gives

$$\begin{aligned}
(dR_1)_x(n(x)) &= N(s+\ell/m, t) = n(R_1(x)),\\
(dR_2)_x(n(x)) &= N(s, t+\ell/m) = n(R_2(x)),\\
(dC_1)_x(n(x)) &= N(-s, t) = n(C_1(x)),\\
(dC_2)_x(n(x)) &= N(s, -t) = n(C_2(x)),\\
(d\sigma)_x(n(x)) &= -N(t, s) = -n(\sigma(x))\\
(dZ)_x(n(x)) &= N(s,t) = n(Z(x)).
\end{aligned}$$

Since $\chi(\sigma) = -1$ and χ equals $+1$ on the other five generators, these computations prove

$$dg_x(n(x)) = \chi(g) n(g(x)) \qquad \forall\, g \in \{R_1, R_2, C_1, C_2, \sigma, Z\}, \quad x \in \mathcal{N}.$$

Indeed, $\chi(\sigma) = -1$, whereas χ equals $+1$ on each of the other five generators. We now show that the same identity holds for every element of Γ_m. Suppose that it holds for two elements $g, h \in \Gamma_m$. Applying first h and then g, the chain rule gives

$$\begin{aligned}
d(g\circ h)_x(n(x)) &= dg_{h(x)}\big(dh_x(n(x))\big)\\
&= \chi(h)\, dg_{h(x)}(n(h(x)))\\
&= \chi(h)\chi(g)\, n(g(h(x)))\\
&= \chi(g\circ h)\, n((g\circ h)(x)) \qquad \forall\, x \in \mathcal{N},
\end{aligned}$$

where the last equality uses that χ is a group homomorphism. Thus the identity is preserved under composition. Moreover, $R_i^{-1} = R_i^{m-1}$ for $i \in \{1,2\}$, and each of C_1, C_2, σ, Z is its own inverse. Every element of Γ_m can therefore be written as a finite composition of these six generators. We conclude that

$$dg_x(n(x)) = \chi(g) n(g(x)) \qquad \forall\, g \in \Gamma_m, \quad x \in \mathcal{N}. \tag{2.16}$$

Let $\Omega \subset \mathcal{N}$ be an open set with smooth boundary $M := \partial_{\mathcal{N}}\Omega$, and assume $g(\Omega) = \Omega$ for every $g \in \Gamma_m$. Use the notation of the preceding subsection for the inclusions $\jmath : M \hookrightarrow \mathcal{N}$ and $\iota : M \hookrightarrow \mathbb{S}^4$, their induced metric g_M, and their second fundamental forms. Choose ν to be the outward unit normal to Ω within $\mathcal{N}$. Each g preserves M, the metric, and the side occupied by Ω. Therefore,

$$dg_x\big(\nu(x)\big) = \nu(g(x)) \qquad \forall\, g \in \Gamma_m, \forall\, x \in M.$$

For $k \in \mathbb{Z}_{\geq 0}$ and $\beta \in (0,1)$, define

$$\begin{aligned}
C^{k,\beta}_{\mathrm{inv}}(M) &:= \{u \in C^{k,\beta}(M) : u \circ g = u \text{ for every } g \in \Gamma_m\},\\
C^{k,\beta}_{\chi}(M) &:= \{b \in C^{k,\beta}(M) : b \circ g = \chi(g) b \text{ for every } g \in \Gamma_m\}
\end{aligned} \tag{2.17}$$

to be the spaces of the *invariant* and χ-*equivariant* functions on M, respectively. In these compositions, g denotes its restriction to M. The same subscripts on $C^\infty(M)$ impose the same conditions on smooth functions. If $V = u\nu + bn$ is a normal vector field along ι, then

$$\begin{aligned}
dg_x\big(V(x)\big) &= u(x)\nu(g(x)) + \chi(g) b(x) n(g(x)),\\
V(g(x)) &= u(g(x))\nu(g(x)) + b(g(x)) n(g(x))
\end{aligned}$$

for every $g \in \Gamma_m$ and $x \in M$. The two vectors $\nu(g(x)), n(g(x))$ are linearly independent. Thus the condition

$$dg_x\big(V(x)\big) = V(g(x)) \qquad \forall\, g \in \Gamma_m, \forall\, x \in M,$$

which is the definition of *equivariance* for a vector field $V \in \Gamma(T_\iota^\perp M)$, is equivalent to $u \circ g = u$ and $b \circ g = \chi(g)b$.

These conditions also explain how the symmetries are preserved by deformations of the surface. For $u, b \in C^\infty(M)$, set $V := u\nu + bn$ and define

$$\begin{aligned} &G_r : M \longrightarrow \mathbb{S}^4, \\ &G_r(x) := \exp^{\mathbb{S}^4}_{\iota(x)}\big(rV(x)\big) \qquad \forall\, x \in M, \quad r \in \mathbb{R}. \end{aligned}$$

These maps are well defined for every $r \in \mathbb{R}$ because the round sphere is complete. Their being immersions requires a restriction on r. Indeed, let $U \subset M$ be an open set whose closure in M is compact. Since $G_0 = \iota$ is an immersion, continuity of the differential of G_r and compactness of $\overline{U}$ give $r_U > 0$ such that $G_r|_U$ is an immersion whenever $|r| < r_U$. If M is compact, this assertion applies with $U = M$.

An isometry maps a geodesic to the geodesic whose initial point and initial velocity are the images of the original initial point and initial velocity. Since ι is the inclusion and g preserves M, one has $\iota(g(x)) = g(\iota(x))$. Consequently, if V is equivariant, then

$$\begin{aligned} G_r(g(x)) &= \exp^{\mathbb{S}^4}_{\iota(g(x))}\big(rV(g(x))\big) \\ &= \exp^{\mathbb{S}^4}_{g(\iota(x))}\big(r\, dg_{\iota(x)}(V(x))\big) \\ &= g\left(\exp^{\mathbb{S}^4}_{\iota(x)}\big(rV(x)\big)\right) \\ &= g(G_r(x)) \end{aligned}$$

for every $g \in \Gamma_m$, $x \in M$ and $r \in \mathbb{R}$. Since $g(M) = M$, this gives $g(G_r(M)) = G_r(M)$. Whenever G_r is an embedding, its image is therefore an embedded surface preserved by Γ_m. The same calculation applies to deformations defined using $\exp^{\mathcal{N}}$, wherever these maps are defined, because each $g|_{\mathcal{N}}$ is an isometry of $(\mathcal{N}, g_{\mathcal{N}})$

We now verify that the linearized operator J preserves invariance and χ-equivariance. Recall that, for a covariant two-tensor h on M and for the previously defined one-form ω, the pullbacks by g are given by

$$\begin{aligned} &(g^*h)_x(X,Y) := h_{g(x)}(dg_x(X), dg_x(Y)), \\ &(g^*\omega)_x(X) := \omega_{g(x)}(dg_x(X)) \qquad \forall\, g \in \Gamma_m, \forall\, x \in M, \forall\, X, Y \in T_xM. \end{aligned}$$

An ambient isometry preserves the Levi-Civita connection and maps the tangent and normal spaces of M at x onto those at $g(x)$. Applying these facts to the definition of A_ι gives

$$(A_\iota)_{g(x)}(dg_x(X), dg_x(Y)) = dg_x\big((A_\iota)_x(X,Y)\big) \qquad \forall\, g \in \Gamma_m, \quad x \in M, \quad X, Y \in T_xM.$$

Pairing this identity with $\nu(g(x)) = dg_x(\nu(x))$ and $n(g(x)) = \chi(g) dg_x(n(x))$ yields

$$\begin{aligned} (g^*h^\nu_\iota)_x(X,Y) &= \big\langle dg_x\big((A_\iota)_x(X,Y)\big), dg_x(\nu(x))\big\rangle = (h^\nu_\iota)_x(X,Y), \\ (g^*h^n_\iota)_x(X,Y) &= \chi(g)\left\langle dg_x\big((A_\iota)_x(X,Y)\big), dg_x(n(x))\right\rangle = \chi(g)(h^n_\iota)_x(X,Y). \end{aligned}$$

Taking traces in an orthonormal frame, whose image under dg_x is again orthonormal, gives

$$H_{\iota,\nu} \circ g = H_{\iota,\nu}, \qquad H_{\iota,n} \circ g = \chi(g) H_{\iota,n} \qquad \forall\, g \in \Gamma_m.$$

Similarly, the one-form ω defined in (2.6) satisfies

$$\begin{aligned} (g^*\omega)_x(X) &= \left\langle \left(\nabla^{\mathbb{S}^4}_{dg_xX}\nu\right)(g(x)), n(g(x))\right\rangle \\ &= \chi(g)\left\langle dg_x\left(\left(\nabla^{\mathbb{S}^4}_X\nu\right)(x)\right), dg_x(n(x))\right\rangle \\ &= \chi(g)\omega_x(X) \qquad \forall\, g \in \Gamma_m, \quad x \in M, \quad X \in T_xM. \end{aligned}$$

Here a covariant derivative in a tangent vector at a point is evaluated using any local extension of that vector.

Isometries commute with contraction, covariant differentiation and the scalar Laplacian. The preceding identities therefore imply

$$\begin{aligned}
\big(\langle h^\nu_\iota, h^n_\iota\rangle\big)\circ g &= \chi(g)\langle h^\nu_\iota, h^n_\iota\rangle,\\
(\operatorname{div}_M \omega)\circ g &= \operatorname{div}_M(g^*\omega) = \chi(g)\operatorname{div}_M\omega,\\
q\circ g &= q,\\
(\Delta_M v)\circ g &= \Delta_M(v\circ g) \qquad \forall\, g\in\Gamma_m,\quad v\in C^\infty(M).
\end{aligned}$$

For the third identity, recall that $q = 2+|S^n_\iota|^2-|\omega|^2$ and $|S^n_\iota| = |h^n_\iota|$, so both squared norms are unchanged by the sign $\chi(g)$. The function $|S^\nu_\jmath|^2+\operatorname{Ric}_{g_\mathcal{N}}(\nu,\nu)$ in $L_{M/\mathcal{N}}$ is invariant as well, since $g|_\mathcal{N}$ is an isometry and preserves ν. Hence both diagonal operators in (2.14) commute with composition by g; here $H^2_{\iota,n}\circ g = H^2_{\iota,n}$.

For completeness, the transformation of the off-diagonal operator B follows directly from its formula in Lemma 2.5. For every $v\in C^\infty(M)$ and $g\in\Gamma_m$,

$$\begin{aligned}
(Bv)\circ g &= -2\langle g^*\omega, d(v\circ g)\rangle - \big((\operatorname{div}_M\omega)\circ g\big)(v\circ g) + \big(\langle h^\nu_\iota, h^n_\iota\rangle\circ g\big)(v\circ g)\\
&= \chi(g)\,(-2\langle\omega, d(v\circ g)\rangle - (\operatorname{div}_M\omega)(v\circ g) + \langle h^\nu_\iota, h^n_\iota\rangle(v\circ g))\\
&= \chi(g)B(v\circ g).
\end{aligned}$$

The same computation with the signs in the definition of B^* gives

$$(B^*v)\circ g = \chi(g)B^*(v\circ g) \qquad \forall\, v\in C^\infty(M),\ \forall\, g\in\Gamma_m.$$

In particular, if $b\circ g = \chi(g)b$ and $u\circ g = u$, then

$$\begin{aligned}
(Bb)\circ g &= \chi(g)B(\chi(g)b) = Bb,\\
(B^*u)\circ g &= \chi(g)B^*u.
\end{aligned}$$

Thus, the off-diagonal operators satisfy

$$\begin{aligned}
B &: C^\infty_\chi(M)\to C^\infty_{\mathrm{inv}}(M),\\
B^* &: C^\infty_{\mathrm{inv}}(M)\to C^\infty_\chi(M).
\end{aligned}$$

The first diagonal entry of J preserves $C^\infty_{\mathrm{inv}}(M)$, and the second preserves $C^\infty_\chi(M)$. Consequently, applying J to $u\nu+bn$ with u invariant and b χ-equivariant produces a normal field whose ν-coefficient is invariant and whose n-coefficient is χ-equivariant.

We finish by describing the consequences near a neck. Choose $\tau>0$ and $\xi_*>0$ such that

$$\tau\cosh\xi_* < \frac{\ell}{2m}, \qquad \tau\xi_* < z_0. \tag{2.18}$$

For each $p=(p_s,p_t)\in L_m$, define

$$\begin{aligned}
F_{\tau,p} : (-\xi_*,\xi_*)\times(\mathbb{R}/2\pi\mathbb{Z}) &\to \mathcal{N},\\
(\xi,\theta) &\mapsto \mathcal{F}\big(p+\tau\cosh\xi(\cos\theta,\sin\theta),\tau\xi\big).
\end{aligned}$$

The first argument of $\mathcal{F}$ is computed in the flat torus $\mathbb{T}$. The first inequality in (2.18) places it in an embedded coordinate disk about p containing no other lattice point, and the second keeps the height inside $(-z_0,z_0)$. In these coordinates, the map is the usual catenoid parametrization, followed by $\mathcal{F}$. Its coordinate derivatives before applying $\mathcal{F}$ are

$$\begin{aligned}
\partial_\xi\big(p+\tau\cosh\xi(\cos\theta,\sin\theta),\tau\xi\big) &= \big(\tau\sinh\xi\cos\theta,\tau\sinh\xi\sin\theta,\tau\big),\\
\partial_\theta\big(p+\tau\cosh\xi(\cos\theta,\sin\theta),\tau\xi\big) &= \big(-\tau\cosh\xi\sin\theta,\tau\cosh\xi\cos\theta,0\big).
\end{aligned}$$

These two coordinate derivatives are linearly independent: the last component of the first is $\tau > 0$, whereas the last component of the second is zero and the second vector is nonzero. For a point $x \in \mathcal{N}$, write

$$\mathcal{F}^{-1}(x) = (q, z), \qquad q = \pi(x) \in \mathbb{T}, \quad z \in (-z_0, z_0).$$

and recall that z is the signed distance coordinate from the central torus within $(\mathcal{N}, g_{\mathcal{N}})$; see Remark 2.1. Therefore, we say that z is the *height* of x. For $x = F_{\tau,p}(\xi, \theta)$, the defining formula gives

$$z = \tau\xi, \qquad q = p + \tau \cosh\xi(\cos\theta, \sin\theta).$$

In particular, we recover the parameters (ξ, θ) from x by

$$\xi = \frac{z}{\tau}, \qquad (\cos\theta, \sin\theta) = \frac{q-p}{|q-p|},$$

where $q - p$ is computed in the flat coordinate disk about p. The denominator is nonzero because $|q - p| = \tau \cosh\xi \geq \tau > 0$. These formulas give a smooth inverse on the image, with θ understood modulo 2π. Together with the linear independence of the coordinate derivatives and the fact that $\mathcal{F}$ is an embedding, this proves that $F_{\tau,p}$ is an embedding. We call the image of $F_{\tau,p}$ the *catenoidal model neck* at p. Note that a catenoidal model need not be minimal in $(\mathcal{N}, g_{\mathcal{N}})$.

For each fixed $\xi \in (-\xi_*, \xi_*)$, varying θ gives a closed curve at height $z = \tau\xi$. Its projection under π is the circle in the flat torus centered at p with radius

$$r(\xi) := \tau \cosh\xi.$$

Since $\cosh\xi \geq 1$, with equality only at $\xi = 0$, this projected radius is smallest when $\xi = 0$. We call the corresponding closed curve the *waist* of the neck. Explicitly,

$$F_{\tau,p}(0, \theta) = f\big(p + \tau(\cos\theta, \sin\theta)\big) \qquad \forall\, \theta \in \mathbb{R}/2\pi\mathbb{Z}.$$

Thus the waist lies in the central torus, and its projection is the circle of radius τ about p. The portions with $\xi > 0$ and $\xi < 0$ lie on opposite sides of the central torus and meet along the waist.

Only group elements fixing p on the base torus preserve this individual neck. Denote this subgroup by

$$\Gamma_{m,p} := \{g \in \Gamma_m : \widehat{g}(p) = p\}.$$

For $p = (j\ell/m, k\ell/m) \in L_m$, set $T_p := R_1^j R_2^k$. Its induced translation sends 0 to p. Define

$$C_{2,p} := T_p C_2 T_p^{-1}, \qquad \sigma_p := T_p \sigma T_p^{-1}, \qquad Q_p := \sigma_p C_{2,p}.$$

These maps and Z belong to $\Gamma_{m,p}$. At $p = 0$, C_2 changes $(r\cos\theta, r\sin\theta)$ to $(r\cos\theta, -r\sin\theta)$, whereas σ changes it to $(r\sin\theta, r\cos\theta)$. Their composition σC_2 changes it to $(-r\sin\theta, r\cos\theta)$. Conjugation by T_p gives the same actions on the displacement from any $p \in L_m$. Finally, $\cosh(-\xi) = \cosh\xi$, so Z changes ξ to $-\xi$ without changing θ. We have therefore proved the following table:

g	map ϑ_g satisfying $g \circ F_{\tau,p} = F_{\tau,p} \circ \vartheta_g$	$\chi(g)$
$C_{2,p}$	$(\xi, \theta) \mapsto (\xi, -\theta)$	$+1$
σ_p	$(\xi, \theta) \mapsto (\xi, \pi/2 - \theta)$	-1
Q_p	$(\xi, \theta) \mapsto (\xi, \theta + \pi/2)$	-1
Z	$(\xi, \theta) \mapsto (-\xi, \theta)$	$+1$

All angular expressions are understood modulo 2π. The signs are unchanged by conjugation, because $\chi(T_p) = 1$.

Let b be a smooth function on this model neck satisfying $b \circ g = \chi(g)b$ for $g \in \Gamma_{m,p}$. For the following calculations, write $b(\xi, \theta)$ for its pullback by $F_{\tau,p}$. The table above gives

$$b(\xi, -\theta) = b(\xi, \theta),$$

$$b(\xi,\pi/2-\theta) = -b(\xi,\theta),$$
$$b(\xi,\theta+\pi/2) = -b(\xi,\theta),$$
$$b(-\xi,\theta) = b(\xi,\theta)$$

for every $\xi \in (-\xi_*,\xi_*)$ and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. To determine the angular Fourier terms, write

$$b(\xi,\theta) = \sum_{k\in\mathbb{Z}} \widehat{b}_k(\xi) e^{ik\theta}, \qquad \widehat{b}_k(\xi) := \frac{1}{2\pi}\int_0^{2\pi} b(\xi,\theta) e^{-ik\theta}\, d\theta.$$

The quarter-turn identity implies $(e^{ik\pi/2}+1)\widehat{b}_k(\xi) = 0$. Thus, a nonzero coefficient requires $k \equiv 2 \pmod 4$. The reflection $\theta \mapsto -\theta$ makes the coefficients at k and $-k$ equal, removing the sine terms. The reflection in ξ makes each remaining coefficient even. Hence

$$b(\xi,\theta) = \sum_{j\geq 0} b_j(\xi)\cos((4j+2)\theta), \qquad b_j(-\xi) = b_j(\xi)$$

for every (ξ,θ) in the parameter domain. Here

$$b_j(\xi) := \frac{1}{\pi}\int_0^{2\pi} b(\xi,\theta)\cos((4j+2)\theta)\, d\theta \qquad \forall\, j \in \mathbb{Z}_{\geq 0}, \quad \xi \in (-\xi_*,\xi_*).$$

The Fourier series and its derivatives converge on compact subsets of this domain, since b is smooth. Orthogonality of the trigonometric functions gives

$$\int_0^{2\pi} |b(\xi,\theta)|^2\, d\theta = \pi \sum_{j\geq 0} |b_j(\xi)|^2,$$
$$\int_0^{2\pi} |\partial_\theta b(\xi,\theta)|^2\, d\theta = \pi \sum_{j\geq 0} (4j+2)^2 |b_j(\xi)|^2$$

for every $\xi \in (-\xi_*,\xi_*)$. Since $(4j+2)^2 \geq 4$,

$$\int_0^{2\pi} |\partial_\theta b(\xi,\theta)|^2\, d\theta \geq 4\int_0^{2\pi} |b(\xi,\theta)|^2\, d\theta \qquad \forall\, \xi \in (-\xi_*,\xi_*). \tag{2.19}$$

The constant 4 depends only on the permitted angular orders. In particular, it is independent of τ and ξ_*.

The same symmetries determine the conditions on the parameter rectangle $[0,\xi_*)\times[0,\pi/4]$. Differentiating $b(\xi,-\theta) = b(\xi,\theta)$ at $\theta = 0$ gives $-\partial_\theta b(\xi,0) = \partial_\theta b(\xi,0)$. Evaluating $b(\xi,\pi/2-\theta) = -b(\xi,\theta)$ at $\theta = \pi/4$ gives $b(\xi,\pi/4) = -b(\xi,\pi/4)$. Differentiating $b(-\xi,\theta) = b(\xi,\theta)$ at $\xi = 0$ gives the corresponding equality for $\partial_\xi b$. Therefore

$$\partial_\theta b(\xi,0) = 0, \quad b(\xi,\pi/4) = 0 \qquad \forall\, \xi \in [0,\xi_*)$$

and

$$\partial_\xi b(0,\theta) = 0 \qquad \forall\, \theta \in [0,\pi/4].$$

The zero-value condition is a homogeneous Dirichlet condition. The zero normal-derivative conditions are homogeneous Neumann conditions. To relate the displayed coordinate derivatives to normal derivatives, direct differentiation of $F_{\tau,p}$ gives

$$F_{\tau,p}^* g_{\mathcal{N}} = \tau^2\big(1+\cos^2(\tau\xi)\sinh^2\xi\big)\, d\xi^2 + \tau^2\cos^2(\tau\xi)\cosh^2\xi\, d\theta^2.$$

This metric is diagonal and positive definite. Consequently, ∂_θ is perpendicular to each angular boundary and ∂_ξ is perpendicular to the boundary $\xi = 0$. These boundary conditions follow from restrictions of smooth functions with the stated symmetries. We impose no boundary condition at the artificial truncation $\xi = \xi_*$. Figure 3 illustrates the three reflection conditions.

For comparison, let u be a smooth function on the model neck invariant under $\Gamma_{m,p}$, and again use its pullback by $F_{\tau,p}$. Its quarter-turn identity is $u(\xi,\theta+\pi/2)=u(\xi,\theta)$. The Fourier calculation above now permits only $k\equiv 0 \pmod 4$. The two evenness conditions give

$$u(\xi,\theta)=u_0(\xi)+\sum_{j\geq 1}u_j(\xi)\cos(4j\theta), \qquad u_j(-\xi)=u_j(\xi) \quad \forall\, j\in\mathbb{Z}_{\geq 0},$$

where

$$u_0(\xi):=\frac{1}{2\pi}\int_0^{2\pi}u(\xi,\theta)\,d\theta,$$
$$u_j(\xi):=\frac{1}{\pi}\int_0^{2\pi}u(\xi,\theta)\cos(4j\theta)\,d\theta \qquad \forall\, j\geq 1, \quad \xi\in(-\xi_*,\xi_*).$$

The expansion holds for every (ξ,θ) in the parameter domain. Invariance under $\theta\mapsto\pi/2-\theta$ gives zero angular derivative at $\theta=\pi/4$, by differentiating that identity at its fixed point. The other two reflections are the same as for b. Thus

$$\partial_\theta u(\xi,0)=0, \quad \partial_\theta u(\xi,\pi/4)=0 \qquad \forall\,\xi\in[0,\xi_*)$$

and

$$\partial_\xi u(0,\theta)=0 \qquad \forall\,\theta\in[0,\pi/4].$$

Both u and b therefore have zero coefficients of $\cos\theta$ and $\sin\theta$, which are the first angular harmonics. However, u_0 need not vanish. For example, a nonzero constant function satisfies all the conditions on u and has zero angular derivative, so no inequality of the form (2.19) can hold for all invariant u. This explains why the later estimates for the correction in the direction n use angular Poincaré inequalities, whereas the analysis within $\mathcal{N}$ must also handle the angular average u_0. The model calculation here concerns a single catenoidal neck; the later analysis also treats its attachment to the two sheets.

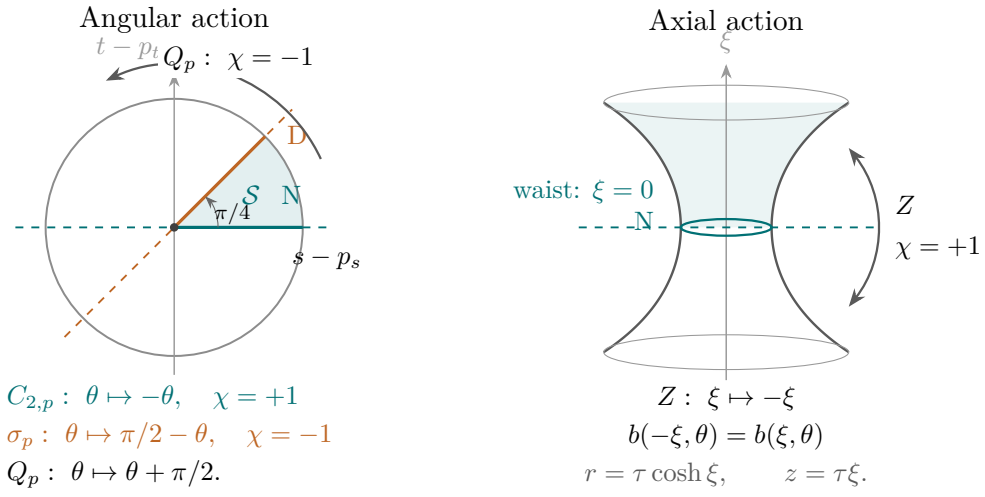


FIGURE 3. The neck symmetries and the boundary conditions they impose on a χ-equivariant coefficient b. Left: the sector $\mathcal{S}=\{0\leq\theta\leq\pi/4\}$ has Neumann data (N) on $\theta=0$ and Dirichlet data (D) on $\theta=\pi/4$. Right: reflection across the waist makes b even in ξ, giving Neumann data at $\xi=0$ on the upper half-neck. Shading only identifies the chosen sector and half-neck.

3. Construction of the equivariant initial surfaces

For each sufficiently large integer m and each $\zeta \in [-1,1]$, we construct a smooth closed embedded surface $M_\zeta \subset \mathcal{N}$. It consists of two sheets on opposite sides of the central torus, joined by one catenoidal neck near each of the m^2 lattice points in L_m. We then establish the estimates needed to deform this surface within $\mathcal{N}$. Throughout this section, scalar mean curvature and normal displacements of a surface refer to its inclusion into $(\mathcal{N}, g_{\mathcal{N}})$.

We first construct and estimate a Green function on a fixed flat torus. After rescaling, it prescribes the height of the sheets away from the lattice points. Its logarithmic terms agree with those of the ends of a catenoid; comparing the constant terms determines the neck radius τ_ζ. We subtract the remaining constant difference and interpolate between these graph heights and the catenoids. The resulting surface is denoted by M_ζ, and we prove its embeddedness, topology, and geometric estimates directly.

The final two subsections define the function spaces and establish the scalar estimates used later. We identify all M_ζ with a fixed reference surface M_0 so that the unknown functions have a fixed domain when ζ varies. Using the linear construction of Kapouleas–McGrath, we solve the scalar linearized equation up to a multiple of one specified function. We retain that coefficient and estimate the nonlinear error for every allowed scalar displacement. Section 4 will add a displacement normal to $\mathcal{N}$, and Section 5 will choose the scalar displacement and ζ so that both components of the spherical mean-curvature vector vanish. No exact minimality is imposed in the present section.

All constants below are independent of m and $\zeta \in [-1,1]$ after the fixed cutoffs and exponents have been chosen. Their dependence on a fixed derivative order or Hölder exponent is allowed. Every assertion for sufficiently large m uses a threshold independent of ζ; we may increase that threshold finitely many times.

3.1. **A Green function on the fixed square torus.** We first construct a function with one logarithmic singularity on a fixed flat torus. Rescaling this function will give a solution of $L_{\mathbb{T}}\Phi_m = 0$ away from L_m, with the same logarithmic singularity at every lattice point. We need estimates for its value and derivatives to construct the initial surfaces. We also prove a lower bound for its Hessian; this bound will later be used to establish negative Gaussian curvature between the necks.

Let $\mathbb{T}_0 := (\mathbb{R}/2\pi\mathbb{Z})^2$, equipped with the flat metric $dX_1^2 + dX_2^2$. Write dX for its area measure, D_0 for its flat covariant derivative, and Δ_0 for its Laplace–Beltrami operator. Thus $\mathrm{Area}(\mathbb{T}_0) = 4\pi^2$. Let δ_0 be the Dirac measure of mass one at the origin of $\mathbb{T}_0$. For every $a \in [0, 1/4]$, define

$$G_a(X) := \sum_{n \in \mathbb{Z}^2 \setminus \{0\}} \frac{e^{in\cdot X}}{a - |n|^2}. \tag{3.1}$$

Here $X = (X_1, X_2)$ and $n \cdot X := n_1X_1 + n_2X_2$ for $n = (n_1, n_2) \in \mathbb{Z}^2$. The series converges in $L^2(\mathbb{T}_0)$ because its coefficients are square summable. We use the Fourier normalization

$$\widehat{v}(n) := \frac{1}{4\pi^2} \int_{\mathbb{T}_0} v(X) e^{-in\cdot X}\, dX \qquad \forall\, n \in \mathbb{Z}^2$$

for integrable functions v. In this normalization, $4\pi^2\delta_0 = \sum_{n\in\mathbb{Z}^2} e^{in\cdot X}$ distributionally. Multiplication of each Fourier coefficient in (3.1) by $a - |n|^2$ therefore gives

$$(\Delta_0 + a)G_a = 4\pi^2\delta_0 - 1, \qquad \int_{\mathbb{T}_0} G_a\, dX = 0.$$

These equations uniquely determine G_a: the mean-zero condition fixes its coefficient at $n = 0$, and $a - |n|^2 \neq 0$ for every $n \neq 0$. The coefficients at n and $-n$ are equal and real, so G_a is real valued. Since they depend only on $|n|^2$, the function is invariant under both coordinate reflections and interchange of the coordinates. By standard elliptic regularity, we have $G_a \in C^\infty(\mathbb{T}_0 \setminus \{0\})$.

To state estimates near the singularity, let $\mathcal{A}_R := \{X : R/2 < |X| < 2R\}$, where $R > 0$ is small enough that this annulus lies in the flat coordinate disk centred at the origin. For a nonnegative integer k, $\beta \in (0,1)$ and $v \in C^{k,\beta}(\mathcal{A}_R)$, define the rescaled Hölder norm

$$\|v\|_{k,\beta;R} := \sum_{j=0}^{k} R^j \|D_0^j v\|_{L^\infty(\mathcal{A}_R)} + R^{k+\beta}[D_0^k v]_{\beta;\mathcal{A}_R},$$

where $D_0^0 v := v$ and, for a tensor field T on the annulus,

$$[T]_{\beta;\mathcal{A}_R} := \sup_{\substack{X,Y\in\mathcal{A}_R \\ X\neq Y}} \frac{|T(X) - T(Y)|}{|X-Y|^\beta}.$$

Tensor components are compared in the parallel orthonormal frame of the flat coordinates, and all pointwise norms use the flat metric. By the chain rule, $\|v\|_{k,\beta;R}$ is the corresponding $C^{k,\beta}$ norm of $Y \mapsto v(RY)$ on $\{Y : 1/2 < |Y| < 2\}$. For annuli in $\mathbb{T}$, we use the same definition in flat coordinates centred at the specified point, with D_0 replaced by the flat covariant derivative D of $g_{\mathbb{T}}$.

Lemma 3.1 (Uniform estimates for G_a). *Set $r_* := 1/4$. There exist constants $a_* \in (0, 1/4]$ and $c, C > 0$ with the following properties. For every $a \in [0, 1/4]$, there are a number $c_a \in \mathbb{R}$ and a smooth function E_a on $B(0, r_*) \setminus \{0\}$ such that*

$$G_a(X) = 2\pi \log r + c_a + E_a(X), \qquad |c_a| \le C, \qquad |c_a - c_0| \le Ca, \tag{3.2}$$

for every $X \in B(0, r_*) \setminus \{0\}$ and $r := |X|$. For every nonnegative integer k and every $\beta \in (0,1)$, there exists $C_{k,\beta} > 0$ such that

$$\|E_a\|_{k,\beta;R} \le C_{k,\beta} R^2 (1 + a|\log R|) \qquad \forall\, a \in [0, 1/4], \quad 0 < R < r_*/2. \tag{3.3}$$

On the whole punctured torus, for every $a \in [0, 1/4]$, $X \in \mathbb{T}_0 \setminus \{0\}$ and $r := d_{\mathbb{T}_0}(X, 0)$, one has

$$|G_a(X) - 2\pi \log r| \le C, \qquad |D_0^j G_a(X)| \le C_j (1 + r^{-j}) \quad \forall\, j \ge 1. \tag{3.4}$$

Finally, for every $a \in [0, a_*]$ and $X \in \mathbb{T}_0 \setminus \{0\}$, with the same definition of r, one has

$$|D_0^2 G_a(X)|^2 \ge c(1 + r^{-4}),$$
$$2a|D_0 G_a(X)|^2 - 2|D_0^2 G_a(X)|^2 \le -c(1 + r^{-4}).$$

All constants are independent of a in the stated ranges.

Proof. We first obtain uniform bounds away from the origin and compare G_a with G_0. For every $n \in \mathbb{Z}^2 \setminus \{0\}$ and $a \in [0, 1/4]$,

$$\left| \frac{1}{a - |n|^2} \right| \le \frac{4}{3|n|^2},$$
$$\widehat{G_a - G_0}(n) = -\frac{a}{|n|^2(|n|^2 - a)}.$$

Parseval's identity and the first inequality give a uniform L^2 bound for G_a. The absolute values in the second line have sum at most $(4a/3)\sum_{n\neq 0} |n|^{-4} \le Ca$. Thus $G_a - G_0$ has a uniformly convergent Fourier series and

$$\|G_a - G_0\|_{C^0(\mathbb{T}_0)} \le Ca.$$

On a fixed compact subset of $\mathbb{T}_0 \setminus \{0\}$, the equation $(\Delta_0 + a)G_a = -1$ and the L^2 bound give uniform bounds for every derivative by interior elliptic estimates.

To identify the logarithmic singularity and its remainder, we construct a radial fundamental solution. The following power series are defined for every $a \geq 0$; only $a \in [0, 1/4]$ is needed for the uniform estimates in this lemma. Set $h_0 := 0$ and $h_j := \sum_{i=1}^{j} i^{-1}$ for every integer $j \geq 1$, and define

$$d_j(a) := \frac{(-a/4)^j}{(j!)^2} \qquad \forall\, j \geq 0,$$

$$\mathcal{B}_a(r) := \sum_{j=0}^{\infty} d_j(a) r^{2j} = J_0(\sqrt{a}\, r),$$

$$W_a(r) := -\sum_{j=1}^{\infty} h_j d_j(a) r^{2j} \qquad \forall\, r \geq 0,$$

where $J_0(x) := \sum_{j=0}^{\infty} (-1)^j (x^2/4)^j/(j!)^2$ is the Bessel function of order zero. These series and all their derivatives converge uniformly when a and r range over compact intervals. For the radial operator $\mathcal{D}_a := \partial_r^2 + r^{-1}\partial_r + a$, the identity

$$4(j+1)^2 d_{j+1}(a) + a d_j(a) = 0 \qquad \forall\, j \geq 0$$

gives $\mathcal{D}_a \mathcal{B}_a = 0$ on $(0, \infty)$. The coefficient of r^{2j} in $\mathcal{D}_a W_a$ is

$$\begin{aligned} -4(j+1)^2 h_{j+1} d_{j+1}(a) - a h_j d_j(a) &= a(h_{j+1} - h_j) d_j(a) \\ &= \frac{a}{j+1} d_j(a) = -4(j+1) d_{j+1}(a) \qquad \forall\, j \geq 0. \end{aligned}$$

These are precisely the coefficients of $-2\mathcal{B}_a'(r)/r$. The product rule also gives

$$\mathcal{D}_a(\mathcal{B}_a \log r) = (\mathcal{D}_a \mathcal{B}_a) \log r + \frac{2\mathcal{B}_a'(r)}{r} = \frac{2\mathcal{B}_a'(r)}{r} \qquad \forall\, r > 0.$$

Consequently the function

$$P_a(r) := 2\pi \left[J_0(\sqrt{a}\, r) \log r + \sum_{j \geq 1} \frac{(-1)^{j+1} h_j}{(j!)^2} \left(\frac{a r^2}{4} \right)^j \right] \qquad \forall\, r > 0 \tag{3.5}$$

satisfies $\mathcal{D}_a P_a = 0$ away from the origin.

The series give, as $r \downarrow 0$ and uniformly for $a \in [0, 1/4]$,

$$\begin{aligned} P_a(r) &= 2\pi \log r + O\big(a r^2 (1 + |\log r|)\big), \\ P_a'(r) &= \frac{2\pi}{r} + O\big(a r (1 + |\log r|)\big). \end{aligned}$$

We also regard $X \mapsto P_a(|X|)$ as a locally integrable function on the flat disk $B(0,1)$. For $\varphi \in C_c^\infty(B(0,1))$, integration by parts on $B(0,1) \smallsetminus B(0,\epsilon)$ gives

$$\int_{B(0,1)\smallsetminus B(0,\epsilon)} P_a(\Delta_0 + a)\varphi \, dX = \int_{\partial B(0,\epsilon)} \big(\varphi P_a' - P_a \partial_r \varphi \big) \, d\sigma,$$

where $d\sigma$ is Euclidean arc length and ∂_r points away from the origin. The boundary term containing $P_a \partial_r \varphi$ tends to zero, since its absolute value is $O(\epsilon |\log \epsilon|)$. The other boundary term tends to $(2\pi)(2\pi)\varphi(0) = 4\pi^2 \varphi(0)$. It follows that $(\Delta_0 + a) P_a = 4\pi^2 \delta_0$ distributionally on $B(0,1)$.

Define $Q_a := G_a - P_a$ on this disk. Subtracting the distributional equations gives $(\Delta_0 + a) Q_a = -1$. Both terms defining Q_a have uniformly bounded L^2 norms on $B(0,1)$. Interior elliptic regularity therefore shows that Q_a is smooth across the origin and, for every nonnegative integer j,

$$\sup_{a \in [0,1/4]} \| D_0^j Q_a \|_{L^\infty(B(0,1/2))} < \infty.$$

Set $c_a := Q_a(0)$. Since $P_a - P_0$ tends to zero at the origin, evaluating the uniformly convergent series for $G_a - G_0$ there gives

$$c_a - c_0 = -a \sum_{n \in \mathbb{Z}^2 \smallsetminus \{0\}} \frac{1}{|n|^2(|n|^2 - a)}.$$

In particular, $|c_a - c_0| \le Ca$ and $|c_a| \le C$. The coordinate reflection symmetries of G_a and the radiality of P_a imply $D_0 Q_a(0) = 0$. Taylor's formula gives

$$Q_a(RY) - Q_a(0) = R^2 \sum_{i,j=1}^{2} Y_i Y_j \int_0^1 (1-t)(\partial_{ij} Q_a)(tRY)\, dt$$

whenever $0 < R < r_*/2$ and $1/2 < |Y| < 2$. Differentiating with respect to Y and using the bounds for the derivatives of Q_a proves

$$\|Q_a - Q_a(0)\|_{k,\beta;R} \le C_{k,\beta} R^2.$$

Substituting $r = R|Y|$ in the power series for P_a similarly gives

$$\|P_a - 2\pi \log r\|_{k,\beta;R} \le C_{k,\beta} a R^2 (1 + |\log R|).$$

Thus (3.2) and (3.3) follow upon setting $E_a := Q_a - Q_a(0) + P_a - 2\pi \log r$. The same estimates give (3.4) near the origin. On the complement of a fixed small disk, they follow from the previously established derivative bounds and the fact that the distance to the origin is bounded above and bounded away from zero. No derivatives of the distance function are taken in (3.4).

It remains to prove the Hessian estimates. The local expansion and its second-derivative bound give

$$D_0^2 G_a(X) = \frac{2\pi}{r^2}(I - 2e_r \otimes e_r) + O(1 + a|\log r|), \qquad e_r := X/r,$$

for $0 < r = |X| < r_*$. Here I is the identity matrix in the flat orthonormal coordinates. The first term has norm $2\sqrt{2}\pi/r^2$. Choose $r_1 \in (0, r_*)$ sufficiently small that the error has at most half this norm for every $0 < r < r_1$ and $a \in [0, 1/4]$. Such a choice is possible because $r^2(1 + |\log r|) \to 0$. Therefore $|D_0^2 G_a|^2 \ge c r^{-4}$ on that disk.

On $\mathbb{T}_0 \smallsetminus B(0, r_1)$, let M_1 be a uniform upper bound for $|G_a|$. Choose $a_* \in (0, 1/4]$ so that $a_* M_1 \le 1/2$. For a symmetric 2×2 matrix H, the inequality $|H|^2 \ge \frac{1}{2}(\operatorname{tr} H)^2$ follows from Cauchy–Schwarz. Applying it to $D_0^2 G_a$ gives

$$|D_0^2 G_a|^2 \ge \frac{1}{2}|\Delta_0 G_a|^2 = \frac{1}{2}|1 + a G_a|^2 \ge \frac{1}{8} \qquad \forall\, a \in [0, a_*]$$

on the complement of the disk. Combining the two regions yields $|D_0^2 G_a|^2 \ge c_1(1 + r^{-4})$ for some $c_1 > 0$. By (3.4), there is also $C_1 > 0$ such that $|D_0 G_a|^2 \le C_1(1 + r^{-4})$. Decrease a_*, if necessary, so that $2a_* C_1 \le c_1$. Then

$$2a|D_0 G_a|^2 - 2|D_0^2 G_a|^2 \le (2aC_1 - 2c_1)(1 + r^{-4}) \le -c_1(1 + r^{-4}),$$

as required. □

3.2. **The lattice solution and the choice of neck radii.** We now construct the function on $\mathbb{T}$ that determines the two sheets away from the necks. We first rescale G_a so that its singularities occur precisely at L_m and its equation becomes $L_{\mathbb{T}} \Phi_m = 0$ away from L_m. We then compare its expansion near a lattice point with the height of a catenoid. This comparison determines the reference neck radius τ_0 and the error represented by ζ.

For an integer $m \geq 2$, set

$$\begin{aligned} a &:= m^{-2}, \qquad \kappa := \sqrt{2}m, \\ U_m(s,t) &:= 1 + aG_a(\kappa s, \kappa t), \\ \Phi_m(s,t) &:= \frac{m^2}{2\pi} U_m(s,t) = \frac{m^2}{2\pi} + \frac{1}{2\pi} G_a(\kappa s, \kappa t), \end{aligned} \tag{3.6}$$

for every $(s,t) \in \mathbb{T} \smallsetminus L_m$. The arguments of G_a are taken modulo 2π in each coordinate. Since $\kappa\ell = 2\pi m$, the map $(s,t) \mapsto (\kappa s, \kappa t)$ is a well-defined covering of $\mathbb{T}_0$ by $\mathbb{T}$. The inverse image of the origin is exactly L_m. The symmetries of G_a imply that U_m and Φ_m are invariant under the action of Γ_m^0 on $\mathbb{T}$.

The normalization in (3.6) and Lemma 3.1 give

$$L_{\mathbb{T}} \Phi_m = 2\pi \sum_{p \in L_m} \delta_p, \qquad \frac{1}{2\pi^2} \int_{\mathbb{T}} \Phi_m \, ds\, dt = \frac{m^2}{2\pi}, \tag{3.7}$$

where the first equality holds distributionally and δ_p is the Dirac measure of mass one at p. Indeed, for $q = (s,t) \notin L_m$ and $X := (\kappa s, \kappa t) \in \mathbb{T}_0$, the chain rule gives

$$\begin{aligned} L_{\mathbb{T}} U_m(q) &= a\kappa^2 \Delta_0 G_a(X) + 2 + 2aG_a(X) \\ &= 2\big(-1 - aG_a(X)\big) + 2 + 2aG_a(X) = 0. \end{aligned}$$

Near each $p \in L_m$, use the flat variable $q - p$ and write $r_p := |q - p|$. The local expansion of G_a gives $\Phi_m(q) = \log r_p + O(1)$, with remainder after subtraction of the constant of order $r_p^2(1 + |\log r_p|)$ for fixed m. The first derivative of this remainder has order $r_p(1 + |\log r_p|)$. Integration by parts around p, as in the preceding proof, therefore gives a point mass $2\pi\delta_p$ and no other distribution supported at p. This proves the first identity in (3.7). The average of $G_a(\kappa s, \kappa t)$ on $\mathbb{T}$ is zero, because the covering consists of m^2 identical square cells. This proves the second identity.

For $q \in \mathbb{T}$, let $\rho(q) := d_{\mathbb{T}}(q, L_m)$. Each point lies within distance $\ell/(\sqrt{2}m) = \pi/m$ of a lattice point. Moreover, the covering and the flat metrics give the exact distance relation

$$d_{\mathbb{T}_0}(X, 0) = \kappa\rho(q) \qquad \forall\, q = (s,t) \in \mathbb{T}, \quad X := (\kappa s, \kappa t).$$

Hence Lemma 3.1 implies

$$\Phi_m(q) = \frac{m^2}{2\pi} + \log(\kappa\rho(q)) + O(1), \tag{3.8}$$

$$|D^j \Phi_m(q)| \leq C_j \rho(q)^{-j} \qquad \forall\, j \geq 1, \tag{3.9}$$

$$|D^2 \Phi_m(q)|^2 \geq c\big(m^4 + \rho(q)^{-4}\big)$$

for every $q \in \mathbb{T} \smallsetminus L_m$ and all sufficiently large m, chosen so that $m^{-2} \leq a_*$. Here and below D is the flat covariant derivative of $g_{\mathbb{T}}$. To see each scaling factor explicitly, in the parallel coordinate frames one has

$$\begin{aligned} D^j \Phi_m(q) &= \frac{\kappa^j}{2\pi} D_0^j G_a(X) \qquad \forall\, j \geq 1, \\ |D^j \Phi_m(q)| &\leq C_j\big(\kappa^j + \rho(q)^{-j}\big) \leq C_j' \rho(q)^{-j}, \\ |D^2 \Phi_m(q)|^2 &= \frac{m^4}{\pi^2} |D_0^2 G_a(X)|^2 \geq c\big(m^4 + \rho(q)^{-4}\big). \end{aligned}$$

The second inequality uses $m\rho(q) \leq \pi$. The derivative bounds also give local Hölder estimates. Precisely, if $B_q := B(q, \rho(q)/8)$, then

$$\sum_{i=0}^{k} \rho(q)^i \|D^{j+i} \Phi_m\|_{L^\infty(B_q)} + \rho(q)^{k+\beta} [D^{j+k} \Phi_m]_{\beta; B_q} \leq C_{j,k,\beta} \rho(q)^{-j}$$

for all integers $j \geq 1$, $k \geq 0$ and $\beta \in (0,1)$. Indeed, ρ is bounded below by $7\rho(q)/8$ on B_q. Apply (3.9) there, and use one more derivative and the mean-value estimate to bound the Hölder seminorm. All tensor comparisons are in the flat frame.

The combination of derivatives that will enter the curvature estimate also has a uniform sign. The chain rule gives $DU_m = (\sqrt{2}/m)D_0G_a$ and $D^2U_m = 2D_0^2G_a$, where the right-hand sides are evaluated at X. Consequently, with $r := \kappa\rho(q)$,

$$|DU_m(q)|^2 - \tfrac{1}{2}|D^2U_m(q)|^2 = 2a|D_0G_a(X)|^2 - 2|D_0^2G_a(X)|^2 \leq -c(1+r^{-4}). \tag{3.10}$$

This inequality is stronger than the assertion that the Hessian does not vanish, as it compares the negative Hessian term directly with the positive gradient term.

We next fix the scales used to join the sheets to the catenoids. Fix $\beta \in (0,1)$ and choose $\alpha > 0$ sufficiently small for Convention 2.10 of [KM23], with the additional restrictions

$$\alpha < \min\left\{\frac{1}{8}, \frac{1}{5}, \frac{1}{9(3+\beta)}\right\}. \tag{3.11}$$

A further restriction on α, depending only on a fixed cutoff, will be specified when we define the diffeomorphisms between the initial surfaces. Every choice of α is made before choosing m. Define

$$\begin{aligned} &\delta := (100m)^{-1}, \qquad A_m := \frac{m^2}{2\pi} + \log(\sqrt{2}m) + \frac{c_{m^{-2}}}{2\pi}, \\ &\tau_0 := 2e^{-A_m}, \qquad \tau_\zeta := \tau_0 e^{\zeta} \quad \forall\, \zeta \in [-1,1], \qquad \varepsilon_\zeta := \frac{m^2\tau_\zeta}{2\pi}. \end{aligned} \tag{3.12}$$

The radius δ is a fixed small fraction of the lattice spacing. The number τ_ζ will be the waist radius of each catenoidal neck. The quantity ε_ζ is the constant term in the expression $\tau_\zeta\Phi_m = \varepsilon_\zeta + (\tau_\zeta/2\pi)G_{m^{-2}}(\kappa s, \kappa t)$. It gives the principal size of the sheet height away from the lattice points.

By (3.2), $c_{m^{-2}} = c_0 + O(m^{-2})$. Therefore

$$\tau_\zeta = \frac{\sqrt{2}}{m}\exp\left(-\frac{m^2}{2\pi} - \frac{c_{m^{-2}}}{2\pi} + \zeta\right) \qquad \forall\, \zeta \in [-1,1].$$

In particular, there are constants $c, C > 0$ such that $cm^2 \leq |\log\tau_\zeta| \leq Cm^2$ for all sufficiently large m and all $\zeta \in [-1,1]$. More generally,

$$\sup_{\zeta\in[-1,1]} m^p\tau_\zeta^q \to 0 \qquad \text{as } m \to \infty$$

for every fixed pair of real numbers $p \geq 0$ and $q > 0$. We will use this fact whenever a fixed power of m multiplies a positive power of τ_ζ.

For $p \in L_m$, write $r_p(q) := d_{\mathbb{T}}(q,p)$ in the flat coordinate disk centred at p. Rescaling the local Green expansion gives

$$\Phi_m(q) = \log r_p(q) + A_m + \mathcal{E}_{m,p}(q), \qquad \|\mathcal{E}_{m,p}\|_{k,\beta;R} \leq C_{k,\beta}R^2(m^2 + |\log R|) \tag{3.13}$$

for $0 < r_p(q) < 2\delta$ and $0 < R < \delta$, with the norm taken on the annulus about p of radii $R/2$ and $2R$. In fact, $\mathcal{E}_{m,p}(q) = (2\pi)^{-1}E_{m^{-2}}(\kappa(q-p))$. Since $\kappa R < \sqrt{2}/100 < r_*/2$, the scaled norm and (3.3) give

$$\|\mathcal{E}_{m,p}\|_{k,\beta;R} \leq C_{k,\beta}(\kappa R)^2\big(1 + m^{-2}|\log(\kappa R)|\big) \leq C'_{k,\beta}R^2(m^2 + |\log R|).$$

In the last step we used $|\log(\kappa R)| \leq |\log R| + \log\kappa$ and $\log\kappa \leq Cm^2$.

Define the proposed upper sheet height before interpolation by

$$\psi_\zeta := \tau_\zeta\Phi_m \quad \text{on } \mathbb{T}\smallsetminus L_m, \qquad \forall\, \zeta \in [-1,1].$$

This function satisfies $L_{\mathbb{T}}\psi_\zeta = 0$ away from L_m and has logarithmic coefficient τ_ζ at each lattice point. Thus, it is a linearized doubling solution in the terminology of [KM23, Definition 3.6]. For $\tau > 0$, the upper half of a Euclidean catenoid of waist radius τ is the radial graph

$$C_\tau(r) := \tau\,\mathrm{arcosh}(r/\tau) \qquad \forall\, r \geq \tau.$$

Using $\operatorname{arcosh} x = \log(x + \sqrt{x^2 - 1})$, we obtain

$$C_\tau(r) = \tau \log r + \tau \log(2/\tau) + O(\tau^3 r^{-2}) \qquad \text{as } r/\tau \to \infty.$$

The coefficients of $\log r$ in $\tau\Phi_m$ and C_τ already agree. Their constant terms agree precisely when $A_m = \log(2/\tau)$, or $\tau = 2e^{-A_m} = \tau_0$. This explains the reference radius in (3.12). Figure 4 compares the catenoidal heights with their logarithmic terms after dividing both radius and height by τ.

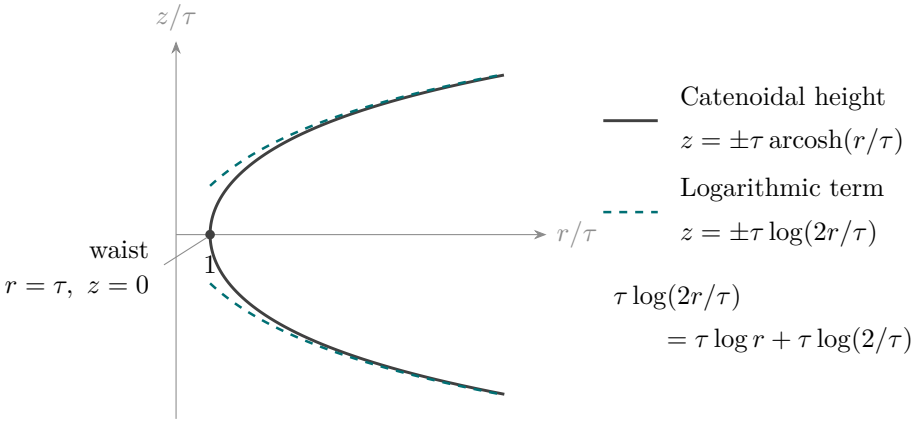


FIGURE 4. The radial section of a catenoid, shown in the dimensionless coordinates r/τ and z/τ, and the logarithmic terms used to match its two ends to the sheets. The solid curve is parametrized by $(r/\tau, z/\tau) = (\cosh\xi, \xi)$. As r/τ tends to infinity, the upper height differs from $\tau \log(2r/\tau)$ by $O(\tau^3 r^{-2})$. The coefficient of $\log r$ fixes the neck radius τ, and matching the remaining constant $\tau \log(2/\tau)$ determines its relation to the Green function.

To relate this comparison to the local matching conditions in the cited construction, we specify both its constant and its first-order part. Suppose that $\tau > 0$ and that u is a smooth solution of $L_{\mathbb{T}} u = 0$ on a punctured flat disk centred at p, with $u - \tau \log r_p = O(1)$ as $q \to p$. There are unique $c_p \in \mathbb{R}$ and $\lambda_p \in T_p^*\mathbb{T}$ such that

$$u(q) = \tau \log(2r_p(q)/\tau) + c_p + \lambda_p(q - p) + O\big(r_p(q)^2 |\log r_p(q)|\big) \qquad \text{as } q \to p.$$

The variable $q - p$ is understood in the flat coordinate disk. In particular, $u - \tau \log(2r_p/\tau)$ extends as a C^1 function across p, with value c_p and differential λ_p. The affine function $q - p \mapsto c_p + \lambda_p(q - p)$ is called the *mismatch jet*. It records the difference between the first-order expansion of u after subtraction of its logarithmic singularity and that of the catenoidal end.

For completeness, the stated regularity follows from the radial construction in (3.5) with parameter $a = 2$. The function $P_2(r)/(2\pi)$ solves $(\Delta + 2)(P_2/(2\pi)) = 0$ for $r > 0$ and differs from $\log r$ by $O(r^2|\log r|)$, with first derivative $O(r|\log r|)$. Thus $h := u - \tau P_2(r_p)/(2\pi)$ is bounded and solves $(\Delta + 2)h = 0$ away from p. To justify removability, extend h as a bounded measurable function on the disk and solve $\Delta w = -2h$ there with zero boundary values. For every finite $p_0 > 2$, local elliptic regularity gives $w \in W^{2,p_0}_{\mathrm{loc}} \subset C^1$. Then $h - w$ is bounded and harmonic on the punctured disk, so its singularity is removable. Hence h solves $(\Delta + 2)h = 0$ distributionally on the whole disk and is smooth by elliptic regularity. Taylor's theorem for h, together with the expansion of P_2, proves the claimed expansion and C^1 extension.

Apply this definition to $u = \psi_\zeta$ and $\tau = \tau_\zeta$. The reflections fixing p force $\lambda_p = 0$, and lattice translations make c_p independent of p. By (3.13), its common value is

$$c_p = \tau_\zeta\big(A_m + \log(\tau_\zeta/2)\big) = \tau_\zeta \zeta \qquad \forall\, p \in L_m, \quad \zeta \in [-1, 1]. \tag{3.14}$$

The last equality uses $\log(\tau_\zeta/2) = -A_m + \zeta$. Thus, the symmetries of the construction allow only one common constant mismatch and no linear mismatch. The parameter ζ varies precisely this

constant. We retain this freedom because the later perturbation argument must also eliminate the one-dimensional scalar error in its linear equation.

3.3. **Joining the two graphs by catenoidal necks.** We now replace the logarithmic singularity of each graph by one end of a catenoid. There are two separate steps. First, we subtract the constant difference $\tau_\zeta \zeta$ between the expansion of ψ_ζ and that of the catenoidal height. We make this subtraction using a function that solves $L_{\mathbb{T}} v = 0$ near each lattice point. The resulting linear error is therefore supported away from the necks. Second, we interpolate between the corrected graph and the catenoid on a smaller annulus. The estimates below show that the resulting surface is smooth, embedded, and has controlled geometry at the scale of its distance from the lattice.

Choose a smooth nondecreasing function $\vartheta : \mathbb{R} \to [0,1]$ such that

$$\begin{cases} \vartheta(x) = 0 & \forall\, x \le -1 \\ \vartheta(x) = 1 & \forall\, x \ge 1 \\ \vartheta(-x) = 1 - \vartheta(x) & \forall\, x \in \mathbb{R}. \end{cases}$$

For $a_1, a_2 \in \mathbb{R}$ with $a_1 \neq a_2$, define

$$\eta_{a_1,a_2}(r) := \vartheta\left(6\frac{r-a_1}{a_2-a_1} - 3\right) \qquad \forall\, r \in \mathbb{R}.$$

Then $\eta_{a_1,a_2} + \eta_{a_2,a_1} = 1$. If $a_1 < a_2$, the function η_{a_1,a_2} is zero for $r \le (2a_1 + a_2)/3$ and one for $r \ge (a_1 + 2a_2)/3$. Thus the change from zero to one occurs strictly between a_1 and a_2. For every integer $j \ge 1$,

$$\left|\frac{d^j}{dr^j}\eta_{a_1,a_2}(r)\right| \le C_j |a_2 - a_1|^{-j} \qquad \forall\, r \in \mathbb{R},$$

where C_j depends only on j and ϑ. This is the cutoff convention in [KM23, Definition 1.5].

Recall that $\delta = (100m)^{-1}$ and that J_0 denotes the Bessel function of order zero, as in the proof of Lemma 3.1. Define smooth functions $\widehat{v}_m, k_m : \mathbb{T} \to \mathbb{R}$ by

$$\widehat{v}_m(q) := \sum_{p \in L_m} \eta_{2\delta,\delta}(r_p(q)) J_0(\sqrt{2} r_p(q)), \qquad k_m := L_{\mathbb{T}} \widehat{v}_m, \tag{3.15}$$

where each summand is defined on $B(p, 2\delta)$ and extended by zero to its complement. The disks $B(p, 2\delta)$, $p \in L_m$, are pairwise disjoint. The cutoff vanishes on a neighborhood of the boundary of each disk, and the power series for $J_0(\sqrt{2} r_p)$ is a smooth power series in r_p^2. These observations justify smoothness both at p and across the boundary of the disk. Moreover,

$$\widehat{v}_m(p) = 1, \qquad d\widehat{v}_m(p) = 0 \quad \forall\, p \in L_m, \qquad \operatorname{supp} k_m \subset \bigcup_{p \in L_m} \{q : \delta \le r_p(q) \le 2\delta\}.$$

Indeed, near p the cutoff is one and

$$J_0(\sqrt{2} r_p) = 1 - \frac{r_p^2}{2} + O(r_p^4), \qquad (\Delta_{\mathbb{T}} + 2) J_0(\sqrt{2} r_p) = 0.$$

The second identity follows from the radial equation $J'' + r^{-1} J' + 2J = 0$ and holds also at p by smoothness. On $B(p, 2\delta)$, the product rule gives

$$L_{\mathbb{T}}\big(\eta_{2\delta,\delta}(r_p) J_0(\sqrt{2} r_p)\big) = J_0(\sqrt{2} r_p) \Delta_{\mathbb{T}} \eta_{2\delta,\delta}(r_p) + 2\left\langle D\eta_{2\delta,\delta}(r_p), D J_0(\sqrt{2} r_p)\right\rangle.$$

Both terms vanish wherever the cutoff is constant, proving the support assertion. In particular, subtracting $c\widehat{v}_m$ subtracts the same constant c from the regular part at every lattice point, changes none of its first derivatives there, and adds the linear error $-ck_m$. The value map $c\widehat{v}_m \mapsto c$ is an isomorphism from $\operatorname{span}\{\widehat{v}_m\}$ to $\mathbb{R}$. In the terminology of the scalar linearized doubling construction, k_m is the substitute function associated with this one-dimensional space.

We apply this correction with $c = \tau_\zeta \zeta$ and set

$$\widetilde{\psi}_\zeta := \tau_\zeta \Phi_m - \tau_\zeta \zeta \widehat{v}_m \qquad \text{on } \mathbb{T} \smallsetminus L_m.$$

Since $L_{\mathbb{T}} \Phi_m = 0$ away from L_m,

$$L_{\mathbb{T}} \widetilde{\psi}_\zeta = -\tau_\zeta \zeta k_m \qquad \text{on } \mathbb{T} \smallsetminus L_m.$$

Thus the correction has removed the constant mismatch, while its linear error is confined to the annuli $\delta \le r_p \le 2\delta$.

For the rest of this subsection, fix $\zeta \in [-1, 1]$ and abbreviate $\tau := \tau_\zeta$ and $r_c := \tau^\alpha$. The radius r_c specifies where we interpolate between the catenoidal height and $\widetilde{\psi}_\zeta$. Since $0 < \alpha < 1$, the estimates for τ_ζ give

$$\frac{\tau}{r_c} = \tau^{1-\alpha} \to 0, \qquad \frac{r_c}{\delta} = 100 m \tau^\alpha \to 0 \qquad \text{as } m \to \infty,$$

uniformly in ζ. Hence, for all sufficiently large m, we have $9\tau < 2r_c < 3r_c < \delta$. The interpolation annuli are therefore much farther from each neck centre than its waist radius τ, and much closer to that centre than the annuli supporting k_m.

On the torus with disks removed,

$$\Omega_\zeta := \mathbb{T} \smallsetminus \bigcup_{p \in L_m} B(p, 9\tau),$$

define the height function

$$h_\zeta(q) := \begin{cases} \eta_{3r_c, 2r_c}(r_p(q)) C_\tau(r_p(q)) + \eta_{2r_c, 3r_c}(r_p(q)) \widetilde{\psi}_\zeta(q), & 9\tau \le r_p(q) \le 3r_c, \quad p \in L_m, \\ \widetilde{\psi}_\zeta(q), & q \notin \bigcup_{p \in L_m} B(p, 3r_c). \end{cases} \tag{3.16}$$

The disks in the first line are disjoint, so the choice of p is unambiguous. The two formulas agree on a neighborhood of $r_p = 3r_c$, where the first cutoff is zero and the second is one. In particular, h_ζ is smooth up to the boundary of Ω_ζ.

For each $p \in L_m$, define

$$\mathcal{D}_\zeta := \{(\xi, \theta) \in \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}) : \tau \cosh \xi < 2r_c\}$$

and the catenoidal parametrization

$$\mathcal{C}_{\zeta, p}(\xi, \theta) := \mathcal{F}\big(p + \tau \cosh \xi (\cos\theta, \sin\theta), \tau\xi\big) \qquad \forall\, (\xi, \theta) \in \mathcal{D}_\zeta. \tag{3.17}$$

Addition to p is taken in its flat coordinate disk. The first component has distance $\tau \cosh \xi$ from p, and the second component is the signed-distance coordinate $z = \tau\xi$ in $\mathcal{N}$. For large m, these distances are less than $2r_c < \delta$ and $|\tau\xi| \le \tau \operatorname{arcosh}(2r_c/\tau) \to 0$, respectively. Thus the whole parametrization lies in the domain of $\mathcal{F}$. Its circle $\xi = 0$ lies at $z = 0$ and projects onto the circle of radius τ about p, which is the waist of the neck.

The height estimates proved below give $|h_\zeta| \le C\varepsilon_\zeta < z_0$ for all sufficiently large m. Thus, the following graphs lie into $\mathcal{N}$. Define the initial surface of parameter ζ by

$$M_\zeta := \{\mathcal{F}(q, h_\zeta(q)) : q \in \Omega_\zeta\} \cup \{\mathcal{F}(q, -h_\zeta(q)) : q \in \Omega_\zeta\} \cup \bigcup_{p \in L_m} \mathcal{C}_{\zeta, p}(\mathcal{D}_\zeta). \tag{3.18}$$

On the overlap $9\tau \le r_p < 2r_c$, the height h_ζ equals $C_\tau(r_p)$. If $r_p = \tau \cosh \xi$, then $C_\tau(r_p) = \tau|\xi|$. Therefore the upper graph agrees exactly with the part of the catenoid where $\xi > 0$, and the lower graph agrees exactly with the part where $\xi < 0$. The graph pieces and catenoidal pieces therefore have exactly the same images on these overlaps.

We write $\jmath_\zeta : M_\zeta \hookrightarrow \mathcal{N}$ and $\iota_\zeta : M_\zeta \hookrightarrow \mathbb{S}^4$ for the inclusions, once the smoothness and embeddedness asserted below have been proved. We retain the notation of Section 2 for their second

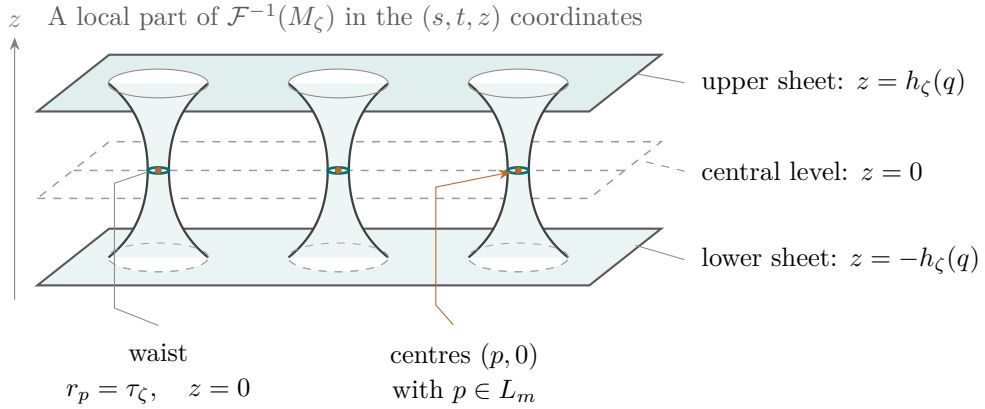


FIGURE 5. The initial surface M_ζ is formed from two graphs over the central torus, joined near each $p \in L_m$ by a catenoidal neck. Three neighbouring necks in one lattice row are shown. The teal circles have radius τ_ζ in the flat base coordinates and lie at $z = 0$; their centres are marked in orange and do not belong to the surface. The dashed lines indicate the reference level $z = 0$, which meets the surface precisely along these circles. The drawing represents the parameter coordinates of $\mathcal{F}$, not an isometric embedding of $\mathcal{N}$ into $\mathbb{R}^3$. It suppresses the interpolation details and exaggerates both the sheet separation and the neck radii. The displayed sheet boundaries are artificial: the complete surface has two punctured toroidal sheets joined by all m^2 necks and has no boundary.

fundamental forms and shape operators. In particular, $S^{\nu_\zeta}_{\jmath_\zeta}$ is the shape operator within $\mathcal{N}$, with respect to the unit normal ν_ζ specified in the lemma. Both inclusions induce the same metric

$$g_\zeta := \jmath_\zeta^* g_{\mathcal{N}} = \iota_\zeta^* g_{\mathbb{S}^4}.$$

Lemma 3.2. *For all sufficiently large m and every $\zeta \in [-1, 1]$, M_ζ is a smooth closed connected embedded Γ_m-invariant surface in $\mathcal{N}$, of genus $m^2 + 1$. It is the boundary of a Γ_m-invariant open subset of $\mathcal{N}$ with compact closure. Let ν_ζ be its outward unit normal within $\mathcal{N}$. For every $x \in M_\zeta$,*

$$|z(x)| \le C\varepsilon_\zeta, \qquad |(\nu_\zeta)_h(x)| \le C\frac{\tau_\zeta}{\rho_\zeta(x)}, \qquad |S^{\nu_\zeta}_{\jmath_\zeta}(x)|_{g_\zeta} \le C\frac{\tau_\zeta}{\rho_\zeta(x)^2}, \tag{3.19}$$

where $(\nu_\zeta)_h$ denotes the orthogonal projection onto $\operatorname{span}\{E_1, E_2\}$ *and*

$$\rho_\zeta(x) := d_{\mathbb{T}}(\pi(x), L_m), \qquad \tau_\zeta \le \rho_\zeta(x) \le \pi/m. \tag{3.20}$$

The norm of $(\nu_\zeta)_h$ is taken with respect to $g_{\mathcal{N}}$. At every $x \in M_\zeta$, coordinates on a neighborhood of radius a fixed positive fraction of $\rho_\zeta(x)$ give uniformly elliptic coefficients for the metric $\rho_\zeta(x)^{-2} g_\zeta$, with bounds for every fixed finite number of their coordinate derivatives. All constants are independent of m, ζ, and x. Moreover,

$$\operatorname{inj}_{g_\zeta}(x) \ge \rho_\zeta(x)/5 \qquad \forall\, x \in M_\zeta. \tag{3.21}$$

Proof. We first check that the two graphs are separated. Write $\rho(q) := d_{\mathbb{T}}(q, L_m)$ for $q \in \mathbb{T}$. From (3.8) and (3.12),

$$\frac{1-\alpha}{4\pi} m^2 \le \Phi_m(q) \le Cm^2 \qquad \text{whenever } \rho(q) \ge r_c, \tag{3.22}$$

for all sufficiently large m. To obtain the lower bound, use $\log \rho \ge \alpha \log \tau$ and $\log \tau = \log 2 - A_m + \zeta$ to write

$$\Phi_m(q) \ge \frac{m^2}{2\pi} + \log \kappa + \alpha \log \tau - C$$

$$= \frac{1-\alpha}{2\pi} m^2 + (1-\alpha)\log\kappa + O(1).$$

The $O(1)$ term is uniform in ζ. The upper bound follows from $\kappa\rho \le \sqrt{2}\pi$. Since $|\widehat{v}_m| \le C$ and $|\zeta| \le 1$, it follows that $\widetilde{\psi}_\zeta > 0$ on $\rho \ge r_c$. Also $C_\tau(r) > 0$ for $r > \tau$. Every value of h_ζ is either one of these positive heights or a convex combination of them. Consequently $h_\zeta > 0$ on Ω_ζ.

We next estimate the difference between the two heights on the interpolation annuli. If $r_p \le \delta$, then $\widehat{v}_m = J_0(\sqrt{2}r_p)$, and the local expansion of Φ_m gives

$$\widetilde{\psi}_\zeta - \tau\log(2r_p/\tau) = \tau\mathcal{E}_{m,p} + \tau\zeta\big(1 - J_0(\sqrt{2}r_p)\big).$$

Here we used $A_m + \log(\tau/2) = \zeta$ to cancel the constant term. On the other hand, for $r > \tau$,

$$\begin{aligned} C_\tau(r) &= \tau\log(2r/\tau) + \tau\log\left(\frac{1+\sqrt{1-\tau^2/r^2}}{2}\right) \\ &= \tau\log(2r/\tau) - \frac{\tau^3}{4r^2} + O(\tau^5 r^{-4}) \qquad \text{as } \tau/r \to 0. \end{aligned}$$

The last expansion follows from the convergent power series in τ^2/r^2 and can be differentiated any fixed number of times on rescaled annuli. Since $m^2 + |\log r_c| \le C|\log\tau|$, (3.13) and the power series for J_0 imply

$$\|\widetilde{\psi}_\zeta - C_\tau\|_{k,\beta;r_c} \le C_{k,\beta}\big(\tau r_c^2|\log\tau| + \tau^3 r_c^{-2}\big). \tag{3.23}$$

In this formula the norm is taken on $\{q : 2r_c \le r_p(q) \le 3r_c\}$ using the same scaled formula as before, with scale r_c. Multiplication by either cutoff in (3.16) preserves this estimate: after the substitution $q = p + r_c Y$, the cutoffs are fixed smooth functions of $|Y|$. In particular, for each positive derivative order, the interpolation error divided by the scale τr_c^{-j} is bounded by

$$C_j\left(r_c^2|\log\tau| + \tau^2 r_c^{-2}\right) = C_j\left(\tau^{2\alpha}|\log\tau| + \tau^{2-2\alpha}\right) \to 0.$$

The graph heights satisfy $|h_\zeta| \le C\tau m^2$. On the part where $h_\zeta = \widetilde{\psi}_\zeta$ this follows from (3.22); on the catenoidal part it follows from $C_\tau(r) \le \tau\log(2r/\tau)$ and $r \le 3r_c$. The interpolated heights satisfy the same estimate. Since $\tau m^2 = 2\pi\varepsilon_\zeta \to 0$, all these graphs lie within the prescribed band $|z| < z_0$ for large m.

To verify regularity of the catenoidal pieces, put $r := \tau\cosh\xi$, $z := \tau\xi$, and $c := \cos z$ at a point of $\mathcal{D}_\zeta$. Along that piece, let

$$E_r := \cos\theta\, E_1 + \sin\theta\, E_2, \qquad E_\theta := -\sin\theta\, E_1 + \cos\theta\, E_2.$$

Differentiating (3.17) gives

$$\begin{aligned} \partial_\xi \mathcal{C}_{\zeta,p} &= \tau c\sinh\xi\, E_r + \tau E_3, \\ \partial_\theta \mathcal{C}_{\zeta,p} &= \tau c\cosh\xi\, E_\theta. \end{aligned}$$

These vectors are orthogonal and nonzero. In particular, the first equals τE_3 at $\xi = 0$. Their squared lengths give the exact induced metric

$$\mathcal{C}_{\zeta,p}^* g_{\mathcal{N}} = \tau^2(1 + c^2\sinh^2\xi)\, d\xi^2 + \tau^2 c^2\cosh^2\xi\, d\theta^2.$$

The height recovers $\xi = z/\tau$ and the direction of the displacement from p recovers $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. Hence each catenoidal parametrization is an embedding. The two graphs are disjoint by positivity of h_ζ. The neck disks are pairwise disjoint, and each neck meets the graphs only where the formulas agree exactly. Since $\mathcal{F}$ is an embedding, the union defining M_ζ is therefore a smooth embedded surface.

For completeness, extend h_ζ to the open set $\{q \in \mathbb{T} : \rho(q) > \tau\}$ by setting $h_\zeta(q) := C_\tau(r_p(q))$ when $\tau < r_p(q) < 9\tau$. This agrees with the existing definition near $r_p = 9\tau$. Define

$$\mathcal{O}_\zeta := \{\mathcal{F}(q,z) : \rho(q) > \tau,\ |z| < h_\zeta(q)\} \subset \mathcal{N}.$$

Its closure is compact and lies in $|z| < z_0$ for large m. Near each waist the condition defining this open set is $r_p > \tau\cosh(z/\tau)$. Its boundary is smooth there because the derivative with respect to r_p of $r_p - \tau\cosh(z/\tau)$ equals one. Consequently $\partial\mathcal{O}_\zeta = M_\zeta$, which defines the outward unit normal ν_ζ. All functions, disks, and parametrized necks used above are preserved by the prescribed symmetries; the reflection $z \mapsto -z$ exchanges the graphs. Thus $\mathcal{O}_\zeta$ and M_ζ are Γ_m-invariant.

The surface is compact and has no boundary. Indeed, the graph pieces have compact domains, and each remaining central portion of a neck is the image of a compact cylinder. All boundaries of these pieces lie in the exact overlaps described above. Topologically, M_ζ consists of two tori with m^2 disks removed from each, joined by m^2 annuli. The first annulus connects the two components; the remaining annuli preserve connectedness. The boundary orientation of $\mathcal{O}_\zeta$ makes M_ζ orientable. Removing the disks gives Euler characteristic $-2m^2$, and attaching the annuli does not change it. Therefore

$$2 - 2\,\mathrm{genus}(M_\zeta) = -2m^2,$$

proving the stated genus. The inequality $\tau \le \rho_\zeta \le \pi/m$ follows from the catenoidal radius $r = \tau\cosh\xi$ and the covering radius π/m of the square lattice.

We now prove the differential estimates. The height estimate has already been established. For each fixed integer $j \ge 1$, the extended graph height satisfies

$$|D^j h_\zeta(q)| \le C_j \tau \rho(q)^{-j} \qquad \text{whenever } \rho(q) \ge 2\tau.$$

On the exterior part, use (3.9). On the support of derivatives of the cutoff defining $\widehat{v}_m$, the quantities ρ and δ are comparable, so the cutoff derivative bounds give the same estimate. Where this cutoff is constant, the power series for J_0 applies. On the interpolation annuli, use (3.23) and the derivatives of C_τ. Finally, on $2\tau \le r_p \le 2r_c$, differentiation of $C_\tau(r_p)$ gives the estimate directly. The same arguments give the corresponding bounds in local scaled Hölder norms.

For a graph with height $w = h_\zeta$ or $w = -h_\zeta$, put $c := \cos w$ and $Q := c^2 + |Dw|^2$, and denote its upward unit normal by ν^+. Its horizontal part has norm $|Dw|/\sqrt{Q}$. Thus $|\nu_h| \le C\tau/\rho$ on $\rho \ge 2\tau$; reversing the normal on the lower graph does not change this estimate. We also give the computation for its second fundamental form. In the coordinates (s,t,z) on the band, the nonzero connection coefficients are

$$\Gamma^z_{ij} = \sin z\cos z\,\delta_{ij}, \qquad \Gamma^i_{jz} = \Gamma^i_{zj} = -\tan z\,\delta^i_j \qquad (i,j \in \{1,2\}).$$

Here and in the following computation, coordinate vectors on the parameter domain denote their images under $d\mathcal{F}$. With graph tangent vectors $T_i := \partial_i + w_i\partial_z$, these identities give

$$\nabla^{\mathcal{N}}_{T_i} T_j = (w_{ij} + \sin w\cos w\,\delta_{ij})\partial_z - \tan w\,(w_j\partial_i + w_i\partial_j).$$

The normal formula for a height graph gives $\langle\partial_z, \nu^+\rangle = c/\sqrt{Q}$ and $\langle\partial_i, \nu^+\rangle = -cw_i/\sqrt{Q}$. Taking the inner product of the preceding identity with ν^+ therefore gives

$$\big\langle \nabla^{\mathcal{N}}_{T_i} T_j, \nu^+\big\rangle = \frac{c}{\sqrt{Q}}\big(w_{ij} + \sin w\cos w\,\delta_{ij} + 2\tan w\,w_iw_j\big) \qquad (i,j \in \{1,2\}).$$

The graph metric has coefficients $c^2\delta_{ij} + w_iw_j$. On $\rho \ge 2\tau$ it is uniformly comparable to the flat metric, because $|w| \to 0$ and $|Dw| \le C$. The preceding formula consequently implies

$$\begin{aligned} |S^{\nu_\zeta}_{\jmath_\zeta}|_{g_\zeta} &\le C\big(|D^2 w| + |w|(1 + |Dw|^2)\big) \\ &\le C\big(\tau\rho^{-2} + \tau m^2\big) \le C\tau\rho^{-2}, \end{aligned}$$

where the last inequality uses $\rho \le \pi/m$.

It remains to obtain these bounds where $\tau \le \rho \le 2\tau$. There the surface is exactly one of the parametrized catenoids and $|\xi|$ is bounded. Its outward unit normal is

$$\nu_\zeta = \frac{-E_r + c\sinh\xi\,E_3}{\sqrt{1 + c^2\sinh^2\xi}}.$$

Since $c \geq \cos z_0$, its horizontal part satisfies

$$|(\nu_\zeta)_h| = \frac{1}{\sqrt{1 + c^2 \sinh^2 \xi}} \leq \frac{\sec z_0}{\cosh \xi} = \sec z_0 \frac{\tau}{\rho}.$$

In coordinates obtained by multiplying the three parameters $(q - p, z)$ by τ^{-1}, the ambient metric, after multiplication by τ^{-2}, is

$$dz'^2 + \cos^2(\tau z')\,(dq_1'^2 + dq_2'^2).$$

On bounded coordinate sets it converges with all derivatives to the Euclidean metric. The parametrized surface in these coordinates is the fixed catenoid $(q', z') = (\cosh \xi(\cos\theta, \sin\theta), \xi)$. Its second fundamental form in the rescaled metrics is therefore uniformly bounded on $|\xi| \leq \operatorname{arcosh} 2$. Returning to the original scale gives $|S^{\nu_\zeta}_{J_\zeta}| \leq C/\tau \leq 4C\tau/\rho^2$. This completes (3.19).

These computations also give the asserted local coordinate estimates. On any fixed region $\rho/\tau \leq B$, with $B > 1$, use the catenoidal coordinates and rescale by τ^{-1}; rescaling instead by ρ^{-1} changes lengths by a factor between B^{-1} and one. On regions where ρ/τ is larger, use the height graph over a flat disk of radius a fixed fraction of ρ. After dividing lengths by ρ, the bounds for $D^j h_\zeta$ above give uniform ellipticity and bounds for any fixed number of derivatives of the metric coefficients. The function ρ itself is never differentiated in this argument: the rescaling uses its value at the centre of each coordinate disk.

Finally, we prove the stated lower bound for the injectivity radius. We first record a bound on the standard Euclidean catenoid. Its horizontal projection to $\mathbb{R}^2 \setminus \{0\}$ is length nonincreasing and induces an isomorphism on fundamental groups. A based loop of length L projects into the Euclidean disk of radius $L/2$ about its base point: each point of the loop can be reached from the base point by one of its two arcs of length at most $L/2$. If the base point has horizontal radius r and $L < 2r$, that disk misses the origin, so the loop is contractible. The complete catenoid has nonpositive Gaussian curvature. Its simply connected covering surface therefore has no cut or conjugate points, and the lower bound $2r$ for lengths of noncontractible based loops implies that its injectivity radius at the base point is at least r.

Fix a large constant $B > 4$. On the initial surface with $\rho \leq 4B\tau$, the parametrization is exactly catenoidal for all sufficiently large m. After rescaling by τ^{-1}, its metric converges smoothly to the Euclidean catenoid metric on this fixed compact portion. If $\rho(x) \leq B\tau$, any path from x to $\rho = 4B\tau$ has length at least $\cos z_0(4B\tau - \rho(x))$. Indeed, for every tangent vector to the band the length of its projection under π is at most $\sec z_0$ times its original length, and distance to the lattice is a 1-Lipschitz function on $\mathbb{T}$. In particular, paths of length at most $\rho(x)/2$ stay in this catenoidal portion. Smooth dependence of the geodesic equation and its differential on the metric, together with the catenoid injectivity bound, then gives $\operatorname{inj}_{g_\zeta}(x) \geq \rho(x)/2$ for large m, uniformly on $\rho(x) \leq B\tau$. For clarity, injectivity here follows as well from this smooth convergence: a sequence of distinct vectors in the closed balls of half the limiting injectivity radius with equal exponential images would have convergent subsequences. Distinct limits contradict injectivity of the limiting exponential map; equal limits contradict its nonsingular differential and uniform local invertibility.

If $\rho(x) \geq B\tau$, consider the flat disk $B_{\mathbb{T}}(\pi(x), \rho(x)/2)$. At every point of this disk the distance to L_m is at least $\rho(x)/2 > \tau$, so the sheet containing x is a single graph over the whole disk. In coordinates rescaled by $\rho(x)^{-1}$, its metric differs from the Euclidean metric in C^2 norm by $O(B^{-2}) + o_m(1)$. This follows from the bounds for the first three derivatives of h_ζ and the metric formula $\cos^2 h_\zeta\, g_{\mathbb{T}} + dh_\zeta \otimes dh_\zeta$. A path leaving this graph neighborhood has length at least $\cos z_0 \rho(x)/2 > \rho(x)/3$. Therefore the exponential map on tangent vectors of length at most $\rho(x)/4$ stays within this neighborhood. After the same rescaling, it is C^1 close to the Euclidean exponential map. Choosing B large and then m large makes it injective and nonsingular on the tangent ball of radius $\rho(x)/5$. This proves (3.21) in the remaining region and completes the proof. □

3.4. **Weighted norms and a fixed domain for the corrections.** The norm of the second fundamental form near a waist is of order τ_ζ^{-1}. The estimates in Lemma 3.2 are uniform only after distances near $x \in M_\zeta$ are divided by $\rho_\zeta(x)$. We use that same scale to measure derivatives of the functions that will deform M_ζ. We also specify how to regard functions on the different surfaces M_ζ as functions on the single surface M_0. These two definitions are needed for the linear estimates below and for the fixed-point argument in Section 5.

Define the positive function $f_\zeta : M_\zeta \to (0,\infty)$ by

$$f_\zeta(x) = \max\{\rho_\zeta(x)^{3/2}, \tau_\zeta^{(1-\alpha)/2}\rho_\zeta(x)^{1/2}\}.$$

The weight f_ζ specifies the allowed size of a normal displacement, apart from its overall norm. The two terms agree when $\rho_\zeta = \tau_\zeta^{(1-\alpha)/2}$. Thus the weight is $\tau_\zeta^{(1-\alpha)/2}\rho_\zeta^{1/2}$ nearer the waist and $\rho_\zeta^{3/2}$ farther from it. Here ρ_ζ is the function in (3.20). For every continuous positive function $w : M_\zeta \to (0,\infty)$, every nonnegative integer k, and every function $u \in C^{k,\beta}(M_\zeta)$, define

$$\|u\|_{C^{k,\beta}_\zeta(w)} = \sup_{x\in M_\zeta} w(x)^{-1}\|u\|_{C^{k,\beta}(B_{g_\zeta}(x,\rho_\zeta(x)/100),\,\rho_\zeta(x)^{-2}g_\zeta)}. \tag{3.24}$$

For each centre x, the factor $\rho_\zeta(x)^{-2}$ is a constant on the whole ball in this formula. In particular, it does not mean that the metric is multiplied by the variable function ρ_ζ^{-2}. In the rescaled metric this ball has radius $1/100$. The $C^{k,\beta}$ norm is the sum of the supremum norms of the covariant derivatives of orders $0,\dots,k$ and the β-Hölder seminorm of the derivative of order k. For that seminorm we compare tensors by parallel transport along the unique short minimizing geodesic and divide by the βth power of the distance in the rescaled metric. Such geodesics are well defined for the pairs of points in these balls. Indeed, ρ_ζ is Lipschitz with constant at most $\sec z_0$, because $|d\pi| \le \sec z_0$, and (3.21) consequently gives an injectivity radius larger than the diameter of the ball at every point of the ball. The functions ρ_ζ and f_ζ need only be continuous; no differentiation of the distance function at a cell boundary is involved in this definition. In particular, a bound $\|u\|_{C^{k,\beta}_\zeta(w)} \le K$ implies

$$|\nabla^j u(x)|_{g_\zeta} \le K w(x)\rho_\zeta(x)^{-j} \qquad \forall\, x \in M_\zeta, \quad 0 \le j \le k,$$

as well as the corresponding Hölder estimates on the indicated balls. Here ∇ is the Levi-Civita connection of g_ζ. In particular, two derivatives of a displacement of size f_ζ have size f_ζ/ρ_ζ^2. The potential term in the scalar Jacobi operator is bounded by $C\rho_\zeta^{-2}$, by (3.19) and the bounded curvature of $\mathcal{N}$. Thus its second-order and zeroth-order terms both send this displacement scale to f_ζ/ρ_ζ^2. We therefore use the following domain and target spaces:

$$X_\zeta = C^{2,\beta}_{\mathrm{inv}}(M_\zeta), \qquad Y_\zeta = C^{0,\beta}_{\mathrm{inv}}(M_\zeta), \qquad \|u\|_{X_\zeta} = \|u\|_{C^{2,\beta}_\zeta(f_\zeta)}, \quad \|E\|_{Y_\zeta} = \|E\|_{C^{0,\beta}_\zeta(f_\zeta/\rho_\zeta^2)}. \tag{3.25}$$

Here $C^{k,\beta}_{\mathrm{inv}}(M_\zeta)$ consists of the $C^{k,\beta}$ functions invariant under every element of Γ_m. For fixed m and ζ, the weights are positive on the compact surface M_ζ, so these norms make X_ζ and Y_ζ Banach spaces.

We explain the relation with the norms used in the cited linear construction. For a function u on a parametrized neck, write $u(\xi,\theta) := u(\mathcal{C}_{\zeta,p}(\xi,\theta))$. Its first angular harmonic is

$$(\Pi_1 u)(\xi,\theta) := \frac{\cos\theta}{\pi}\int_0^{2\pi} u(\xi,\vartheta)\cos\vartheta\,d\vartheta + \frac{\sin\theta}{\pi}\int_0^{2\pi} u(\xi,\vartheta)\sin\vartheta\,d\vartheta.$$

The half-turn about the lattice point belongs to the symmetry group and preserves the axial coordinate. Thus an invariant function satisfies $u(\xi,\theta+\pi) = u(\xi,\theta)$. Changing variables by $\vartheta \mapsto \vartheta+\pi$ in each integral shows that $\Pi_1 u = 0$. Thus the additional first-harmonic summand in Definition 4.2 of [KM23] vanishes identically, and the norms above are uniformly equivalent to the restrictions of its two-weight norms with $\gamma = 3/2$ and $\gamma' = 1/2$. The possible change in the fixed radius used in the local norm changes only a constant independent of m and ζ.

To use the prescribed error k_m in an equation on M_ζ, define

$$\mathcal{J}_\zeta = k_m \circ \pi\big|_{M_\zeta}. \tag{3.26}$$

This is a function on M_ζ. It is smooth, vanishes on the necks, and on each of the two sheets equals k_m at the projected point. We use the outward unit normal on both sheets. The isometry $z \mapsto -z$ exchanges the sheets and preserves their outward normals as vector fields under its differential; hence their scalar mean curvatures have the same transformation rule. This explains why the same function, rather than its negative, is used on the lower sheet. In the notation of [KM23, Notation 3.25], it is exactly $J_{M_\zeta}(k_m, k_m)$.

The surfaces themselves depend on ζ, so a function on M_ζ does not have a fixed domain when ζ varies. We remove this difficulty by constructing explicit diffeomorphisms $F_\zeta : M_0 \to M_\zeta$. The subscript 0 here refers to $\zeta = 0$; the integer m remains fixed. Near a waist these maps rescale the catenoidal axial coordinate, while away from the necks they preserve the base point on $\mathbb{T}$. Set, for every $\zeta \in [-1, 1]$,

$$r_0 = \tfrac{1}{2}\tau_0^\alpha, \qquad s_0 = \operatorname{arcosh}(r_0/\tau_0), \qquad s_\zeta = \operatorname{arcosh}(\tau_\zeta^{\alpha-1}/2), \qquad a_\zeta = s_\zeta/s_0.$$

For sufficiently large m, both arguments of arcosh exceed one, so $s_0, s_\zeta > 0$. The circles $|\xi| = s_0$ on the reference neck have radius r_0, and the circles $|\xi| = s_\zeta$ on the target neck have radius $\tau_\zeta^\alpha/2$. Choose a fixed smooth cutoff $\chi_0 : [0, \infty) \to [0, 1]$ equal to one on $[0, 1]$ and zero on $[2, \infty)$, and for $r \geq \tau_0$ set

$$\begin{aligned} A_\zeta(r) &= \tau_\zeta \cosh\big(a_\zeta \operatorname{arcosh}(r/\tau_0)\big), \\ R_\zeta(r) &= r + \chi_0(r/r_0)(A_\zeta(r) - r). \end{aligned}$$

The function A_ζ is the change in radius induced by the linear change $\xi \mapsto a_\zeta \xi$ in the axial coordinate. The function R_ζ makes this change equal to the identity once $r \geq 2r_0$. Near each neck define $F_\zeta : M_0 \to M_\zeta$ by preserving the polar angle and the choice of upper or lower sheet, replacing the base radius r by $R_\zeta(r)$, and taking the corresponding point on M_ζ. Outside the radius-$2r_0$ disks it preserves the base point and sheet sign. On the region $\rho_0 \leq r_0$, the definition is equivalently the smooth formula

$$F_\zeta(\mathcal{C}_{0,p}(\xi, \theta)) = \mathcal{C}_{\zeta,p}(a_\zeta \xi, \theta), \qquad |\xi| \leq s_0. \tag{3.27}$$

Lemma 3.3 (Comparison of the surfaces and their function spaces). *After taking α sufficiently small in terms of χ_0, the maps F_ζ are symmetry-preserving smooth diffeomorphisms for all sufficiently large m. They depend smoothly on ζ and F_0 is the identity. There is a constant C_F, independent of m, ζ, such that*

$$\begin{aligned} C_F^{-1}\|u\|_{X_\zeta} \leq \|u \circ F_\zeta\|_{X_0} \leq C_F\|u\|_{X_\zeta}, \\ C_F^{-1}\|E\|_{Y_\zeta} \leq \|E \circ F_\zeta\|_{Y_0} \leq C_F\|E\|_{Y_\zeta}. \end{aligned} \tag{3.28}$$

The inequalities in the first line hold for every $u \in X_\zeta$, and those in the second line hold for every $E \in Y_\zeta$.

Proof. Set $L_0 := |\log \tau_0|$. The expansion $\operatorname{arcosh} x = \log(2x) + O(x^{-2})$ gives

$$\begin{aligned} s_0 &= (1-\alpha)L_0 + O(\tau_0^{2(1-\alpha)}), \\ s_\zeta - s_0 &= -(1-\alpha)\zeta + O(\tau_0^{2(1-\alpha)}), \\ a_\zeta - 1 &= O(L_0^{-1}), \end{aligned}$$

uniformly for $\zeta \in [-1, 1]$. If $r \in [r_0, 2r_0]$ and $\xi := \operatorname{arcosh}(r/\tau_0)$, then $\xi = s_0 + \log(r/r_0) + o_m(1)$. Therefore

$$\log \frac{A_\zeta(r)}{r} = \zeta + \log\cosh(a_\zeta \xi) - \log\cosh \xi$$

$$= \zeta + (a_\zeta - 1)\xi + o_m(1) = \alpha\zeta + O(L_0^{-1}).$$

Here $o_m(1)$ denotes a quantity tending uniformly to zero as $m \to \infty$. Differentiation gives

$$\begin{aligned} A'_\zeta(r) &= \frac{\tau_\zeta}{\tau_0} a_\zeta \frac{\sinh(a_\zeta \xi)}{\sinh \xi} \\ &= a_\zeta \frac{A_\zeta(r)}{r} \frac{\tanh(a_\zeta \xi)}{\tanh \xi}. \end{aligned}$$

Consequently, on $[r_0, 2r_0]$,

$$A_\zeta(r)/r - 1 = O(\alpha) + o_m(1), \qquad A'_\zeta(r) - 1 = O(\alpha) + o_m(1),$$

with constants uniform for α in a fixed small interval. The derivative of the interpolating radius is

$$R'_\zeta(r) = 1 + \chi_0(r/r_0)(A'_\zeta(r) - 1) + \frac{\chi'_0(r/r_0)}{r_0}(A_\zeta(r) - r).$$

Choose α so small, in terms of the fixed cutoff, that the sum of the $O(\alpha)$ terms has absolute value less than $1/4$. Then choose m large enough that the remaining terms have absolute value less than $1/4$. This gives $R'_\zeta > 1/2$ on the transition annulus. Outside it, R_ζ equals either the increasing function A_ζ or the identity. Since $R_\zeta(\tau_0) = \tau_\zeta$, it maps the allowed radii bijectively onto the corresponding radii of M_ζ.

Both the transition annulus and its image lie in the portions defined by (3.17). Indeed, their outer radii satisfy

$$\max\{2r_0, R_\zeta(2r_0)\} \le \tau_0^\alpha(1 + O(\alpha) + o_m(1)) < 2\tau_\zeta^\alpha.$$

Smoothness at the waist follows from (3.27), which is the map $(\xi, \theta) \mapsto (a_\zeta \xi, \theta)$ between regular catenoidal coordinates. Elsewhere the graph descriptions and $R'_\zeta > 0$ give smoothness and a smooth inverse. All formulas depend smoothly on ζ, and radial changes of coordinates commute with the rotations, reflections, and factor interchange used in Γ_m. The sheet signs are preserved and height reflection exchanges them on both sides. Thus F_ζ commutes with every element of Γ_m. When $\zeta = 0$, $a_0 = 1$ and $R_0(r) = r$, so F_0 is the identity.

For the norm estimates, the ratios $\rho_\zeta(F_\zeta(x))/\rho_0(x)$ are bounded above and below uniformly. On $\rho_0 \le r_0$, this follows from

$$\frac{\rho_\zeta(F_\zeta(x))}{\rho_0(x)} = e^\zeta \frac{\cosh(a_\zeta \xi)}{\cosh \xi}, \qquad |\xi| \le s_0,$$

because $|(a_\zeta - 1)\xi|$ is uniformly bounded. The transition-annulus estimates give the same conclusion there, and outside those annuli the radii agree exactly. Since $\tau_\zeta/\tau_0 \in [e^{-1}, e]$, the ratios of the weights $f_\zeta \circ F_\zeta$ and f_0 are also bounded above and below uniformly.

Differentiating the formulas for A_ζ and R_ζ in rescaled graph coordinates gives uniform derivative bounds of every fixed order on the transition annuli. In the catenoidal coordinates at the waist this follows from the linear map in ξ and the uniform local geometry of Lemma 3.2. The same statements hold for the inverses. Hence the pullback metrics, after rescaling by the corresponding radii, are uniformly comparable with uniformly bounded derivatives. The chain rule then compares the derivative terms in the local Hölder norms. Distances are uniformly comparable as well, giving the comparison of the Hölder seminorms. An image of one of the balls in (3.24) is covered by a uniformly bounded number of such balls in the target. Taking the supremum over the centers proves (3.28). □

3.5. **The scalar linear equation and the remaining corrections.** We now construct a first correction of the scalar mean curvature within $\mathcal{N}$, and estimate every further correction allowed in Sections 4 and 5. We first solve the linearized equation with an additional term $c\mathcal{J}_\zeta$, where $c \in \mathbb{R}$ is part of the output. We apply that solution operator to the initial mean-curvature error to obtain a specified function u_ζ. Finally, we allow an additional function v on M_0 and estimate the total displacement on M_ζ.

Write $H_\zeta := H_{\jmath_\zeta,\nu_\zeta}$ for the scalar mean curvature of the inclusion into $\mathcal{N}$, measured with its outward unit normal. Its linearization under a normal variation within $\mathcal{N}$ is the operator $\mathcal{L}_\zeta : C^{2,\beta}(M_\zeta) \to C^{0,\beta}(M_\zeta)$ given by

$$\mathcal{L}_\zeta u := \Delta_{g_\zeta} u + \big(|S^{\nu_\zeta}_{\jmath_\zeta}|^2_{g_\zeta} + \mathrm{Ric}_{g_\mathcal{N}}(\nu_\zeta,\nu_\zeta)\big)u \qquad \forall\, u \in C^{2,\beta}(M_\zeta).$$

No minimality assumption on M_ζ is needed for this linearization formula. We use the construction of [KM23] to solve $\mathcal{L}_\zeta u = E$ up to a multiple of $\mathcal{J}_\zeta$. The extra term allows the linear construction to adjust the common value of its auxiliary solution on the central torus at the lattice points while maintaining the weighted estimates. The assertion below is a right inverse for the augmented map $(u,c) \mapsto \mathcal{L}_\zeta u - c\mathcal{J}_\zeta$; it does not assert invertibility of $\mathcal{L}_\zeta$ alone. The total coefficient of $\mathcal{J}_\zeta$ in the remaining equation will be set equal to zero by choosing ζ in Section 5.

The estimates are needed not just for one correction, but for all scalar functions considered in that argument. Using the maps F_ζ to express those functions on M_0, define

$$\mathcal{B}_m := \{(v,\zeta) \in X_0 \times [-1,1] : \|v\|_{X_0} \le \tau_0^{1+\alpha}\}. \tag{3.29}$$

The integer m is fixed when this set is used in a fixed-point argument. The function v always has domain M_0; $v \circ F_\zeta^{-1}$ has domain M_ζ. The total displacement, denoted by ϕ in the proposition, is the sum of this function and the preliminary correction u_ζ. Thus v and ϕ are not independent unknowns.

Proposition 3.4 (Uniform scalar estimates). *For all sufficiently large m, the following hold uniformly for $|\zeta| \le 1$.*

(1) There is a linear map $\mathscr{R}_\zeta : Y_\zeta \to X_\zeta \times \mathbb{R}$, $\mathscr{R}_\zeta E = (u_E, c_E)$, such that

$$\mathcal{L}_\zeta u_E = E + c_E \mathcal{J}_\zeta, \qquad \|u_E\|_{X_\zeta} + |c_E|\|k_m\|_{C^{0,\beta}(\mathbb{T})} \le C\delta^{-4-2\beta}\|E\|_{Y_\zeta}. \tag{3.30}$$

In particular,

$$|c_E| \le C\delta^{-2-\beta}\|E\|_{Y_\zeta}. \tag{3.31}$$

The solution operators can be chosen with the following continuity property. Fix m and $0 < \beta' < \beta$. Suppose $\zeta_j \to \zeta$, $E_j \in Y_{\zeta_j}$ have uniformly bounded Y_{ζ_j} norms, $E \in Y_\zeta$, and $E_j \circ F_{\zeta_j} \to E \circ F_\zeta$ in $C^{0,\beta'}(M_0)$. If $(u_j, c_j) = \mathscr{R}_{\zeta_j} E_j$ and $(u,c) = \mathscr{R}_\zeta E$, then $u_j \circ F_{\zeta_j} \to u \circ F_\zeta$ in $C^{2,\beta'}(M_0)$ and $c_j \to c$.

(2) Define the scalar error D_ζ and the preliminary correction u_ζ by

$$D_\zeta = H_\zeta + \tau_\zeta \zeta \mathcal{J}_\zeta, \qquad (u_\zeta, c_{H,\zeta}) = -\mathscr{R}_\zeta D_\zeta. \tag{3.32}$$

Then

$$\|D_\zeta\|_{Y_\zeta} \le C\tau_\zeta^{1+\alpha/3}, \tag{3.33}$$

$$\mathcal{L}_\zeta u_\zeta + H_\zeta = (-\tau_\zeta\zeta + c_{H,\zeta})\mathcal{J}_\zeta. \tag{3.34}$$

(3) For every $(v,\zeta) \in \mathcal{B}_m$, define the total scalar displacement $\phi : M_\zeta \to \mathbb{R}$ by

$$\phi = u_\zeta + v \circ F_\zeta^{-1}.$$

Then

$$\|\phi\|_{X_\zeta} + |c_{H,\zeta}|\|k_m\|_{C^{0,\beta}(\mathbb{T})} \le \tau_\zeta^{1+\alpha/4}. \tag{3.35}$$

The map

$$G_{\zeta,\phi} : M_\zeta \to \mathcal{N},$$
$$G_{\zeta,\phi}(x) := \exp^{\mathcal{N}}_{\jmath_\zeta(x)}(\phi(x)\nu_\zeta(x)) \qquad \forall\, x \in M_\zeta$$

is well defined and is an embedding of class $C^{2,\beta}$. The same conclusion holds with ϕ replaced by $t\phi$ for every $t \in [0,1]$. Write $M_{\zeta,\phi} := G_{\zeta,\phi}(M_\zeta)$. Choose the unit normal $\nu_{\zeta,\phi}$ along $G_{\zeta,\phi}$ by continuity from ν_ζ along the family $G_{\zeta,t\phi}$, $0 \le t \le 1$, and set $H_{\zeta,\phi} := H_{G_{\zeta,\phi},\nu_{\zeta,\phi}}$. This is a function on the fixed domain M_ζ, computed using the induced metric $G^*_{\zeta,\phi} g_{\mathcal{N}}$. The nonlinear remainder

$$Q_\zeta(\phi) = H_{\zeta,\phi} - H_\zeta - \mathcal{L}_\zeta \phi \tag{3.36}$$

satisfies

$$\|Q_\zeta(\phi)\|_{Y_\zeta} \le C\tau_\zeta^{-\alpha/2}\|\phi\|^2_{X_\zeta} \le C\tau_\zeta^2. \tag{3.37}$$

In particular, on every member of this iteration set,

$$|\nabla^j \phi(x)|_{g_\zeta} \le C\tau_\zeta^{1+\alpha/4} f_\zeta(x)\rho_\zeta(x)^{-j}, \qquad x \in M_\zeta, \quad j \in \{0,1,2\}, \tag{3.38}$$

with the corresponding local scaled Hölder estimates.

Proof. The analytic statements we use from [KM23] are the linear construction in Proposition 4.18 and Corollary 4.21, the initial mean-curvature estimate in Lemma 4.6, and the normal-graph estimate in Lemma 5.1. We first verify their geometric hypotheses and explain the restriction to invariant functions. We then derive the estimates in the present notation. The background three-manifold is the fixed open band $\mathcal{N}$, and the base is the closed two-sided minimal torus. Completeness of the background is not required in its Convention 2.1. Its Fermi map agrees with $\mathcal{F}$ because the curves $z \mapsto \mathcal{F}(q,z)$ are unit-speed geodesics normal to the central torus.

We first check the invertibility required on the central torus. On the full torus, the eigenvalues of $L_{\mathbb{T}}$ on $e^{i\sqrt{2}(js+kt)}$ are $2 - 2(j^2+k^2)$. Indeed, translating s by ℓ/m multiplies this Fourier mode by $e^{2\pi i j/m}$, while translating t by ℓ/m multiplies it by $e^{2\pi i k/m}$. Invariance under both translations therefore forces $j,k \in m\mathbb{Z}$. For $m \ge 2$ none of these eigenvalues is zero. Thus the Γ^0_m-invariant kernel is trivial. More explicitly, the full kernel is spanned by $\cos(\sqrt{2}s)$, $\sin(\sqrt{2}s)$, $\cos(\sqrt{2}t)$, and $\sin(\sqrt{2}t)$. Every invariant source is orthogonal to these functions. The inverse of the fixed operator on their orthogonal complement has a fixed $C^{0,\beta} \to C^{2,\beta}$ bound, and uniqueness makes it commute with all the isometries under consideration. This supplies the invariant version of Assumption 4.1 of the cited paper; the unrestricted kernel is not being assumed trivial.

We next verify that the values and differentials prescribed at the lattice points can be corrected with the required bounds. Before imposing symmetry, at each p take the cutoff in (3.15) times the three functions

$$J_0(\sqrt{2}r), \qquad \sqrt{2}J_1(\sqrt{2}r)\cos\theta, \qquad \sqrt{2}J_1(\sqrt{2}r)\sin\theta,$$

where $J_1(x) = \sum_{j\ge 0}(-1)^j (x/2)^{2j+1}/(j!(j+1)!)$. Write $x_1 = s - s(p)$, $x_2 = t - t(p)$ in the flat disk, $r = (x_1^2 + x_2^2)^{1/2}$, and $(x_1,x_2) = r(\cos\theta, \sin\theta)$. The Bessel equations give $(\Delta + 2)u = 0$ for all three functions. Their expansions are

$$J_0(\sqrt{2}r) = 1 - r^2/2 + O(r^4),$$
$$\sqrt{2}J_1(\sqrt{2}r)\cos\theta = x_1 + O(r^3),$$
$$\sqrt{2}J_1(\sqrt{2}r)\sin\theta = x_2 + O(r^3).$$

Hence their values and differentials at p form a basis for $\mathbb{R} \oplus T_p^*\mathbb{T}$. Their supports and the supports of their Jacobi images obey Assumption 3.11 of the cited paper. If the prescribed value and differential have norm at most one, the corresponding combination of these three cutoff functions has $C^{2,\beta}$ norm

at most $C\delta^{-2-\beta}$. To see this, every derivative of a cutoff contributes one factor δ^{-1}; taking a Hölder quotient contributes at most $\delta^{-\beta}$. The uncut Bessel functions have uniformly bounded derivatives on the fixed small disks. The difference from each prescribed affine jet is $O(r^2)$, with all fixed scaled derivatives, by the displayed Bessel series. Disjoint supports give the same estimates for arbitrary collections of jets, with the maximum norm on their coefficients; there is no factor equal to the number of necks. Invariance under the reflections removes the linear jets, and lattice transitivity identifies all constant jets. The invariant space is therefore exactly $\mathbb{R}\widehat{v}_m$, with Jacobi image $\mathbb{R}k_m$.

We check the separation and smallness assumptions with the scales already chosen. Convention 3.8 holds: the 9δ disks are disjoint because $18\delta < \ell/m$; the Fermi injectivity radius has a fixed positive lower bound; and the Dirichlet operator $\Delta + 2$ on every disk of radius $\delta' \le 3\delta$ is invertible. For the last assertion its first $-\Delta$ eigenvalue is $j_{0,1}^2/(\delta')^2 > 2$ for large m, where $j_{0,1}$ is the first positive zero of J_0. The same convergent radial series (3.5) with $a = 2$, divided by 2π, supplies a local Green function $P_2/(2\pi)$ with logarithmic coefficient one on these small flat disks, together with the required regularity.

All logarithmic coefficients are equal to $\tau = \tau_\zeta$. For large m,

$$9\tau^\alpha < \tau^{\alpha/100} < \delta, \qquad \tau \le \tau^{1-\alpha/100}.$$

The boundary norm in Convention 3.15(iv) satisfies

$$\delta^{-2}\|\psi_\zeta\|_{C^{2,\beta}(\partial B(p,\delta))} \le C\tau m^{4+\beta} = o(\tau^{1-\alpha/9}).$$

On $\rho \ge \tau^\alpha$, the local derivative estimates and a two-point estimate for the Hölder seminorm give

$$\|\psi_\zeta\|_{C^{3,\beta}} \le C\tau^{1-\alpha(3+\beta)} = o(\tau^{8/9}).$$

For the two-point estimate, use a scaled local estimate when the points are within a fixed fraction of τ^α, and the supremum bound otherwise. If needed near the boundary, these local estimates are taken on the larger region $\rho \ge \frac{1}{2}\tau^\alpha$. Finally (3.22) gives $\psi_\zeta \ge c\tau m^2 \ge \tau^{1+\alpha/5}$ there. This verifies all parts of Convention 3.15 uniformly in ζ. The inequalities in Convention 4.5 are precisely $1/4 > 2\alpha$ and $(1-\alpha)/2 > 2\alpha$, which follow from (3.11). The same convention fixes a large constant separating the neck region $\rho \le B\tau$ from its graphical ends. Choose that constant once and for all before choosing m.

We now check the definition of the initial surface and record the parameter conditions needed in the later existence argument. The lattice, local disks and jet spaces do not move with ζ; the base diffeomorphisms are identities. The normalized mismatch map is $c \mapsto c/\tau_\zeta$, so (3.14) makes the normalized mismatch exactly ζ. The neck elevation and tilt parameters are zero. The bounded ratio $\tau_\zeta/\tau_0 = e^\zeta$ and the preceding estimates give the invariant specialization of Assumption 5.2. Definitions (3.16) and (3.18) are Definition 3.17 of the cited paper with these choices. Since the supports of the Jacobi images k_m lie in $\delta \le r_p \le 2\delta$, they are disjoint from all regions $\rho \le 3\tau^\alpha$ for large m; its lifted substitute function is therefore exactly (3.26).

With these verifications, we apply the linear construction and explain why its error term has only one coefficient here. Proposition 4.18 and Corollary 4.21 of the cited paper construct a linear solution operator whose equation has one substitute-function output on each of the two sheets. We use their proofs on the invariant spaces, as in the symmetry restriction in the proof of their Theorem 6.17. In particular, the torus equation in their equation (4.13) can be solved explicitly within the invariant spaces. Given a Γ_m^0-invariant source $E' \in C^{0,\beta}(\mathbb{T})$, let u^{base} be the unique invariant solution of

$$L_{\mathbb{T}} u^{\mathrm{base}} = E'.$$

Lattice transitivity makes its value at every $p \in L_m$ equal to a common number c, and the coordinate reflections fixing p imply $du^{\mathrm{base}}(p) = 0$. Using the cutoff function already constructed, set

$$\widehat{v} := c\widehat{v}_m, \qquad u' := u^{\mathrm{base}} - \widehat{v}, \qquad w' := -L_{\mathbb{T}}\widehat{v} = -ck_m.$$

Then all these functions are invariant and satisfy

$$L_{\mathbb{T}}u' = E' + w', \qquad u'(p) = 0, \qquad du'(p) = 0 \qquad \forall\, p \in L_m.$$

The invariant torus inverse and the cutoff estimates above provide exactly the bounds for this step; no inverse on the unrestricted torus space is needed. The symmetry restriction is likewise legitimate at every other step: the base inverse just described, the radial cutoffs, the local neck solutions, the jet maps, and the sums over the lattice commute with Γ_m^0. If a local solution operator is not initially invariant, average its conjugates over the finite group fixing the corresponding neck. For a local group G and operator R, this average is $|G|^{-1}\sum_{g\in G} g^* R(g^{-1})^*$. Since the group acts isometrically and preserves the radial weights, the average has no larger norm and satisfies the same local equation. Translating the chosen operators to every lattice point then gives an invariant global construction. The fixed-torus inverse used in that proof is replaced by the invariant inverse justified above; all other local estimates are unchanged. One then imposes Z-invariance. Height reflection exchanges the two exterior substitute outputs with the same sign, so both are $c_E k_m$. The two-weight norms are uniformly equivalent to those in (3.25), because the first-harmonic projection vanishes. Corollary 4.21 therefore gives (3.30), with loss $C\delta^{-4-2\beta}$.

For the coefficient estimate, the fixed cutoff satisfies

$$c\delta^{-2-\beta} \le \|k_m\|_{C^{0,\beta}(\mathbb{T})} \le C\delta^{-2-\beta}. \tag{3.39}$$

The upper bound follows by scaling. For the lower bound, write $r = \delta x$ on a single annulus and let $\eta(x) = \eta_{2,1}(x)$. Then

$$k_m(p + \delta x e_\theta) = \delta^{-2}\big(\eta''(x) + x^{-1}\eta'(x)\big) + O(1).$$

The leading function is not identically zero, since a radial harmonic function that is constant near each boundary circle cannot change from one to zero. Compare a point where this function is nonzero with a point in the inner region where $k_m = 0$. Their distance is comparable to δ, giving the required Hölder quotient. Combining (3.39) with (3.30) proves (3.31).

We next apply the initial mean-curvature estimate and compute the equation satisfied by u_ζ. The two substitute terms in Lemma 4.6 of the cited paper are both $-\tau_\zeta\zeta k_m$, because the same height correction $-\tau_\zeta\zeta\widehat{v}_m$ is used on the two sheets. That lemma therefore gives

$$\|H_\zeta + \tau_\zeta\zeta\mathcal{J}_\zeta\|_{Y_\zeta} \le C\tau_\zeta^{1+\alpha/3},$$

which is (3.33). By the linearity of $\mathscr{R}_\zeta$, the definition $(u_\zeta, c_{H,\zeta}) = -\mathscr{R}_\zeta D_\zeta$ gives

$$\begin{aligned}\mathcal{L}_\zeta u_\zeta &= -D_\zeta + c_{H,\zeta}\mathcal{J}_\zeta \\ &= -H_\zeta + (-\tau_\zeta\zeta + c_{H,\zeta})\mathcal{J}_\zeta.\end{aligned}$$

This proves (3.34) with the stated signs.

The solution operators may be chosen continuously in the precise sense stated in the proposition. The right inverse is the one constructed by the local solution operators and convergent correction series in Proposition 4.18 and Corollary 4.21.

We retain the normalizations in equations (4.13) and (4.16) of [KM23]. Each auxiliary torus solution has value and differential zero at every point of L_m, as in the construction of u' above. On each catenoid, the solutions for the constant and first angular harmonics have both their value and their ξ-derivative equal to zero at $\xi = 0$, the waist. The remaining angular modes are solved using the inverse of $\Delta_{\mathbb{S}^2} + 2$ on functions with angular Fourier orders at least two, after the conformal change of metric given by the catenoid Gauss map. The restriction to these angular orders excludes every nonzero element of the kernel of $\Delta_{\mathbb{S}^2} + 2$. In the invariant spaces used here, the first angular harmonics vanish identically.

After conjugation by F_ζ, each finite stage of that construction is jointly continuous in the parameter and the source with any loss from exponent β to β': the cutoffs and coefficients vary smoothly, the

local elliptic inverses vary continuously, and the moving smooth pullbacks are continuous on bounded stronger-Hölder sets. The tails of the correction series are uniformly small in the β norm, hence also in the β' norm. Applying these observations first to a finite stage and then letting the number of stages tend to infinity proves the precise sequential continuity asserted above.

It remains to estimate the total displacement ϕ and its nonlinear mean-curvature error on all of $\mathcal{B}_m$. The preceding bounds give

$$\|u_\zeta\|_{X_\zeta} + |c_{H,\zeta}|\|k_m\|_{C^{0,\beta}} \le C\delta^{-4-2\beta}\tau_\zeta^{1+\alpha/3}.$$

Its ratio to $\tau_\zeta^{1+\alpha/4}$ tends to zero uniformly because $\delta^{-4-2\beta}\tau_\zeta^{\alpha/12} \to 0$. Also (3.28) gives

$$\|v \circ F_\zeta^{-1}\|_{X_\zeta} \le C_F\tau_0^{1+\alpha} \le C\tau_\zeta^{1+\alpha},$$

whose ratio to $\tau_\zeta^{1+\alpha/4}$ also tends to zero. In fact we have proved the stronger uniform bound

$$\|\phi\|_{X_\zeta} + |c_{H,\zeta}|\|k_m\|_{C^{0,\beta}(\mathbb{T})} = o\big(\tau_\zeta^{1+\alpha/4}\big).$$

The little-o estimate is uniform for $(v,\zeta) \in \mathcal{B}_m$ as $m \to \infty$. This proves (3.35) and also absorbs the fixed constants in the comparison with the norms of [KM23]. The smallness hypothesis of its Lemma 5.1 is therefore satisfied. That lemma gives the embeddedness of $G_{\zeta,\phi}$ and the first inequality in (3.37). We explain the scaling of this estimate, since it will be used again for the second normal direction. On a coordinate neighborhood of radius comparable to $\rho = \rho_\zeta(x)$, divide lengths by ρ. The rescaled normal displacement is ϕ/ρ, and its $C^{2,\beta}$ norm is at most $C\|\phi\|_{X_\zeta} f_\zeta(x)/\rho$. The local geometry established above gives uniform bounds for the coefficients of the mean-curvature map in these coordinates. Taylor's formula at zero therefore bounds its remainder, after subtracting its value and derivative, by the square of this displacement norm. Returning to the original metric contributes the factor ρ^{-1} to mean curvature, so

$$\|Q_\zeta(\phi)\|_{C^{0,\beta}(B_{g_\zeta}(x,\rho/100),\rho^{-2}g_\zeta)} \le C\rho^{-3}f_\zeta(x)^2\|\phi\|_{X_\zeta}^2.$$

Dividing by the target weight $f_\zeta(x)/\rho^2$ leaves the factor

$$\frac{f_\zeta(x)}{\rho} = \max\{\rho^{1/2}, \tau_\zeta^{(1-\alpha)/2}\rho^{-1/2}\} \le C\tau_\zeta^{-\alpha/2},$$

since $\tau_\zeta \le \rho \le \pi/m$. This is the factor in the cited nonlinear estimate. In addition, the rescaled displacement is small uniformly, because $\|\phi\|_{X_\zeta} f_\zeta/\rho \le C\tau_\zeta^{1-\alpha/4} \to 0$. Finally,

$$\tau_\zeta^{-\alpha/2}\|\phi\|_{X_\zeta}^2 \le \tau_\zeta^{-\alpha/2}\tau_\zeta^{2+\alpha/2} = \tau_\zeta^2,$$

proving the second inequality in (3.37). Evaluating the derivative terms in the definition of X_ζ at the center of each ball gives (3.38), and the seminorm term gives its local Hölder counterpart. □

Remark 3.5. All estimates in Proposition 3.4 hold on the entire scalar iteration set; none assumes that an iterate is minimal in $\mathcal{N}$ or in $\mathbb{S}^4$. In particular, the available bounds control the scalar correction and its first two covariant derivatives, together with the local β-Hölder seminorm of its second derivative. Subsequent estimates on arbitrary iterates will use this regularity without differentiating the correction a third time.

4. Correction normal to the suspension band

Section 3 provides a family of embeddings $G_{\zeta,\phi} : M_\zeta \to \mathcal{N}$, where $\phi = u_\zeta + v \circ F_\zeta^{-1}$ and $(v,\zeta) \in \mathcal{B}_m$. Each image has two independent normal directions in $\mathbb{S}^4$: its unit normal ν within $\mathcal{N}$, and the unit normal n to $\mathcal{N}$. Its mean-curvature vector has a scalar component in each direction. To obtain a minimal surface in $\mathbb{S}^4$, both components must vanish.

In this section we deform each of these surfaces along the spherical geodesics with initial velocity n. The displacement is a scalar function b on M_ζ. We choose b so that one scalar component of

the mean-curvature vector of the displaced surface vanishes. Since the tangent plane changes under the deformation, the normal direction in which we test this component must also be specified; we do this by projecting the parallel transport of n onto the new normal plane. We then estimate the change in the other scalar component. Section 5 will use that estimate to choose v and ζ so that the remaining component vanishes as well. No minimality assumption is imposed on $G_{\zeta,\phi}$ here.

The argument has three parts. First, we obtain geometric estimates valid for every $(v,\zeta) \in \mathcal{B}_m$. Next, we use the symmetries of the coefficient b to invert the scalar operator that describes the first-order change of the component in the direction n. Finally, we estimate the nonlinear terms, solve for b, and bound its effect on the remaining component. All estimates must be uniform on $\mathcal{B}_m$, because the choice of (v,ζ) is made only in the next section.

Fix $(v,\zeta) \in \mathcal{B}_m$, with $\mathcal{B}_m$ as in (3.29), and abbreviate

$$\tau := \tau_\zeta, \qquad \varepsilon := \varepsilon_\zeta, \qquad \rho := \rho_\zeta, \qquad f := f_\zeta, \qquad \phi := u_\zeta + v \circ F_\zeta^{-1}.$$

Recall that $\jmath_\zeta : M_\zeta \hookrightarrow \mathcal{N}$ is the inclusion, and set

$$\begin{aligned} G = G_{\zeta,\phi} : M_\zeta &\to \mathcal{N}, \\ G(x) &:= \exp^{\mathcal{N}}_{\jmath_\zeta(x)}\big(\phi(x)\nu_\zeta(x)\big) \qquad \forall\, x \in M_\zeta, \\ g_\phi &:= G^* g_{\mathcal{N}}. \end{aligned}$$

Proposition 3.4 shows that G is a $C^{2,\beta}$ embedding with image $M_{\zeta,\phi}$. To distinguish the two ambient manifolds in the notation for its second fundamental forms, let

$$\jmath := G : M_\zeta \to \mathcal{N}, \qquad \iota := \iota_{\mathcal{N}} \circ G : M_\zeta \to \mathbb{S}^4,$$

where $\iota_{\mathcal{N}}$ is the inclusion of $\mathcal{N}$ into $\mathbb{S}^4$. Both maps induce g_ϕ on M_ζ. We write $\nu := \nu_{\zeta,\phi}$ for the unit normal along $\jmath$ chosen in Proposition 3.4, by continuity from ν_ζ along $G_{\zeta,t\phi}$, $0 \le t \le 1$. We also write n for the field along ι given by

$$n(x) := N\big(\pi(G(x))\big) \qquad \forall\, x \in M_\zeta.$$

Thus (ν, n) is an orthonormal frame of $T_\iota^\perp M_\zeta$. With the notation of Section 2,

$$h^\nu_\iota = h^\nu_\jmath, \qquad S^\nu_\iota = S^\nu_\jmath, \qquad H_{\iota,\nu} = H_{\jmath,\nu} = H_{\zeta,\phi}, \qquad \mathbf{H}_\iota = H_{\zeta,\phi}\nu + H_{\iota,n} n.$$

The first three identities hold because the difference between the spherical and the $\mathcal{N}$ covariant derivatives is in the direction n, perpendicular to ν. All the displayed tensors are defined on the domain M_ζ. In particular, they are the geometric quantities of the image $M_{\zeta,\phi}$ pulled back by G; they are not the corresponding quantities of the initial inclusion $\jmath_\zeta$.

The normal-connection one-form ω is the one in (2.6), now computed along ι in the frame (ν, n). The field ν is $C^{1,\beta}$. The field n is $C^{2,\beta}$, since it is obtained by composing G with the fixed smooth normal field on $\mathcal{N}$. We continue to use the smooth metric g_ζ and the function $\rho = \rho_\zeta$ on the initial surface to define all weighted norms. In particular, ρ is not replaced by a distance function on the deformed surface.

Unless a dependence is explicitly stated, every constant below is independent of m and of $(v,\zeta) \in \mathcal{B}_m$. It may depend on the fixed band, cutoffs, α, and β. Continuity in (v,ζ) always refers to fixed m. We write $o_m(1)$ for a quantity tending to zero as $m \to \infty$, uniformly over all parameters and points under consideration.

4.1. **Uniform geometry after the deformation within $\mathcal{N}$.** We first specify the norms for the new displacement b and the scalar component it must cancel. Unlike the displacement ϕ, whose allowed size varies according to f_ζ, the new displacement will have size $O(\tau_\zeta^2)$ everywhere. Its first

and second derivatives are still measured at the local length scale ρ_ζ. Using (3.24), define

$$\begin{aligned} \mathcal{X}_\zeta &:= C^{2,\beta}_\chi(M_\zeta), & \|b\|_{\mathcal{X}_\zeta} &:= \|b\|_{C^{2,\beta}_\zeta(1)}, \\ \mathcal{Z}_\zeta &:= C^{0,\beta}(M_\zeta), & \|E\|_{\mathcal{Z}_\zeta} &:= \|E\|_{C^{0,\beta}_\zeta(\rho^{-2})}, \\ \mathcal{Y}_\zeta &:= C^{0,\beta}_\chi(M_\zeta), & \|E\|_{\mathcal{Y}_\zeta} &:= \|E\|_{\mathcal{Z}_\zeta}. \end{aligned} \tag{4.1}$$

Here the subscript χ imposes the transformation rule defined in (2.17), with the character χ of (2.16). The space $\mathcal{Z}_\zeta$ has no symmetry restriction; we will also use its norm for the change in the invariant scalar component. It differs from the scalar source space Y_ζ in (3.25), whose weight is f_ζ/ρ_ζ^2. All three spaces in (4.1) are Banach spaces for fixed m and ζ.

For example, if $b \in \mathcal{X}_\zeta$ and $K_b := \|b\|_{\mathcal{X}_\zeta}$, then

$$\begin{aligned} |b(x)| &\le K_b, \\ |\nabla^{g_\zeta} b(x)|_{g_\zeta} &\le K_b \rho(x)^{-1}, \\ |(\nabla^{g_\zeta})^2 b(x)|_{g_\zeta} &\le K_b \rho(x)^{-2} \qquad \forall\, x \in M_\zeta. \end{aligned}$$

The same norm includes the corresponding Hölder estimates on the balls in (3.24). Similarly,

$$|E(x)| \le \|E\|_{\mathcal{Z}_\zeta} \rho(x)^{-2} \qquad \forall\, x \in M_\zeta, \quad E \in \mathcal{Z}_\zeta.$$

Thus a second derivative of a function of size one has the size allowed by the source weight ρ^{-2}. The estimate $|H_{\iota,n}| \le C\tau^2\rho^{-2}$ below will consequently give a source of norm $O(\tau^2)$.

We also need a precise convention for the local Hölder estimates on geometric tensors. Given $x \in M_\zeta$, put $r := \rho(x)$ and $U_x := B_{g_\zeta}(x, r/100)$. When we assert an estimate $|T| \le Cw(\rho)$ with local scaled $C^{0,\beta}$ control, we mean that

$$\sup_{U_x} |T| + r^\beta [T]_{\beta; U_x} \le Cw(r) \qquad \forall\, x \in M_\zeta.$$

In the seminorm, points are compared using their g_ζ distance, tangent tensors are transported by the g_ζ connection, and ambient vector components are compared in $\mathbb{R}^5$. Tangent tensor norms use g_ϕ, unless another metric is displayed. These are norms in the original metric, not in the rescaled metric $r^{-2}g_\zeta$; the factor r^β scales only the Hölder quotient. The same estimates hold on balls with any other fixed sufficiently small multiple of r as radius, after adjusting the constant.

Lemma 4.1 (Geometry after the deformation within $\mathcal{N}$). *There exists $C > 0$, independent of m and of $(v, \zeta) \in \mathcal{B}_m$, such that the following estimates hold on M_ζ, with local scaled $C^{0,\beta}$ control:*

$$\begin{aligned} |z \circ G| &\le C\varepsilon, & |\nu_h| &\le C\tau/\rho, & |S^\nu_\iota| &\le C\tau/\rho^2, \\ |h^n_\iota| &\le C, & |H_{\iota,n}| &\le C\tau^2/\rho^2, & |\omega| &\le C\tau/\rho, \\ |\nabla^{g_\phi}\omega| &\le C\tau/\rho^2, & |q| &\le C, & q &:= 2 + |S^n_\iota|^2 - |\omega|^2. \end{aligned} \tag{4.2}$$

Here $z : \mathcal{N} \to (-z_0, z_0)$ is the height coordinate, and

$$\nu_h(x) := \sum_{a=1}^{2} \big\langle \nu(x), E_a(G(x)) \big\rangle E_a(G(x)) \qquad \forall\, x \in M_\zeta.$$

Norms and contractions of tangent tensors use g_ϕ; ambient vector norms and inner products use $g_{\mathbb{S}^4}$. Moreover,

$$d_{\mathbb{T}}\big(\pi(G(x)), L_m\big) = \big(1 + o_m(1)\big)\rho(x) \qquad \forall\, x \in M_\zeta, \tag{4.3}$$

uniformly for $(v, \zeta) \in \mathcal{B}_m$.

Proof. We first compare the size of ϕ with the local length scale ρ and with the neck radius τ. Put $B_\tau := \tau^{1+\alpha/4}$. By (3.35), the local scaled $C^{2,\beta}$ norm of ϕ on U_x is at most $B_\tau f(x)$. The definition of f gives

$$\frac{B_\tau f(x)}{\rho(x)} = \max\left\{\tau^{1+\alpha/4}\rho(x)^{1/2},\, \tau^{3/2-\alpha/4}\rho(x)^{-1/2}\right\} \qquad \forall\, x \in M_\zeta.$$

Since $\tau \le \rho(x) \le \pi/m$, it follows that

$$\sup_{x\in M_\zeta} \frac{B_\tau f(x)}{\rho(x)} \le C\left(\tau^{1-\alpha/4} + \tau^{1+\alpha/4}m^{-1/2}\right) = o_m(1). \tag{4.4}$$

A second estimate will ensure that the change in the normal and second fundamental form is smaller than the bounds already proved for the initial surface. Both terms defining f are increasing functions of ρ. At $\rho = \pi/m$, the ratio of the second term to the first is

$$\frac{\tau^{(1-\alpha)/2}(\pi/m)^{1/2}}{(\pi/m)^{3/2}} = \frac{m}{\pi}\tau^{(1-\alpha)/2} = o_m(1).$$

Therefore

$$\sup_{x\in M_\zeta} \frac{B_\tau f(x)}{\tau} \le C\tau^{\alpha/4}m^{-3/2} = o_m(1). \tag{4.5}$$

The limits are uniform in ζ, since $e^{-1}\tau_0 \le \tau_\zeta \le e\tau_0$ and (3.12) implies that every positive power of τ_0 tends to zero faster than every fixed inverse power of m.

We next explain how these two estimates control the geometric quantities. Fix $x \in M_\zeta$ and put $r := \rho(x)$. Use one of the charts supplied by Lemma 3.2, with its independent variables rescaled so that they range over a fixed disk $D \subset \mathbb{R}^2$. Denote this chart by $\kappa_x : D \to M_\zeta$, with $\kappa_x(0) = x$, and write $I := \jmath_\zeta \circ \kappa_x$. The coefficients of $r^{-2}\kappa_x^* g_\zeta$ are uniformly elliptic with uniformly bounded derivatives. The explicit graph and catenoid parametrizations in the proof of that lemma also bound every fixed number of derivatives of $(I - I(0))/r$ and $\nu_\zeta \circ \kappa_x$. By using a smaller fixed disk if necessary, the estimates include the ball U_x.

Define

$$\widehat{\phi}(y) := r^{-1}\phi(\kappa_x(y)),$$
$$\mathcal{E}_r(y,t) := r^{-1}\left[\exp^{\mathcal{N}}_{I(y)}\bigl(rt\nu_\zeta(\kappa_x(y))\bigr) - I(0)\right]$$

for $y \in D$ and $|t|$ less than a fixed positive constant. The subtraction is in $\mathbb{R}^5$. The initial height tends uniformly to zero and $r \le \pi/m$, so these exponential maps are defined inside the fixed band for large m. On a smaller fixed disk and a fixed interval of t, all derivatives of $\mathcal{E}_r$ needed below are uniformly bounded. This follows by differentiating the exponential map of the fixed smooth metric, using the stated bounds for I and ν_ζ; each differentiation with respect to t contributes a factor r before the overall division by r. The rescaled deformed immersion is $y \mapsto \mathcal{E}_r(y, \widehat{\phi}(y))$, and

$$\|\widehat{\phi}\|_{C^{2,\beta}(D)} \le C\frac{B_\tau f(x)}{r} = o_m(1),$$

after restricting the disk once more if necessary. A fixed finite covering by the balls used in (3.24) gives this estimate; ρ and f are comparable to their central values on each such chart.

For $i, j \in \{1, 2\}$, the first two coordinate derivatives are

$$\partial_i[\mathcal{E}_r(y,\widehat{\phi}(y))] = (\mathcal{E}_r)_i + (\mathcal{E}_r)_t\partial_i\widehat{\phi},$$
$$\partial_{ij}[\mathcal{E}_r(y,\widehat{\phi}(y))] = (\mathcal{E}_r)_{ij} + (\mathcal{E}_r)_{it}\partial_j\widehat{\phi} + (\mathcal{E}_r)_{jt}\partial_i\widehat{\phi} + (\mathcal{E}_r)_{tt}\partial_i\widehat{\phi}\partial_j\widehat{\phi} + (\mathcal{E}_r)_t\partial_{ij}\widehat{\phi},$$

where the derivatives of $\mathcal{E}_r$ on the right are evaluated at $(y, \widehat{\phi}(y))$. The induced metric depends on the first derivatives in these formulas. The unit normal is obtained by solving its orthogonality equations and normalizing, and the second fundamental form is obtained by pairing the second

covariant derivatives of the immersion with this normal. These are smooth functions of the displayed derivatives, the ambient metric, and its first derivatives, as long as the induced metric is uniformly positive definite. Estimate (4.4) preserves that positive definiteness.

Applying the mean value theorem to these smooth functions, and using $[ab]_\beta \le \|a\|_\infty [b]_\beta + \|b\|_\infty [a]_\beta$, bounds the change in the rescaled normal and second fundamental form by $C\|\widehat{\phi}\|_{C^{2,\beta}}$ in $C^{0,\beta}$. The same argument bounds the change in the rescaled metric coefficients in $C^{1,\beta}$. In particular, g_ϕ and g_ζ are uniformly equivalent. Returning to the original length scale, the changes in the normal and shape operator are bounded by

$$C\frac{B_\tau f(x)}{r}, \qquad C\frac{B_\tau f(x)}{r^2},$$

respectively, with their local scaled $C^{0,\beta}$ bounds. Normals may be compared in $\mathbb{R}^5$; shape operators are compared as endomorphisms of the same tangent bundle TM_ζ.

In view of (4.5), these changes are $o_m(1)\tau/r$ and $o_m(1)\tau/r^2$. The horizontal projection also changes because it is evaluated at $G(x)$ instead of $\jmath_\zeta(x)$. The fields E_1, E_2 are smooth with bounded derivatives on the fixed band, so this contributes at most $C|\phi|$, with the corresponding local scaled bound. Since $r < 1$ for large m, this is bounded by $CB_\tau f(x)/r$. The initial estimates (3.19) therefore give

$$|\nu_h| \le C\tau/\rho, \qquad |S^\nu_\iota| \le C\tau/\rho^2,$$

with the asserted Hölder control. Here $S^\nu_\iota = S^\nu_\jmath$, as explained above.

The coordinate z has $|\nabla^{\mathcal{N}} z| = 1$. The geodesic segment defining $G(x)$ has length $|\phi(x)|$, and hence

$$|z(G(x))| \le |z(\jmath_\zeta(x))| + |\phi(x)| \le C\varepsilon + B_\tau f(x) \le C\varepsilon \qquad \forall\, x \in M_\zeta.$$

The last step uses (4.5) and $\varepsilon = m^2\tau/(2\pi)$. On the initial charts, the explicit parametrizations in Lemma 3.2 give local scaled Hölder bounds of size $C\varepsilon$ for the initial height. Composing the smooth height function with the exponential map above changes these bounds by at most $CB_\tau f(x)$. This proves the same estimate for $z \circ G$.

The other quantities follow from the identities of Section 2. Written on the domain M_ζ, they are

$$\begin{aligned} h^n_\iota(Y_1, Y_2) &= \big\langle S(dG(Y_1)), dG(Y_2)\big\rangle, \\ H_{\iota,n} &= -\big\langle S(\nu_h), \nu_h\big\rangle, \\ \omega(Y_1) &= \big\langle S(dG(Y_1)), \nu_h\big\rangle \end{aligned}$$

for $Y_1, Y_2 \in \Gamma(TM_\zeta)$, with S evaluated at $G(x)$ in each pointwise identity. Since dG is an isometry from (TM_ζ, g_ϕ) onto the tangent plane of the image and $|S| \le C$, we obtain

$$\begin{aligned} |h^n_\iota| &\le C, \\ |H_{\iota,n}| &\le C|\nu_h|^2 \le C\tau^2/\rho^2, \\ |\omega| &\le C|\nu_h| \le C\tau/\rho. \end{aligned}$$

The local scaled Hölder bounds follow from the same formulas and the product inequality. In particular, the estimate for $H_{\iota,n}$ uses both factors of ν_h; replacing it by a bound linear in $|\nu_h|$ would lose the factor of τ needed for the displacement estimate.

To estimate the covariant derivative of ω, use (2.9) in the explicit pullback form

$$(\nabla^{g_\phi}_{Y_1}\omega)(Y_2) = \big\langle (\nabla^{\mathcal{N}}_{dG(Y_1)} S)(dG(Y_2)), \nu\big\rangle + h^\nu_\iota(Y_1, Y_2)\langle S(\nu), \nu\rangle - \big\langle S(dG(Y_2)), dG(S^\nu_\iota(Y_1))\big\rangle$$

for $Y_1, Y_2 \in \Gamma(TM_\zeta)$. It follows that

$$|\nabla^{g_\phi}\omega| \le |(\nabla^{\mathcal{N}} S) \circ G| + 2|S \circ G|\,|S^\nu_\iota| \le C\big(|z \circ G| + |S^\nu_\iota|\big) \le C\big(m^2\tau + \tau/\rho^2\big) \le C\tau/\rho^2.$$

The last inequality uses $\rho \le \pi/m$. The explicit connection and shape-operator formulas in Lemma 2.2 show that $\nabla^{\mathcal{N}} S$ is a smooth tensor with a factor $\tan z$. Its composition with G therefore has local

scaled $C^{0,\beta}$ norm at most $C\varepsilon$. Applying the product inequality to the preceding identity proves the claimed Hölder bound for $\nabla^{g_\phi}\omega$. This argument differentiates neither S_ι^ν nor the second derivatives of ϕ. The bound for q now follows from $|S_\iota^n| = |h_\iota^n| \le C$ and $|\omega| \le C$, since $\rho \ge \tau$.

Finally, $|d\pi| \le \sec z_0$ on the fixed band. Projecting the geodesic segment used to define $G(x)$ gives

$$d_{\mathbb{T}}\big(\pi(G(x)), \pi(\jmath_\zeta(x))\big) \le \sec z_0\, |\phi(x)|.$$

Distance to the set L_m is a 1-Lipschitz function on $\mathbb{T}$. Recalling $\rho(x) = d_{\mathbb{T}}(\pi(\jmath_\zeta(x)), L_m)$, we consequently obtain

$$\big|d_{\mathbb{T}}\big(\pi(G(x)), L_m\big) - \rho(x)\big| \le \sec z_0\, |\phi(x)| \le CB_\tau f(x) = o_m(1)\rho(x) \qquad \forall\, x \in M_\zeta.$$

This proves (4.3). No differentiability of the distance to L_m at a boundary between adjacent lattice cells is needed. □

4.2. **An auxiliary metric and a uniform inverse for its Laplacian.** The linear operator for the correction in the direction n contains a Laplace operator and a bounded zeroth-order term. We first study a Laplace operator for which the symmetry conditions give estimates independent of the neck radius. For this purpose we define a smooth metric g_* on M_ζ. This metric agrees with the Euclidean catenoid metric near every waist and with the flat metric pulled back from $\mathbb{T}$ away from the necks. We will prove a uniform inverse estimate for Δ_{g_*} on functions with character χ. The next subsection compares this operator with the operator on the deformed surface. The metric g_* is used only for these estimates; it is not asserted to be induced by an immersion into $\mathbb{S}^4$.

Set

$$a_m := \frac{\ell}{2m}, \qquad r_c := \tau^\alpha, \qquad \theta_* := \frac{\pi}{4}.$$

Thus a_m is half the side length of a square lattice cell. Since $\tau/r_c \to 0$ and $r_c/a_m \to 0$ as $m \to \infty$, all the changes to the flat metric described below occur strictly inside each cell. Choose a fixed smooth function $\eta_* : [0,\infty) \to [0,1]$ that equals one on $[0,1/4]$ and zero on $[1/2,\infty)$. Define

$$R(\xi) := \frac{\tau}{2}e^\xi, \qquad r_\tau(\xi) := R(\xi) + \eta_*(R(\xi)/r_c)\frac{\tau^2}{4R(\xi)} \qquad \forall\, \xi \in [0,\infty). \tag{4.6}$$

When $R(\xi) \le r_c/4$, the cutoff equals one, and

$$r_\tau(\xi) = \frac{\tau}{2}(e^\xi + e^{-\xi}) = \tau\cosh\xi.$$

When $R(\xi) \ge r_c/2$, it equals zero, so $r_\tau(\xi) = R(\xi)$. In the remaining interval, differentiation gives

$$\begin{aligned} r_\tau'(\xi) &= R(\xi) + \frac{\tau^2}{4R(\xi)}\left[\frac{R(\xi)}{r_c}\eta_*'(R(\xi)/r_c) - \eta_*(R(\xi)/r_c)\right] \\ &= R(\xi)\big(1 + O((\tau/r_c)^2)\big). \end{aligned}$$

The constants in this estimate depend only on the fixed cutoff. Consequently, for sufficiently large m, $r_\tau(0) = \tau$, $r_\tau'(0) = 0$, and $r_\tau'(\xi) > 0$ for every $\xi > 0$. The variable ξ therefore determines the distance $r_\tau(\xi)$ from the lattice point on the upper half of a neck and on its adjoining sheet.

To specify the parametrization on this entire region, extend the upper height h_ζ in (3.16) to $r_p > \tau$ by setting it equal to $C_\tau(r_p)$ for $\tau < r_p \le 9\tau$. The extension is smooth because the original height already equals C_τ on the overlap. For $p \in L_m$, define

$$\Theta_{\zeta,p}(\xi,\theta) := \mathcal{F}\big(p + r_\tau(\xi)(\cos\theta,\sin\theta), h_\zeta(p + r_\tau(\xi)(\cos\theta,\sin\theta))\big), \tag{4.7}$$

on the domain

$$\mathcal{W}_{m,\tau} := \left\{(\xi,\theta) : 0 < \theta < \theta_*, \quad 0 < \xi < \log\frac{2a_m}{\tau\cos\theta}\right\}. \tag{4.8}$$

Here the displacement from p is taken in the flat coordinate disk containing its square cell. The image is the part of the upper sheet over the triangular sector $0 < \theta < \pi/4$ of that cell. Its outer

boundary is the cell wall $r_\tau(\xi)\cos\theta = a_m$. This wall lies outside the cutoff region, where $r_\tau = R$, and hence its equation is

$$\xi = d_m - \log\cos\theta, \qquad d_m := \log\frac{2a_m}{\tau}.$$

This explains the upper bound for ξ in (4.8). Near $\xi = 0$, the height in (4.7) is $C_\tau(\tau\cosh\xi) = \tau\xi$. Thus this parametrization extends smoothly to the waist, even though the height as a function of the base radius has an unbounded derivative there.

We define g_* on this parametrized sector by

$$\Theta_{\zeta,p}^* g_* := r_\tau(\xi)^2(d\xi^2 + d\theta^2), \tag{4.9}$$

and extend the definition using the symmetries of Section 2. We check that this procedure defines a smooth metric on all of M_ζ. Near the waist, extending ξ to negative values gives the signed catenoidal parametrization, and the metric is

$$\tau^2\cosh^2\xi\,(d\xi^2 + d\theta^2).$$

This expression is smooth across $\xi = 0$ and is unchanged by $\xi \mapsto -\xi$. Across the sides $\theta = 0$ and $\theta = \pi/4$, smoothness follows because the coefficient depends only on ξ. Finally, outside the radial cutoff we have $r_\tau = R$ and $dr_\tau = r_\tau\,d\xi$, so

$$r_\tau^2(d\xi^2 + d\theta^2) = dr_\tau^2 + r_\tau^2 d\theta^2 = \Theta_{\zeta,p}^*(\pi^* g_{\mathbb{T}}).$$

In particular, the metrics defined in adjacent cells agree on a neighborhood of their common wall, where both are $\pi^* g_{\mathbb{T}}$. These observations also apply at intersections of the sides. The resulting metric is smooth and Γ_m-invariant.

The scale function ρ remains the one already defined on M_ζ. Since its value is the distance of the base point from L_m, the present coordinates give

$$\rho\circ\Theta_{\zeta,p} = r_\tau(\xi), \qquad R(\xi) \le r_\tau(\xi) \le 2R(\xi), \qquad \tau \le \rho \le \pi/m.$$

The middle inequalities follow from $0 \le \eta_* \le 1$ and $R \ge \tau/2$. The last upper bound is the distance from the center of a square cell to one of its corners.

A subscript $*$ on the spaces and norms in (4.1) means that g_ζ is replaced by g_*. More generally, $C^{k,\beta}_{*,\rho}(w)$ denotes the norm (3.24) with metric g_*, scale ρ, and weight w. The scale is frozen at the center of each ball, exactly as in that definition. In particular,

$$\|b\|_{\mathcal{X}_{\zeta,*}} = \sup_{x\in M_\zeta} \|b\|_{C^{2,\beta}(B_{g_*}(x,\rho(x)/100),\,\rho(x)^{-2}g_*)},$$
$$\|E\|_{\mathcal{Y}_{\zeta,*}} = \sup_{x\in M_\zeta} \rho(x)^2\|E\|_{C^{0,\beta}(B_{g_*}(x,\rho(x)/100),\,\rho(x)^{-2}g_*)}.$$

Both spaces consist of functions with character χ.

We record the uniform local estimates for g_* needed below. For every fixed integer $j \ge 1$, the jth derivative of $\log r_\tau$ with respect to ξ is uniformly bounded. On the cutoff interval this follows by differentiating

$$\frac{r_\tau(\xi)}{R(\xi)} - 1 = \eta_*(R(\xi)/r_c)\frac{\tau^2}{4R(\xi)^2}, \qquad \left|\partial_\xi^j\left(\frac{r_\tau}{R} - 1\right)\right| \le C_j(\tau/r_c)^2.$$

On the other intervals it follows from $r_\tau = \tau\cosh\xi$ or $r_\tau = (\tau/2)e^\xi$. These formulas also give $|d\rho|_{g_*} \le 2$ wherever ρ is smooth, for sufficiently large m. At cell walls, ρ is the minimum of the adjacent distance functions, so it is globally 2-Lipschitz. In particular, its values are uniformly comparable on balls whose radii are a sufficiently small fixed multiple of $\rho(x)$.

For completeness, there are uniform coordinate disks on these balls, including the balls that meet a waist or a cell wall. To see this near a waist, fix $B > 4$. For sufficiently large m, the region $\rho \le 4B\tau$ has exactly the Euclidean catenoid metric. At a point with $\rho(x) \le B\tau$, any path reaching

$\rho = 4B\tau$ has length at least $(4B\tau - \rho(x))/2 > \rho(x)$, by the Lipschitz bound just proved. The catenoid injectivity estimate used in the proof of (3.21) therefore applies on a ball of radius $\rho(x)/2$.

For $\rho(x) \geq B\tau$, use the base coordinates on the sheet over $B_{\mathbb{T}}(\pi(x), \rho(x)/2)$. The distance from every point of this disk to L_m is at least $\rho(x)/2$, so the disk avoids all the waists. In a cell, writing $r = r_\tau(\xi)$ gives

$$g_* = \left(\frac{r_\tau(\xi)}{r'_\tau(\xi)}\right)^2 dr^2 + r^2 d\theta^2.$$

In the catenoidal region, the radial coefficient is $(1-\tau^2/r^2)^{-1}$. In the cutoff region it is $1+O((\tau/r_c)^2)$, with the same bounds for its fixed-order derivatives after multiplication by the corresponding powers of r_c. Elsewhere it equals one. Consequently, after distances in the base disk are divided by $\rho(x)$, the metric differs from the Euclidean metric in C^2 norm by $O(B^{-2}) + o_m(1)$. The same conclusion holds across cell walls, where the metric is already flat. Choosing B large and then m large gives

$$\mathrm{inj}_{g_*}(x) \geq \rho(x)/10 \qquad \forall\, x \in M_\zeta.$$

Indeed, a path leaving the indicated base disk has length at least $\rho(x)/3$, by the metric comparison, and the exponential map on vectors of length at most $\rho(x)/10$ is C^1 close, after rescaling, to the Euclidean exponential map. The argument by local invertibility and injectivity is the same as in the proof of (3.21). The derivative bounds above give uniform bounds of every fixed order for the metric coefficients in these scaled coordinate disks. Thus the constants in the local elliptic estimates used below do not depend on m or ζ.

The boundary of the sector has a different role from these local coordinate disks. It is used only to prove a global estimate for functions with character χ. Figure 6 shows its domain in the (θ, ξ) plane. At fixed ξ, the angular interval ends at θ_*. It begins at 0 for $0 < \xi \leq d_m$, and at $\arccos(e^{d_m - \xi})$ for $d_m < \xi < d_m + \log\sqrt{2}$. Thus its length is always at most $\pi/4$. This length bound, including where the interval is shortened by the cell wall, is what gives the estimate below.

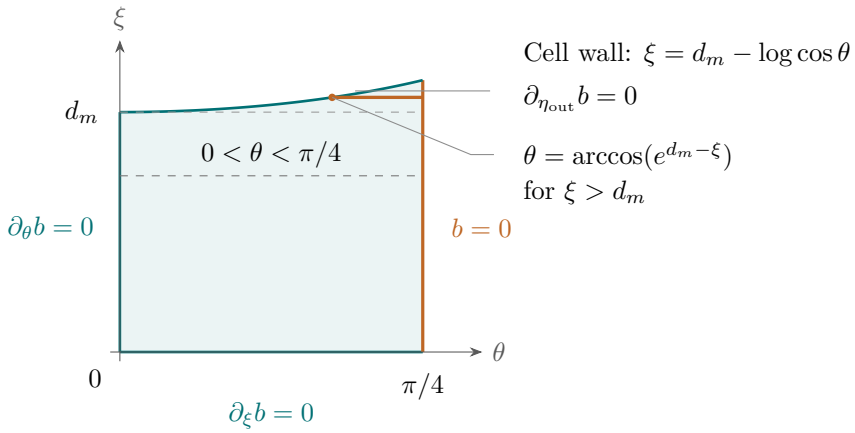


FIGURE 6. The domain $\mathcal{W}_{m,\tau}$ used for the model metric, with $d_m = \log(2a_m/\tau)$. The boundary conditions shown are those for the coefficient b. Below $\xi = d_m$, a horizontal interval has endpoints 0 and $\pi/4$. Above it, the interval begins on the cell wall and is shorter, but still has zero value at its endpoint $\theta = \pi/4$. Its length is therefore at most $\pi/4$, which is the fact used in the one-dimensional Poincaré estimate. The outward derivative on the curved boundary is taken in the model metric.

For the weak formulation used below, let $H^1(M_\zeta)$ denote the Sobolev space of functions with square-integrable value and weak differential, computed with respect to g_*. Define

$$H^1_\chi(M_\zeta) := \left\{ b \in H^1(M_\zeta) : b \circ g = \chi(g) b \text{ almost everywhere on } M_\zeta \text{ for every } g \in \Gamma_m \right\}.$$

Each $g \in \Gamma_m$ preserves g_*, so the map $b \mapsto b \circ g$ is an isometry of $H^1(M_\zeta)$. The defining conditions therefore make $H^1_\chi(M_\zeta)$ a closed linear subspace of $H^1(M_\zeta)$.

Lemma 4.2 (Uniform inverse for the auxiliary Laplacian). *The operator*

$$\Delta_{g_*} : \mathcal{X}_{\zeta,*} \to \mathcal{Y}_{\zeta,*}$$

is an isomorphism. There is a constant $C > 0$, independent of m and ζ, such that

$$\|b\|_{\mathcal{X}_{\zeta,*}} \le C\|\Delta_{g_*} b\|_{\mathcal{Y}_{\zeta,*}} \qquad \forall\, b \in \mathcal{X}_{\zeta,*}. \tag{4.10}$$

Proof. We first describe the restrictions of a smooth function b with character χ to one sector. Interchange of the two coordinates fixes its diagonal side and has character -1. Hence $b = 0$ on $\theta = \theta_*$. Reflection across the axis $\theta = 0$ and reflection across the waist both have character $+1$. The same is true of reflection across the cell wall: on the base it is the composition of a lattice translation with a coordinate reflection. These three reflections preserve g_*, and the corresponding evenness of b implies zero outward normal derivative on the axis, waist, and cell wall. Thus the boundary conditions in Figure 6 follow from the globally imposed symmetry. They are not additional conditions on b.

At a fixed value of ξ, denote the angular interval by $(a, a+L)$, where $a + L = \theta_*$ and $0 < L \le \theta_*$. If $v \in H^1(a, a+L)$ has trace zero at $a+L$, its even extension $\widetilde{v}(t) := v(a+|t|)$ belongs to $H^1_0(-L, L)$. The Dirichlet Poincaré inequality on this interval gives

$$2\int_a^{a+L} |v'|^2\, d\theta = \int_{-L}^{L} |\widetilde{v}'|^2\, dt \ge \frac{\pi^2}{4L^2}\int_{-L}^{L} |\widetilde{v}|^2\, dt = \frac{\pi^2}{2L^2}\int_a^{a+L} |v|^2\, d\theta.$$

Dividing by two and using $L \le \pi/4$ yields $\int |v'|^2 \ge 4\int |v|^2$. No condition at the left endpoint was used. This is why the same inequality applies above $\xi = d_m$, where that endpoint lies on the curved cell wall.

Apply this inequality to the angular slices of b. In the coordinates (4.9), the area measure and the energy density satisfy

$$d\mu_* = \rho^2\, d\xi\, d\theta,$$
$$|\nabla b|^2_{g_*}\, d\mu_* = \left(|\partial_\xi b|^2 + |\partial_\theta b|^2\right) d\xi\, d\theta.$$

Integration over ξ therefore gives

$$4\int_{\mathcal{W}_{m,\tau}} \rho^{-2} b^2\, d\mu_* \le \int_{\mathcal{W}_{m,\tau}} |\nabla b|^2_{g_*}\, d\mu_*. \tag{4.11}$$

Here, and in the computations on the sector below, functions, the metric, and the area measure are pulled back by $\Theta_{\zeta,p}$. The argument proves (4.11) for every H^1 function with zero trace on the diagonal side. Summing over all reflected sectors gives the same inequality on M_ζ for functions in $H^1_\chi(M_\zeta)$. In particular,

$$\frac{4m^2}{\pi^2}\int_{M_\zeta} b^2\, d\mu_* \le \int_{M_\zeta} |\nabla b|^2_{g_*}\, d\mu_*.$$

This uses $\rho \le \pi/m$ and excludes a nonzero constant function from the space under consideration.

We next solve the equation before proving the uniform Hölder estimate. For $E \in \mathcal{Y}_{\zeta,*}$, the preceding inequality makes the Dirichlet bilinear form coercive on the closed subspace $H^1_\chi(M_\zeta) \subset H^1(M_\zeta)$. The Lax–Milgram theorem therefore gives a unique $b \in H^1_\chi(M_\zeta)$ satisfying

$$\int_{M_\zeta} \langle \nabla b, \nabla w\rangle_{g_*}\, d\mu_* = -\int_{M_\zeta} Ew\, d\mu_* \qquad \forall\, w \in H^1_\chi(M_\zeta).$$

This is also the weak equation against arbitrary H^1 test functions. Indeed, for such a function w, set

$$P_\chi w := \frac{1}{|\Gamma_m|}\sum_{g\in\Gamma_m}\chi(g)\, w\circ g.$$

Then $P_\chi w \in H^1_\chi(M_\zeta)$. Since b, E have character χ and each g preserves g_*, change of variables gives

$$\int_{M_\zeta} E(P_\chi w)\, d\mu_* = \int_{M_\zeta} Ew\, d\mu_*,$$
$$\int_{M_\zeta} \langle \nabla b, \nabla(P_\chi w)\rangle_{g_*}\, d\mu_* = \int_{M_\zeta} \langle \nabla b, \nabla w\rangle_{g_*}\, d\mu_*.$$

Elliptic regularity on the smooth closed surface gives $b \in C^{2,\beta}_\chi(M_\zeta)$. At this step m is fixed. The uniform estimate is established next.

To bound the supremum of b, set

$$A_0 := \sup_{x\in M_\zeta} \rho(x)^2|E(x)|, \qquad \Phi(\theta) := \theta_*^2 - \theta^2 \quad \forall\, \theta\in[0,\theta_*].$$

The conformal formula for the Laplacian gives

$$-\Delta_{g_*}\Phi = -\rho^{-2}(\partial_\xi^2\Phi + \partial_\theta^2\Phi) = 2\rho^{-2}.$$

The function Φ vanishes on the diagonal. Its outward derivative is zero on the axis and waist, because $\partial_\theta\Phi(0) = 0$ and $\partial_\xi\Phi = 0$. On the outer wall, a defining function whose positive side is outside the sector is $R(\xi)\cos\theta - a_m$. Its differential is $R\cos\theta\, d\xi - R\sin\theta\, d\theta$. Thus the outward unit normal to the outer wall, computed within the sector with respect to g_*, is

$$\eta_{\mathrm{out}} = \rho^{-1}\big(\cos\theta\,\partial_\xi - \sin\theta\,\partial_\theta\big), \qquad \partial_{\eta_{\mathrm{out}}}\Phi = \frac{2\theta\sin\theta}{\rho} \geq 0.$$

Consider either of the two functions $w := A_0\Phi \pm b$. Since $\Delta_{g_*} b = E$,

$$-\Delta_{g_*} w = 2A_0\rho^{-2} \mp E \geq A_0\rho^{-2} \geq 0.$$

Moreover, $w = 0$ on the diagonal and $\partial_{\eta_{\mathrm{out}}} w \geq 0$ on every other side. Let $w^- := \max\{-w, 0\}$. Its trace vanishes on the diagonal. Integration by parts therefore gives

$$\begin{aligned} -\int_{\mathcal{W}_{m,\tau}} |\nabla w^-|^2_{g_*}\, d\mu_* &= \int_{\mathcal{W}_{m,\tau}} \langle \nabla w, \nabla w^-\rangle_{g_*}\, d\mu_* \\ &= \int_{\mathcal{W}_{m,\tau}} (-\Delta_{g_*} w) w^-\, d\mu_* + \int_{\partial\mathcal{W}_{m,\tau}} (\partial_{\eta_{\mathrm{out}}} w) w^-\, d\sigma_* \geq 0. \end{aligned}$$

Here $d\sigma_*$ is the boundary length measure. The boundary term on the diagonal is zero, and the terms on the other sides are nonnegative. The identity is valid for the H^1 test function w^- on this piecewise smooth domain; the corner points have boundary measure zero. It follows that $\nabla w^- = 0$. Applying (4.11) to w^- gives $w^- = 0$. Performing this argument for both signs yields $|b| \leq A_0\Phi$ on the sector. Absolute values are preserved under all reflections, so

$$\|b\|_{L^\infty(M_\zeta)} \leq \theta_*^2 A_0 \leq \theta_*^2 \|E\|_{\mathcal{Y}_{\zeta,*}}. \tag{4.12}$$

Finally, fix $x \in M_\zeta$ and put $\rho_x := \rho(x)$ and $\widehat{g}_x := \rho_x^{-2} g_*$. This is a constant rescaling of the metric, so

$$\Delta_{\widehat{g}_x} b = \rho_x^2 E.$$

In the uniform coordinate disks constructed above, the coefficients of $\widehat{g}_x$ are uniformly elliptic and have uniform $C^{1,\beta}$ bounds. Interior Schauder estimates on two concentric disks, with radii bounded below independently of x, m, ζ in the rescaled metric, give

$$\|b\|_{C^{2,\beta}(B_{g_*}(x,\rho_x/100),\widehat{g}_x)} \leq C\left(\|b\|_{L^\infty(M_\zeta)} + \rho_x^2\|E\|_{C^{0,\beta}(B_{g_*}(x,\rho_x/20),\widehat{g}_x)}\right).$$

The larger ball is covered by a uniformly bounded number of balls used in the definition of $\mathcal{Y}_{\zeta,*}$. On these balls ρ is comparable to ρ_x. The source term on the right is therefore bounded by $C\|E\|_{\mathcal{Y}_{\zeta,*}}$. This includes its Hölder seminorm: sufficiently close pairs lie in one of the smaller coordinate balls, while pairs separated by a fixed positive rescaled distance are controlled by the supremum bound. Taking the supremum over x and using (4.12) proves (4.10). All these estimates are interior estimates on the full smooth surface. The sides of the sector introduce no boundary regularity problem. □

Remark 4.3. The source in Lemma 4.2 is a globally defined $C^{0,\beta}$ function with character χ. A Hölder function specified only on the sector need not have this property: for example, its trace on the diagonal must vanish in order that its odd reflection be continuous. The sources used below are defined on the full surface and have the required character by equivariance of the geometric operators.

4.3. **An inverse for the linearized operator in the direction n.** The preceding lemma gives an inverse for Δ_{g_*}. We now transfer that inverse to $\Delta_{g_\phi}+q$, which is the linearization of the mean-curvature component in the direction n. There are two differences to estimate: the change from the auxiliary metric g_* to the induced metric g_ϕ, and the addition of the function q. The metric difference will tend to zero in local rescaled coordinates. The function q is uniformly bounded, so its contribution to the weighted operator norm is bounded by a constant times $\sup_{M_\zeta}\rho^2 \le \pi^2/m^2$.

To state the metric comparison precisely, for a $C^{1,\beta}$ covariant two-tensor k on M_ζ define

$$\mathfrak{M}_*(k) := \sup_{x\in M_\zeta} \|\rho(x)^{-2}k\|_{C^{1,\beta}(B_{g_*}(x,\rho(x)/100),\,\rho(x)^{-2}g_*)}.$$

At each centre x, the number $\rho(x)$ is held fixed throughout the ball. Both the tensor and the metric are rescaled: if $r=\rho(x)$, then

$$|r^{-2}k|_{r^{-2}g_*} = |k|_{g_*}, \qquad |\nabla^{r^{-2}g_*}(r^{-2}k)|_{r^{-2}g_*} = r|\nabla^{g_*}k|_{g_*}.$$

Thus $\mathfrak{M}_*(k)$ controls the relative size of the metric coefficients, their first derivatives at scale r, and the corresponding Hölder seminorms.

Lemma 4.4 (Comparison of the induced and auxiliary metrics). *For $\zeta\in[-1,1]$, set*

$$\eta_m(\zeta) := \varepsilon^2 + \tau^{2-2\alpha} + \tau^{1-\alpha/4} + \tau^{1+\alpha/4}m^{-1/2}.$$

Then $\eta_m(\zeta)$ tends to zero uniformly in ζ as $m\to\infty$. Uniformly for $(v,\zeta)\in\mathcal{B}_m$, one has

$$\begin{aligned} \mathfrak{M}_*(g_\zeta - g_*) &\le C(\varepsilon^2+\tau^{2-2\alpha}),\\ \mathfrak{M}_*(g_\phi - g_*) &\le C\eta_m(\zeta). \end{aligned} \tag{4.13}$$

The norms based on g_* and g_ζ defining $\mathcal{X}_\zeta$ and $\mathcal{Z}_\zeta$ are uniformly equivalent. Moreover, for every $b\in\mathcal{X}_\zeta$,

$$\|(\Delta_{g_\phi}-\Delta_{g_*})b\|_{\mathcal{Z}_\zeta} \le C\eta_m(\zeta)\|b\|_{\mathcal{X}_\zeta}. \tag{4.14}$$

Proof. We first compare g_ζ with g_* in the three regions used to define the auxiliary metric. On the part of each neck where $r_\tau(\xi)=\tau\cosh\xi$, the initial immersion is the catenoidal parametrization with $z=\tau\xi$. Its induced metric is

$$\Theta_{\zeta,p}^* g_\zeta = \tau^2\big(\cos^2 z\sinh^2\xi+1\big)d\xi^2 + \tau^2\cos^2 z\cosh^2\xi\, d\theta^2,$$

where the pullback is by (4.7). Since $\Theta_{\zeta,p}^* g_* = \tau^2\cosh^2\xi(d\xi^2+d\theta^2)$, the relative coefficient matrix of $g_\zeta - g_*$ in these coordinates is

$$(\cos^2 z-1)\begin{pmatrix}\tanh^2\xi & 0\\ 0 & 1\end{pmatrix}.$$

For $j \in \{0,1,2\}$, its jth derivative in ξ has norm at most

$$C\big(z^2 + \tau|z| + \tau^2\big) \le C\varepsilon^2.$$

Here $|z| \le C\varepsilon$ by Lemma 3.2, $\partial_\xi z = \tau$, and $\tau \le C\varepsilon$. The coordinates rescaled at a centre of radius ρ have uniformly bounded coefficients, as proved above for g_*. The displayed derivative bounds therefore give a local rescaled $C^{1,\beta}$ bound $C\varepsilon^2$. The same calculation holds in the signed coordinate ξ through the waist.

Next consider the radial cutoff region $r_c/4 \le R \le r_c/2$ in (4.6). Put

$$a(\xi) := \eta_*(R(\xi)/r_c), \qquad e(\xi) := \frac{\tau^2}{4R(\xi)^2}.$$

The notation in this paragraph is local to this calculation. Since $R' = R$ and $e' = -2e$, one has

$$r_\tau = R(1+ae), \qquad r_\tau' = R\big(1 + e(a'-a)\big), \qquad r_\tau^2 - \tau^2 = R^2\big((1+ae)^2 - 4e\big).$$

Every fixed-order derivative of a is uniformly bounded on this region, whereas $|\partial_\xi^j e| \le C_j(\tau/r_c)^2$. Consequently,

$$\frac{r_\tau'^2}{r_\tau^2 - \tau^2} = \frac{\big(1 + e(a'-a)\big)^2}{(1+ae)^2 - 4e} = 1 + O\big((\tau/r_c)^2\big),$$

and the same bound holds after any fixed positive number of ξ derivatives of the difference from 1. The denominator is bounded away from zero for all sufficiently large m.

The initial surface in this region is still the exact catenoidal parametrization. On its upper half, $z = \tau\,\mathrm{arcosh}(r_\tau/\tau)$. The radial coefficient of its metric, divided by r_τ^2, can be written as

$$\frac{r_\tau'^2}{r_\tau^2}\left(\cos^2 z + \frac{\tau^2}{r_\tau^2 - \tau^2}\right) = \frac{r_\tau'^2}{r_\tau^2 - \tau^2} + (\cos^2 z - 1)\frac{r_\tau'^2}{r_\tau^2}.$$

Its angular coefficient, divided by r_τ^2, is $\cos^2 z$. In addition to $|z| \le C\varepsilon$, differentiation of the height gives

$$z' = \frac{\tau r_\tau'}{\sqrt{r_\tau^2 - \tau^2}}, \qquad |\partial_\xi^j z| \le C_j\tau \quad \text{for every fixed } j \ge 1.$$

These formulas bound the relative metric error and its first two coordinate derivatives by $C(\varepsilon^2 + \tau^2/r_c^2)$. They give the corresponding local $C^{1,\beta}$ estimate, including the derivatives of the radial cutoff. The lower half has the same metric because its height is $-z$.

On the rest of each sheet, where $R \ge r_c/2$, the auxiliary metric is the pullback of the flat metric on $\mathbb{T}$. In the base coordinates the initial metric is

$$g_\zeta = \cos^2 h_\zeta\, g_{\mathbb{T}} + dh_\zeta \otimes dh_\zeta.$$

The graph estimates in the proof of Lemma 3.2, including the matching estimate (3.23), give

$$|h_\zeta| \le C\varepsilon, \qquad \rho^j|D^j h_\zeta| \le C_j\tau \quad \text{for every fixed } j \ge 1,$$

together with their local scaled Hölder bounds. For example, the first scaled derivative of $\cos^2 h_\zeta - 1$ is bounded by $C|h_\zeta|\rho|Dh_\zeta| \le C\varepsilon\tau$, while that of $dh_\zeta \otimes dh_\zeta$ is bounded by $C\rho|Dh_\zeta||D^2h_\zeta| \le C\tau^2/\rho^2$. The product estimate gives the same orders in the scaled Hölder norms. Since $\rho \ge r_c/2$ here, the total relative $C^{1,\beta}$ error is at most

$$C\left(\varepsilon^2 + \frac{\tau^2}{\rho^2}\right) \le C\left(\varepsilon^2 + \frac{\tau^2}{r_c^2}\right).$$

The three regions cover M_ζ, and the coefficient estimates agree on their overlaps. Because $r_c = \tau^\alpha$, this proves the first inequality in (4.13).

We next include the deformation ϕ within $\mathcal{N}$. Let $B_\tau := \tau^{1+\alpha/4}$ and set

$$d_\phi := \sup_{x \in M_\zeta} \frac{B_\tau f(x)}{\rho(x)}.$$

In an initial chart of scale $r = \rho(x)$, write $\widehat{I}$ and $\widehat{G}$ for the initial and deformed immersions into $\mathbb{R}^5$, after subtracting the initial chart centre and dividing by r. The chain-rule calculation in the proof of Lemma 4.1 gives

$$\|\widehat{G} - \widehat{I}\|_{C^{2,\beta}} \le C d_\phi, \qquad \|\widehat{I}\|_{C^{2,\beta}} \le C.$$

Put $W := \widehat{G} - \widehat{I}$. The coefficients of the two rescaled induced metrics satisfy the exact identity

$$(\widehat{g}_\phi - \widehat{g}_\zeta)_{ij} = \langle \partial_i \widehat{I}, \partial_j W \rangle + \langle \partial_i W, \partial_j \widehat{I} \rangle + \langle \partial_i W, \partial_j W \rangle \qquad i, j \in \{1, 2\}.$$

The inner products are Euclidean. Differentiating this identity once uses at most two derivatives of W, hence at most two derivatives of ϕ. The $C^{1,\beta}$ product estimate therefore yields

$$\mathfrak{M}_*(g_\phi - g_\zeta) \le C d_\phi (1 + d_\phi).$$

By (4.4),

$$d_\phi \le C \left(\tau^{1-\alpha/4} + \tau^{1+\alpha/4} m^{-1/2} \right) = o_m(1).$$

Adding the initial metric error proves the second inequality in (4.13).

The first metric comparison gives uniform ellipticity and uniform $C^{1,\beta}$ bounds for the coefficients of g_ζ in the scaled g_* charts. In particular, the balls used in the two definitions of the weighted norms have uniformly comparable radii. A ball in one definition is covered by a uniformly bounded number of slightly smaller balls for the other definition. The scale ρ is comparable on these balls. Finally, for a scalar function b, the first covariant derivative is db, and in local coordinates its second covariant derivative is

$$(\nabla^g)^2_{ij} b = \partial_{ij} b - \Gamma(g)^k_{ij} \partial_k b.$$

The $C^{0,\beta}$ bounds for the Christoffel symbols thus give uniform equivalence of the two $C^{2,\beta}$ function norms. The same argument for order zero, with the common weight ρ^{-2}, gives equivalence of the source norms.

To compare the Laplacians, fix a scaled chart with centre x and $r = \rho(x)$. In its coordinates the difference is

$$r^2(\Delta_{g_\phi} - \Delta_{g_*}) b = (\widehat{g}_\phi^{ij} - \widehat{g}_*^{ij}) \partial_{ij} b - \left(\widehat{g}_\phi^{ij} \Gamma(\widehat{g}_\phi)^k_{ij} - \widehat{g}_*^{ij} \Gamma(\widehat{g}_*)^k_{ij} \right) \partial_k b.$$

Here b denotes its coordinate expression, and the repeated indices range from 1 to 2. Inverting a uniformly positive definite matrix and forming its Christoffel symbols are smooth operations on the metric coefficients and their first derivatives. Therefore the two coefficients on the right have $C^{0,\beta}$ norms at most $C\eta_m(\zeta)$. The product estimate, followed by the supremum over chart centres, proves (4.14). □

Proposition 4.5 (Uniform inverse in the direction n). *Let q be the function defined in* (4.2)*, and set*

$$P_{\zeta,\phi} := \Delta_{g_\phi} + q.$$

For all sufficiently large m and every $(v, \zeta) \in \mathcal{B}_m$, this operator is an isomorphism from $\mathcal{X}_\zeta$ to $\mathcal{Y}_\zeta$. Its inverse $\mathscr{G}_{\zeta,\phi}$ satisfies, for every $E \in \mathcal{Y}_\zeta$,

$$\|\mathscr{G}_{\zeta,\phi} E\|_{\mathcal{X}_\zeta} \le C \|E\|_{\mathcal{Y}_\zeta}, \tag{4.15}$$

with C independent of (v,ζ) and m. Furthermore,

$$\begin{aligned} \|H_{\iota,n}\|_{\mathcal{Y}_\zeta} &\le C\tau^2, \\ \|Bb\|_{\mathcal{Z}_\zeta} &\le C\tau\|b\|_{\mathcal{X}_\zeta} \qquad \forall\, b \in \mathcal{X}_\zeta, \end{aligned} \tag{4.16}$$

where B is the off-diagonal operator in (2.13)*, evaluated along ι.*

Proof. Equivariance of G implies that g_ϕ is Γ_m-invariant. The expression for q in (4.2) is invariant as well. Hence $P_{\zeta,\phi}$ sends functions with character χ to functions with the same character.

Write $\mathcal{D} := P_{\zeta,\phi} - \Delta_{g_*}$. The preceding lemma bounds the difference of the Laplacians. For the potential term, fix $x \in M_\zeta$ and put $r = \rho(x)$. The local scaled $C^{0,\beta}$ bound for q from Lemma 4.1 and the Hölder product estimate give

$$r^2\|qb\|_{0,\beta;r} \le Cr^2\|b\|_{\mathcal{X}_\zeta} \le Cm^{-2}\|b\|_{\mathcal{X}_\zeta}.$$

Here $\|\cdot\|_{0,\beta;r}$ denotes the local Hölder norm after distances are divided by r; tensor sizes are measured in the unrescaled metric, as specified at the beginning of this section. Taking the supremum over x proves

$$\|\mathcal{D}b\|_{\mathcal{Y}_\zeta} \le C\bigl(\eta_m(\zeta) + m^{-2}\bigr)\|b\|_{\mathcal{X}_\zeta} \qquad \forall\, b \in \mathcal{X}_\zeta.$$

Let $K_* := (\Delta_{g_*})^{-1}$ be the inverse in Lemma 4.2, now regarded as an operator $\mathcal{Y}_\zeta \to \mathcal{X}_\zeta$ by the equivalence of norms just proved. Its norm is uniformly bounded. Since $\eta_m(\zeta) + m^{-2} \to 0$ uniformly, we may choose the lower threshold for m so that

$$\|K_*\mathcal{D}\|_{\mathcal{X}_\zeta \to \mathcal{X}_\zeta} \le \frac{1}{2} \qquad \forall\, (v,\zeta) \in \mathcal{B}_m.$$

The identity $P_{\zeta,\phi} = \Delta_{g_*}(I + K_*\mathcal{D})$ then gives

$$\mathscr{G}_{\zeta,\phi} = (I + K_*\mathcal{D})^{-1}K_* = \sum_{j=0}^{\infty}(-K_*\mathcal{D})^j K_*.$$

The series converges in operator norm. It is an inverse on both sides, and its norm is at most $2\|K_*\|$. This proves both the isomorphism assertion and (4.15).

All operators in this argument act on the same spaces of functions on the closed surface M_ζ, with the same prescribed character. Their restrictions to a reflection sector therefore have the same boundary conditions. More explicitly, at a fixed curve of a reflection, the $+1$ and -1 tangent eigenspaces are orthogonal for every invariant metric. An even function has zero derivative in the -1 direction, which is the metric-normal direction to that curve; an odd function has zero value on it. This applies to both g_* and g_ϕ, including the reflection through the waist. At intersections of reflection curves the argument is made on the full surface in its ordinary local coordinates.

It remains to estimate the source and the off-diagonal term. The transformation rule for n and equivariance of the mean-curvature vector give $H_{\iota,n} \in \mathcal{Y}_\zeta$. At a centre x with $r = \rho(x)$, the geometric estimates of Lemma 4.1 give

$$\|H_{\iota,n}\|_{0,\beta;r} \le C\tau^2/r^2.$$

Multiplication by r^2 and the supremum over $x \in M_\zeta$ yield the first inequality in (4.16).

For the second inequality, the exact formula for the off-diagonal operator is

$$Bb = -2\langle\omega, db\rangle_{g_\phi} - (\operatorname{div}_{g_\phi}\omega)b + \langle h_\iota^\nu, h_\iota^n\rangle_{g_\phi} b.$$

Set $K_b := \|b\|_{\mathcal{X}_\zeta}$ and keep the same centre and scale. The definition of this norm gives local scaled bounds K_b for b and CK_b/r for db. The geometric lemma gives bounds $C\tau/r$ for ω, $C\tau/r^2$ for $\nabla^{g_\phi}\omega$ and h_ι^ν, and C for h_ι^n. Each of these statements includes its local scaled $C^{0,\beta}$ bound. Thus

$$\|2\langle\omega, db\rangle_{g_\phi}\|_{0,\beta;r} \le C\frac{\tau}{r}\frac{K_b}{r},$$

$$\|(\operatorname{div}_{g_\phi}\omega)b\|_{0,\beta;r} \le C\frac{\tau}{r^2}K_b,$$
$$\|\langle h^\nu_\iota, h^n_\iota\rangle_{g_\phi} b\|_{0,\beta;r} \le C\frac{\tau}{r^2}K_b.$$

Contraction by g_ϕ^{-1} preserves these bounds because its coefficients have uniform local scaled $C^{0,\beta}$ bounds. Multiplication by r^2 and the supremum over chart centres prove the second inequality in (4.16). The output Bb is invariant, as proved in the symmetry discussion in Section 2; this is why its norm is taken in $\mathcal{Z}_\zeta$, which imposes no character condition. □

The first source estimate allows $H_{\iota,n}$ to have size independent of τ at a waist, where $\rho = \tau$. Indeed, the local estimate there is only $|H_{\iota,n}| \le C\tau^2/\rho^2 = C$. Nevertheless, the solution of the linear correction equation $P_{\zeta,\phi}b = -H_{\iota,n}$ satisfies

$$\|b\|_{\mathcal{X}_\zeta} \le C\tau^2, \qquad |b(x)| + \rho(x)|db(x)|_{g_\zeta} \le C\tau^2 \quad \forall\, x \in M_\zeta.$$

At a waist, the displacement divided by the neck radius and the size of its first derivative are therefore both $O(\tau)$. The next subsection estimates the nonlinear error so that this linear correction can be replaced by one for which the chosen mean-curvature component vanishes exactly.

4.4. **The displaced surface and its two mean-curvature components.** A function b will specify how far to move each point of $M_{\zeta,\phi}$ along the spherical geodesic normal to $\mathcal{N}$. We first write this new immersion explicitly and compute its tangent vectors and induced metric. We then define two scalar functions whose simultaneous vanishing means that the new immersion is minimal in $\mathbb{S}^4$. This requires choosing normal vectors along the new immersion: parallel transport of ν and n alone does not in general give vectors perpendicular to its new tangent plane.

We will project the transported vectors onto that normal plane and normalize them. This choice has a useful property: at $b = 0$, the derivatives of the two scalar functions are exactly B and $P_{\zeta,\phi}$, including when $M_{\zeta,\phi}$ is not minimal. The final estimate in this subsection bounds the error after subtracting these linear terms. It will allow us to solve the equation with derivative $P_{\zeta,\phi}$ by contraction.

Keep $(v,\zeta) \in \mathcal{B}_m$ fixed, and write $X := \iota : M_\zeta \to \mathbb{S}^4 \subset \mathbb{R}^5$ and $g := g_\phi$. Thus X and G have the same values as maps into $\mathbb{R}^5$, but the specified target of X is $\mathbb{S}^4$. For $b \in \mathcal{X}_\zeta$, define $X_b : M_\zeta \to \mathbb{S}^4$ and a vector field V_b along it by

$$\begin{aligned} X_b(x) &:= \exp^{\mathbb{S}^4}_{X(x)}(b(x)n(x)) = \cos(b(x))X(x) + \sin(b(x))n(x), \\ V_b(x) &:= -\sin(b(x))X(x) + \cos(b(x))n(x) \qquad \forall\, x \in M_\zeta. \end{aligned} \tag{4.17}$$

These expressions define a map for every real-valued b. Below we restrict its norm so that the map is an immersion and the normal projections we use are nonzero. For each $x \in M_\zeta$, set

$$\gamma_x(t) := \cos t\, X(x) + \sin t\, n(x) \qquad \forall\, t \in \mathbb{R}.$$

Then γ_x is a unit-speed spherical geodesic, $X_b(x) = \gamma_x(b(x))$, and $V_b(x) = \dot\gamma_x(b(x))$. Indeed, $\ddot\gamma_x = -\gamma_x$, and the spherical connection gives

$$\nabla^{\mathbb{S}^4}_{\dot\gamma_x}\dot\gamma_x = \ddot\gamma_x + |\dot\gamma_x|^2\gamma_x = 0.$$

Thus $V_b(x)$ is the parallel transport of $n(x)$ from $t = 0$ to $t = b(x)$. The vector $\nu(x)$, kept constant in $\mathbb{R}^5$, is perpendicular to $\gamma_x(t)$ and $\dot\gamma_x(t)$ for every t, since it is perpendicular to both $X(x)$ and $n(x)$. Consequently

$$\nabla^{\mathbb{S}^4}_{\dot\gamma_x}\nu(x) = 0 + \langle\dot\gamma_x, \nu(x)\rangle\gamma_x = 0.$$

We therefore use the same vector $\nu(x)$ for its parallel transport to $X_b(x)$. All inner products in the explicit $\mathbb{R}^5$ formulas below are Euclidean; on vectors tangent to $\mathbb{S}^4$ they equal the spherical metric.

Fix a coordinate chart on M_ζ, write ∂_i for its coordinate vector fields, and use indices $i, j \in \{1, 2\}$, summing over repeated indices. Set $X_i := dX(\partial_i)$, $b_i := \partial_i b$ and $\omega_i := \omega(\partial_i)$. The Weingarten equation is

$$\partial_i n = \nabla^{\mathbb{S}^4}_{\partial_i} n = -dX\big(S^n_\iota(\partial_i)\big) - \omega_i \nu.$$

The first equality holds because $\langle X_i, n\rangle = 0$, and the sign in the last term is the one fixed by $\nabla^\perp_{\partial_i}\nu = \omega_i n$. Define the tangent-bundle endomorphism

$$\mathsf{P}_b := \cos b \ \mathrm{Id}_{TM_\zeta} - \sin b\, S^n_\iota.$$

Differentiating (4.17) now gives

$$\begin{aligned}\partial_i X_b &= \cos b\, X_i + \sin b\, \partial_i n + b_i(-\sin b\, X + \cos b\, n)\\ &= \cos b\, dX(\partial_i) - \sin b\, dX\big(S^n_\iota(\partial_i)\big) - \sin b\, \omega_i \nu + b_i V_b.\end{aligned}$$

Equivalently,

$$\partial_i X_b = dX\big(\mathsf{P}_b(\partial_i)\big) - \sin b\, \omega_i \nu + b_i V_b. \tag{4.18}$$

The three terms in (4.18) lie in mutually orthogonal subspaces of $\mathbb{R}^5$: the first is tangent to X, the second is proportional to ν, and the third belongs to $\mathrm{span}\{X, n\}$. Thus the pullback metric $g_b := X_b^* g_{\mathbb{S}^4}$ has coefficients

$$(g_b)_{ij} = g\big(\mathsf{P}_b(\partial_i), \mathsf{P}_b(\partial_j)\big) + \sin^2 b\, \omega_i\omega_j + b_i b_j. \tag{4.19}$$

There is a constant $c_0 > 0$, independent of (v, ζ) and m, such that X_b is an immersion whenever $\|b\|_{\mathcal{X}_\zeta} \le c_0\tau$. To see this, the norm definition, $\rho \ge \tau$, and metric equivalence give

$$|b| \le c_0\tau, \qquad |db|_g \le C c_0\tau/\rho \le C c_0.$$

Since $|S^n_\iota|_g \le C$, decreasing c_0 ensures $|\mathsf{P}_b(Y)|_g \ge |Y|_g/2$ for every tangent vector Y. Formula (4.19) then gives $g_b(Y,Y) \ge g(Y,Y)/4$. It also gives a uniform upper bound $g_b \le Cg$. The small constant c_0 will be decreased below to satisfy the other stated requirements.

We next project the transported vectors V_b and ν onto the normal bundle of X_b within $\mathbb{S}^4$. If $(g_b)^{ij}$ denotes the inverse matrix to $(g_b)_{ij}$, the tangential projection of a vector Z along X_b is $(g_b)^{ij}\langle Z, \partial_i X_b\rangle \partial_j X_b$. In the present case, (4.18) gives

$$\langle V_b, \partial_i X_b\rangle = b_i, \qquad \langle \nu, \partial_i X_b\rangle = -\sin b\, \omega_i.$$

The required normal projections are therefore

$$\begin{aligned}U_b &= V_b - (g_b)^{ij} b_i \partial_j X_b,\\ W_b &= \nu + \sin b\, (g_b)^{ij}\omega_i \partial_j X_b.\end{aligned} \tag{4.20}$$

Both projections remain tangent to $\mathbb{S}^4$ and are perpendicular to $dX_b(TM_\zeta)$. They are defined independently of the coordinate chart because orthogonal projection is intrinsic. To calculate their lengths, set $T_b := (g_b)^{ij} b_i \partial_j X_b$ and $T_\omega := (g_b)^{ij}\omega_i \partial_j X_b$. Then $|T_b|^2 = \langle V_b, T_b\rangle = |db|^2_{g_b}$, $|T_\omega|^2 = |\omega|^2_{g_b}$, and $\langle \nu, T_\omega\rangle = -\sin b\, |\omega|^2_{g_b}$. Also $\langle V_b, T_\omega\rangle = \langle T_b, T_\omega\rangle = \langle db, \omega\rangle_{g_b}$ and $\langle T_b, \nu\rangle = -\sin b \langle db, \omega\rangle_{g_b}$. For example,

$$\begin{aligned}|U_b|^2 = |V_b - T_b|^2 &= 1 - 2\langle V_b, T_b\rangle + |T_b|^2 = 1 - |db|^2_{g_b},\\ \langle U_b, W_b\rangle &= \sin b\, \langle V_b, T_\omega\rangle - \langle T_b, \nu\rangle - \sin b\, \langle T_b, T_\omega\rangle = \sin b\, \langle db, \omega\rangle_{g_b}.\end{aligned}$$

Together with the same expansion for W_b, this proves

$$\begin{aligned}|U_b|^2 &= 1 - |db|^2_{g_b},\\ |W_b|^2 &= 1 - \sin^2 b\, |\omega|^2_{g_b},\\ \langle U_b, W_b\rangle &= \sin b\, \langle db, \omega\rangle_{g_b}.\end{aligned} \tag{4.21}$$

On the indicated ball, $|db|_{g_b} \le Cc_0$ and $|\sin b|\,|\omega|_{g_b} \le Cc_0\tau$. It follows that

$$|U_b|^2|W_b|^2 - \langle U_b, W_b\rangle^2 \ge 1 - Cc_0^2 \ge \frac{1}{2}$$

after decreasing c_0. Hence $U_b(x)$ and $W_b(x)$ are linearly independent and span the normal plane of X_b at every $x \in M_\zeta$. Define the unit normal fields

$$n_b(x) := \frac{U_b(x)}{|U_b(x)|}, \qquad \nu_b(x) := \frac{W_b(x)}{|W_b(x)|} \qquad \forall\, x \in M_\zeta. \tag{4.22}$$

They form a basis of the normal plane, but need not be orthogonal. At $b = 0$ they reduce to n and ν.

Let $\mathbf{H}(X_b)$ denote the mean-curvature vector of X_b in $\mathbb{S}^4$, as a vector field along the map with domain M_ζ. We retain the trace convention from Section 2. For $b \in \mathcal{X}_\zeta$ with $\|b\|_{\mathcal{X}_\zeta} \le c_0\tau$, define functions on M_ζ by

$$\mathscr{P}_{\zeta,\phi}(b) := \langle \mathbf{H}(X_b), n_b\rangle, \qquad \mathscr{T}_{\zeta,\phi}(b) := \langle \mathbf{H}(X_b), \nu_b\rangle.$$

These are scalar products with the two unit normals; they are not being used as coefficients of a decomposition in the generally nonorthogonal basis (n_b, ν_b). Since that basis spans the normal plane,

$$\mathbf{H}(X_b) = 0 \quad \Leftrightarrow \quad \mathscr{P}_{\zeta,\phi}(b) = 0 \text{ and } \mathscr{T}_{\zeta,\phi}(b) = 0.$$

At zero the two functions are $H_{\iota,n}$ and $H_{\zeta,\phi}$, respectively. In particular the latter agrees with the scalar mean curvature defined in Proposition 3.4. Figure 7 illustrates the displacement defining X_b.

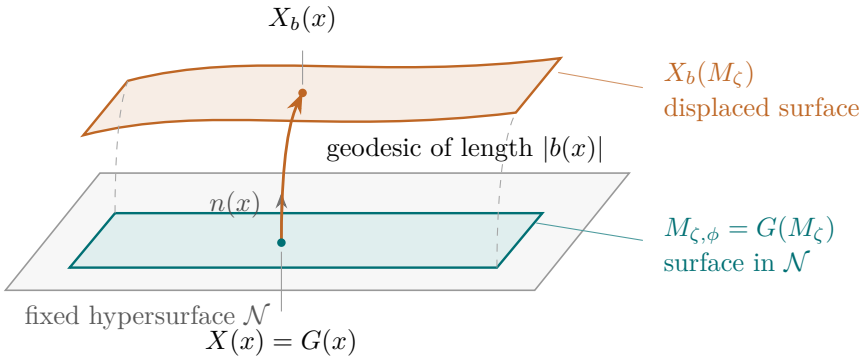


FIGURE 7. The displacement normal to the fixed hypersurface $\mathcal{N}$. The teal patch belongs to the surface $M_{\zeta,\phi}$ obtained by the preceding deformation within $\mathcal{N}$. Each point $X(x) = G(x)$ moves along the spherical geodesic with initial unit velocity $n(x)$ to $X_b(x) = \cos(b(x))X(x) + \sin(b(x))n(x)$. A point with $b(x) > 0$ is shown; the function b may have either sign. The gray plane represents $\mathcal{N}$ schematically, and the displacement is exaggerated. The drawing does not represent the spherical metric or all three dimensions of $\mathcal{N}$.

Lemma 4.6 (Linearization without a minimality assumption). *For every $h \in \mathcal{X}_\zeta$,*

$$D\mathscr{P}_{\zeta,\phi}(0)h = P_{\zeta,\phi}h, \qquad D\mathscr{T}_{\zeta,\phi}(0)h = Bh.$$

These formulas require no minimality of $M_{\zeta,\phi}$.

Proof. Fix $h \in \mathcal{X}_\zeta$ and consider $b = th$ for t in an interval about zero satisfying $|t|\|h\|_{\mathcal{X}_\zeta} < c_0\tau$. At $t = 0$ the velocity of X_{th} is hn. The ordinary derivative of V_{th} is $-hX$, so its spherical covariant derivative is $-hX + \langle hn, n\rangle X = 0$.

In the tangential term of U_{th} there is a factor $t\partial_i h$. At zero, its derivative is consequently $g^{ij}(\partial_i h)X_j = dX(\nabla^g h)$; the derivatives of the inverse metric and of $\partial_j X_{th}$ are multiplied by $t = 0$. Likewise, differentiating the tangential term of W_{th} differentiates only $\sin(th)$ at zero. Thus

$$\nabla_t^{\mathbb{S}^4} U_{th}\Big|_{t=0} = -dX(\nabla^g h), \qquad \nabla_t^{\mathbb{S}^4} W_{th}\Big|_{t=0} = h\, dX(\omega^{\sharp_g}).$$

Here $\omega^{\sharp_g}$ is defined by $g(\omega^{\sharp_g}, Y) = \omega(Y)$ for every tangent vector Y. Both displayed derivatives are tangent to X. Hence, differentiating the squared lengths gives

$$\frac{d}{dt}\Big|_{t=0} |U_{th}|^2 = 2\langle -dX(\nabla^g h), n\rangle = 0,$$
$$\frac{d}{dt}\Big|_{t=0} |W_{th}|^2 = 2\langle h\, dX(\omega^{\sharp_g}), \nu\rangle = 0.$$

Since $U_0 = n$ and $W_0 = \nu$ are unit vectors, normalization does not change these first derivatives. Differentiating each scalar component now yields

$$D\mathscr{P}_{\zeta,\phi}(0)h = \Big\langle \nabla_t^{\mathbb{S}^4}\Big|_{t=0} \mathbf{H}(X_{th}), n\Big\rangle - \langle \mathbf{H}(X), dX(\nabla^g h)\rangle,$$
$$D\mathscr{T}_{\zeta,\phi}(0)h = \Big\langle \nabla_t^{\mathbb{S}^4}\Big|_{t=0} \mathbf{H}(X_{th}), \nu\Big\rangle + \langle \mathbf{H}(X), h\, dX(\omega^{\sharp_g})\rangle.$$

The last term in each line is zero because the mean-curvature vector is normal to X. The first terms are the n and ν components of $J(hn)$. Lemma 2.5 gives $J(hn) = (Bh)\nu + (P_{\zeta,\phi}h)n$, proving both formulas. No step has used $\mathbf{H}(X) = 0$. □

We now estimate the terms left after subtracting the value and derivative at $b = 0$. Define

$$\begin{aligned} Q_n(b) &= \mathscr{P}_{\zeta,\phi}(b) - H_{\iota,n} - P_{\zeta,\phi}b, \\ Q_\nu(b) &= \mathscr{T}_{\zeta,\phi}(b) - H_{\zeta,\phi} - Bb. \end{aligned} \tag{4.23}$$

The fixed pair (ζ, ϕ) is suppressed in this notation. Both remainders vanish at zero and have zero derivative there. The next estimate quantifies their quadratic dependence on b, uniformly over the scalar deformations allowed in $\mathcal{B}_m$.

Lemma 4.7 (Quadratic estimates for the two components). *There are constants $c_0, C > 0$ such that, for every $b, \widetilde{b} \in \mathcal{X}_\zeta$ with $\|b\|_{\mathcal{X}_\zeta}, \|\widetilde{b}\|_{\mathcal{X}_\zeta} \le c_0\tau$,*

$$\|Q_\eta(b) - Q_\eta(\widetilde{b})\|_{\mathcal{Z}_\zeta} \le C\tau^{-1}\big(\|b\|_{\mathcal{X}_\zeta} + \|\widetilde{b}\|_{\mathcal{X}_\zeta}\big)\|b - \widetilde{b}\|_{\mathcal{X}_\zeta}, \qquad \eta \in \{n, \nu\}. \tag{4.24}$$

The constants are uniform over $\mathcal{B}_m$. Only the $C^{2,\beta}$ bound for the scalar correction ϕ is used. In particular, setting $\widetilde{b} = 0$ gives

$$\|Q_\eta(b)\|_{\mathcal{Z}_\zeta} \le C\tau^{-1}\|b\|_{\mathcal{X}_\zeta}^2 \qquad \forall\, \eta \in \{n, \nu\}.$$

Proof. Fix $x \in M_\zeta$ and set $r := \rho(x)$. Choose one of the uniformly controlled charts for the initial surface from Lemma 3.2. After rescaling its independent variables, denote its map by $\kappa_x : D \to M_\zeta$, where $D \subset \mathbb{R}^2$ is a fixed disk and $\kappa_x(0) = x$. We use a smaller concentric disk for the local norm; all estimates are obtained on a slightly larger disk of the same fixed size. Set

$$\overline{X} := \frac{X \circ \kappa_x - X(x)}{r}, \qquad \overline{X}_b := \frac{X_b \circ \kappa_x - X(x)}{r}, \qquad \overline{g}_b := r^{-2}\kappa_x^* g_b.$$

The rescaled immersion $\overline{X}$ has bounded $C^{2,\beta}$ norm and uniformly nondegenerate first derivative. This follows from the initial geometry and (4.4). In the computations in this chart, write X, n, b also for their compositions with κ_x, and let ∂_i denote differentiation with respect to its coordinates. The field n is obtained by composing X with a fixed smooth field on $\mathcal{N}$. Extend that field smoothly to a fixed neighborhood in $\mathbb{R}^5$ of the smaller band containing all the images of X, and denote the extension by $\widetilde{n}$. The chain rule gives

$$\partial_i n = d\widetilde{n}_X(\partial_i X),$$

$$\partial_{ij} n = d^2\widetilde{n}_X(\partial_i X, \partial_j X) + d\widetilde{n}_X(\partial_{ij} X).$$

Thus the value and derivatives of n through order two are controlled by those of X. The derivatives of $\widetilde{n}$ are bounded because the extension and neighborhood are fixed. These formulas compute the derivatives of the original normal field along the parametrized surface and do not depend on the choice of extension. In particular, no second derivative of the other normal ν is needed.

Differentiating (4.17) twice gives

$$\begin{aligned}
\partial_i X_b &= \cos b\, \partial_i X + \sin b\, \partial_i n + b_i V_b,\\
\partial_{ij} X_b &= \cos b\, \partial_{ij} X + \sin b\, \partial_{ij} n + b_{ij} V_b - b_i b_j X_b\\
&\quad + b_i(-\sin b\, \partial_j X + \cos b\, \partial_j n) + b_j(-\sin b\, \partial_i X + \cos b\, \partial_i n).
\end{aligned}$$

These expressions involve no derivative of b or X of order greater than two. The coefficients of the metric involve only first derivatives of X_b. The vector ν can be computed from $X, n, \partial_1\overline{X}, \partial_2\overline{X}$ and the chosen orientation: it is the oriented unit vector orthogonal to these four vectors. Their Gram matrix is uniformly nondegenerate, because X, n are orthonormal and perpendicular to the rescaled tangent vectors, whose Gram matrix is the uniformly elliptic rescaled induced metric. Thus this algebraic formula for ν has uniformly bounded derivatives with respect to its entries. The one-form ω occurring in W_b can also be computed from first derivatives, using $\omega_i = \langle S(X_i), \nu\rangle$. Thus the testing normals in (4.22) are algebraic functions of these first derivatives, the fixed ambient tensors, and the values of b, db.

For either $\eta_b = n_b$ or $\eta_b = \nu_b$,

$$\langle \mathbf{H}(X_b), \eta_b\rangle = (g_b)^{ij}\langle \partial_{ij} X_b, \eta_b\rangle. \tag{4.25}$$

Indeed, the Euclidean formula for the spherical mean-curvature vector is

$$\mathbf{H}(X_b) = (g_b)^{ij}\big(\partial_{ij} X_b - \Gamma(g_b)^k_{ij}\partial_k X_b\big) + 2X_b.$$

The terms containing the Christoffel symbols are tangent to X_b, and η_b is perpendicular to X_b itself. Pairing with η_b therefore gives (4.25). In the rescaled chart it becomes

$$r\langle \mathbf{H}(X_b), \eta_b\rangle = \overline{g}_b^{ij}\langle \partial_{ij}\overline{X}_b, \eta_b\rangle.$$

Thus the rescaled scalar functions are obtained by smooth algebraic expressions in the values and derivatives of X and b through order two, and the first derivatives determining ν. The determinant of $\overline{g}_b$ and the lengths of U_b, W_b are uniformly bounded away from zero, by (4.19) and (4.21).

More explicitly, write $b = r\widehat{b}$ on this chart and multiply mean curvature by r. Substituting $X_b = \cos(r\widehat{b})X + \sin(r\widehat{b})n$ into (4.25) yields a smooth map of $\widehat{b}$ from $C^{2,\beta}$ to $C^{0,\beta}$ whose second Fréchet derivative is uniformly bounded on a fixed small ball. To see that dilation introduces no singular coefficient, write the rescaled graph difference as

$$\frac{X_b - X}{r} = \frac{\cos(r\widehat{b}) - 1}{r}X + \frac{\sin(r\widehat{b})}{r}n.$$

The two scalar coefficients extend smoothly to $r = 0$, with limits 0 and $\widehat{b}$. One can see this without dividing a Taylor remainder by r: for a scalar a,

$$\begin{aligned}
\frac{\sin(ra)}{r} &= a\int_0^1 \cos(tra)\, dt,\\
\frac{\cos(ra) - 1}{r} &= -ra^2\int_0^1 (1-t)\cos(tra)\, dt.
\end{aligned}$$

Both formulas and every fixed derivative in a are bounded uniformly for bounded r, a. All other coefficients are the rescaled derivatives of X, n already estimated, or inverses and square roots of uniformly positive quantities.

To state the resulting Taylor estimate precisely, let $\mathcal{M}_{r,\eta}(\widehat{b})$ denote r times the scalar product with η_b, where $\eta \in \{n, \nu\}$. The formulas above express it as a function $\mathcal{F}_{r,\eta}(y, \mathbf{z}_{\widehat{b}}(y))$, where $y \in D$ and

$$\mathbf{z}_{\widehat{b}} := (\widehat{b}, \partial_1\widehat{b}, \partial_2\widehat{b}, \partial_{11}\widehat{b}, \partial_{12}\widehat{b}, \partial_{22}\widehat{b}).$$

The function $\mathcal{F}_{r,\eta}$ is smooth in these six variables. Its derivatives through order three in those variables have uniformly bounded local $C^{0,\beta}$ norms on the fixed small ball. This follows from the explicit formulas: the operations are products, trigonometric functions, inverses of nondegenerate metric matrices, and square roots of quantities bounded away from zero. The Hölder product and composition estimates consequently give

$$\|D^2\mathcal{M}_{r,\eta}(\widehat{b})[h,k]\|_{C^{0,\beta}} \le C\|h\|_{C^{2,\beta}}\|k\|_{C^{2,\beta}}.$$

For example, set $(\mathcal{D}_1, \dots, \mathcal{D}_6) := (\mathrm{Id}, \partial_1, \partial_2, \partial_{11}, \partial_{12}, \partial_{22})$. The second Fréchet derivative is exactly

$$D^2\mathcal{M}_{r,\eta}(\widehat{b})[h,k] = \sum_{a,c=1}^{6} (\partial_{z_a z_c}\mathcal{F}_{r,\eta})(y, \mathbf{z}_{\widehat{b}}(y))\mathcal{D}_a h\, \mathcal{D}_c k.$$

Estimating each of the finitely many products proves the preceding bound. The derivatives in this formula are derivatives with respect to the six variables, not additional spatial derivatives of X or ϕ.

If $\mathcal{Q}_{r,\eta}$ is the remainder obtained by subtracting the value and derivative of $\mathcal{M}_{r,\eta}$ at zero, then

$$\mathcal{Q}_{r,\eta}(a) - \mathcal{Q}_{r,\eta}(c) = \int_0^1 \big(D\mathcal{M}_{r,\eta}(c + t(a-c)) - D\mathcal{M}_{r,\eta}(0)\big)(a-c)\,dt.$$

Applying the second-derivative bound along the line segment from zero to $c + t(a-c)$ bounds the norm of this expression by $C(\|a\|_{C^{2,\beta}} + \|c\|_{C^{2,\beta}})\|a-c\|_{C^{2,\beta}}$. Take $a = b/r$ and $c = \widetilde{b}/r$. Since $\mathcal{Q}_{r,\eta}(b/r) = rQ_\eta(b)$, multiplication by r gives on the original chart

$$r^2\|Q_\eta(b) - Q_\eta(\widetilde{b})\|_{0,\beta;r} \le \frac{C}{r}\big(\|b\|_{2,\beta;r} + \|\widetilde{b}\|_{2,\beta;r}\big)\|b - \widetilde{b}\|_{2,\beta;r}.$$

Here $\|\cdot\|_{k,\beta;r}$ denotes the ordinary $C^{k,\beta}$ norm after rescaling the independent variable by r^{-1}, with no rescaling of function values. Passing to the frozen-scale norms and using $r \ge c\tau$ proves (4.24).

The normal ν is used only algebraically in (4.20); it is not differentiated twice. No bound on ∇S^ν_ι, $\nabla^2\nu$, or $\nabla^3\phi$ has entered the argument. The normalization of the testing fields is also included: because $\mathbf{H}(X_b)$ is normal, its two scalar products equal

$$\frac{\langle \mathbf{H}(X_b), V_b\rangle}{\sqrt{1 - |db|^2_{g_b}}}, \qquad \frac{\langle \mathbf{H}(X_b), \nu\rangle}{\sqrt{1 - \sin^2 b\,|\omega|^2_{g_b}}}.$$

Their normalization corrections start at quadratic order and are covered by the same Taylor estimate. □

We also need the nonlinear maps to preserve the symmetry conditions of the linear problem. Fix $a \in \Gamma_m$ and let $b \circ a = \chi(a)b$. Equivariance of X and the transformation rule for n give $X_b \circ a = a \circ X_b$: the cosine is even, and the signs from $\sin(b\circ a)$ and $n \circ a$ cancel. The same computation in the formula for V_b gives

$$\begin{aligned} V_b(a(x)) &= \chi(a)\, da_{X_b(x)}(V_b(x)), \\ \nu(a(x)) &= da_{X_b(x)}(\nu(x)) \qquad \forall\, x \in M_\zeta. \end{aligned}$$

In the second line ν denotes the transported field along X_b. Since a is an isometry and X_b is equivariant, its differential maps the tangent and normal planes of X_b at x onto the corresponding planes at $a(x)$. Orthogonal projection and normalization therefore give the same two transformation rules for n_b and ν_b, respectively. The mean-curvature vector itself transforms by da. Taking scalar products proves

$$\mathscr{P}_{\zeta,\phi}(b) \circ a = \chi(a)\mathscr{P}_{\zeta,\phi}(b), \qquad \mathscr{T}_{\zeta,\phi}(b) \circ a = \mathscr{T}_{\zeta,\phi}(b). \tag{4.26}$$

for every $a \in \Gamma_m$. The constant and linear terms subtracted in (4.23) have the same symmetries by Section 2. Thus $Q_n(b) \in \mathcal{Y}_\zeta$, whereas $Q_\nu(b)$ is an invariant element of $\mathcal{Z}_\zeta$.

4.5. **The normal correction and the remaining scalar equation.** For each fixed $(v, \zeta) \in \mathcal{B}_m$, we now solve $\mathscr{P}_{\zeta,\phi}(b) = 0$. Its source at $b = 0$ has weighted size $O(\tau^2)$, its linearization has a uniformly bounded inverse, and its quadratic remainder has Lipschitz constant $O(\tau)$ on a ball of radius $O(\tau^2)$. A contraction therefore selects a unique small function $b(v, \zeta)$. We then retain the change in the other component of mean curvature as a function of the original choice (v, ζ).

Proposition 4.8 (Correction in the direction n). *There exist constants $K > 0$ and m_0 such that for every $m \geq m_0$ and every $(v, \zeta) \in \mathcal{B}_m$ there is a unique solution $b = b(v, \zeta)$ in the ball*

$$\{b \in \mathcal{X}_\zeta : \|b\|_{\mathcal{X}_\zeta} \leq K\tau_\zeta^2\} \tag{4.27}$$

of

$$\mathscr{P}_{\zeta,\phi}(b) = 0. \tag{4.28}$$

The change in the other scalar mean-curvature component caused by this displacement is the following invariant function on M_ζ:

$$\mathcal{R}_\zeta(\phi) = \mathscr{T}_{\zeta,\phi}(b(v, \zeta)) - H_{\zeta,\phi}. \tag{4.29}$$

It obeys

$$\|\mathcal{R}_\zeta(\phi)\|_{\mathcal{Z}_\zeta} \leq C\tau_\zeta^3, \qquad \|\mathcal{R}_\zeta(\phi)\|_{Y_\zeta} \leq C\tau_\zeta^{2+\alpha/2}. \tag{4.30}$$

In particular $\mathcal{R}_\zeta(\phi)$ has zero first angular harmonic on every neck.

For fixed m, these outputs are continuous with any loss of Hölder exponent. Precisely, if $0 < \beta' < \beta$, $(v_j, \zeta_j), (v, \zeta) \in \mathcal{B}_m$, $\zeta_j \to \zeta$ and $v_j \to v$ in $C^{2,\beta'}(M_0)$, then

$$\begin{aligned} b(v_j, \zeta_j) \circ F_{\zeta_j} &\to b(v, \zeta) \circ F_\zeta \qquad \text{in } C^{2,\beta'}(M_0), \\ \mathcal{R}_{\zeta_j}(\phi_j) \circ F_{\zeta_j} &\to \mathcal{R}_\zeta(\phi) \circ F_\zeta \qquad \text{in } C^{0,\beta'}(M_0), \end{aligned} \tag{4.31}$$

where $\phi_j = u_{\zeta_j} + v_j \circ F_{\zeta_j}^{-1}$.

Proof. Let C_L be the inverse bound in (4.15), and let C_H be the forcing constant in (4.16). Fix $K > 2C_L C_H$. Consider

$$\mathcal{A}_{\zeta,\phi}(b) = -\mathscr{G}_{\zeta,\phi}\big(H_{\iota,n} + Q_n(b)\big).$$

Choose m_0 large enough that $K\tau \leq c_0$ for every allowed ζ and $m \geq m_0$. The ball (4.27) is then contained in the domain of Lemma 4.7. Since $Q_n(0) = 0$, that lemma gives

$$\|\mathcal{A}_{\zeta,\phi}(b)\|_{\mathcal{X}_\zeta} \leq C_L C_H \tau^2 + CC_L K^2 \tau^3.$$

The first term is less than $K\tau^2/2$ by the choice of K. Increasing m_0 makes the second at most $K\tau^2/2$. Thus $\mathcal{A}_{\zeta,\phi}$ maps the closed ball into itself. For any two points of the ball, the same estimates give

$$\|\mathcal{A}_{\zeta,\phi}(b) - \mathcal{A}_{\zeta,\phi}(\widetilde{b})\|_{\mathcal{X}_\zeta} \leq 2CC_L K\tau \|b - \widetilde{b}\|_{\mathcal{X}_\zeta}.$$

This factor is at most $1/2$ for large m. Equation (4.26) ensures that $H_{\iota,n} + Q_n(b) \in \mathcal{Y}_\zeta$, so the inverse in the definition of $\mathcal{A}_{\zeta,\phi}$ is applied to its proper domain. The closed ball is a complete metric space, and the contraction theorem gives a unique fixed point in it. Applying $P_{\zeta,\phi}$ to the fixed-point identity gives

$$P_{\zeta,\phi} b = -H_{\iota,n} - Q_n(b),$$

which is precisely (4.28). Conversely, any solution in this ball satisfies the fixed-point identity, so the uniqueness is uniqueness for that equation as well.

By (4.23), the remaining component is

$$\mathcal{R}_\zeta(\phi) = Bb + Q_\nu(b).$$

Using (4.16) and (4.24), we obtain

$$\|\mathcal{R}_\zeta(\phi)\|_{\mathcal{Z}_\zeta} \le C\tau\|b\|_{\mathcal{X}_\zeta} + C\tau^{-1}\|b\|^2_{\mathcal{X}_\zeta} \le C\tau^3.$$

This proves the first bound in (4.30). To convert it to the scalar source norm, use the definitions with the same frozen-scale balls:

$$\|\mathcal{R}_\zeta(\phi)\|_{Y_\zeta} \le C\tau^3 \sup_{M_\zeta} f(\rho)^{-1}.$$

The weight f is increasing and $\rho \ge \tau$; since $0 < \tau < 1$,

$$f(\tau) = \max\{\tau^{3/2}, \tau^{1-\alpha/2}\} = \tau^{1-\alpha/2}.$$

This gives the second bound. The function $\mathcal{R}_\zeta(\phi)$ is invariant by (4.26). In particular, a quarter-turn around each neck preserves the axial coordinate and the function. If $a_1(\xi), a_2(\xi)$ were its coefficients of $\cos\theta, \sin\theta$, this invariance would require the vector (a_1, a_2) to be fixed by a rotation through $\pi/2$, and hence to be zero. Thus its first angular harmonic vanishes. It therefore belongs to the precise scalar source space used in Proposition 3.4.

It remains to prove continuity. Fix m and $\beta' < \beta$. After pullback by F_{ζ_j}, the scalar corrections ϕ_j converge in $C^{2,\beta'}$ to the correction at (v, ζ), by the continuity statement in Proposition 3.4 and the smooth parameter dependence of the initial surfaces. They are uniformly bounded in $C^{2,\beta}$, since m is fixed and the norms in Section 3 are uniformly equivalent over the compact parameter interval. The solutions for the displacement in the direction n have the same uniform stronger bound by (4.27).

Every subsequence of $b(v_j, \zeta_j) \circ F_{\zeta_j}$ therefore has a further subsequence converging in $C^{2,\beta'}(M_0)$. The explicit formulas (4.20) and (4.25) allow passage to the limit in the equation $\mathscr{P}_{\zeta,\phi}(b) = 0$. The limit retains the bound $\|b\|_{\mathcal{X}_\zeta} \le K\tau_\zeta^2$: the stronger Hölder seminorm is lower semicontinuous, and the smooth parameter identifications and metrics converge. Uniqueness in the contraction ball identifies this limit with $b(v, \zeta) \circ F_\zeta$. Thus the whole sequence converges. Applying the same formulas, which use derivatives only through order two, to the other scalar component proves the second convergence in (4.31). □

Proposition 4.9 (The remaining equation). *For every $(v, \zeta) \in \mathcal{B}_m$ and all sufficiently large m, the map*

$$\widehat{X}_{v,\zeta} : M_\zeta \to \mathbb{S}^4,$$
$$\widehat{X}_{v,\zeta}(x) := \cos(b(v,\zeta)(x))G_{\zeta,\phi}(x) + \sin(b(v,\zeta)(x))n(x) \qquad \forall\, x \in M_\zeta$$

is a $C^{2,\beta}$ embedding. Its image is a closed connected surface, and it is minimal in $\mathbb{S}^4$ if and only if

$$H_{\zeta,\phi} + \mathcal{R}_\zeta(\phi) = 0 \quad \text{on } M_\zeta. \tag{4.32}$$

Its displacement in the direction n satisfies

$$|(\nabla^{g_\zeta})^j b(v,\zeta)(x)|_{g_\zeta} \le C\tau_\zeta^2 \rho_\zeta(x)^{-j} \qquad \forall\, x \in M_\zeta, \quad j \in \{0,1,2\}, \tag{4.33}$$

with the corresponding local scaled Hölder bounds.

Proof. Write $b := b(v, \zeta)$ and $G := G_{\zeta,\phi}$. By Proposition 3.4, the map G is an embedding. Consider

$$x \mapsto \big(\mathcal{F}^{-1}(G(x)), b(x)\big) \quad \text{from } M_\zeta \text{ into } \mathbb{T} \times (-z_0, z_0) \times \mathbb{R}.$$

This map is an embedding: projection onto the first factor $\mathbb{T} \times (-z_0, z_0)$ gives the embedding $\mathcal{F}^{-1} \circ G$. Its injectivity and injective differential follow immediately from this projection, and its domain is compact.

The bounds $|z \circ G| \le C\varepsilon$ and $\|b\|_\infty \le K\tau^2$ tend uniformly to zero. For large m, the displayed graph therefore lies in the domain of the fixed tubular embedding Ψ from Lemma 2.6. The definition of Ψ and $n(x) = N(\pi(G(x)))$ give the exact identity

$$\widehat{X}_{v,\zeta}(x) = \Psi\big(\mathcal{F}^{-1}(G(x)), b(x)\big) \qquad \forall\, x \in M_\zeta.$$

Thus $\widehat{X}_{v,\zeta}$ is a composition of two embeddings and is itself an embedding. Its image is closed and connected because M_ζ is closed and connected.

The component tested against n_b vanishes by construction. Both n_b and ν_b belong to the normal plane and are linearly independent. A vector in that plane that is orthogonal to both must be zero, whether or not this pair is orthogonal. Thus $\mathbf{H}(\widehat{X}_{v,\zeta}) = 0$ is equivalent to the vanishing of its component tested against ν_b. By definition that component is the left side of (4.32). Finally, (4.33) is the pointwise and local Hölder content of the bound (4.27). □

5. Construction of the final minimal surfaces

For each pair $(v, \zeta) \in \mathcal{B}_m$, the preceding section constructs an embedded surface in $\mathbb{S}^4$ in two steps. First, the initial surface M_ζ is displaced within $\mathcal{N}$ by the function $\phi = u_\zeta + v \circ F_\zeta^{-1}$. Then its image is displaced along the spherical geodesics initially normal to $\mathcal{N}$ by the function $b(v, \zeta)$. The second displacement is chosen so that the mean-curvature vector of the resulting surface has zero inner product with the normal field n_b defined in Section 4. Its inner product with the other normal field ν_b is $H_{\zeta,\phi} + \mathcal{R}_\zeta(\phi)$, expressed as a function on M_ζ. By Proposition 4.9, the resulting surface is minimal if and only if this function vanishes.

We now choose v and ζ to obtain that vanishing. The function b is determined by these choices, so it is not an additional unknown in this section. We also prove that the final embedding is smooth, identify the topology of its image, and record the estimates for both displacements needed in the curvature argument of Section 6.

The choice of ϕ and ζ is made by a fixed-point argument. We work on the set $\mathcal{B}_m$ defined in (3.29), whose function coordinate belongs to the fixed reference surface M_0. Thus the function space does not change when the parameter ζ changes. The constants α, β and the cutoff functions are fixed as in the preceding sections. All constants denoted by C in this section are independent of m and of $(v, \zeta) \in \mathcal{B}_m$. Statements concerning compactness and continuity are made with m fixed; the equivalence constants between weighted and unweighted Hölder norms in those arguments may depend on m.

The definitions in (3.12) and the bound for $c_{m^{-2}}$ in Lemma 3.1 give

$$\begin{aligned} \delta &= (100m)^{-1}, \qquad e^{-1}\tau_0 \le \tau_\zeta \le e\tau_0 \quad \forall\, \zeta \in [-1, 1], \\ \log \tau_0 &= -\frac{m^2}{2\pi} - \log m + O(1) \qquad \text{as } m \to \infty. \end{aligned} \tag{5.1}$$

In particular, for every fixed $p \ge 0$ and $q > 0$,

$$\delta^{-p}\tau_0^q \to 0 \qquad \text{as } m \to \infty.$$

Indeed, the logarithm of this expression is $p \log(100m) + q \log \tau_0$, which tends to $-\infty$. Consequently, a fixed power of δ^{-1} multiplied by a positive power of τ_0 still tends to zero. We will use this fact when applying the scalar inverse estimates, whose constants contain powers of δ^{-1}.

5.1. The remaining equation and the fixed-point map.

For $(v, \zeta) \in \mathcal{B}_m$, recall the displacement $\phi(v, \zeta)$ and define the combined nonlinear remainder $E_\zeta(v)$ by

$$\phi(v, \zeta) := u_\zeta + v \circ F_\zeta^{-1}, \qquad E_\zeta(v) := Q_\zeta(\phi(v, \zeta)) + \mathcal{R}_\zeta(\phi(v, \zeta)).$$

Here u_ζ is the preliminary correction defined in (3.32). The function $Q_\zeta(\phi)$ is the nonlinear remainder in the scalar mean curvature of the graph within $\mathcal{N}$, defined in (3.36). The function $\mathcal{R}_\zeta(\phi)$,

defined in (4.29), is the change in the remaining mean-curvature component caused by the subsequent displacement $b(v,\zeta)$ normal to $\mathcal{N}$. Thus $E_\zeta(v)$ contains exactly the terms not included in $H_\zeta + \mathcal{L}_\zeta\phi$:

$$H_{\zeta,\phi} + \mathcal{R}_\zeta(\phi) = H_\zeta + \mathcal{L}_\zeta\phi + E_\zeta(v) \qquad \text{on } M_\zeta, \quad \phi := \phi(v,\zeta).$$

The two summands defining $E_\zeta(v)$ are Γ_m-invariant and belong to Y_ζ. In particular, their projections onto the angular functions $\cos\theta$ and $\sin\theta$ on every neck vanish, as required for the scalar estimates of Proposition 3.4.

The geometric meaning of ϕ is shown in Figure 8. For fixed ζ, the map $G_{\zeta,\phi}$ moves each point of M_ζ along a geodesic of $\mathcal{N}$ whose initial velocity is normal to M_ζ within $\mathcal{N}$. Changing ζ also changes the initial surface and the neck radius τ_ζ. For every such choice, the second displacement $b(v,\zeta)$ is the one constructed in the preceding section.

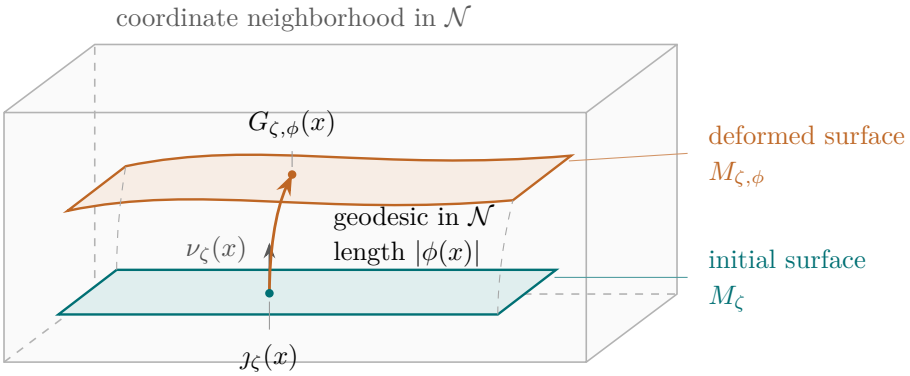


FIGURE 8. For a fixed ζ, the deformation within $\mathcal{N}$ sends $\jmath_\zeta(x)$ to $G_{\zeta,\phi}(x) = \exp^{\mathcal{N}}_{\jmath_\zeta(x)}(\phi(x)\nu_\zeta(x))$. Both surface patches and the connecting geodesics lie in $\mathcal{N}$. The geodesic starts in a direction normal to M_ζ within $\mathcal{N}$; this direction need not be the coordinate direction E_3. A point with $\phi(x) > 0$ is shown. The box is a schematic coordinate neighborhood, and the drawing does not represent the metric of $\mathcal{N}$. The patch boundaries are artificial and the displacement is exaggerated. In this section both ϕ and ζ are chosen; for each choice, the correction $b(v,\zeta)$ from Section 4 is recomputed and applied afterwards. That second deformation, which generally leaves $\mathcal{N}$, is not shown here.

Apply the linear map $\mathscr{R}_\zeta$ from Proposition 3.4 with a minus sign, and write

$$(w_{v,\zeta}, d_{v,\zeta}) := -\mathscr{R}_\zeta E_\zeta(v) \in X_\zeta \times \mathbb{R}. \tag{5.2}$$

By (3.30), these quantities satisfy

$$\mathcal{L}_\zeta w_{v,\zeta} = -E_\zeta(v) + d_{v,\zeta}\mathcal{J}_\zeta \qquad \text{on } M_\zeta. \tag{5.3}$$

Thus $w_{v,\zeta}$ cancels $E_\zeta(v)$ in the linearized equation up to a multiple of the prescribed function $\mathcal{J}_\zeta$. The preliminary correction already satisfies

$$H_\zeta + \mathcal{L}_\zeta u_\zeta = (-\tau_\zeta\zeta + c_{H,\zeta})\mathcal{J}_\zeta.$$

Substituting $\phi = u_\zeta + v\circ F_\zeta^{-1}$ and $E_\zeta(v) = -\mathcal{L}_\zeta w_{v,\zeta} + d_{v,\zeta}\mathcal{J}_\zeta$ into the remaining component therefore gives the exact identity

$$\begin{aligned} H_{\zeta,\phi} + \mathcal{R}_\zeta(\phi) = \mathcal{L}_\zeta\big(v\circ F_\zeta^{-1} - w_{v,\zeta}\big) \\ + (-\tau_\zeta\zeta + c_{H,\zeta} + d_{v,\zeta})\mathcal{J}_\zeta \qquad \text{on } M_\zeta. \end{aligned} \tag{5.4}$$

It is thus enough to choose (v,ζ) so that the function in parentheses in the first line vanishes and the real coefficient in the second line is zero. The nonlinear terms used to construct $w_{v,\zeta}$ are evaluated at

the input $\phi(v,\zeta)$. We must therefore require that the returned correction agrees with the correction in that same input. This is the reason for using a fixed point.

Define the map, initially with values in $X_0 \times \mathbb{R}$, by

$$\mathcal{T}_m(v,\zeta) := \left(w_{v,\zeta} \circ F_\zeta, \frac{c_{H,\zeta} + d_{v,\zeta}}{\tau_\zeta} \right) \qquad \forall\, (v,\zeta) \in \mathcal{B}_m. \tag{5.5}$$

The first coordinate is expressed on M_0 using the diffeomorphism $F_\zeta : M_0 \to M_\zeta$. At a fixed point, it gives $v \circ F_\zeta^{-1} = w_{v,\zeta}$. The second coordinate gives $\tau_\zeta \zeta = c_{H,\zeta} + d_{v,\zeta}$. These are precisely the two equalities that make the right-hand side of (5.4) vanish. We first verify that $\mathcal{T}_m$ takes values in $\mathcal{B}_m$, so that it can be used in a fixed-point argument.

Lemma 5.1. *For every sufficiently large integer m, the map $\mathcal{T}_m$ takes $\mathcal{B}_m$ into itself. Uniformly for $(v,\zeta) \in \mathcal{B}_m$, one has*

$$\begin{aligned} \|E_\zeta(v)\|_{Y_\zeta} &\le C\tau_\zeta^2, \\ \|w_{v,\zeta}\|_{X_\zeta} &\le C\delta^{-4-2\beta}\tau_\zeta^2, \\ |d_{v,\zeta}| &\le C\delta^{-2-\beta}\tau_\zeta^2, \\ |c_{H,\zeta}| &\le C\delta^{-2-\beta}\tau_\zeta^{1+\alpha/3}. \end{aligned} \tag{5.6}$$

Writing $\mathcal{T}_m = (\mathcal{T}_m^1, \mathcal{T}_m^2)$, it follows that

$$\frac{\|\mathcal{T}_m^1(v,\zeta)\|_{X_0}}{\tau_0^{1+\alpha}} \le C\delta^{-4-2\beta}\tau_0^{1-\alpha} \to 0, \tag{5.7}$$

$$|\mathcal{T}_m^2(v,\zeta)| \le C\delta^{-2-\beta}(\tau_0^{\alpha/3} + \tau_0) \to 0 \tag{5.8}$$

as $m \to \infty$, uniformly on $\mathcal{B}_m$.

Proof. By (3.37) and (4.30),

$$\begin{aligned} \|E_\zeta(v)\|_{Y_\zeta} &\le \|Q_\zeta(\phi(v,\zeta))\|_{Y_\zeta} + \|\mathcal{R}_\zeta(\phi(v,\zeta))\|_{Y_\zeta} \\ &\le C\tau_\zeta^2 + C\tau_\zeta^{2+\alpha/2} \le C\tau_\zeta^2, \end{aligned}$$

where the last inequality uses $0 < \tau_\zeta < 1$. Apply the function estimate in (3.30) and the coefficient estimate (3.31) to (5.2). They give

$$\begin{aligned} \|w_{v,\zeta}\|_{X_\zeta} &\le C\delta^{-4-2\beta}\|E_\zeta(v)\|_{Y_\zeta} \le C\delta^{-4-2\beta}\tau_\zeta^2, \\ |d_{v,\zeta}| &\le C\delta^{-2-\beta}\|E_\zeta(v)\|_{Y_\zeta} \le C\delta^{-2-\beta}\tau_\zeta^2. \end{aligned}$$

Similarly, from $(u_\zeta, c_{H,\zeta}) = -\mathscr{R}_\zeta D_\zeta$ and (3.33),

$$|c_{H,\zeta}| \le C\delta^{-2-\beta}\|D_\zeta\|_{Y_\zeta} \le C\delta^{-2-\beta}\tau_\zeta^{1+\alpha/3}.$$

This proves the four asserted bounds.

The comparison of norms in (3.28) and the scale comparison (5.1) imply

$$\begin{aligned} \|\mathcal{T}_m^1(v,\zeta)\|_{X_0} &= \|w_{v,\zeta} \circ F_\zeta\|_{X_0} \\ &\le C_F\|w_{v,\zeta}\|_{X_\zeta} \le C\delta^{-4-2\beta}\tau_0^2. \end{aligned}$$

Dividing by the radius $\tau_0^{1+\alpha}$ of the function ball gives (5.7). For the second coordinate, dividing the estimates for $c_{H,\zeta}$ and $d_{v,\zeta}$ by τ_ζ gives

$$\begin{aligned} |\mathcal{T}_m^2(v,\zeta)| &\le \frac{|c_{H,\zeta}| + |d_{v,\zeta}|}{\tau_\zeta} \\ &\le C\delta^{-2-\beta}(\tau_\zeta^{\alpha/3} + \tau_\zeta) \le C\delta^{-2-\beta}(\tau_0^{\alpha/3} + \tau_0). \end{aligned}$$

Because $0 < \alpha < 1$, all powers of τ_0 in (5.7) and (5.8) are positive. Both right-hand sides therefore tend to zero uniformly. For sufficiently large m, they are at most $1/2$. In particular,

$$\|\mathcal{T}_m^1(v,\zeta)\|_{X_0} \le \tau_0^{1+\alpha}, \qquad |\mathcal{T}_m^2(v,\zeta)| \le 1,$$

which are precisely the defining inequalities for membership in $\mathcal{B}_m$.

The same calculation isolates the contribution of the displacement normal to $\mathcal{N}$. Its contribution to the function coordinate is at most $C\delta^{-4-2\beta}\tau_\zeta^{2+\alpha/2}$. Relative to the radius $\tau_0^{1+\alpha}$, this is bounded by

$$C\delta^{-4-2\beta}\tau_0^{1-\alpha/2} \to 0.$$

Thus the estimate in the preceding section is strong enough to be included in the same scalar fixed-point argument. □

5.2. **Compactness and continuity in a weaker Hölder topology.** The preceding lemma gives the bounds needed for the fixed-point argument. We must also specify a topology in which its domain is compact and the map is continuous. The bound defining $\mathcal{B}_m$ controls two derivatives with Hölder exponent β. We retain that bound and use convergence with a smaller exponent. All arguments in this subsection are for a fixed m; none requires a compactness estimate uniform as $m \to \infty$.

Fix $\beta' \in (0,\beta)$ and let

$$\mathcal{E}_m := C^{2,\beta'}_{\mathrm{inv}}(M_0) \times \mathbb{R},$$
$$\|(v,\zeta)\|_{\mathcal{E}_m} := \|v\|_{C^{2,\beta'}(M_0,g_0)} + |\zeta|.$$

Here g_0 is the induced metric of the initial surface M_0, and the subscript inv has the same meaning as in (3.25): the functions are invariant under Γ_m. Invariance is preserved under uniform convergence, so this is a closed subspace of the ordinary Hölder space. In particular, $\mathcal{E}_m$ is a Banach space. We regard $\mathcal{B}_m$ as a subset of $\mathcal{E}_m$, with its original defining bound $\|v\|_{X_0} \le \tau_0^{1+\alpha}$.

Lemma 5.2. *For every fixed sufficiently large m, the set $\mathcal{B}_m$ is a nonempty compact convex subset of $\mathcal{E}_m$. The map $\mathcal{T}_m : \mathcal{B}_m \to \mathcal{B}_m$ is continuous in the topology induced by $\mathcal{E}_m$.*

Proof. The pair $(0,0)$ belongs to $\mathcal{B}_m$. Its function coordinate ranges over a norm ball in the linear space X_0, and its real coordinate ranges over $[-1,1]$. Both sets are convex, hence so is $\mathcal{B}_m$.

We first prove compactness. With m fixed, the surface M_0 is compact and the continuous weights ρ_0, f_0 are strictly positive. Their minima are therefore positive and their maxima finite. Consequently, the radii of the local balls in (3.24) are bounded below by a positive number, and all the constant metric-rescaling factors and function weights are bounded above and below. A finite covering by smaller such balls shows that $\|\cdot\|_{X_0}$ is equivalent to the ordinary $C^{2,\beta}(M_0,g_0)$ norm. The constants in this equivalence may depend on m.

Thus the function coordinates of $\mathcal{B}_m$ are bounded in $C^{2,\beta}(M_0,g_0)$. The compact inclusion into $C^{2,\beta'}(M_0,g_0)$, together with compactness of $[-1,1]$, shows that every sequence in $\mathcal{B}_m$ has a subsequence converging in $\mathcal{E}_m$. To prove compactness of $\mathcal{B}_m$ itself, we must check that the limit still satisfies the original weighted bound.

Suppose $(v_j,\zeta_j) \in \mathcal{B}_m$ and $(v_j,\zeta_j) \to (v,\zeta)$ in $\mathcal{E}_m$. Fix a centre $x \in M_0$ and the corresponding ball in (3.24). On this fixed ball, the covariant derivatives of v_j of orders zero, one and two converge uniformly to those of v. For each pair of distinct points, the β-Hölder quotient of the second derivative converges as well: the metric, the distance and the parallel transport used to compare the tensors are all fixed. Taking the supremum over these pairs gives lower semicontinuity of the Hölder seminorm. The three supremum-norm terms converge. Multiplying by $f_0(x)^{-1}$ and then taking the supremum over x therefore gives

$$\|v\|_{X_0} \le \liminf_{j\to\infty} \|v_j\|_{X_0} \le \tau_0^{1+\alpha}.$$

In particular, the finite local Hölder bounds show that $v \in C^{2,\beta}(M_0)$. Invariance passes to the limit pointwise, and $\zeta \in [-1,1]$. Hence the limit belongs to $\mathcal{B}_m$. This proves compactness.

We next prove continuity of $\mathcal{T}_m$. Let $(v_j,\zeta_j) \to (v,\zeta)$ in $\mathcal{E}_m$, with all these pairs in $\mathcal{B}_m$. Functions on different initial surfaces will be compared on M_0 by composition with the corresponding F_ζ. We follow the order of the operations defining (5.5): first the preliminary correction, then the nonlinear source, and finally the scalar solution operator.

The initial embeddings, their induced metrics and the diffeomorphisms F_ζ depend smoothly on ζ. Therefore the functions $D_\zeta = H_\zeta + \tau_\zeta \zeta \mathcal{J}_\zeta$ satisfy

$$D_{\zeta_j} \circ F_{\zeta_j} \to D_\zeta \circ F_\zeta \qquad \text{in } C^{0,\beta'}(M_0).$$

Their stronger source norms are bounded by (3.33). We may therefore apply the sequential continuity assertion in Proposition 3.4 to $(u_\zeta, c_{H,\zeta}) = -\mathscr{R}_\zeta D_\zeta$. It gives

$$\begin{aligned} u_{\zeta_j} \circ F_{\zeta_j} &\to u_\zeta \circ F_\zeta \qquad \text{in } C^{2,\beta'}(M_0), \\ c_{H,\zeta_j} &\to c_{H,\zeta}. \end{aligned} \tag{5.9}$$

Setting $\phi_j := \phi(v_j,\zeta_j)$ and $\phi := \phi(v,\zeta)$, we obtain

$$\begin{aligned} \phi_j \circ F_{\zeta_j} &= u_{\zeta_j} \circ F_{\zeta_j} + v_j \\ &\to u_\zeta \circ F_\zeta + v = \phi \circ F_\zeta \qquad \text{in } C^{2,\beta'}(M_0). \end{aligned}$$

The scalar mean curvature of a graph is, in local coordinates, a smooth function of the graph height and its derivatives of orders one and two, on the set where the induced metric is positive definite. The embeddings supplied by Proposition 3.4 have this property. With m fixed, their metrics are uniformly comparable to the initial metrics, and the latter form a smooth family on the compact parameter interval. The Hölder product and composition estimates therefore apply to the preceding convergence. Using also the smooth dependence of H_ζ and $\mathcal{L}_\zeta$, the definition (3.36) gives

$$Q_{\zeta_j}(\phi_j) \circ F_{\zeta_j} \to Q_\zeta(\phi) \circ F_\zeta \qquad \text{in } C^{0,\beta'}(M_0).$$

Proposition 4.8 gives convergence in the same space for $\mathcal{R}_{\zeta_j}(\phi_j) \circ F_{\zeta_j}$. Adding these two convergences proves

$$E_{\zeta_j}(v_j) \circ F_{\zeta_j} \to E_\zeta(v) \circ F_\zeta \qquad \text{in } C^{0,\beta'}(M_0).$$

The bound (5.6) also gives a uniform bound for their Y_{ζ_j} norms, with m fixed. These are precisely the two hypotheses needed to apply the sequential continuity of $\mathscr{R}_\zeta$ again, this time to $(w_{v,\zeta}, d_{v,\zeta}) = -\mathscr{R}_\zeta E_\zeta(v)$. We obtain

$$\begin{aligned} w_{v_j,\zeta_j} \circ F_{\zeta_j} &\to w_{v,\zeta} \circ F_\zeta \qquad \text{in } C^{2,\beta'}(M_0), \\ d_{v_j,\zeta_j} &\to d_{v,\zeta}. \end{aligned}$$

Finally, $\tau_\zeta = \tau_0 e^\zeta$ is positive and continuous. Together with (5.9), the last two limits prove convergence of both coordinates of $\mathcal{T}_m(v_j,\zeta_j)$ in $\mathcal{E}_m$. □

5.3. **Existence, regularity, and bounds for the deformations.** The two preceding lemmas provide a fixed point. We now verify that its defining identities make the entire mean-curvature vector vanish. We also record stronger estimates for this fixed point than those required to define $\mathcal{B}_m$, since the curvature comparison in the next section will use those stronger estimates.

Proposition 5.3. *There exists $m_0 \in \mathbb{N}$ such that, for every integer $m \geq m_0$, there is a pair $(v_m,\zeta_m) \in \mathcal{B}_m$ with the following properties. Define*

$$\tau_m := \tau_{\zeta_m}, \qquad \phi_m := u_{\zeta_m} + v_m \circ F_{\zeta_m}^{-1}, \qquad b_m := b(v_m,\zeta_m).$$

The displacement within $\mathcal{N}$ is given by

$$G_m : M_{\zeta_m} \to \mathcal{N},$$
$$G_m(x) := \exp^{\mathcal{N}}_{\jmath_{\zeta_m}(x)}\big(\phi_m(x)\nu_{\zeta_m}(x)\big) \qquad \forall\, x \in M_{\zeta_m}.$$

Here $\jmath_{\zeta_m} : M_{\zeta_m} \hookrightarrow \mathcal{N}$ is the inclusion. Define the unit normal to $\mathcal{N}$ along G_m by

$$n_m(x) := N\big(\pi(G_m(x))\big) \qquad \forall\, x \in M_{\zeta_m}.$$

The map $\widehat{X}_m : M_{\zeta_m} \to \mathbb{S}^4$ describing the subsequent displacement normal to $\mathcal{N}$, given by

$$\widehat{X}_m(x) := \exp^{\mathbb{S}^4}_{G_m(x)}\big(b_m(x)n_m(x)\big) = \cos\big(b_m(x)\big)G_m(x) + \sin\big(b_m(x)\big)n_m(x) \quad \forall\, x \in M_{\zeta_m}, \tag{5.10}$$

is a smooth embedding. Its image $\Sigma_m := \widehat{X}_m(M_{\zeta_m})$ is a closed, connected, Γ_m-invariant minimal surface in $\mathbb{S}^4$, with

$$\operatorname{genus}(\Sigma_m) = m^2 + 1.$$

The following estimates hold with constants independent of m:

$$|\zeta_m| \le C\delta^{-2-\beta}\tau_m^{\alpha/3}, \tag{5.11}$$
$$\|v_m\|_{X_0} \le C\delta^{-4-2\beta}\tau_m^2,$$
$$\|\phi_m\|_{X_{\zeta_m}} \le C\delta^{-4-2\beta}\tau_m^{1+\alpha/3}, \tag{5.12}$$
$$\|b_m\|_{\mathcal{X}_{\zeta_m}} \le C\tau_m^2. \tag{5.13}$$

In particular, $\zeta_m \to 0$ and

$$\frac{\tau_m}{\tau_0} = 1 + O\big(\delta^{-2-\beta}\tau_0^{\alpha/3}\big) \to 1 \qquad \text{as } m \to \infty. \tag{5.14}$$

Write $\rho_m := \rho_{\zeta_m}$ and $f_m := f_{\zeta_m}$, and take covariant derivatives in the following estimates with respect to g_{ζ_m}. For every $x \in M_{\zeta_m}$ and $j \in \{0,1,2\}$,

$$\begin{aligned} |\nabla^j \phi_m(x)|_{g_{\zeta_m}} &\le C\delta^{-4-2\beta}\tau_m^{1+\alpha/3} f_m(x)\rho_m(x)^{-j}, \\ |\nabla^j b_m(x)|_{g_{\zeta_m}} &\le C\tau_m^2 \rho_m(x)^{-j}. \end{aligned} \tag{5.15}$$

The corresponding local scaled Hölder estimates hold on the balls and with the rescaled metrics specified in (3.24)*. All estimates in Sections 3 and 4 that hold for every $(v,\zeta) \in \mathcal{B}_m$ apply to this pair.*

Proof. By Lemmas 5.1 and 5.2, the Schauder fixed-point theorem applies to $\mathcal{T}_m$ on the nonempty compact convex subset $\mathcal{B}_m$ of the Banach space $\mathcal{E}_m$. Choose a fixed point (v_m, ζ_m).

We first verify minimality. In this calculation, write $(v,\zeta) := (v_m, \zeta_m)$, $w := w_{v,\zeta}$, $d := d_{v,\zeta}$, and $\phi := \phi_m$. Equality of the two coordinates in $\mathcal{T}_m(v,\zeta) = (v,\zeta)$ means

$$v \circ F_\zeta^{-1} = w, \qquad \tau_\zeta \zeta = c_{H,\zeta} + d. \tag{5.16}$$

In particular, $\phi = u_\zeta + w$. Using in order the definition of Q_ζ, the definition of $E_\zeta(v)$, (3.34), and (5.3), we obtain

$$\begin{aligned} H_{\zeta,\phi} + \mathcal{R}_\zeta(\phi) &= H_\zeta + \mathcal{L}_\zeta\phi + Q_\zeta(\phi) + \mathcal{R}_\zeta(\phi) \\ &= H_\zeta + \mathcal{L}_\zeta u_\zeta + \mathcal{L}_\zeta w + E_\zeta(v) \\ &= (-\tau_\zeta\zeta + c_{H,\zeta})\mathcal{J}_\zeta + \big(-E_\zeta(v) + d\mathcal{J}_\zeta\big) + E_\zeta(v) \\ &= (-\tau_\zeta\zeta + c_{H,\zeta} + d)\mathcal{J}_\zeta \\ &= 0. \end{aligned}$$

This is the mean-curvature component tested against ν_{b_m} on the final surface. The component tested against n_{b_m} is zero by (4.28). Those two vectors span the normal plane at every point. Since the mean-curvature vector is normal to the surface and has zero inner product with both, it vanishes identically.

Proposition 4.9 already gives a $C^{2,\beta}$ embedding in (5.10). The exact relation with the tubular map of Lemma 2.6 is

$$\widehat{X}_m(x) = \Psi\big(\mathcal{F}^{-1}(G_m(x)), b_m(x)\big) \qquad \forall\, x \in M_{\zeta_m}.$$

The graph $x \mapsto (\mathcal{F}^{-1}(G_m(x)), b_m(x))$ is embedded because its projection onto the first factor is the embedding $\mathcal{F}^{-1} \circ G_m$. Section 4 ensures that this graph lies in the domain on which Ψ is an embedding. Thus its image is closed and connected, and is diffeomorphic as a $C^{2,\beta}$ surface to M_{ζ_m}.

We also check the symmetries at the level of these maps. Fix $g \in \Gamma_m$ and $x \in M_{\zeta_m}$. The isometry g preserves $\mathcal{N}$ and hence commutes with its exponential map. Invariance of ϕ_m and equivariance of ν_{ζ_m} therefore give

$$G_m(g(x)) = g(G_m(x)).$$

The transformation rules for b_m and n give

$$\begin{aligned} b_m(g(x)) &= \chi(g) b_m(x), \\ n_m(g(x)) &= \chi(g)\, dg_{G_m(x)}(n_m(x)). \end{aligned}$$

Since $\chi(g)^2 = 1$, their product consequently satisfies

$$b_m(g(x)) n_m(g(x)) = dg_{G_m(x)}\big(b_m(x) n_m(x)\big).$$

Applying equivariance of the spherical exponential map now gives

$$\widehat{X}_m(g(x)) = \exp^{\mathbb{S}^4}_{g(G_m(x))}\big(dg_{G_m(x)}(b_m(x) n_m(x))\big) = g\left(\exp^{\mathbb{S}^4}_{G_m(x)}\big(b_m(x) n_m(x)\big)\right) = g(\widehat{X}_m(x)).$$

This proves Γ_m-invariance of Σ_m. Its genus is the genus of M_{ζ_m}, namely m^2+1 by Lemma 3.2.

We next prove the estimates. The second identity in (5.16) gives

$$|\zeta_m| \le \tau_m^{-1}\big(|c_{H,\zeta_m}| + |d_{v_m,\zeta_m}|\big) \le C\delta^{-2-\beta}(\tau_m^{\alpha/3} + \tau_m) \le C\delta^{-2-\beta}\tau_m^{\alpha/3}.$$

The last inequality uses $0 < \tau_m < 1$ and $\alpha/3 < 1$. This proves (5.11), and its right-hand side tends to zero by (5.1). Since $\tau_m/\tau_0 = e^{\zeta_m}$ and $|\zeta_m| \le 1$,

$$\left|\frac{\tau_m}{\tau_0} - 1\right| = |e^{\zeta_m} - 1| \le e|\zeta_m| \le C\delta^{-2-\beta}\tau_0^{\alpha/3},$$

which proves (5.14).

The first identity in (5.16) and the norm comparison give

$$\|v_m\|_{X_0} = \|w_{v_m,\zeta_m} \circ F_{\zeta_m}\|_{X_0} \le C\delta^{-4-2\beta}\tau_m^2.$$

For the preliminary correction, the scalar inverse estimate and (3.33) give

$$\|u_{\zeta_m}\|_{X_{\zeta_m}} \le C\delta^{-4-2\beta}\|D_{\zeta_m}\|_{Y_{\zeta_m}} \le C\delta^{-4-2\beta}\tau_m^{1+\alpha/3}.$$

Adding the estimate for w_{v_m,ζ_m} proves (5.12), because $\phi_m = u_{\zeta_m} + w_{v_m,\zeta_m}$ and $2 > 1 + \alpha/3$. The estimate (5.13) is the bound in Proposition 4.8.

To obtain (5.15), evaluate the local norms defining X_{ζ_m} and $\mathcal{X}_{\zeta_m}$ at a center x. Multiplication of the metric by $\rho_m(x)^{-2}$ multiplies the norm of the j-th covariant derivative of a function by $\rho_m(x)^j$. The function weight is $f_m(x)$ in X_{ζ_m} and is 1 in $\mathcal{X}_{\zeta_m}$. Thus the norm bounds just proved give exactly the two pointwise inequalities in (5.15). The Hölder terms in the same local norms give the stated local Hölder estimates. Since the fixed point lies in $\mathcal{B}_m$, it also retains all bounds proved on that set, including $\|\phi_m\|_{X_{\zeta_m}} \le \tau_m^{1+\alpha/4}$.

It remains to prove smoothness. We first prove that the image Σ_m is smooth, and then that its given parametrization is smooth. Throughout this part of the argument m is fixed; no uniform estimate for derivatives of order greater than two is claimed.

Near each point of the $C^{2,\beta}$ embedded surface Σ_m, choose smooth ambient coordinates (y^1, y^2, y^3, y^4) such that the surface is represented by

$$Z(y^1, y^2) = \big(y^1, y^2, V^1(y^1, y^2), V^2(y^1, y^2)\big),$$

with $V = (V^1, V^2) \in C^{2,\beta}$ on an open subset of $\mathbb{R}^2$. Write $\overline{g}_{BC}$ and $\overline{\Gamma}^A_{BC}$ for the metric components and Christoffel symbols of $g_{\mathbb{S}^4}$ in these coordinates, with ambient indices in $\{1, 2, 3, 4\}$. For $i, j \in \{1, 2\}$, set

$$T^B_i := \partial_i Z^B, \qquad g_{ij} := \overline{g}_{BC}(Z) T^B_i T^C_j, \qquad (a^{ij}) := (g_{ij})^{-1}.$$

Repeated indices in these local formulas are summed over their indicated ranges. Let $\Gamma^k_{ij}(g)$ be the Christoffel symbols of the induced metric. Vanishing of the mean-curvature vector means

$$a^{ij}\big(\partial_{ij} Z^A + \overline{\Gamma}^A_{BC}(Z) T^B_i T^C_j - \Gamma^k_{ij}(g)\partial_k Z^A\big) = 0 \qquad \forall\, A \in \{1, 2, 3, 4\}.$$

Since $Z^k = y^k$ for $k \in \{1, 2\}$, the first two equations give

$$a^{ij}\Gamma^k_{ij}(g) = a^{ij}\overline{\Gamma}^k_{BC}(Z) T^B_i T^C_j \qquad \forall\, k \in \{1, 2\}.$$

Substituting these identities into the remaining two equations yields, for $\mu \in \{1, 2\}$,

$$a^{ij}\partial_{ij} V^\mu = a^{ij}\big(\partial_k V^\mu\, \overline{\Gamma}^k_{BC}(Z) - \overline{\Gamma}^{\mu+2}_{BC}(Z)\big) T^B_i T^C_j.$$

Both a^{ij} and the right-hand side are smooth functions of (y, V, DV). The matrix (a^{ij}) is positive definite and uniformly so on a smaller relatively compact coordinate neighborhood. Since $V \in C^{2,\beta}$, its coefficients and right-hand sides belong to $C^{1,\beta}$. Interior Schauder regularity applied to each component gives $V \in C^{3,\beta}$ on smaller neighborhoods. Repeating this argument raises the regularity by one derivative each time. Thus Σ_m is smooth.

Smoothness of the image alone does not yet assert smoothness of the original parametrization. To recover it, use the smooth inverse of the fixed tubular map (2.15). Define its two relevant components by

$$\begin{aligned} P &: \operatorname{im}\Psi \to \mathcal{N}, & P(\Psi(s, t, z, b)) &:= \mathcal{F}(s, t, z), \\ t_\perp &: \operatorname{im}\Psi \to (-b_0, b_0), & t_\perp(\Psi(s, t, z, b)) &:= b, \end{aligned}$$

for every (s, t, z, b) in the domain of Ψ. By construction,

$$P \circ \widehat{X}_m = G_m, \qquad t_\perp \circ \widehat{X}_m = b_m.$$

The smooth map $P|_{\Sigma_m}$ is injective because G_m is injective. It is also an immersion: differentiating $P \circ \widehat{X}_m = G_m$ shows that its differential has rank two, since dG_m has rank two and $d\widehat{X}_m$ maps onto the tangent space of Σ_m. An injective immersion of a compact smooth manifold into a Hausdorff manifold is an embedding. Consequently,

$$M_{\zeta_m, \phi_m} = P(\Sigma_m) = G_m(M_{\zeta_m})$$

is a smooth embedded surface in $\mathcal{N}$.

To recover the original parametrization of M_{ζ_m, ϕ_m}, we use normal exponential coordinates based on the smooth initial surface M_{ζ_m}. Define

$$\mathfrak{E}_m(x, t) := \exp^{\mathcal{N}}_{\jmath_{\zeta_m}(x)}\big(t\nu_{\zeta_m}(x)\big)$$

for $x \in M_{\zeta_m}$ and for every $t \in \mathbb{R}$ for which the indicated geodesic is defined. This is a smooth map on its open domain. We will prove that its differential is invertible at $(x, \phi_m(x))$ for every x. Invertibility at $(x, 0)$ alone would not justify this claim, because the surfaces and their normal displacements depend on m.

Fix $x \in M_{\zeta_m}$ and put $r := \rho_m(x)$. For large m, the bound $|z(x)| \le C\varepsilon_{\zeta_m}$ places x in the smaller band $|z| < z_0/2$. Since $|\nabla^{\mathcal{N}} z| = 1$ and $r \le \pi/m$, the unit-speed normal geodesic

$$c_x(t) := \mathfrak{E}_m(x,t)$$

is defined for $|t| \le c_* r$, with any fixed $c_* \in (0,1)$ once m is sufficiently large. Indeed, it cannot reach the boundary of the band in such a time, and the metric is smooth on a neighborhood of the closure of every smaller band. The curvature tensor of $g_{\mathcal{N}}$ is uniformly bounded there by the Gauss equation and (2.3).

In the following calculation, vector norms and inner products in $T\mathcal{N}$ are those of $g_{\mathcal{N}}$.

Let $\mathsf{P}_t : T_{\jmath_{\zeta_m}(x)}\mathcal{N} \to T_{c_x(t)}\mathcal{N}$ denote parallel transport along c_x. For $\xi \in T_x M_{\zeta_m}$, differentiating the family of normal geodesics in its initial point gives the Jacobi field

$$J_\xi(t) := d\big(\mathfrak{E}_m(\cdot,t)\big)_x(\xi).$$

Set $\xi_0 := d\jmath_{\zeta_m,x}(\xi)$ and $T_x := S^{\nu_{\zeta_m}}_{\jmath_{\zeta_m}}(x)$. The initial conditions and the Jacobi equation are

$$J_\xi(0) = \xi_0, \qquad \nabla_t J_\xi(0) = -d\jmath_{\zeta_m,x}(T_x(\xi)), \qquad \nabla_t^2 J_\xi = -R^{\mathcal{N}}(J_\xi, \dot c_x)\dot c_x.$$

Here $R^{\mathcal{N}}$ is the Riemann curvature tensor of $g_{\mathcal{N}}$, and ∇_t denotes covariant differentiation along c_x. The formula for the initial derivative is the Weingarten equation for $M_{\zeta_m} \subset \mathcal{N}$. By (3.19) and $\tau_m \le r$, one has $|T_x| \le C/r$.

Parallel transport reduces this equation to an ordinary differential equation in the fixed vector space $T_{\jmath_{\zeta_m}(x)}\mathcal{N}$. More precisely, set

$$\begin{aligned} K_\xi(t) &:= \mathsf{P}_t^{-1} J_\xi(t), \\ \mathcal{K}_t(w) &:= \mathsf{P}_t^{-1}\big(R^{\mathcal{N}}(\mathsf{P}_t w, \dot c_x(t))\dot c_x(t)\big) \qquad \forall\, w \in T_{\jmath_{\zeta_m}(x)}\mathcal{N}. \end{aligned}$$

The uniform curvature bound gives $|\mathcal{K}_t| \le C$. Integrating the Jacobi equation twice gives

$$K_\xi(t) = \xi_0 - t\, d\jmath_{\zeta_m,x}(T_x(\xi)) - \int_0^t (t-s)\mathcal{K}_s(K_\xi(s))\, ds.$$

Since $\jmath_{\zeta_m}$ is isometric, it follows that

$$\sup_{|t|\le c_* r} |K_\xi(t)| \le (1 + Cc_*)|\xi|_{g_{\zeta_m}} + Cc_*^2 r^2 \sup_{|t|\le c_* r} |K_\xi(t)|.$$

Choose c_* small enough to absorb the last term; this choice is independent of m and x, since $r \le 1$ for large m. Substituting the resulting bound back into the integral identity yields

$$|K_\xi(t) - \xi_0| \le C\left(\frac{|t|}{r} + t^2\right)|\xi|_{g_{\zeta_m}} \qquad \forall\, |t| \le c_* r.$$

Decreasing c_* once more makes the right-hand side at most $|\xi|_{g_{\zeta_m}}/2$. Thus

$$|J_\xi(t)|_{g_{\mathcal{N}}} \ge \tfrac{1}{2}|\xi|_{g_{\zeta_m}} \qquad \forall\, |t| \le c_* r.$$

The function $t \mapsto \langle J_\xi(t), \dot c_x(t)\rangle$ has zero second derivative by the Jacobi equation and the symmetries of $R^{\mathcal{N}}$. Its value and first derivative at zero vanish by the initial conditions. Hence $J_\xi(t)$ is orthogonal to the unit vector $\dot c_x(t) = \partial_t \mathfrak{E}_m(x,t)$. The lower bound just obtained therefore shows that

$$d\mathfrak{E}_{m,(x,t)}(\xi, a) = J_\xi(t) + a\dot c_x(t) \qquad \forall\, (\xi, a) \in T_x M_{\zeta_m} \times \mathbb{R}$$

is an isomorphism for $|t| \le c_* r$. Estimate (4.4) implies $\sup_x |\phi_m(x)|/\rho_m(x) \to 0$, so this applies at $t = \phi_m(x)$ for every x once m is large. The smooth inverse function theorem consequently makes $\mathfrak{E}_m$ a smooth coordinate map on a neighborhood of each $(x, \phi_m(x))$.

After shrinking these coordinate neighborhoods, the inverse image of the already smooth surface M_{ζ_m,ϕ_m} under $\mathfrak{E}_m$ is the graph of the $C^{2,\beta}$ function ϕ_m. This graph is therefore a smooth submanifold of $M_{\zeta_m} \times \mathbb{R}$. Projection of this graph to the x coordinate has invertible differential: in the known parametrization $x \mapsto (x, \phi_m(x))$, that projection is the identity. The smooth inverse function theorem

on this graph therefore gives a smooth inverse to this projection, whose last coordinate is ϕ_m. Thus ϕ_m is smooth, and so is G_m. Finally,

$$\widehat{X}_m = (P|_{\Sigma_m})^{-1} \circ G_m, \qquad b_m = t_\perp \circ \widehat{X}_m$$

are smooth. This proves the asserted regularity of the parametrization and completes the proof. □

6. Strictly negative Gaussian curvature and the area limit

Let $\Sigma_m = \widehat{X}_m(M_{\zeta_m})$ be the smooth embedded minimal surface obtained in Proposition 5.3. We prove that its Gaussian curvature is strictly negative at every point when m is sufficiently large, and that its area converges to $4\pi^2$. We use different coordinates in two parts of the surface. Away from the circles of smallest neck radius, we first express $G_m(M_{\zeta_m}) \subset \mathcal{N}$ as local height graphs over $\mathbb{T}$. We compare their heights with the Green function used in Section 3, compute their Gaussian curvatures, and estimate the change caused by the displacement b_m normal to $\mathcal{N}$. Near each waist, the height graph coordinates are not valid. There we rescale lengths by the neck radius and compare the final surface with a fixed portion of the Euclidean catenoid. After proving negativity in both regions, we estimate the areas of the initial surfaces and compare their metrics with those of the final surfaces.

The first comparison must be quantitative. On the portions of the sheets furthest from the lattice, the negative curvature that we obtain tends to zero as $m \to \infty$. Consequently, an error in curvature that merely tends to zero would not suffice. We must bound that error by a small multiple of the positive function $\mathcal{W}$ defined below, and prove a negative upper bound of the form $-c\mathcal{W}$, with $c > 0$ independent of m.

For the rest of the section, abbreviate

$$\zeta := \zeta_m, \qquad \tau := \tau_m, \qquad \varepsilon := \frac{m^2\tau}{2\pi}, \qquad r_c := \tau^\alpha, \qquad \psi := \tau\Phi_m.$$

Let $h := h_\zeta$ be the upper initial height from (3.16), extended to the domain $\{q \in \mathbb{T} : d_{\mathbb{T}}(q, L_m) > \tau\}$ by $h(q) := C_\tau(r_p(q))$ wherever $\tau < r_p(q) \le 9\tau$, for each $p \in L_m$. Here $r_p(q) = d_{\mathbb{T}}(q,p)$ and $C_\tau(r) = \tau \operatorname{arcosh}(r/\tau)$ for $r > \tau$. This extension agrees with the existing height on its overlap with the exact catenoidal region. It describes the upper half of the initial surface arbitrarily close to a waist, but not the waist itself. For every $q \in \mathbb{T} \setminus L_m$, put

$$\rho(q) := d_{\mathbb{T}}(q, L_m), \qquad \mathcal{W}(q) := \varepsilon^2 + \tau^2\rho(q)^{-4}.$$

The two sides of each local graph comparison must be evaluated at the same point of $\mathbb{T}$. The deformation G_m can change that point. Accordingly, for every $x \in M_\zeta$, we use the previously defined initial radius $\rho_\zeta(x)$ and denote the radius after deformation by $\widehat{\rho}(x)$:

$$\rho_\zeta(x) = \rho(\pi(x)), \qquad \widehat{\rho}(x) := \rho(\pi(G_m(x))).$$

By (4.3), $\widehat{\rho}(x) = (1 + o_m(1))\rho_\zeta(x)$ uniformly in x. Here $o_m(1)$ denotes a quantity tending to zero as $m \to \infty$, uniformly over the points under consideration. Although the distance function ρ need not be smooth, we use it only as a pointwise scale and never differentiate it. Each square lattice cell has side length ℓ/m, so its points have distance at most $\ell/(\sqrt{2}m) = \pi/m$ from the lattice. Thus

$$\frac{\varepsilon^2}{\tau^2\rho(q)^{-4}} = \frac{m^4\rho(q)^4}{4\pi^2} \le \frac{\pi^2}{4} \qquad \forall\, q \in \mathbb{T} \setminus L_m,$$

and hence

$$\tau^2\rho^{-4} \le \mathcal{W} \le \left(1 + \frac{\pi^2}{4}\right)\tau^2\rho^{-4} \qquad \text{on } \mathbb{T} \setminus L_m.$$

We will first choose a large constant R, independently of m, and then choose m sufficiently large in terms of R. There is a fixed lower threshold $R_0 > 10$ such that all constants in the graph estimates below are independent of $R \ge R_0$. Constants in the comparison on a fixed rescaled neck are allowed to depend on R. The two regions will be specified on the initial surface by $\rho_\zeta \ge 2R\tau$ and $\rho_\zeta \le 2R\tau$.

In the first region, the comparison between $\widehat{\rho}$ and ρ_ζ gives $\widehat{\rho} \geq R\tau$ for large m. In the second, the rescaled neck radius is at most $2R$. These two conditions cover the initial surface, and hence their images cover Σ_m.

6.1. **Height coordinates and comparison with the Green function.** The estimates in Proposition 5.3 are stated on the initial surface M_ζ. The curvature calculation below instead uses the height of $G_m(M_\zeta)$ over the flat torus. We must therefore estimate the change of the torus coordinate caused by the normal displacement ϕ_m, invert that change, and express both ϕ_m and b_m in the resulting coordinates. We then compare the new height, together with its first two derivatives, with $\psi = \tau\Phi_m$. These are precisely the derivatives that enter the second fundamental form.

For every $r > 0$, define

$$f_\tau(r) := \max\{r^{3/2}, \tau^{(1-\alpha)/2} r^{1/2}\}, \qquad \mathfrak{a}_m := \delta^{-4-2\beta}\tau^{1+\alpha/3}, \qquad \delta := (100m)^{-1}.$$

Thus $f_\zeta(x) = f_\tau(\rho_\zeta(x))$ for every $x \in M_\zeta$. By (3.12), there are constants $c, C > 0$ such that $cm^2 \leq |\log\tau| \leq Cm^2$ for all sufficiently large m. In particular, $m^p\tau^q \to 0$ for every fixed $p \geq 0$ and $q > 0$. Both terms in the maximum defining f_τ are increasing functions of r. Moreover,

$$\frac{\tau^{(1-\alpha)/2}(\pi/m)^{1/2}}{(\pi/m)^{3/2}} = \frac{m}{\pi}\tau^{(1-\alpha)/2} \to 0.$$

Consequently, for all sufficiently large m, the supremum of f_τ on $[\tau, \pi/m]$ equals $(\pi/m)^{3/2}$. Set

$$\sigma_m := \frac{\mathfrak{a}_m}{\tau} \sup_{\tau \leq r \leq \pi/m} f_\tau(r) \leq C\delta^{-4-2\beta}\tau^{\alpha/3}m^{-3/2} \to 0. \tag{6.1}$$

In particular, the scalar displacement has size at most $C\sigma_m\tau$, and its first two derivatives have the corresponding bounds after multiplication by the appropriate powers of ρ_ζ. This will make the change of torus coordinate small relative to ρ_ζ.

Lemma 6.1 (Height coordinates away from the neck waists). *Fix $R \geq R_0$. For all sufficiently large m and every $x \in M_\zeta$ with $\rho_\zeta(x) \geq 2R\tau$, the surface $G_m(M_\zeta)$ is locally a height graph over $\mathbb{T}$ near $G_m(x)$. After applying the reflection Z if x lies on the lower sheet, there are an open neighborhood $U \subset \{q' \in \mathbb{T} : \rho(q') > R\tau\}$ of $q := \pi(G_m(x))$ and smooth functions z, b, η on U such that the corresponding portions of $G_m(M_\zeta)$ and Σ_m are parametrized by*

$$\begin{aligned} X(q') &:= \mathcal{F}(q', z(q')), \\ X_b(q') &:= \cos\big(b(q')\big)X(q') + \sin\big(b(q')\big)N(q'), \\ z(q') &= h(q') + \eta(q') \qquad \forall\, q' \in U. \end{aligned} \tag{6.2}$$

For every $q' \in U$ and $j \in \{0, 1, 2\}$,

$$\begin{aligned} |D^j\eta(q')| &\leq C\mathfrak{a}_m f_\tau(\rho(q'))\rho(q')^{-j}, \\ |D^j b(q')| &\leq C\tau^2\rho(q')^{-j}. \end{aligned} \tag{6.3}$$

Here D is the flat covariant derivative of $g_{\mathbb{T}}$, and the derivative norms are induced by $g_{\mathbb{T}}$. On the same neighborhood,

$$\begin{aligned} |z(q')| &\leq C\varepsilon, \\ |Dz(q')| &\leq C\tau/\rho(q'), \\ |D^2 z(q')| &\leq C\tau/\rho(q')^2 \qquad \forall\, q' \in U. \end{aligned} \tag{6.4}$$

The constants are independent of $R \geq R_0$, m, and x.

Proof. We first record the derivatives of the initial upper height. Increasing the fixed threshold R_0 if necessary, we have

$$|D^k h(p)| \leq C_k\tau\rho(p)^{-k} \qquad \text{whenever } \rho(p) \geq R_0\tau/2, \tag{6.5}$$

for every $p \in \mathbb{T}$ in the stated region and every fixed integer $k \geq 1$. To verify this, consider the three parts of the definition of h. On a catenoidal part, $h = C_\tau(r)$, where r is distance to the corresponding lattice point and

$$C_\tau(r) = \tau \operatorname{arcosh}(r/\tau),$$
$$C'_\tau(r) = \frac{\tau}{\sqrt{r^2 - \tau^2}}, \qquad C''_\tau(r) = -\frac{\tau r}{(r^2 - \tau^2)^{3/2}} \qquad \forall\, r > \tau.$$

Differentiating these expressions, or the logarithmic formula for C_τ, gives $|D^k C_\tau| \leq C_k \tau r^{-k}$ whenever $r \geq R_0\tau/2$. On the exterior part, $h = \widetilde{\psi}_\zeta = \tau\Phi_m - \tau\zeta\widehat{v}_m$. The estimate for Φ_m is (3.9). The cutoff derivatives in $\widehat{v}_m$ have size at most $C_k\delta^{-k}$ and are supported where ρ is comparable to δ; where the cutoff is constant, the derivatives of $J_0(\sqrt{2}r)$ satisfy the same required bounds. Finally, on an interpolation annulus $2r_c \leq r \leq 3r_c$, the estimates in Section 3 give

$$|D^k(C_\tau - \widetilde{\psi}_\zeta)| \leq C_k\big(\tau r_c^2 |\log\tau| + \tau^3 r_c^{-2}\big) r_c^{-k} = C_k \tau r_c^{-k}\big(r_c^2 |\log\tau| + \tau^2 r_c^{-2}\big)$$

for every fixed $k \geq 0$. The last factor tends to zero because $r_c = \tau^\alpha$ and $0 < \alpha < 1$. Each derivative of an interpolation cutoff contributes at most $C_j r_c^{-j}$. The product rule therefore proves (6.5) on these annuli too.

Suppose now that x lies on the upper initial sheet. Put $p_0 := \pi(x)$ and $r := \rho(p_0)$, so that $r \geq 2R\tau$. The map $X_+ : \{p \in \mathbb{T} : \rho(p) > \tau\} \to M_\zeta$ given by

$$X_+(p) := \mathcal{F}(p, h(p)) \qquad \forall\, p \in \mathbb{T} \text{ with } \rho(p) > \tau$$

parametrizes this sheet away from the waists. When an ambient point is needed below, we view $X_+(p)$ in $\mathcal{N}$ through the inclusion $\jmath_\zeta$. Choose a fixed small $c_1 > 0$ and a flat coordinate disk $B(p_0, 4c_1 r)$ that does not meet L_m. On this disk, $\rho(p)$ is comparable to r because ρ is 1-Lipschitz. The powers in the definition of f_τ then show that $f_\tau(\rho(p))$ is comparable to $f_\tau(r)$ as well. Write

$$A := \mathfrak{a}_m f_\tau(r), \qquad \widetilde{\phi} := \phi_m \circ X_+, \qquad \widetilde{b} := b_m \circ X_+.$$

In the flat coordinates, the induced metric of X_+ is

$$(X_+^* g_\zeta)_{ij} = \cos^2 h\, \delta_{ij} + \partial_i h\, \partial_j h \qquad \forall\, i, j \in \{1, 2\}.$$

It is uniformly equivalent to the flat metric, since $|h| \to 0$ and $|Dh| \leq C\tau/r$. Its first derivatives have size at most C/r, so its Christoffel symbols have the same bound. Applying (5.15) gives

$$|D^j \widetilde{\phi}| \leq CAr^{-j}, \qquad |D^j \widetilde{b}| \leq C\tau^2 r^{-j} \qquad \forall\, j \in \{0, 1, 2\}.$$

For $j = 2$, this uses the coordinate identity $\partial_{ij} v = (\nabla^2 v)_{ij} + \Gamma_{ij}^k \partial_k v$ for a scalar function v, applied with the metric $X_+^* g_\zeta$.

We next estimate the change of coordinates caused by $\widetilde{\phi}$. For p in a smaller concentric disk and $|t| < c_2 r$, where $c_2 > 0$ is fixed and small, define

$$\mathcal{E}(p, t) := \mathcal{F}^{-1}\Big(\exp^{\mathcal{N}}_{X_+(p)}\big(t\nu_\zeta(X_+(p))\big)\Big).$$

The values of $\mathcal{E}$ are expressed in the chosen flat coordinates and the coordinate z. These geodesics remain in the coordinate neighborhood for this range of t: their speed is one, the metric is uniformly equivalent to the Euclidean coordinate metric, and their initial points lie in a smaller disk. Their heights also remain in $(-z_0, z_0)$ because $|h| \to 0$ and $r \leq \pi/m$.

The required dependence on p and t can be read directly from the geodesic equations. If $\mathcal{E}(p, t) = (P_p(t), Z_p(t))$, then

$$\ddot{P}_p = 2 \tan Z_p\, \dot{Z}_p\, \dot{P}_p,$$
$$\ddot{Z}_p = -\sin Z_p \cos Z_p\, |\dot{P}_p|^2.$$

Here dots denote derivatives in t, and the norm in the second equation is the flat norm on the base. The initial position is $(p, h(p))$. Writing $c(p) := \cos h(p)$, the formula for the upward unit normal of a height graph gives the initial velocity

$$\big(\dot P_p(0), \dot Z_p(0)\big) = \left(-\frac{Dh(p)}{c(p)\sqrt{c(p)^2 + |Dh(p)|^2}}, \frac{c(p)}{\sqrt{c(p)^2 + |Dh(p)|^2}} \right).$$

After the changes of variables $p = p_0 + ry$ and $t = ru$, and division of the coordinate displacement by r, the equations have uniformly bounded smooth coefficients on fixed coordinate neighborhoods. The initial position and velocity have bounded derivatives in y through order two by (6.5) with $k \le 3$. Differentiating the first-order system for position and velocity in y once and twice gives linear ordinary differential equations for the corresponding derivatives; the inhomogeneous terms in the second equation are products of bounded first derivatives. Integration on the fixed interval $|u| < c_2$ gives uniform bounds for these derivatives. Derivatives in u are then bounded by the system itself.

More explicitly, set

$$\mathcal{V}(p,t) := \int_0^1 \partial_t \mathcal{E}(p, \theta t)\, d\theta.$$

The preceding bounds, after returning to the original coordinates, imply

$$|D_p^j \partial_t^k \mathcal{V}(p,t)| \le C_{j,k} r^{-j-k} \qquad \forall\, j,k \ge 0 \text{ with } j+k \le 2, \quad |t| < c_2 r.$$

The fundamental theorem of calculus gives $\mathcal{E}(p,t) = (p, h(p)) + t\mathcal{V}(p,t)$. Consequently, writing

$$\begin{aligned} G_m(X_+(p)) &= \mathcal{F}(Q(p), H(p)), \\ Q(p) &:= p + e(p), \qquad H(p) := h(p) + a(p), \end{aligned}$$

we have the vector identity

$$(e(p), a(p)) = \widetilde\phi(p) \mathcal{V}(p, \widetilde\phi(p)).$$

The product and chain rules now give

$$\begin{aligned} |e| + |a| &\le CA, \\ |De| + |Da| &\le C(A/r + A^2/r^2), \\ |D^2 e| + |D^2 a| &\le C(A/r^2 + A^2/r^3 + A^3/r^4). \end{aligned}$$

For example, the first derivative of $\widetilde\phi \mathcal{V}(p, \widetilde\phi)$ is

$$D\widetilde\phi\, \mathcal{V} + \widetilde\phi\big(D_p \mathcal{V} + \partial_t \mathcal{V}\, D\widetilde\phi\big),$$

with every occurrence of $\mathcal{V}$ and its derivatives evaluated at $(p, \widetilde\phi(p))$. Differentiating this expression once more gives the stated second derivative bound; no third derivative of $\widetilde\phi$ is required. Since

$$A \le \sigma_m \tau, \qquad \frac{A}{r} \le \sigma_m \to 0,$$

the substitution $t = \widetilde\phi(p)$ is allowed for large m, and the last three estimates imply

$$|D^j e| + |D^j a| \le C \mathfrak{a}_m f_\tau(r) r^{-j} \qquad \forall\, j \in \{0,1,2\}.$$

In particular, $DQ = I + De$ is invertible for large m. The inverse function theorem provides a local inverse Q^{-1} near $q = Q(p_0)$. Shrinking its domain if necessary, differentiation of $Q \circ Q^{-1} = \mathrm{id}$ gives

$$\begin{aligned} |DQ^{-1}| &\le 2, \\ D^2 Q^{-1} &= -(DQ \circ Q^{-1})^{-1} (D^2 Q \circ Q^{-1})[DQ^{-1}, DQ^{-1}], \\ |D^2 Q^{-1}| &\le CA/r^2. \end{aligned}$$

Moreover,

$$|\rho(Q(p)) - \rho(p)| \le |Q(p) - p| = |e(p)| \le CA.$$

Thus $\rho(Q(p))/\rho(p) = 1 + o_m(1)$ uniformly on the smaller disk. At its center, $\rho(q) \ge \frac{3}{4}r \ge \frac{3}{2}R\tau$ for large m. We may therefore choose the domain U of the local inverse inside $\{\rho > R\tau\}$.

On U, define

$$z := H \circ Q^{-1}, \qquad b := \widetilde{b} \circ Q^{-1}, \qquad \eta := z - h.$$

The definition of $\widehat{X}_m$ gives exactly (6.2) in these coordinates. To estimate η, put $k := h - h \circ Q$, so that $\eta \circ Q = a + k$. The mean-value formula and (6.5) give

$$\begin{aligned} |k| &\le C\tau A/r, \\ |Dk| &\le C\tau A/r^2, \\ |D^2k| &\le C\tau A/r^3 + C\tau A^2/r^4. \end{aligned}$$

Indeed, the first inequality follows by integrating Dh along the segment from p to $Q(p)$. These segments lie in the larger disk chosen above. For the other two inequalities, use

$$\begin{aligned} Dk &= Dh - (Dh \circ Q)(I + De), \\ D^2k &= D^2h - (D^2h \circ Q)[I + De, I + De] - (Dh \circ Q)D^2e. \end{aligned}$$

The differences $Dh - Dh \circ Q$ and $D^2h - D^2h \circ Q$ are bounded by integrating D^2h and D^3h, respectively, along the same segments. Since $\tau/r \le 1$ and $A/r \le 1$, it follows that $|D^j(a + k)| \le CAr^{-j}$ for $j \in \{0, 1, 2\}$. The chain rule for $(a + k) \circ Q^{-1}$ now gives

$$\begin{aligned} |\eta| &\le CA, \qquad |D\eta| \le CA/r, \\ |D^2\eta| &\le CA/r^2 + C(A/r)(A/r^2) \le CA/r^2. \end{aligned}$$

The same calculation for $b = \widetilde{b} \circ Q^{-1}$ gives

$$|b| \le C\tau^2, \qquad |Db| \le C\tau^2/r, \qquad |D^2b| \le C\tau^2/r^2 + C(\tau^2/r)(A/r^2) \le C\tau^2/r^2.$$

On U, $\rho(q')$ is comparable to r, and $f_\tau(\rho(q'))$ is comparable to $f_\tau(r)$. These estimates prove (6.3). The bound $|z| \le C\varepsilon$ follows from Lemma 4.1. The derivative bounds in (6.4) follow by adding the estimates for h and η, using $\mathfrak{a}_m f_\tau(\rho) \le \sigma_m \tau$.

Finally, Z is an isometry preserving both surfaces, and G_m is equivariant under Z. It sends the lower initial sheet to the upper one and leaves the torus coordinate unchanged. The preceding argument therefore applies after this reflection to every point on the lower sheet. □

The next lemma estimates the difference between h and ψ on every part of the sheet, including the catenoidal part where they are different functions. The term R^{-2} comes from the difference between a catenoidal height and its logarithmic approximation at distances at least $R\tau$ from a lattice point. It becomes small by choosing R large. The other errors become small as $m \to \infty$ with R fixed.

Lemma 6.2 (Comparison of the glued and Green heights). *For every $R \ge R_0$ and all sufficiently large m,*

$$\sum_{j=0}^{2} \rho(q)^j |D^j(h - \psi)(q)| \le C\tau\big(|\zeta| + m^2\tau^{2\alpha} + R^{-2}\big) \tag{6.6}$$

for every $q \in \mathbb{T}$ with $\rho(q) \ge R\tau$. Consequently, on every upper graph neighborhood U from Lemma 6.1,

$$\begin{aligned} \sum_{j=0}^{2} \rho(q)^j |D^j(z - \psi)(q)| &\le C\mu_m(R)\tau \qquad \forall\, q \in U, \\ \mu_m(R) &:= |\zeta| + m^2\tau^{2\alpha} + \sigma_m + R^{-2}. \end{aligned} \tag{6.7}$$

The constant C is independent of $R \geq R_0$ and m. For every fixed R, $\mu_m(R) \to R^{-2}$ as $m \to \infty$.

Proof. Since $\widetilde{\psi}_\zeta = \psi - \tau\zeta\widehat{v}_m$, the definition of the glued height and its catenoidal extension gives

$$h - \psi = -\tau\zeta\widehat{v}_m + \sum_{p\in L_m} \eta_{3r_c,2r_c}(r_p)(C_\tau(r_p) - \widetilde{\psi}_\zeta) \qquad \text{on } \{\rho > \tau\}. \tag{6.8}$$

Each summand is defined where $\tau < r_p < 3r_c$ and extended by zero where $r_p \geq 3r_c$. This extension is smooth because the cutoff vanishes in a neighborhood of its outer endpoint. The disks of radius $3r_c$ about distinct lattice points are disjoint for large m. Thus at most one summand is nonzero at each point, and estimating the sum does not introduce a factor m^2.

Fix $p \in L_m$ and a point with $R\tau \leq r := r_p \leq 3r_c$. On this disk, $\rho = r$ and $\widehat{v}_m = J_0(\sqrt{2}r)$. The local expansion (3.13) and the identity $\log\tau = \log 2 - A_m + \zeta$ yield

$$\begin{aligned}\widetilde{\psi}_\zeta - \tau\log(2r/\tau) &= \tau\big(\log r + A_m + \mathcal{E}_{m,p} - \zeta J_0(\sqrt{2}r) - \log r - \log 2 + \log\tau\big)\\ &= \tau\mathcal{E}_{m,p} + \tau\zeta(1 - J_0(\sqrt{2}r)).\end{aligned}$$

For every $j \in \{0,1,2\}$, the Green remainder estimate and the power series for J_0 give

$$\begin{aligned} r^j|D^j\mathcal{E}_{m,p}| &\leq Cr^2(m^2 + |\log r|),\\ r^j|D^j(1 - J_0(\sqrt{2}r))| &\leq Cr^2.\end{aligned}$$

The derivatives here are flat derivatives at the selected point. The annular estimate (3.13), used at scale r, gives the first line. In the second line, the series begins with a constant multiple of r^2. Since $\tau \leq r < 1$, $|\log r| \leq |\log\tau| \leq Cm^2$.

To compare the catenoidal height with the same logarithm, define

$$F(s) := \log\left(\frac{1 + \sqrt{1-s}}{2}\right) \qquad \forall\, s \in [0, R_0^{-2}].$$

Taking $R_0 > 1$, this is a smooth function with $F(0) = 0$ and bounded first and second derivatives. The exact catenoidal formula gives

$$C_\tau(r) - \tau\log(2r/\tau) = \tau F(\tau^2/r^2).$$

If $s(r) := \tau^2/r^2$, then $s'(r) = -2s/r$ and $s''(r) = 6s/r^2$. Therefore the last expression, its first radial derivative, and its second radial derivative have respective bounds $C\tau s$, $C\tau s/r$, and $C\tau s/r^2$. For a radial function $v = v(r)$ in two flat dimensions,

$$\begin{aligned} Dv &= v'(r)e_r,\\ D^2v &= v''(r)e_r \otimes e_r + \frac{v'(r)}{r}(I - e_r \otimes e_r), \qquad e_r := \frac{q-p}{|q-p|}.\end{aligned}$$

Here the calculation is in the flat disk about p, and I is the identity matrix in its parallel orthonormal coordinates. It follows that

$$r^j\left|D^j\big(C_\tau - \tau\log(2r/\tau)\big)\right| \leq C\tau^3 r^{-2} \qquad \forall\, j \in \{0,1,2\}.$$

Combining the last bounds with $|\zeta| \leq 1$ gives

$$r^j|D^j(C_\tau - \widetilde{\psi}_\zeta)| \leq C\tau\left(m^2r^2 + \frac{\tau^2}{r^2}\right) \qquad \forall\, j \in \{0,1,2\}. \tag{6.9}$$

Derivatives of $\eta_{3r_c,2r_c}(r)$ are supported where r is comparable to r_c and satisfy $|D^j\eta_{3r_c,2r_c}| \leq C_j r_c^{-j}$. In a term where i of the j derivatives fall on the cutoff, multiplication by r^j leaves a factor $r^i r_c^{-i}$ times the scaled derivative of order $j - i$ in (6.9). This factor is uniformly bounded. Thus multiplication by the cutoff preserves the bound in that equation. On its support, $r \leq 3r_c = 3\tau^\alpha$ and $r \geq R\tau$, so every such term is bounded by

$$C\tau\big(m^2\tau^{2\alpha} + R^{-2}\big).$$

It remains to estimate the first term of (6.8). The definition of $\widehat{v}_m$ gives

$$\sum_{j=0}^{2} \rho(q)^j |D^j \widehat{v}_m(q)| \le C \qquad \forall\, q \in \mathbb{T} \smallsetminus L_m.$$

Inside a disk where its cutoff is one, the assertion follows from the bounded derivatives of $J_0(\sqrt{2} r_p)$. On each cutoff annulus, $\rho = r_p$ is comparable to δ, and the factors ρ^j cancel the factors δ^{-j} from cutoff differentiation. Outside these disks the function is zero. The first term in (6.8) is consequently bounded by $C\tau|\zeta|$ in the stated sum of scaled derivatives. Adding the estimates proves (6.6).

Finally, on an upper graph neighborhood, (6.3) gives

$$\sum_{j=0}^{2} \rho(q)^j |D^j (z-h)(q)| \le C \mathfrak{a}_m f_\tau(\rho(q)) \le C \sigma_m \tau \qquad \forall\, q \in U.$$

The triangle inequality proves (6.7). By (5.11), (6.1), and the exponential decay of τ, the quantities $|\zeta|$, σ_m, and $m^2 \tau^{2\alpha}$ all tend to zero. This proves the last assertion. □

6.2. **Negative curvature away from the neck waists.** The preceding comparison controls the height of the surface $G_m(M_\zeta) \subset \mathcal{N}$ and its first two derivatives. We now relate those quantities to its Gaussian curvature. We first compute the curvature of an arbitrary height graph inside $\mathcal{N}$. We then use the equation satisfied by ψ to prove that the graphs supplied by Lemma 6.1 have strictly negative curvature. At this stage we consider the parametrization X in (6.2); the effect of replacing X by the final parametrization X_b will be estimated separately.

For an open set $U \subset \mathbb{T}$ and a function $z \in C^2(U, (-z_0, z_0))$, put

$$\mathcal{K}_2(z) := \det D^2 z + z \Delta_{\mathbb{T}} z + |Dz|^2.$$

Here the Hessian $D^2 z$ and its determinant are computed with respect to the flat metric $g_{\mathbb{T}}$, and $\Delta_{\mathbb{T}} z = \operatorname{tr}_{g_{\mathbb{T}}} D^2 z$. The quantity $\mathcal{K}_2(z)$ consists of the terms of order two in the height and its first two derivatives that occur in the curvature formula. The exact curvature also contains higher-order terms and a positive denominator. The following lemma bounds their contribution by a small multiple of the weight $\mathcal{W}$. This is the comparison needed to obtain a strict sign from the Green estimates.

Lemma 6.3 (Exact curvature formula for a height graph). *Let $U \subset \mathbb{T}$ be open, let $z \in C^2(U, (-z_0, z_0))$, and define the embedding $X : U \to \mathcal{N}$ by $X(q) := \mathcal{F}(q, z(q))$ for every $q \in U$. Denote by K_X the Gaussian curvature of the induced metric $g_X := X^* g_{\mathcal{N}}$, regarded as a function on U. At each point of U, write*

$$c := \cos z, \qquad s_z := \sin z, \qquad p := Dz, \qquad H := D^2 z, \qquad Q := c^2 + |p|^2.$$

Then

$$Q^2 K_X = \det H + c s_z \operatorname{tr} H + |p|^2 + |p|^4 + 2 \tan z \operatorname{cof}(H)(p,p). \tag{6.10}$$

Here $\operatorname{cof}(H) := (\operatorname{tr} H) I - H$ is the cofactor matrix of the symmetric 2×2 matrix H, I is the identity matrix, and $\operatorname{cof}(H)(p,p) := p^T \operatorname{cof}(H) p$. Suppose, in addition, that $R \ge R_0$, that $\rho(q) \ge R\tau$ for every $q \in U$, and that (6.4) holds on U. For all sufficiently large m, one then has

$$|K_X - \mathcal{K}_2(z)| \le C(\varepsilon^2 + R^{-2}) \mathcal{W} \qquad \text{on } U. \tag{6.11}$$

The constant C depends on the constants in (6.4) and on z_0, but is independent of m and $R \ge R_0$.

Proof. Fix $q \in U$, and choose flat coordinates near q. Write $\partial_1 := \partial_s$ and $\partial_2 := \partial_t$, and put $p_i := \partial_i z$ and $H_{ij} := \partial_{ij} z$ for $i, j \in \{1, 2\}$. All scalar quantities in the following computation are evaluated at q, and the frame fields E_1, E_2, E_3 are evaluated at $X(q)$. Differentiating X gives the tangent vectors

$$T_i := dX_q(\partial_i) = c E_i + p_i E_3 \qquad \forall\, i \in \{1, 2\}.$$

Consequently the induced metric has matrix

$$g_{ij} := g_{\mathcal{N}}(T_i, T_j) = c^2\delta_{ij} + p_i p_j,$$
$$\det(g_{ij}) = (c^2 + p_1^2)(c^2 + p_2^2) - p_1^2 p_2^2 = c^2(c^2 + |p|^2) = c^2 Q.$$

In particular, X is an immersion. Its injectivity follows from $\pi \circ X = \mathrm{id}_U$, and the same identity gives a continuous inverse on its image. Thus X is an embedding.

The unit normal to this graph whose E_3 component is positive is

$$\nu = \frac{-p_1 E_1 - p_2 E_2 + cE_3}{\sqrt{Q}}.$$

Indeed, its numerator has squared norm Q and is orthogonal to both T_1 and T_2. In the next calculation, h^ν_X is the scalar second fundamental form of X as an immersion into $\mathcal{N}$. The connection identities from Lemma 2.2 give

$$\nabla^{\mathcal{N}}_{T_i} E_j = c\tan z\, \delta_{ij} E_3,$$
$$\nabla^{\mathcal{N}}_{T_i} E_3 = -c\tan z\, E_i \qquad \forall\, i, j \in \{1, 2\}.$$

Applying the product rule to $T_j = cE_j + p_j E_3$, with the connection taken along X, yields

$$\begin{aligned}\nabla^{\mathcal{N}}_{T_i} T_j &= -s_z p_i E_j + c\nabla^{\mathcal{N}}_{T_i} E_j + H_{ij} E_3 + p_j \nabla^{\mathcal{N}}_{T_i} E_3 \\ &= -s_z(p_i E_j + p_j E_i) + (H_{ij} + cs_z\delta_{ij})E_3.\end{aligned}$$

Taking the inner product with ν, we obtain

$$\begin{aligned}h^\nu_X(\partial_i, \partial_j) &= \frac{2s_z p_i p_j + c(H_{ij} + cs_z\delta_{ij})}{\sqrt{Q}} \\ &= \frac{c}{\sqrt{Q}}\big(H_{ij} + cs_z\delta_{ij} + 2\tan z\, p_i p_j\big).\end{aligned}$$

Define the symmetric matrix $B := H + cs_z I + 2\tan z\, p \otimes p$. Since the matrix of the shape operator S^ν_X is the product of $(g_{ij})^{-1}$ with the matrix of h^ν_X, it follows that

$$\det S^\nu_X = \frac{\det\big(h^\nu_X(\partial_i, \partial_j)\big)}{\det(g_{ij})} = \frac{(c^2/Q)\det B}{c^2 Q} = \frac{\det B}{Q^2}.$$

We also need the sectional curvature of the tangent plane $\mathrm{span}\{T_1, T_2\}$ in $\mathcal{N}$. The formulas for the shape operator S of $\mathcal{N} \subset \mathbb{S}^4$ give

$$S(T_1) = -E_1,$$
$$S(T_2) = E_2,$$

and hence

$$g_{\mathcal{N}}(S(T_1), T_1) = -c,$$
$$g_{\mathcal{N}}(S(T_2), T_2) = c,$$
$$g_{\mathcal{N}}(S(T_1), T_2) = 0.$$

The Gauss equation for $\mathcal{N} \subset \mathbb{S}^4$ therefore gives

$$\sec_{g_{\mathcal{N}}}(\mathrm{span}\{T_1, T_2\}) = 1 + \frac{(-c)c}{c^2 Q} = 1 - \frac{1}{Q} = \frac{|p|^2 - s_z^2}{Q}.$$

Applying the Gauss equation once more, now to $X : U \to \mathcal{N}$, yields

$$Q^2 K_X = Q(|p|^2 - s_z^2) + \det B.$$

To expand $\det B$, recall that for a symmetric 2×2 matrix A, a vector $p \in \mathbb{R}^2$ and a scalar $t \in \mathbb{R}$,

$$\begin{aligned}\det(A + tp \otimes p) &= \det A + t(A_{22}p_1^2 - 2A_{12}p_1 p_2 + A_{11}p_2^2) \\ &= \det A + t\,\mathrm{cof}(A)(p, p).\end{aligned}$$

The terms of order two in t cancel in the determinant. Apply this identity with $A = H + cs_zI$ and $t = 2\tan z$. Since

$$\begin{aligned}\det(H + cs_zI) &= \det H + cs_z \operatorname{tr} H + c^2 s_z^2,\\ \operatorname{cof}(H + cs_zI) &= \operatorname{cof}(H) + cs_zI,\end{aligned}$$

we obtain

$$\det B = \det H + cs_z \operatorname{tr} H + c^2 s_z^2 + 2\tan z \operatorname{cof}(H)(p,p) + 2s_z^2|p|^2.$$

Finally,

$$Q(|p|^2 - s_z^2) = c^2|p|^2 + |p|^4 - c^2 s_z^2 - s_z^2|p|^2.$$

Adding these two identities cancels $c^2 s_z^2$ and leaves $(c^2 + s_z^2)|p|^2 = |p|^2$ as the terms of degree two in p. This proves (6.10).

We now assume the bounds in the second part of the lemma. For large m, $\rho \le \pi/m < 1$ and

$$|p| \le C\tau\rho^{-1} \le C/R, \qquad |H| \le C\tau\rho^{-2}, \qquad |z| \le C\varepsilon \le C\tau\rho^{-2}.$$

The last inequality uses $m^2\rho^2 \le \pi^2$. Consequently

$$\begin{aligned}|H|^2 &\le C\tau^2\rho^{-4} \le C\mathcal{W},\\ |z||H| &\le C\tau^2\rho^{-4} \le C\mathcal{W},\\ |p|^2 &\le C\tau^2\rho^{-2} \le C\mathcal{W}.\end{aligned}$$

Since $|\det H| \le |H|^2/2$ and $|\operatorname{tr} H| \le \sqrt{2}|H|$, these bounds imply $|\mathcal{K}_2(z)| \le C\mathcal{W}$. Moreover Q is bounded above and below by positive constants independent of m and $R \ge R_0$. Indeed $Q \ge \cos^2 z_0 > 0$, and $|p| \le C/R_0$. The identities $Q - 1 = -\sin^2 z + |p|^2$ and $Q^{-2} - 1 = (1-Q)(1+Q)/Q^2$ give

$$|Q^{-2} - 1| \le C(z^2 + |p|^2).$$

The difference between the numerator in (6.10) and $\mathcal{K}_2(z)$ is exactly

$$E := (cs_z - z)\operatorname{tr} H + |p|^4 + 2\tan z \operatorname{cof}(H)(p,p).$$

For $|z| < z_0$, Taylor's theorem gives $|\cos z \sin z - z| \le C|z|^3$ and $|\tan z| \le C|z|$. Moreover, $|\operatorname{cof}(H)(p,p)| \le |H||p|^2$. Hence

$$\begin{aligned}|E| &\le C\big(|z|^3|H| + |p|^4 + |z||H||p|^2\big)\\ &\le C(z^2 + |p|^2)\mathcal{W}.\end{aligned}$$

Each term in the second line is controlled by one of the three preceding bounds: for example, $|z|^3|H| = z^2(|z||H|)$ and $|z||H||p|^2 = |p|^2(|z||H|)$. It follows that

$$\begin{aligned}|K_X - \mathcal{K}_2(z)| &\le |Q^{-2} - 1|\,|\mathcal{K}_2(z)| + Q^{-2}|E|\\ &\le C(z^2 + |p|^2)\mathcal{W} \le C(\varepsilon^2 + R^{-2})\mathcal{W},\end{aligned}$$

as required. □

We apply this formula first to the Green height ψ. Because $\Delta_{\mathbb{T}}\psi = -2\psi$ away from the lattice, the terms involving ψ^2 cancel in $\mathcal{K}_2(\psi)$. What remains is $|D\psi|^2 - \frac{1}{2}|D^2\psi|^2$, whose negative upper bound was proved in Section 3. We then estimate the effect of replacing ψ by the actual graph height z. The comparison must be made relative to $\mathcal{W}$, since this negative upper bound tends to zero on the parts of the sheets furthest from the necks.

Lemma 6.4 (Negative curvature of the graphs inside $\mathcal{N}$). *There exists $c_* > 0$ such that, after choosing R sufficiently large and then m sufficiently large, every scalar graph $X : U \to \mathcal{N}$ from Lemma 6.1 satisfies*

$$K_X(q) \le -c_* \mathcal{W}(q)$$

at each point in its domain with $\rho(q) \ge R\tau$.

Proof. On $\mathbb{T} \smallsetminus L_m$ we have $\Delta_{\mathbb{T}}\psi = -2\psi$. For any symmetric 2×2 matrix H, $\det H = ((\operatorname{tr} H)^2 - |H|^2)/2$. Therefore

$$\begin{aligned}\mathcal{K}_2(\psi) &= \tfrac{1}{2}\big((\Delta_{\mathbb{T}}\psi)^2 - |D^2\psi|^2\big) + \psi\Delta_{\mathbb{T}}\psi + |D\psi|^2 \\ &= 2\psi^2 - \tfrac{1}{2}|D^2\psi|^2 - 2\psi^2 + |D\psi|^2 \\ &= |D\psi|^2 - \tfrac{1}{2}|D^2\psi|^2.\end{aligned}$$

Recall that $\psi = \varepsilon U_m$. At each point of $\mathbb{T} \smallsetminus L_m$, set $r := \sqrt{2}m\rho$. Since ε is constant on $\mathbb{T}$, multiplication of (3.10) by ε^2 gives

$$\mathcal{K}_2(\psi) = \varepsilon^2\big(|DU_m|^2 - \tfrac{1}{2}|D^2U_m|^2\big) \le -c\varepsilon^2(1 + r^{-4}) = -c\left(\varepsilon^2 + \frac{\tau^2\rho^{-4}}{16\pi^2}\right).$$

In the last equality we used $\varepsilon^2 r^{-4} = \tau^2\rho^{-4}/(16\pi^2)$. Since $\mathcal{W} = \varepsilon^2 + \tau^2\rho^{-4}$, reducing the positive constant if necessary gives

$$\mathcal{K}_2(\psi) \le -c_0\mathcal{W}, \tag{6.12}$$

where $c_0 > 0$ is independent of m and R. On $\rho \ge R\tau$, the Green estimates also give

$$|\psi| \le C\varepsilon \le C\tau\rho^{-2}, \qquad |D\psi| \le C\tau\rho^{-1}, \qquad |D^2\psi| \le C\tau\rho^{-2}.$$

The derivative bounds follow from the Green estimates in Section 3. For the value of ψ, use $\Phi_m = m^2/(2\pi) + \log(\sqrt{2}m\rho) + O(1)$. Since $R \ge R_0 > 1$, $\rho \ge R\tau$ and $\rho \le \pi/m < 1$, we have $|\log\rho| \le |\log\tau| \le Cm^2$ for large m. Thus $|\psi| \le C\tau m^2 = C(2\pi)\varepsilon$. The further bound $\varepsilon \le C\tau\rho^{-2}$ follows from $m^2\rho^2 \le \pi^2$.

Consider an upper graph neighborhood. All estimates below are at its points with $\rho \ge R\tau$. Put $e := z - \psi$ and $\mu := \mu_m(R)$. By Lemma 6.2,

$$|e| \le C\mu\tau, \qquad |De| \le C\mu\tau\rho^{-1}, \qquad |D^2e| \le C\mu\tau\rho^{-2}.$$

Using $\det(H + E) - \det H = \operatorname{cof}(H) : E + \det E$ for symmetric 2×2 matrices, we obtain

$$\begin{aligned}\mathcal{K}_2(z) - \mathcal{K}_2(\psi) = {} & \operatorname{cof}(D^2\psi) : D^2e + \det D^2e + \psi\Delta_{\mathbb{T}}e + e\Delta_{\mathbb{T}}\psi + e\Delta_{\mathbb{T}}e \\ & + 2\langle D\psi, De\rangle + |De|^2.\end{aligned}$$

The colon denotes the sum of the products of corresponding matrix entries. For symmetric 2×2 matrices, $|\operatorname{cof}(H)| = |H|$, and $|\det H| \le |H|^2/2$. The Hessian terms therefore satisfy

$$\begin{aligned}|\operatorname{cof}(D^2\psi) : D^2e| &\le C\mu\tau^2\rho^{-4}, \\ |\det D^2e| &\le C\mu^2\tau^2\rho^{-4}.\end{aligned}$$

For the terms involving the heights, the preceding bounds give

$$\begin{aligned}|\psi\Delta_{\mathbb{T}}e| &\le C(\tau\rho^{-2})(\mu\tau\rho^{-2}) = C\mu\tau^2\rho^{-4}, \\ |e\Delta_{\mathbb{T}}\psi| &\le C(\mu\tau)(\tau\rho^{-2}) = C\mu\tau^2\rho^{-2}, \\ |e\Delta_{\mathbb{T}}e| &\le C(\mu\tau)(\mu\tau\rho^{-2}) = C\mu^2\tau^2\rho^{-2}.\end{aligned}$$

Finally, the gradient terms satisfy

$$2|\langle D\psi, De\rangle| + |De|^2 \le C(\mu + \mu^2)\tau^2\rho^{-2}.$$

Since $\rho < 1$, every occurrence of ρ^{-2} is bounded by ρ^{-4}. Summing these estimates and using $\tau^2\rho^{-4} \le \mathcal{W}$ yields

$$|\mathcal{K}_2(z) - \mathcal{K}_2(\psi)| \le C(\mu + \mu^2)\mathcal{W}.$$

Combining this with (6.12) and (6.11) yields

$$K_X \le \big[-c_0 + C(\mu + \mu^2 + \varepsilon^2 + R^{-2})\big]\mathcal{W}.$$

To make the choice of parameters explicit, write $\mu = t_m + R^{-2}$, where $t_m := |\zeta| + m^2\tau^{2\alpha} + \sigma_m \to 0$. First choose $R \ge R_0$ so large that $C(2R^{-2} + R^{-4}) < c_0/4$. Keeping this R fixed, choose m so large that $C(t_m + t_m^2 + 2t_mR^{-2} + \varepsilon^2) < c_0/4$. We then have $K_X \le -(c_0/2)\mathcal{W}$, proving the claim with $c_* := c_0/2$. Reflection by Z preserves the induced metric, so the same curvature bound holds on the lower graphical parts. □

We now estimate the change in Gaussian curvature produced by b, which moves the graph X in the direction normal to $\mathcal{N}$ within $\mathbb{S}^4$. The bound $|b| \le C\tau^2$ alone does not control this change: curvature also depends on the first and second derivatives of the parametrization. In fact, the bound for $|D^2b|$ is only C/R^2 when $\rho = R\tau$, and need not tend to zero for fixed R. The following estimate compares the curvature change directly with the positive function $\mathcal{W}$ in the preceding lemma. It will therefore allow us to preserve the negative upper bound for K_X.

Lemma 6.5 (Curvature change under the deformation normal to $\mathcal{N}$). *Let $X, X_b : U \to \mathbb{S}^4$ be the parametrizations in Lemma 6.1. For every $q \in U$ with $\rho(q) \ge R\tau$, one has*

$$|K_{X_b}(q) - K_X(q)| \le C(m^{-2} + \tau)\mathcal{W}(q). \tag{6.13}$$

Here K_X and K_{X_b} are the Gaussian curvatures of the metrics induced on U by the indicated parametrizations. The constant C is independent of $R \ge R_0$ and of sufficiently large m.

Proof. We use local flat coordinates (s,t) on U and regard X and X_b as maps into $\mathbb{R}^5$. Derivatives and matrix norms in this proof are Euclidean. All estimates are evaluated at a point $q \in U$ as in the statement, and we write ρ for $\rho(q)$. We may assume $R_0 \ge 1$, so that

$$\tau \le \rho \le \frac{\pi}{m} < 1.$$

Put $\Lambda := |D^2z(q)|$. Since all derivatives of the fixed maps f and N are bounded, differentiating $X = \cos z\, f + \sin z\, e_5$ and using (6.4) gives

$$|DX| \le C, \qquad |D^2X| \le C(1+\Lambda), \qquad \Lambda \le C\tau\rho^{-2}.$$

The matrix $g_X := (DX)^T DX$ of the induced metric satisfies the exact identity

$$g_X = \cos^2 z\, I + Dz \otimes Dz.$$

In particular, $g_X \ge \cos^2 z_0\, I$, and its eigenvalues are bounded above and below by positive constants independent of m and R.

We first estimate the change in the first two derivatives of the parametrization. Write $\partial_1 := \partial_s$ and $\partial_2 := \partial_t$. For every $i, j \in \{1,2\}$, differentiating $X_b = \cos b\, X + \sin b\, N$ gives

$$\begin{aligned}
\partial_i X_b - \partial_i X &= (\cos b - 1)\partial_i X + \sin b\, \partial_i N + b_i(-\sin b\, X + \cos b\, N),\\
\partial_{ij} X_b - \partial_{ij} X &= (\cos b - 1)\partial_{ij} X + \sin b\, \partial_{ij} N\\
&\quad + b_i(-\sin b\, \partial_j X + \cos b\, \partial_j N) + b_j(-\sin b\, \partial_i X + \cos b\, \partial_i N)\\
&\quad + b_{ij}(-\sin b\, X + \cos b\, N) - b_i b_j X_b,
\end{aligned}$$

where $b_i := \partial_i b$ and $b_{ij} := \partial_{ij} b$. Define the nonnegative quantities

$$E_1 := \tau^2\rho^{-1}, \qquad E_2 := \tau^2\rho^{-2}.$$

The coordinate estimates (6.3) imply

$$|DX_b - DX| \le CE_1, \qquad |D^2X_b - D^2X| \le CE_2.$$

To check the second bound, the first term in the second derivative formula is bounded by

$$C|b|^2(1+\Lambda) \le C\tau^4(1+\tau\rho^{-2}) \le C\tau^2\rho^{-2}.$$

Each of the remaining terms is bounded by a constant times one of $|b|$, $|Db|$, $|D^2b|$, or $|Db|^2$. Their sum is at most CE_2, since $\tau < 1$ and $\rho < 1$. The first derivative formula similarly gives

$$|DX_b - DX| \le C\big(|b| + |Db|\big) \le C(\tau^2 + \tau^2\rho^{-1}) \le CE_1.$$

Moreover, $E_1 \le \tau/R \le \tau \to 0$. Thus the eigenvalues of g_{X_b} also have uniform positive lower and upper bounds for large m.

We next express curvature in terms of these derivatives. For either immersion $Y = X$ or $Y = X_b$, let P_Y be the Euclidean orthogonal projection onto the normal space of Y in $\mathbb{R}^5$. In matrix notation,

$$P_Y = I - DY(g_Y)^{-1}(DY)^T, \qquad g_Y := (DY)^T DY.$$

Its Euclidean second fundamental form has coordinate components $P_Y\partial_{ij}Y$. The Gauss equation therefore gives

$$K_Y = \frac{\langle P_Y\partial_{ss}Y, P_Y\partial_{tt}Y\rangle - |P_Y\partial_{st}Y|^2}{\det g_Y}. \tag{6.14}$$

This computes the Gaussian curvature of the metric induced by Y also when Y takes values in $\mathbb{S}^4$: the Euclidean normal projection includes the radial direction of the sphere, so that no additional ambient-curvature term is to be added to this formula.

Write $g := g_X$ and $\widetilde{g} := g_{X_b}$. Expanding the products of first derivatives in the metric matrices gives $|\widetilde{g} - g| \le CE_1$. Their uniform positive lower bounds and the identity

$$\widetilde{g}^{-1} - g^{-1} = \widetilde{g}^{-1}(g - \widetilde{g})g^{-1}$$

then give $|\widetilde{g}^{-1} - g^{-1}| \le CE_1$. Substitution into the projection formula yields the first of the following estimates; the second follows by expanding the 2×2 determinants and using their positive lower bounds:

$$|P_{X_b} - P_X| \le CE_1,$$
$$|(\det g_{X_b})^{-1} - (\det g_X)^{-1}| \le CE_1.$$

For $i, j \in \{1, 2\}$, put

$$T_{ij} := P_X\partial_{ij}X, \qquad \widetilde{T}_{ij} := P_{X_b}\partial_{ij}X_b.$$

Since orthogonal projections have operator norm one,

$$|T_{ij}| \le C(1+\Lambda),$$
$$|\widetilde{T}_{ij} - T_{ij}| \le |P_{X_b}(\partial_{ij}X_b - \partial_{ij}X)| + |(P_{X_b} - P_X)\partial_{ij}X| \le C\big(E_2 + E_1(1+\Lambda)\big).$$

Define $\mathcal{Q} := \langle T_{11}, T_{22}\rangle - |T_{12}|^2$ and let $\widetilde{\mathcal{Q}}$ be the corresponding expression with $\widetilde{T}_{ij}$. Subtracting these quadratic expressions gives

$$|\widetilde{\mathcal{Q}} - \mathcal{Q}| \le C\big[(1+\Lambda)\big(E_2 + E_1(1+\Lambda)\big) + \big(E_2 + E_1(1+\Lambda)\big)^2\big].$$

For large m we have $E_1 \le 1$, and hence

$$\big(E_2 + E_1(1+\Lambda)\big)^2 \le 2E_2^2 + 2E_1(1+\Lambda)^2.$$

Using also $|\mathcal{Q}| \le C(1+\Lambda)^2$, we obtain from (6.14)

$$|K_{X_b} - K_X| \le \frac{|\widetilde{\mathcal{Q}} - \mathcal{Q}|}{\det\widetilde{g}} + |\mathcal{Q}|\,|(\det\widetilde{g})^{-1} - (\det g)^{-1}| \le C\big((1+\Lambda)E_2 + E_1(1+\Lambda)^2 + E_2^2\big).$$

Substituting the definitions of E_1, E_2 and $\Lambda \le C\tau\rho^{-2}$ gives

$$|K_{X_b} - K_X| \le C\left(\frac{\tau^2}{\rho^2} + \frac{\tau^3}{\rho^4} + \frac{\tau^2}{\rho} + \frac{\tau^3}{\rho^3} + \frac{\tau^4}{\rho^5} + \frac{\tau^4}{\rho^4}\right).$$

We compare each term with $\mathcal{W} \ge \tau^2\rho^{-4}$. Division by this lower bound yields

$$\frac{|K_{X_b} - K_X|}{\mathcal{W}} \le C\left(\rho^2 + \tau + \rho^3 + \tau\rho + \frac{\tau^2}{\rho} + \tau^2\right) \le C(m^{-2} + \tau).$$

In the last inequality we used $\rho \le \pi/m < 1$, $\tau \le \rho$, and $\tau < 1$. All constants depend only on the fixed geometry and the constants in Lemma 6.1, and are therefore independent of $R \ge R_0$. □

6.3. **Curvature near the neck waists and the area limit.** At a neck waist, the projection of the initial surface onto $\mathbb{T}$ is not a local diffeomorphism. Thus the height coordinates used in the preceding subsections do not cover that circle. We instead use the catenoidal parametrization from Section 3. Dividing lengths by τ transforms the region $\rho_\zeta \le 2R\tau$ into a fixed portion of the unit catenoid. We will prove that the final immersions converge to this catenoid in C^2 in the rescaled coordinates. Its curvature is bounded above by a strictly negative constant on this fixed portion, which gives the remaining curvature estimate. We then combine the two regions and compare the area elements before and after both deformations.

Lemma 6.6 (Negative curvature near the neck waists). *For every fixed $R \ge R_0$, there exists $m_R \in \mathbb{N}$ such that, whenever $m \ge m_R$,*

$$K_{\Sigma_m}(\widehat{X}_m(x)) \le -\frac{1}{2(2R)^4\tau^2} \qquad \forall\, x \in M_\zeta \text{ such that } \rho_\zeta(x) \le 2R\tau. \tag{6.15}$$

Proof. Fix $R \ge R_0$. Since $\tau^{1-\alpha} \to 0$, we have $3R\tau < 2r_c$ for all sufficiently large m. The part of M_ζ with $\rho_\zeta \le 3R\tau$ therefore lies entirely in the catenoidal pieces defined by (3.17).

We first specify the ambient coordinates and the metric after rescaling. Fix $p \in L_m$, let $B_R := (-5R, 5R)^4 \subset \mathbb{R}^4$, and define

$$\mathcal{P}_{p,\tau}(y_1, y_2, v, w) := \Psi\big(p + \tau(y_1, y_2), \tau v, \tau w\big) \qquad \forall\,(y_1, y_2, v, w) \in B_R.$$

The addition to p is in flat coordinates on $\mathbb{T}$. This map is defined and is a diffeomorphism onto its image for sufficiently large m. Indeed, the horizontal displacement is at most $5\sqrt{2}R\tau < \delta$ for large m, and the remaining two coordinates have absolute value at most $5R\tau$. Thus its values lie in the domain of the tubular embedding Ψ. The metric that measures lengths in units of τ is

$$\overline{g}_{p,\tau} := \tau^{-2}\mathcal{P}_{p,\tau}^* g_{\mathbb{S}^4}.$$

Write $\widetilde{g}_{ab}$ for the coefficients of $\Psi^* g_{\mathbb{S}^4}$ in the coordinates (s, t, z, b). The chain rule gives

$$(\overline{g}_{p,\tau})_{ab}(y_1, y_2, v, w) = \widetilde{g}_{ab}\big(p + \tau(y_1, y_2), \tau v, \tau w\big) \qquad \forall\, a, b \in \{1, 2, 3, 4\}.$$

At $(p, 0, 0)$, the four coordinate vectors of Ψ are E_1, E_2, E_3, n at $f(p)$, so $\widetilde{g}_{ab}(p, 0, 0) = \delta_{ab}$. The coefficients $\widetilde{g}_{ab}$ and all their derivatives are bounded on a fixed smaller tubular band containing the coordinate images under consideration. Taylor's theorem consequently bounds the difference of the displayed coefficients from δ_{ab} by $C_R\tau$ on B_R. A derivative of positive order j in the rescaled variables introduces a factor τ^j. Thus $\overline{g}_{p,\tau}$ converges to the Euclidean metric in every fixed C^k norm on B_R, uniformly for $p \in L_m$.

For $A \in \{2R, 3R\}$, put

$$D_A := [-\operatorname{arcosh} A, \operatorname{arcosh} A] \times \big(\mathbb{R}/(2\pi\mathbb{Z})\big).$$

In the coordinates just defined, the initial catenoidal parametrization is the fixed map

$$c(\xi, \theta) := \big(\cosh\xi\cos\theta, \cosh\xi\sin\theta, \xi, 0\big) \qquad \forall\,(\xi, \theta) \in D_{3R}.$$

More precisely, if $\mathcal{C}_{\zeta,p}$ denotes the parametrization in (3.17), then

$$\mathcal{C}_{\zeta,p} = \mathcal{P}_{p,\tau} \circ c, \qquad \rho_\zeta(\mathcal{C}_{\zeta,p}(\xi,\theta)) = \tau\cosh\xi \qquad \forall\,(\xi,\theta) \in D_{3R}.$$

The restriction of $\overline{g}_{p,\tau}$ to $w = 0$ is

$$dv^2 + \cos^2(\tau v)(dy_1^2 + dy_2^2).$$

Therefore the metric induced on D_{3R} is exactly

$$\tau^{-2}\mathcal{C}_{\zeta,p}^* g_\zeta = \big(1 + \cos^2(\tau\xi)\sinh^2\xi\big)d\xi^2 + \cos^2(\tau\xi)\cosh^2\xi\, d\theta^2.$$

Its coefficients converge smoothly to those of $\cosh^2\xi(d\xi^2 + d\theta^2)$, and their eigenvalues are bounded above and below by positive constants depending only on R.

We next estimate both displacement functions in these fixed parameters. For a smooth function u on the initial catenoidal piece, the preceding metric formula implies

$$\left\| \frac{u \circ \mathcal{C}_{\zeta,p}}{\tau} \right\|_{C^2(D_{3R})} \le \frac{C_R}{\tau} \sup_{\mathcal{C}_{\zeta,p}(D_{3R})} \big(|u| + \tau|\nabla u|_{g_\zeta} + \tau^2|\nabla^2 u|_{g_\zeta}\big).$$

Here the C^2 norm is the sum of the supremum norms of the ordinary (ξ,θ) derivatives through order two. To see the second-order assertion explicitly, the Hessian in these parameters satisfies

$$\partial_{ij}(u \circ \mathcal{C}_{\zeta,p}) = (\mathcal{C}_{\zeta,p}^* \nabla^2 u)_{ij} + \Gamma_{ij}^k \partial_k (u \circ \mathcal{C}_{\zeta,p}), \qquad i,j \in \{1,2\},$$

with summation over $k \in \{1,2\}$. The Christoffel symbols Γ_{ij}^k are those of the displayed rescaled metric and are uniformly bounded on D_{3R}. Its comparison with the product metric bounds the two terms by $C_R\tau^2|\nabla^2 u|_{g_\zeta}$ and $C_R\tau|\nabla u|_{g_\zeta}$, respectively.

Since $\rho_\zeta = \tau\cosh\xi$ on this piece,

$$f_\tau(\rho_\zeta) = \max\{\tau^{3/2}\cosh^{3/2}\xi, \tau^{1-\alpha/2}\cosh^{1/2}\xi\} \le C_R\tau^{1-\alpha/2}.$$

Define functions on D_{3R} by

$$\overline{\phi}_{p,m} := \frac{\phi_m \circ \mathcal{C}_{\zeta,p}}{\tau}, \qquad \overline{b}_{p,m} := \frac{b_m \circ \mathcal{C}_{\zeta,p}}{\tau}.$$

Applying (5.15) in the preceding norm comparison gives

$$\|\overline{\phi}_{p,m}\|_{C^2(D_{3R})} \le C_R\delta^{-4-2\beta}\tau^{1+\alpha/3}\tau^{1-\alpha/2}\tau^{-1} = C_R\delta^{-4-2\beta}\tau^{1-\alpha/6} \to 0,$$
$$\|\overline{b}_{p,m}\|_{C^2(D_{3R})} \le C_R\tau \to 0.$$

Both limits are uniform in p. The first follows because $1 - \alpha/6 > 0$ and every positive power of τ tends to zero faster than every fixed inverse power of m.

These are estimates for scalar functions on the initial parametrization. We now relate them to the final immersion. Let $\overline{g}_{p,\tau}^{\mathcal{N}}$ be the restriction of $\overline{g}_{p,\tau}$ to $w = 0$, and let $\overline{\nu}_{p,\tau}$ be the unit normal to c in that metric corresponding to ν_ζ. Equivalently,

$$d\mathcal{P}_{p,\tau}|_{c(\xi,\theta)}\big(\overline{\nu}_{p,\tau}(\xi,\theta)\big) = \tau\nu_\zeta(\mathcal{C}_{\zeta,p}(\xi,\theta)).$$

Define the normal exponential map in these coordinates by

$$\mathcal{E}_{p,\tau}(y,t) := \exp_{c(y)}^{\overline{g}_{p,\tau}^{\mathcal{N}}}\big(t\overline{\nu}_{p,\tau}(y)\big), \qquad y = (\xi,\theta) \in D_{3R}.$$

This map is defined for $|t|$ less than a fixed positive constant depending only on R, on a slightly larger parameter strip as well. Its derivatives needed below are uniformly bounded, because the metrics converge smoothly and the initial normal is determined smoothly by its unit-length and orthogonality conditions. Multiplication of a metric by a positive constant does not change its Levi-Civita connection or exponential map. The definitions of G_m and $\mathcal{P}_{p,\tau}$ therefore give the exact formula

$$\mathcal{P}_{p,\tau}^{-1} \circ G_m \circ \mathcal{C}_{\zeta,p}(y) = \mathcal{E}_{p,\tau}\big(y, \overline{\phi}_{p,m}(y)\big).$$

Both sides lie in $w = 0$.

For completeness, write $a := \overline{\phi}_{p,m}$ and $\mathcal{E} := \mathcal{E}_{p,\tau}$. The first two derivatives of this composition are

$$\begin{aligned}\partial_i[\mathcal{E}(y, a(y))] &= \mathcal{E}_i + \mathcal{E}_t \partial_i a,\\ \partial_{ij}[\mathcal{E}(y, a(y))] &= \mathcal{E}_{ij} + \mathcal{E}_{it}\partial_j a + \mathcal{E}_{jt}\partial_i a + \mathcal{E}_{tt}\partial_i a \partial_j a + \mathcal{E}_t \partial_{ij} a,\end{aligned}$$

for $i, j \in \{1, 2\}$, where all derivatives of $\mathcal{E}$ on the right are evaluated at $(y, a(y))$. Since $\mathcal{E}(y, 0) = c(y)$, subtracting the same formulas at $a = 0$ and applying the mean value theorem yields

$$\|\mathcal{E}(\,\cdot, a(\cdot)) - c\|_{C^2(D_{3R})} \le C_R \|a\|_{C^2(D_{3R})}\big(1 + \|a\|_{C^2(D_{3R})}\big)^2 \to 0.$$

In tubular coordinates, the second deformation leaves these three coordinates unchanged and sets the fourth coordinate equal to $\overline{b}_{p,m}$. Thus, writing $\mathcal{E}^1, \mathcal{E}^2, \mathcal{E}^3$ for the first three components of $\mathcal{E}$, we have

$$\mathcal{P}_{p,\tau}^{-1} \circ \widehat{X}_m \circ \mathcal{C}_{\zeta,p}(y) = \big(\mathcal{E}^1(y, a(y)), \mathcal{E}^2(y, a(y)), \mathcal{E}^3(y, a(y)), \overline{b}_{p,m}(y)\big).$$

It follows that these final rescaled immersions converge to c in C^2, uniformly for $p \in L_m$. The larger strip D_{3R} provides neighborhoods of all the points in the smaller strip D_{2R}, including its boundary.

We compute the curvature of the limiting immersion. Its first derivatives satisfy

$$\langle c_\xi, c_\xi\rangle = \langle c_\theta, c_\theta\rangle = \cosh^2\xi, \qquad \langle c_\xi, c_\theta\rangle = 0.$$

A Euclidean unit normal in the hyperplane $w = 0$ is

$$\nu_c(\xi, \theta) := \left(-\frac{\cos\theta}{\cosh\xi}, -\frac{\sin\theta}{\cosh\xi}, \tanh\xi, 0\right).$$

Differentiation gives

$$\begin{aligned}c_{\xi\xi} &= (\cosh\xi\cos\theta, \cosh\xi\sin\theta, 0, 0),\\ c_{\theta\theta} &= (-\cosh\xi\cos\theta, -\cosh\xi\sin\theta, 0, 0),\\ c_{\xi\theta} &= (-\sinh\xi\sin\theta, \sinh\xi\cos\theta, 0, 0).\end{aligned}$$

Pairing these vectors with ν_c gives -1, 1, and 0, respectively. The second normal is the constant vector $(0, 0, 0, 1)$, and all three pairings with it vanish. Hence the Euclidean Gauss equation gives

$$K_c = \frac{(-1)\cdot 1 - 0^2}{\cosh^4\xi} = -\cosh^{-4}\xi.$$

All these formulas hold for every $(\xi, \theta) \in D_{3R}$.

By the Gauss equation, Gaussian curvature depends continuously on the immersion through its first two derivatives and on the ambient metric through its first two derivatives, provided the induced metric remains positive definite. The limiting induced metric above has this property uniformly on D_{3R}. The C^2 convergence of the immersions and of the ambient metrics therefore implies uniform convergence of their Gaussian curvatures on D_{2R}. The rescaled induced metric is τ^{-2} times the pullback of the final induced metric. Its Gaussian curvature is consequently $\tau^2 K_{\Sigma_m}$. We have proved

$$\sup_{p\in L_m}\ \sup_{(\xi,\theta)\in D_{2R}} \left|\tau^2 K_{\Sigma_m}\big(\widehat{X}_m(\mathcal{C}_{\zeta,p}(\xi,\theta))\big) + \cosh^{-4}\xi\right| \to 0.$$

On D_{2R}, one has $\cosh^{-4}\xi \ge (2R)^{-4}$. Choosing m large enough that the displayed supremum is at most $(2R)^{-4}/2$ proves (6.15). □

Proof of Theorem 1.1. Choose $R \ge R_0$ as required by Lemma 6.4, and keep it fixed. Then choose m sufficiently large that all the estimates above hold for this R and $C(m^{-2} + \tau) < c_*/2$ in (6.13). If $x \in M_\zeta$ satisfies $\rho_\zeta(x) \ge 2R\tau$, put $q := \pi(G_m(x))$ and use the parametrizations X, X_b of Lemma 6.1.

Reflection by Z, when needed to use the upper sheet, preserves the induced metric and Gaussian curvature. The two curvature comparisons therefore give

$$\begin{aligned} K_{\Sigma_m}(\widehat{X}_m(x)) &= K_{X_b}(q) \\ &\le K_X(q) + |K_{X_b}(q) - K_X(q)| \\ &\le \big[-c_* + C(m^{-2}+\tau)\big]\mathcal{W}(q) \\ &\le -\frac{c_*}{2}\mathcal{W}(q) < 0. \end{aligned}$$

If instead $\rho_\zeta(x) \le 2R\tau$, Lemma 6.6 gives strict negativity. These two conditions use the same function ρ_ζ on the initial surface and cover all of M_ζ. Thus no region is omitted because of the change from $\pi(x)$ to $\pi(G_m(x))$ during the first deformation. Since $\widehat{X}_m$ is an embedding, we have proved $K_{\Sigma_m} < 0$ everywhere. Proposition 5.3 already gives smoothness, closedness, connectedness, embeddedness, minimality, and genus m^2+1.

We next compute the area of the initial surface. Set

$$\mathcal{U}_m := \mathbb{T} \smallsetminus \bigcup_{p\in L_m} B(p, 3r_c),$$

where the balls are taken in the flat metric $g_{\mathbb{T}}$. Over this set, M_ζ consists of the two graphs $q \mapsto \mathcal{F}(q, h(q))$ and $q \mapsto \mathcal{F}(q, -h(q))$. The initial height estimates give

$$|h(q)| \le C\varepsilon, \qquad |Dh(q)| \le C\tau/r_c \qquad \forall\, q \in \mathcal{U}_m.$$

The metric of either graph has matrix $\cos^2 h\, I + Dh \otimes Dh$ in the flat coordinates. Its determinant is $\cos^4 h + \cos^2 h\, |Dh|^2$, so the area density relative to $d\,\mathrm{vol}_{g_{\mathbb{T}}}$ is

$$J_h = \cos h \sqrt{\cos^2 h + |Dh|^2}.$$

We used $|h| < z_0 < \pi/4$ to take the positive square root of $\cos^2 h$. For $|h| + |Dh|$ small,

$$\begin{aligned} |J_h - 1| &\le |\cos^2 h - 1| + \cos h \left| \sqrt{\cos^2 h + |Dh|^2} - \cos h \right| \\ &= \sin^2 h + \frac{\cos h\, |Dh|^2}{\sqrt{\cos^2 h + |Dh|^2} + \cos h} \\ &\le C(h^2 + |Dh|^2) \le C(\varepsilon^2 + \tau^2 r_c^{-2}). \end{aligned}$$

The last bound tends to zero, because $\varepsilon = m^2\tau/(2\pi) \to 0$ and $\tau/r_c = \tau^{1-\alpha} \to 0$. For sufficiently large m, the m^2 disks removed from $\mathbb{T}$ are disjoint embedded flat disks. Their total area is

$$\mathrm{Area}(\mathbb{T} \smallsetminus \mathcal{U}_m) = 9\pi m^2 r_c^2 = 9\pi m^2 \tau^{2\alpha} \to 0.$$

Integrating the estimate for J_h over $\mathcal{U}_m$ shows that the sum of the areas of the two graph pieces tends to $2\,\mathrm{Area}(\mathbb{T}) = 4\pi^2$.

It remains to bound the initial pieces above the removed disks. For each $p \in L_m$, these consist of a catenoid with horizontal radius at most $2r_c$ and two graphs over the annulus $2r_c \le r_p \le 3r_c$. The metric $dz^2 + \cos^2 z\,(ds^2 + dt^2)$ is bounded above, as a quadratic form, by $dz^2 + ds^2 + dt^2$. Thus the area of the catenoidal piece is no larger than its Euclidean area. Put $a_c := \operatorname{arcosh}(2r_c/\tau)$. The Euclidean metric induced by this catenoid is $\tau^2 \cosh^2\xi(d\xi^2 + d\theta^2)$, and hence

$$\begin{aligned} \mathrm{Area}_{\mathrm{Eucl}} &= \int_{-a_c}^{a_c}\int_0^{2\pi} \tau^2 \cosh^2\xi\, d\theta\, d\xi \\ &= 2\pi\tau^2(a_c + \sinh a_c \cosh a_c) \\ &= 2\pi\left(2r_c\sqrt{4r_c^2 - \tau^2} + \tau^2 \operatorname{arcosh}(2r_c/\tau)\right) \\ &\le Cr_c^2. \end{aligned}$$

Indeed, $2r_c\sqrt{4r_c^2-\tau^2} \le 4r_c^2$, and $\operatorname{arcosh} u \le u^2$ for $u \ge 1$ gives the same bound for the second term in parentheses. On each of the two annuli, $|Dh| \le C\tau/r_c$, so the graph area density is bounded by a fixed constant. Their areas are consequently at most Cr_c^2 as well. Summing over the m^2 lattice points bounds the total omitted area by $Cm^2r_c^2 = Cm^2\tau^{2\alpha} \to 0$. We obtain

$$\operatorname{Area}(M_\zeta) \to 4\pi^2. \tag{6.16}$$

Finally, we prove that the two deformations change area by a factor tending uniformly to one. Write $\widetilde{g}_m := G_m^* g_{\mathcal{N}}$ for the metric on M_ζ after the deformation within $\mathcal{N}$. We will use the following dimensionless bound for the displacement and its first derivatives:

$$\begin{aligned}\vartheta_m &:= \sup_{\tau \le r \le \pi/m} \frac{\tau^{1+\alpha/4} f_\tau(r)}{r} \\ &= \tau^{1+\alpha/4} \sup_{\tau \le r \le \pi/m} \max\{r^{1/2}, \tau^{(1-\alpha)/2} r^{-1/2}\} \\ &\le C\left(\tau^{1+\alpha/4} m^{-1/2} + \tau^{1-\alpha/4}\right) \to 0.\end{aligned}$$

In the last inequality, the maximum of $r^{1/2}$ is attained at $r = \pi/m$, and that of $r^{-1/2}$ at $r = \tau$. The use of the bound $\tau^{1+\alpha/4} f_\tau$ for ϕ_m is justified by the uniform estimates on $\mathcal{B}_m$; the displayed bound is (4.4) applied to the fixed point.

To obtain a metric estimate, fix $x \in M_\zeta$, set $r := \rho_\zeta(x)$, and use a rescaled local parametrization $\kappa : D \to M_\zeta$ as in the proof of Lemma 4.1. Regard the initial and deformed maps as maps into $\mathbb{R}^5$, and put

$$\begin{aligned} I(y) &:= r^{-1}\big(\jmath_\zeta(\kappa(y)) - \jmath_\zeta(x)\big), \\ I_\phi(y) &:= r^{-1}\big(G_m(\kappa(y)) - \jmath_\zeta(x)\big) \qquad \forall\, y \in D.\end{aligned}$$

The chain-rule estimates for the exponential map in that proof give $\|dI_\phi - dI\|_{C^0(D)} \le C\vartheta_m$. The initial derivatives dI are uniformly bounded, and $r^{-2}\kappa^* g_\zeta$ has a uniform positive lower bound. The coefficients of the two rescaled induced metrics are Euclidean inner products of the derivatives of I and I_ϕ. For $i, j \in \{1, 2\}$, their difference is

$$\begin{aligned}\langle \partial_i I_\phi, \partial_j I_\phi\rangle - \langle \partial_i I, \partial_j I\rangle &= \langle \partial_i(I_\phi - I), \partial_j I\rangle + \langle \partial_i I, \partial_j(I_\phi - I)\rangle \\ &\quad + \langle \partial_i(I_\phi - I), \partial_j(I_\phi - I)\rangle.\end{aligned}$$

It is bounded by $C\vartheta_m(1 + \vartheta_m)$. The lower bound for the initial metric converts this coefficient estimate into a relative estimate. Since x was arbitrary, we obtain the quadratic-form bounds

$$(1 - C\vartheta_m(1 + \vartheta_m))g_\zeta \le \widetilde{g}_m \le (1 + C\vartheta_m(1 + \vartheta_m))g_\zeta.$$

We estimate the second change of metric directly. Regard $\iota_{\mathcal{N}} \circ G_m$ as an immersion into $\mathbb{S}^4$ with induced metric $\widetilde{g}_m$, where $\iota_{\mathcal{N}}$ is the inclusion of $\mathcal{N}$ into $\mathbb{S}^4$. Its unit normal in the direction orthogonal to $\mathcal{N}$ is n_m. Let $\mathcal{S}_m := S^{n_m}_{\iota_{\mathcal{N}} \circ G_m}$ be its shape operator in that direction, and let ω_m be the normal-connection one-form for the frame consisting of its chosen unit normal within $\mathcal{N}$ and n_m. The bounds in (4.2), (5.15), and the first metric comparison give

$$\begin{aligned} |\mathcal{S}_m|_{\widetilde{g}_m} + |\omega_m|_{\widetilde{g}_m} &\le C, \\ |b_m| &\le C\tau^2, \\ |db_m|_{\widetilde{g}_m} &\le C\tau^2/\rho_\zeta \le C\tau\end{aligned}$$

everywhere on M_ζ. The exact metric formula (4.19) gives, for every $x \in M_\zeta$ and $v \in T_x M_\zeta$,

$$(\widehat{X}_m^* g_{\mathbb{S}^4})_x(v, v) = \big|\cos(b_m(x))v - \sin(b_m(x))\mathcal{S}_m(v)\big|^2_{\widetilde{g}_m} + \sin^2(b_m(x))\omega_m(v)^2 + db_m(v)^2.$$

Expanding the first square and subtracting $|v|^2_{\widetilde{g}_m}$ yields the terms

$$\begin{aligned}&-\sin^2(b_m(x))|v|^2_{\widetilde{g}_m} - 2\cos(b_m(x))\sin(b_m(x))\widetilde{g}_m(v,\mathcal{S}_m(v))\\&\quad+\sin^2(b_m(x))|\mathcal{S}_m(v)|^2_{\widetilde{g}_m} + \sin^2(b_m(x))\omega_m(v)^2 + db_m(v)^2.\end{aligned}$$

Using the preceding bounds and $|\sin t| \le |t|$, their sum has absolute value at most

$$C\big(|b_m(x)| + |b_m(x)|^2 + |db_m(x)|^2_{\widetilde{g}_m}\big)|v|^2_{\widetilde{g}_m} \le C\tau^2|v|^2_{\widetilde{g}_m}.$$

Consequently,

$$(1 - C\tau^2)\widetilde{g}_m \le \widehat{X}_m^* g_{\mathbb{S}^4} \le (1 + C\tau^2)\widetilde{g}_m.$$

Combining the two comparisons gives numbers $\epsilon_m \to 0$, with $0 \le \epsilon_m < 1$ for large m, such that

$$(1 - \epsilon_m)g_\zeta \le \widehat{X}_m^* g_{\mathbb{S}^4} \le (1 + \epsilon_m)g_\zeta.$$

At each point, the two eigenvalues of the final metric relative to g_ζ lie in $[1 - \epsilon_m, 1 + \epsilon_m]$. The ratio of the area elements is the square root of their product, so it lies in the same interval. Since $\widehat{X}_m$ is an embedding, integration gives

$$(1 - \epsilon_m)\operatorname{Area}(M_\zeta) \le \operatorname{Area}(\Sigma_m) \le (1 + \epsilon_m)\operatorname{Area}(M_\zeta).$$

The area limit follows from (6.16), completing the proof. □

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